



# OPERATOR'S MANUAL

## eECONIC

Mercedes-Benz  
Trucks you can trust



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## **Editors**

You are welcome to forward any queries or suggestions you may have regarding these operating instructions to the documentation team at the following address:

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## **Vehicle manufacturer**

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**Germany**

**Item number:** A9565845519

## Welcome to the world of Mercedes-Benz Trucks

Before you first drive off, read these operating instructions carefully and familiarise yourself with your vehicle. Please adhere to the information and warning notes in these operating instructions for your own safety, and to ensure a longer operating life for your vehicle. Disregarding them may lead to vehicle damage or personal injury.

The standard equipment and product description of your vehicle may vary, and depends on the following factors:

- Model
- Order
- National version
- Availability

The illustrations in these operating instructions show a left-hand drive vehicle. The locations of vehicle parts and controls for right-hand drive vehicles differ accordingly.

Daimler Truck AG is continuously improving its vehicles.

Daimler Truck AG therefore reserves the right to introduce changes in the following areas:

- Design
- Equipment
- Technical features

Therefore the descriptions and illustrations provided may occasionally differ from your own vehicle.

The following documents are integral parts of the vehicle:

- Operating instructions
- Equipment-dependent supplementary instructions

To keep the language of the text simple, often only the masculine form is used. Semantically, people of any gender identity are always meant.

Always keep these documents in the vehicle. If you sell the vehicle, always pass on all documents to the new owner.

## Vehicle data and certifications for body manufacturers

The certifications can be found in the "Confirmations" section.





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In these operating instructions, you will find the following symbols:



**Danger**

## **Danger due to not observing the warning notices**

Warning notes make you aware of risks which may endanger your life or health or the life or health of other people.

- Observe the warning notes.



**ENVIRONMENTAL NOTE**

## **Environmental damage due to failure to observe environmental notes**

Environmental notes include information on environmentally responsible behaviour or environmentally responsible disposal.

- Observe environmental notes.



**NOTE**

## **Damage to property due to failure to observe notes on material damage**

Notes on material damage inform you of risks which may lead to your vehicle being damaged.

- Observe notes on material damage.



**Note**

These symbols indicate useful instructions or further information that could be helpful to you.

|         |                                                            |
|---------|------------------------------------------------------------|
| →       | Further information on a topic                             |
| Display | Messages on the display of the instrument cluster          |
| ►       | Instructions                                               |
| ◁       | Expected system response after an action                   |
| ▽       | Unexpected system response after an action                 |
| ●       | List in any order                                          |
| ↪       | Highest menu level to be selected in the multimedia system |



Corresponding submenus to be selected in the multimedia system

## General notes

### Environmental protection

#### Environmental protection



#### ENVIRONMENTAL NOTE

##### **Environmental damage due to operating conditions and personal driving style**

Operate the vehicle in an environmentally responsible manner to make a contribution to environmental protection. To do this, observe the following recommendations on operating conditions and your personal driving style.

##### **Operating conditions:**

- ▶ Make sure that the tyre pressures are always correct.
- ▶ Do not carry any unnecessary weight.
- ▶ Monitor energy consumption.
- ▶ Observe the service intervals.
  - ◁ A regularly serviced vehicle will contribute to environmental protection.
- ▶ Always have maintenance work carried out at a qualified specialist workshop.

#### Personal driving style:

- ▶ Drive carefully and maintain a sufficient distance to other vehicles.
- ▶ Avoid frequent, sudden acceleration and braking.
- ▶ Drive in a way that conserves energy.



#### ENVIRONMENTAL NOTE

##### **Environmental pollution caused by irresponsible disposal of the high-voltage battery**

A high-voltage battery contains materials which are harmful to the environment.

- ▶ Dispose of faulty high-voltage batteries at a qualified specialist workshop.

#### Mercedes-Benz Genuine Parts



#### ENVIRONMENTAL NOTE

##### **Environmental damage caused by not using recycled, reconditioned components**

Daimler Truck AG offers recycled and reconditioned components and parts having the same quality as new parts. The same entitlement from the implied warranty is valid as for new parts.

- ▶ Use recycled reconditioned components and parts from Daimler Truck AG.

If you use parts, tyres, wheels or safety-critical accessories that have not been approved by Daimler Truck AG, the operating safety of the vehicle may be jeopardised. Safety-relevant systems, e.g. the brake system, may malfunction. Only use Mercedes-Benz Genuine Parts or equivalent quality. Use tyres, wheels and accessories approved for your vehicle model only.

Daimler Truck AG tests Genuine Parts and conversion parts and accessories that

## Information about attachments, add-on equipment, installations, and conversions

have been specifically approved for your vehicle model for their reliability, safety and suitability. Despite ongoing market research, Daimler Truck is unable to assess other parts. Daimler Truck AG therefore accepts no responsibility for the use of such parts in Mercedes-Benz trucks, even if they have been independently or officially approved by a technical testing centre.

Certain parts are only officially approved for installation or modification if they comply with legal requirements. All Mercedes-Benz Genuine Parts meet the approval requirements. The use of non-approved parts may invalidate the vehicle's general operating permit.

This is the case in the following situations:

- The vehicle type is different from that for which the vehicle's general operating permit was granted.
- Other road users may be endangered.
- The noise level increases.

Always specify the Vehicle Identification Number (VIN) when ordering Mercedes-Benz Genuine Parts (→ Page 268).

### Information about attachments, add-on equipment, installations, and conversions

#### Information on body/equipment mounting directives

For safety reasons, have bodies produced and fitted in accordance with applicable Mercedes-Benz Trucks body/equipment mounting directives. These body/equipment mounting directives ensure that the chassis and add-on equipment form one unit and that the greatest possible level of operational and driving safety is achieved.

For safety reasons, Daimler Truck AG recommends the following:

- Do not make any other changes to the vehicle.
- Obtain approval from Daimler Truck AG in the event of deviations from the

approved body/equipment mounting directives.

Acceptance tests performed by public test bodies or official approvals do not rule out safety risks.

Refer to the information about Mercedes-Benz Genuine parts (→ Page 2).

You can find the Mercedes-Benz Trucks body/equipment mounting directives online at <https://bb-portal.mercedes-benz-trucks.com>

You can also find information on PIN assignment and changing fuses there.



#### Note

Further information can be obtained at a qualified workshop.



#### Danger

#### Risk of accident and injury due to improper vehicle conversions or modifications

Conversions or modifications to the vehicle may cause systems or compon-

## General notes

### Operating Instructions

ents to no longer function as intended and/or jeopardise the operating safety of the vehicle.

- ▶ Always have conversions or modifications to the vehicle carried out at a qualified workshop.

Confirmation of the work carried out can be found in the "Confirmations" chapter (→ Page 338).

#### Notes on cameras and sensors

If cameras, sensors and their detection ranges are covered or their covers are damaged, sensors, cameras or driving and safety systems may malfunction or fail.

Observe the following notes on cameras and sensors, especially distance sensors and radar sensors:

- Do not install attachments or accessories in front of the camera or the sensor, e.g. bullbars or advertising.
- Do not paint over or cover radar sensors, for example with stickers or decals.

- The load or the load safety equipment must not protrude into the detection range of the cameras and sensors.
- Immediately have any damage to sensor covers or damage around the camera repaired at a qualified specialist workshop.

#### Notes on vehicle radiator

Even seemingly minor changes to the vehicle such as attaching a radiator grille in winter operation are not permitted. Do not cover up the vehicle radiator. Do not use thermal mats, insect protection covers, or anything similar.

Otherwise, the values produced by the diagnostic system will be distorted. In some countries, the recording of vehicle diagnostics data is prescribed by law and must be verifiable and correct at all times.

#### Operating Instructions

These operating instructions describe all possible models and standard and special equipment of your vehicle. Note that your vehicle may not be fitted with

all features described. This is also the case for systems and functions relevant to safety. Therefore, the equipment on your vehicle may differ from that in some descriptions and illustrations.

The original purchase agreement for your vehicle contains a list of all of the systems in your vehicle.

Contact an authorised Mercedes-Benz Trucks Service Centre if you have any questions about equipment or operation.

The operating instructions are an important document and should be kept in the vehicle.

#### Operating safety



**Danger**

#### **Risk of accident due to malfunctions or system failures**

If you do not have the prescribed service/maintenance work or any required repairs carried out, this may result in malfunctions or system failures.

- ▶ Always have the prescribed service and maintenance work, as well any required repairs, carried out at a qualified specialist workshop.

**Danger**

**Risk of accident or injury due to incorrect modifications to electronic component parts**

Carrying out modifications to electronic components, their software, or wiring, may impair their function and/or the function of other interconnected components or safety-relevant systems.

This may jeopardise the operating safety of the vehicle.

- ▶ Do not tamper with the wiring and electronic component parts or their software.
- ▶ Always have work on electrical and electronic devices carried out at a qualified specialist workshop.

Electromagnetic radiation from two-way radios can interfere with the onboard electronics if the two-way radios are ma-

nipulated or retrofitted incorrectly. This may jeopardise the operating safety of the vehicle. There is a risk of accident! Always have work on electrical and electronic devices carried out at a qualified specialist workshop.

If you make any changes to the onboard electronics, the general operating permit is rendered invalid.

**NOTE**

**Damage to the vehicle due to driving too fast and impacts to the vehicle underbody and suspension components**

In the following situations, in particular, there is a risk of damage to the vehicle:

- The vehicle bottoms out, e.g. on a high kerb or unpaved surfaces.
- The vehicle is driven too fast over an obstacle, e.g. a kerb, speed bumps or a pothole.
- A heavy object strikes the underbody or suspension components.

In situations such as these, damage to the body, underbody, suspension components, wheels, tyres or parts of the high-voltage battery may not be visible. Components damaged in this way can unexpectedly fail or, in the case of an accident, may not absorb the resulting force as intended.

- ▶ Have the vehicle checked and repaired immediately at a qualified specialist workshop.

or

- ▶ If driving safety is impaired while continuing your journey, pull over and stop the vehicle immediately, while paying attention to road and traffic conditions, and contact a qualified specialist workshop.

Electric vehicles have an electric motor. The electric motor's voltage supply is provided by the high-voltage on-board electrical system.

## General notes

### Operating safety



#### Warning

#### **Risk of death and fire due to modified and/or damaged components of the high-voltage on-board electrical system**

The vehicle's high-voltage on-board electrical system is under high voltage, regardless of the operating mode. If you modify component parts in the high-voltage on-board electrical system or touch damaged component parts, you may be electrocuted. In addition, altered and/or damaged components can cause a fire.

The component parts in the vehicle's high-voltage on-board electrical system may be damaged in an accident or if the vehicle underside bottoms out, although the damage is not visible.

- ▶ Never make any modifications to the high-voltage on-board electrical system.
- ▶ Do not switch on or use the vehicle if its high-voltage on-board electrical

system components have been modified or damaged.

- ▶ Never touch damaged components of the high-voltage on-board electrical system.
- ▶ After an accident, do not touch any components of the high-voltage on-board electrical system.
- ▶ After an accident, have the vehicle transported away.
- ▶ Have the components of the high-voltage on-board electrical system checked at a qualified specialist workshop and replaced if necessary.
- ▶ Have the safety of the vehicle ensured by an appropriately qualified electrician.

The components of the vehicle's high-voltage on-board electrical system are marked with yellow warning stickers. The cables of the vehicle's high-voltage on-board electrical system are orange.



Vehicles with an electric motor generate significantly less noise than vehicles with combustion engines. As a result, your vehicle may not be heard by other road users in certain situations. This can occur, for example, when you are manoeuvring and your vehicle cannot be seen by other road users. This requires you to adopt a particularly anticipatory driving style, because you must allow for the possibility that other road users may behave unpredictably.

Furthermore, the vehicle is equipped with a sound generator, which serves as an acoustic vehicle alerting system (AVAS) (→ Page 117).



### Declarations of conformity and national information

#### Declarations of conformity and national information

##### Electromagnetic compatibility

The electromagnetic compatibility of the vehicle components has been checked and certified according to the currently valid version of Directive UN-R 10.

#### Jack

##### EU and EFTA countries only:

Copy and translation of the original declaration of conformity:

EC declaration of conformity  
2006/42/EC

We, WEBER-HYDRAULIK GMBH, Heilbronner Str. 30, 74363 Güglingen, declare that the product "Weber hydraulic bottle jack"

Models:

- A 5-212
- AD 5-217
- AD 10-220
- AD 12-270
- ADX 10-370

- AH 25-240
- ATN 10-175
- ATPX 12-190
- ATPX 12-230
- ATPX 16-240
- ATQ 12-230

Capacity: 2,000 to 100,000 kg

Serial no.: as of year of manufacture  
01/2010

complies with the relevant essential health and safety requirements of the EC Machinery Directive.

This EC declaration of conformity becomes invalid:

- In the event of modifications or repairs performed by an unqualified person,
- if the products are not used in accordance with the application specified in the operating instructions.
- if the required regular checks are not performed.

Relevant EC directives: EC Machinery  
Directive 2006/42/EC

Applied standards: ISO 11530:1993

EN ISO 12100:2010

Quality assurance: DIN EN ISO 9001:2000

Güglingen, 15th December 2021

Signed by:

Manager, WEBER-HYDRAULIK GmbH

Executive officer for technical documentation, WEBER-HYDRAULIK GMBH

Heilbronner Straße 30, 74363 Güglingen

##### For the United Kingdom only:

Copy and translation of the original declaration of conformity:

UKCA declaration of conformity

We, WEBER-HYDRAULIK GMBH, Heilbronner Str. 30, 74363 Güglingen, declare that the product "Weber hydraulic bottle jack"

Models:

- A
- AD
- ADX
- AH
- AHX

## General notes

### Diagnostics connection

- AL
- AT
- ATD
- ATDX
- ATG
- ATN
- ATGX
- ATPX
- ATQ
- AX

Capacity: 2,000 to 100,000 kg

Serial no.: From year of manufacture  
01/2022

Complies with the relevant essential  
health and safety requirements of the  
Supply of Machinery (Safety) Regulation  
2008.

This UKCA declaration of conformity be-  
comes invalid:

- In the event of modifications or re-  
pairs performed by an unqualified  
person,
- if the products are not used according  
to the operating instructions,

- if the required regular checks are not  
performed.

Relevant UK guidelines: Supply of Ma-  
chinery/Regulations 2008

Applied standards: ISO 11530:1993

BS EN ISO 12100:2010-12-31

Safety of Machinery

General principles for design

Risk assessment and risk reduction

Güglingen, 1st February 2022

Signed by:

Manager (CEO), WEBER-HYDRAULIK  
GmbH

Executive officer for technical documenta-  
tion

WEBER-HYDRAULIK GMBH

Heilbronner Straße 30, 74363 Güglingen

### Diagnostics connection

The diagnostics connection is a technical  
interface in the vehicle. It is used, for ex-  
ample, in the context of repair and main-

tenance work or for reading out vehicle  
data in a specialist workshop. Diagnostic  
devices should therefore be connected  
only at a qualified specialist workshop.



**Danger**

#### **Risk of accident due to connecting devices to the diagnostics connection**

If you connect devices to the vehicle's  
diagnostics connection, the operation  
of vehicle systems and operating safety  
may be impaired.

- For safety reasons, we recommend  
that you only use and connect  
products approved by an authorised  
Mercedes-Benz Trucks Service  
Centre.



**Danger**

#### **Risk of accident due to objects in the driver's footwell**

Objects in the driver's footwell may im-  
pede pedal travel or block a depressed  
pedal.

### Qualified specialist workshop

This jeopardises the operating and road safety of the vehicle.

- ▶ Stow all objects in the vehicle securely so that they cannot get into the driver's footwell.
- ▶ Always fit the floor mats securely and as prescribed in order to ensure that there is always sufficient room for the pedals.
- ▶ Do not use loose floor mats and do not place floor mats on top of one another.

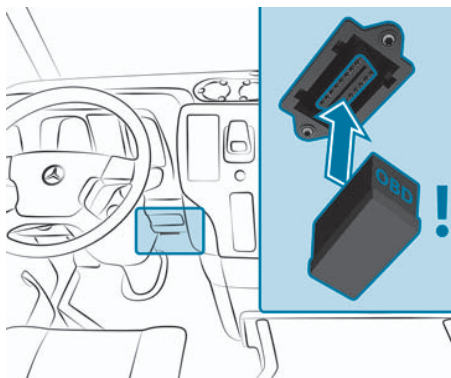


#### NOTE

#### Battery discharging from using devices connected to the diagnostics connection

Using devices at the diagnostics connection drains the battery.

- ▶ Check the charge level of the battery.
- ▶ Charge the battery if the charge level is low.



The connection and use of another device on the diagnostics connection can have the following effects:

- Malfunctions in the vehicle systems
- Permanent damage to the vehicle components

### Qualified specialist workshop

A qualified specialist workshop has the necessary special skills, tools and qualifications to correctly carry out any necessary work on your vehicle. This particularly applies to safety-relevant works.

Always have the following work on the vehicle carried out at a qualified specialist workshop:

- Safety-relevant work
- Service and maintenance work
- Repair work
- Modifications, installations and conversions
- Work on electronic component parts

Daimler Truck AG recommends that you use an authorised Mercedes-Benz Trucks Service Centre for this purpose.

### Vehicle registration

Daimler Truck AG may ask its authorised service centres to carry out technical inspections on certain vehicles. The quality or safety of the vehicle is improved as a result of the inspection.

Daimler Truck AG can only inform you about vehicle checks if it has your registration data.

## General notes

### Correct use of the vehicle

It is possible that your vehicle has not yet been registered in your name in the following cases:

- If you have not purchased your vehicle from an authorised specialist dealer.
- If your vehicle has not yet been examined at an authorised Mercedes-Benz Trucks Service Centre.

It is advisable to register your vehicle with an authorised Mercedes-Benz Trucks Service Centre. Inform Daimler Truck AG as soon as possible about any change in address or vehicle ownership. You can do this, for example, at a Mercedes-Benz Trucks Service Centre.

### Correct use of the vehicle

If you remove any warning stickers, you or others could fail to recognise certain dangers. Leave warning stickers in position.

When using the vehicle, take particular account of the following information:

- the safety notes in this manual

- Technical data of the vehicle
- Traffic rules and regulations –
- laws and safety standards pertaining to motor vehicles

### Information on the REACH regulation

#### For EU and EFTA countries only:

The REACH Regulation (Regulation (EC) No. 1907/2006, Article 33) provides for an information obligation for substances of very high concern (SVHC).

Daimler Truck AG acts to the best of its knowledge to avoid the use of these SVHCs and to enable the customer to handle these substances safely. According to supplier information and Daimler Truck AG's internal product information, SVHCs are known that account for more than 0.1 percent by weight of individual products in this vehicle.

Further information can be obtained from the following addresses:

- <https://reach.daimlertruck.com/de>

- <https://reach.daimlertruck.com/en>

### Notes for persons with electronic medical aids

Daimler Truck AG cannot, despite careful development, completely rule out the interaction of vehicle systems with electronic medical aids such as cardiac pacemakers.

In addition, there are components built into the vehicle that, regardless of the operating status of your vehicle, can generate magnetic fields on a par with permanent magnets. These fields can be found, for example, in the area around the multimedia and sound system or also in the seating area, depending on the vehicle equipment.

For this reason, the following can occur in isolated cases, depending on the aids used:

- medical aids malfunctioning
- adverse health effects

Observe the notes and warnings of the manufacturer of the medical aids; if in

## Liability for material defects

doubt, contact the device manufacturer and/or your doctor. Daimler Truck AG recommends using only few electrical vehicle systems and/or maintaining a distance from the components if there is continuing uncertainty concerning the possibility of medical aids malfunctioning.

When charging the high-voltage battery, keep a distance of at least an arm's length between the medical aid and the following components:

- the power supply equipment  
This includes charging stations in the form of a wallbox or a public charging point, for example.
- vehicle components carrying live voltage  
This includes the charging cable and the charging control box, for example.

Only have repairs and maintenance work in the area of the following components carried out by a qualified specialist workshop:

- vehicle components carrying live voltage

- transmission aerals
- multimedia system and sound system

If you have any queries or suggestions, consult a qualified specialist workshop.

## Liability for material defects

Warranty claims on new vehicles, the scope of which is based on the purchase agreement and the applicable general terms and conditions, can be filed with any authorised Mercedes-Benz service partner.



## NOTE

**Damage to the vehicle arising from violation of these operating instructions**

Damage to the vehicle can arise from violation of these operating instructions. Such damage is not covered by the implied warranty.

- ▶ Follow the instructions in these operating instructions on proper

operation of your vehicle as well as on possible vehicle damage.

According to the purchase agreement, extended limitation periods apply to the factory-installed lorry component parts of electric drive axle(s) with drive inverter, within the time and mileage limit specified therein. For the installed high-voltage battery(ies), the extended limitation periods apply within the specified time and mileage limit, up to the specified number of charging cycles at the longest and insofar as the defined State of Health (SoH) percentage is not reached. The State of Health (SoH) percentage is the measure of the age of the battery cells.

Please follow the instructions in these operating instructions, even if you hand the vehicle over to a third party for use or care. In this way, you ensure that your warranty claims are preserved.

Have the prescribed service work and maintenance work for your vehicle, in particular during the implied warranty period, carried out as follows:

## General notes

### Data storage

- Regularly
- In good time
- At a qualified specialist workshop that has the necessary special skills and tools to carry out the work required

Daimler Truck AG recommends that you use a Mercedes-Benz Trucks Service Centre for this purpose.

If the specified service and maintenance work is not carried out, the decision as to whether a warranty claim can be accepted or not can be made only after the manufacturer has submitted a report on the damaged part.

Proof of service and maintenance work performed at a qualified specialist workshop is stored for you in our central Mercedes-Benz Trucks database. You are provided with a printout of the service report.

Maintenance jobs do not include repair work. A separate order must be placed for such work.

The claim shall also lapse in particular if the fault has arisen due to one of the following causes:

- The purchaser was aware of a malfunction or damage and failed to either indicate such immediately or to arrange for said malfunction/damage to be repaired.
- The vehicle has been handled improperly, damaged or subjected to excessive demands (e.g. vehicle tuning, exceeding the permissible gross vehicle weight, misuse of the high-voltage battery).
- The high-voltage battery has been handled improperly. In particular, low states of charge (State of Charge (SoC) of 0 %) over a longer period of time (more than seven days) must be avoided at all costs.
- Parts not approved by the manufacturer were installed or the vehicle was converted in a manner not approved by the manufacturer.
- The purchaser failed to observe the manufacturer's specifications for handling, maintenance and care of

the vehicle (e.g. the operating instructions). This includes the use of unsuitable lubricants and operating fluids and disregard for charging and storage requirements in these operating instructions.

- The high-voltage on-board electrical system was incorrectly repaired and/or repaired at a workshop not authorised by the manufacturer.

### Data storage

### Data processing in the vehicle

### Electronic control units

Electronic control units are fitted in your vehicle. Control units process data that they receive from vehicle sensors, for example, generate themselves or exchange with each other. Some control units are required for the safe operation of your vehicle, some assist you when driving, such as driver assistance systems, while others enable convenience or infotainment functions.

The following provides you with general information regarding data processing in the vehicle. Additional information regarding exactly which data in your vehicle is collected, saved and transmitted to third parties, and for what purpose, can be found in the information directly related to the functional characteristics in question in their respective operating instructions. This information is also available online and, depending on the vehicle equipment, digitally.

You can find data protection information regarding how Daimler Truck AG collects and uses personal data from the vehicle under the "Data protection information" heading (→ Page 16) in these operating instructions.

### Personal data

Every vehicle is identified by a unique vehicle identification number. Depending on the country, this vehicle identification number can be used by, for example, governmental authorities to determine the identity of the owner. There are other possibilities for using data collected from

the vehicle to identify the owner or driver, such as the licence plate number.

Therefore, data generated or processed by control units may be attributable to a person or, under certain conditions, become attributable to a person. Depending on which vehicle data is available, it may be possible to make inferences about, for example, your driving behaviour, your location, your route or your use patterns.

### Legal requirements regarding the disclosure of data

If legally required to do so, manufacturers are, in individual cases, legally obliged to provide governmental entities, upon request and to the extent required, data stored by the manufacturer. For example, this may be the case during the investigation of a criminal offence.

Governmental entities are themselves, in individual cases and within the applicable legal framework, authorised to read out data from the vehicle. In the case of an accident, information that could help with an investigation can, therefore, be

read out from the airbag control unit, for example.

### Operational data in the vehicle

This is data regarding the operation of the vehicle, which has been processed by control units.

This includes the following data, for example:

- Vehicle status information such as the speed, longitudinal acceleration, lateral acceleration, number of wheel revolutions or the fastened seat belts display
- Ambient conditions, such as temperature, rain sensor or distance sensor

Generally, the use of these data is temporary; it will not be stored beyond the period of operation and will only be processed within the vehicle itself. Control units, vehicle keys for example, often contain data memories. Their use permits the temporary or permanent documentation of technical information about the vehicle's operating state, component

## General notes

### Data storage

loads, maintenance requirements and technical events or faults.

Depending on the technical equipment, the following data is stored:

- Operating status of system components, such as fill levels, tyre pressure or battery status
- Malfunctions or faults in important system components, such as lights or brakes
- System reactions in special driving situations, such as airbag deployment or the intervention of stability control systems
- Information on events leading to vehicle damage
- Charge level of the high-voltage battery, estimated range

In certain cases, it may be required to store data that would have otherwise been used only temporarily. This may be the case if the vehicle has detected a malfunction, for example.

If you use services, such as repair services and maintenance work, stored oper-

ational data as well as the vehicle identification number can be read out and used, if necessary. They can be read out by service network employees, such as workshops and manufacturers or third parties, such as breakdown services. The same is true in the case of warranty claims and quality assurance measures.

In general, the read-out is performed via the legally prescribed port for the diagnostics connection in the vehicle. The operational data that is read out documents technical states of the vehicle or of individual components and assist in the diagnosis of malfunctions, compliance with warranty obligations and quality improvement. To that end, this data, in particular information about component loads, technical events, malfunctions and other faults may be transmitted along with the vehicle identification number to the manufacturer. Furthermore, the manufacturer is subject to product liability. For this reason, the manufacturer also uses operational data from the vehicle, for example, for recalls. This data can also be

used to examine the customer's warranty claims.

Fault memories in the vehicle can be reset by a service outlet or, at your request, as part of repair or maintenance work.

### Convenience and infotainment functions

You can store convenience settings and individual settings in the vehicle and change or reset them at any time.

Depending on the vehicle equipment, this includes the following settings, for example:

- Seat and steering wheel positions
- Suspension and climate control settings
- Individual settings, such as interior lighting

Depending on the selected equipment, you can import data into vehicle infotainment functions yourself.

Depending on the vehicle equipment, this includes the following data, for example:



- Multimedia data, such as music, films or photos for playback in an integrated multimedia system
- Address book data for use in an integrated hands-free system or an integrated navigation system
- Entered navigation destinations
- Data on the use of Internet services

This data for convenience and infotainment functions may be saved locally in the vehicle or they may be located on a device that you have connected to the vehicle, such as a smartphone, USB flash drive or an MP3 player. If you have entered this data yourself, you can delete it at any time.

This data is transmitted from the vehicle to third parties only at your request. This applies, in particular, when you use online services in accordance with the settings that you have selected.

#### **Smartphone integration (e.g. Android Auto or Apple CarPlay®)**

If your vehicle is accordingly equipped, you can connect your smartphone or an-

other mobile end device to the vehicle. You can then control them by means of the controls integrated in the vehicle. Images and audio from the smartphone can be output via the multimedia system. Certain information is simultaneously transferred to your smartphone. Depending on the type of integration, this includes position data, day/night mode and other general vehicle statuses. For more information, please consult the vehicle/infotainment system operating instructions.

This integration allows the use of selected smartphone apps, such as navigation or music player apps. There is no further interaction between the smartphone and the vehicle; in particular, vehicle data is not directly accessible. The type of additional data processing is determined by the provider of the app being used. Which settings you can make, if any, depends on the specific app and the operating system of your smartphone.

#### **Information on service providers**

##### **Wireless network connection**

If your vehicle has a wireless network connection, data can be exchanged between your vehicle and other systems. The wireless network connection is made possible by the vehicle's own transceiver or by a mobile end device that you have brought into the vehicle, for example, a smartphone. Online functions can be used via the wireless network connection. This includes online services and applications/apps provided to you by the manufacturer or by other providers.

##### **Manufacturer's services**

Regarding the manufacturer's online services, the individual functions are described by the manufacturer in a suitable place, for example, in the operating instructions or on the manufacturer's website.

You can have services and functions, some of which are subject to a fee, activated or deactivated. This excludes legally

## General notes

### Data storage

prescribed functions and services, such as an emergency call system.

At a workshop, for example, with the support of a specialist and possibly for a fee, you can have data read out which is only stored locally in the vehicle.

### Third-party services

If you use online services from other providers (third parties), these services are the responsibility of the provider in question and subject to that provider's data protection conditions and terms of use. As a general rule, the manufacturer has no influence on the content exchanged.

For this reason, when services are provided by third parties, please ask the service provider in question for information about the type, extent and purpose of the collection and use of personal data.

### Data protection information

Daimler Truck AG offers various digital services for your vehicles and is constantly working to further develop your products, services and process, to improve their quality and to adapt to customer requirements.

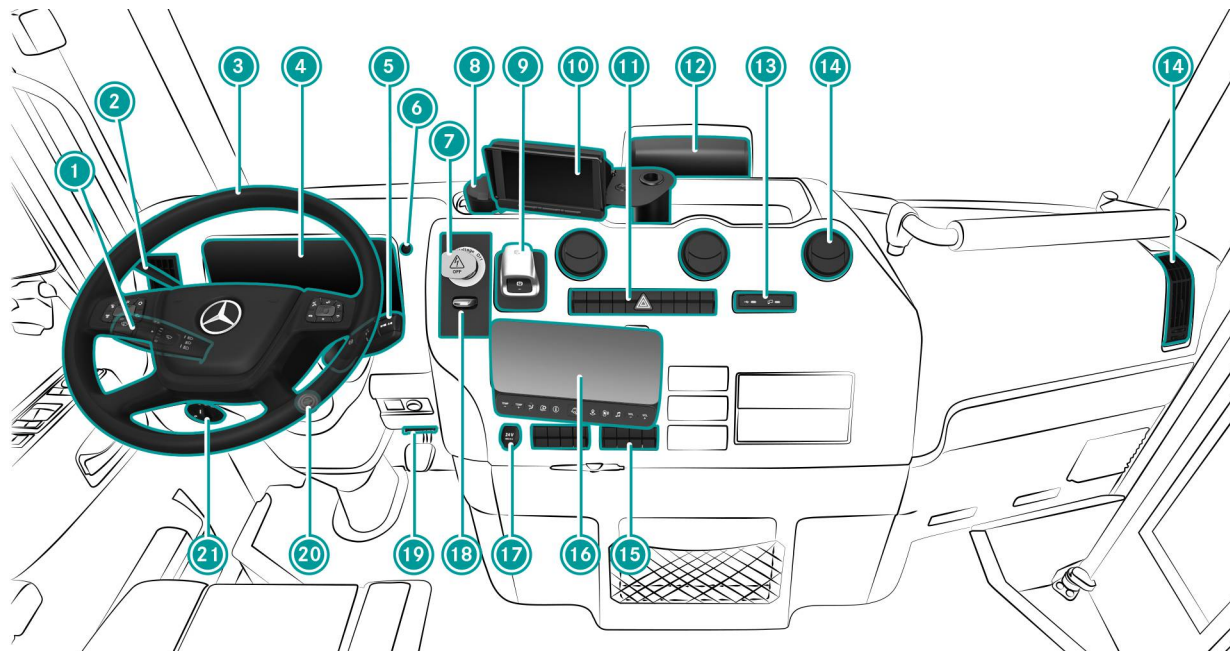
Daimler Truck AG processes vehicle data for these purposes. The vehicle data is technical information. However, it can sometimes identify the owner or driver of a vehicle and is therefore subject to data protection legislation.

You can find further information about how Daimler Truck AG collects and uses personal data for the aforementioned purposes in accordance with Art. 13, 14 of the EU General Data Protection Regulation at:

<https://www.mercedes-benz-trucks.com/gdpr-vehicle-user>



Cockpit



Overview (example using a left-hand drive vehicle)

## At a glance

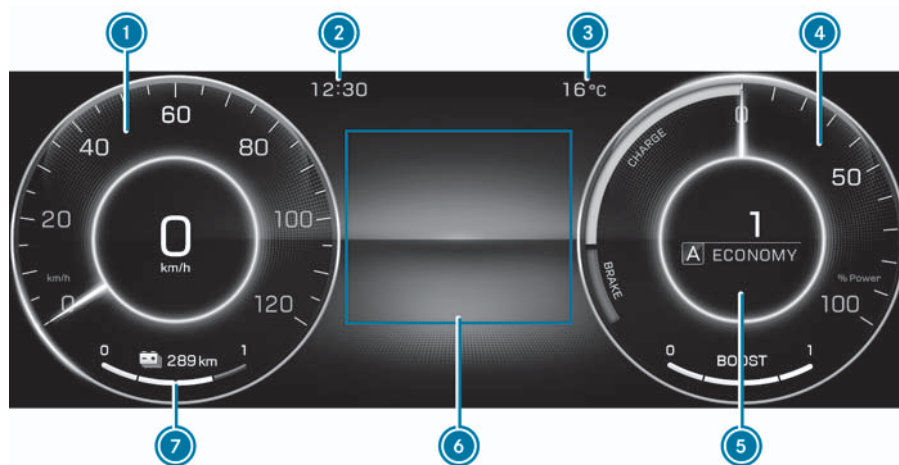
### Cockpit

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### Instrument cluster

### Displays on the instrument cluster

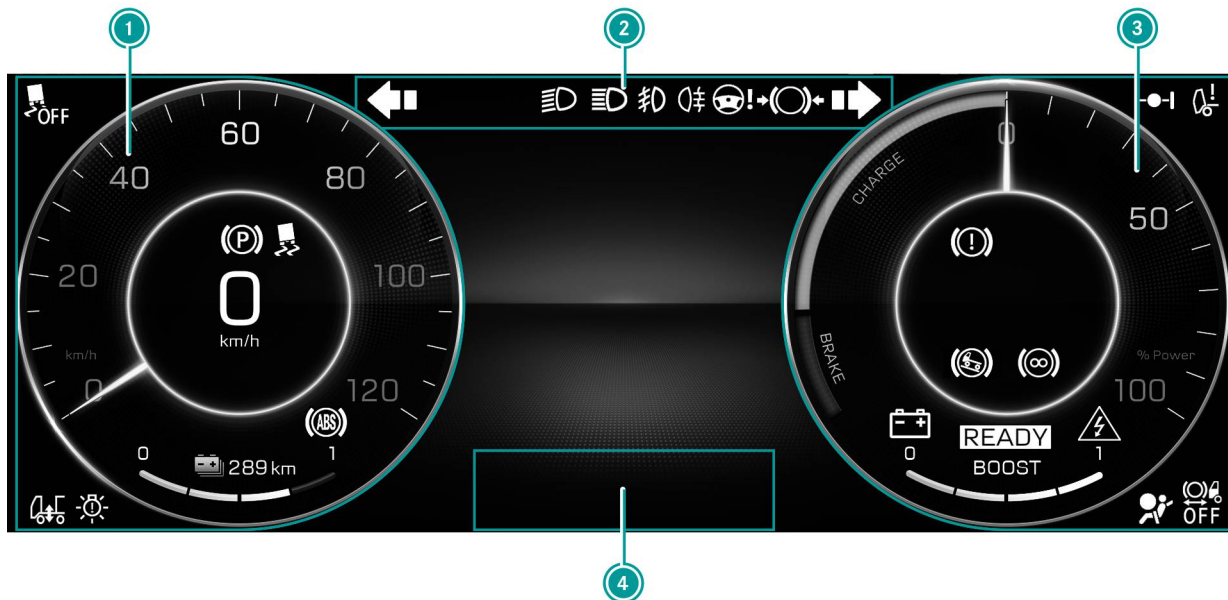


| Item | Description                             | Page |
|------|-----------------------------------------|------|
| ①    | Speedometer                             |      |
| ②    | Time                                    | 79   |
| ③    | Outside temperature                     | 79   |
| ④    | Power display                           | 77   |
| ⑤    | Selected gear, selected driving program | 140  |
| ⑥    | Menu display section                    | 80   |
| ⑦    | Charge level display                    | 78   |

## At a glance

### Instrument cluster

Warning/indicator lamps on the instrument cluster



Instrument cluster (example)

| Item                                                                                | Description                                             | Page       |
|-------------------------------------------------------------------------------------|---------------------------------------------------------|------------|
| ①                                                                                   | Left-hand warning/indicator lamps                       |            |
|    | Control intervention by Stability Control Assist or ASR | 147<br>146 |
|    | Stability Control Assist or ASR deactivated             | 147<br>146 |
|    | ASR (acceleration skid control system)                  | 146        |
|    | Parking brake                                           | 132        |
|    | ABS (anti-lock braking system)                          | 130        |
|    | (yellow) High-voltage battery reserve                   | 82         |
|    | (grey) Charging cable connected, not charging           | 81         |
|    | (blue) Charging cable connected, charging               | 81         |
|    | Level control system outside driving level              | 190        |
|    | Lamp check                                              | 52         |
| ②                                                                                   | Top warning/indicator lamps                             |            |
|   | Turn signal indicators, left/right                      | 56         |
|  | Parking lights                                          | 52         |

| Item                                                                                | Description                                                                                  | Page |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------|
|    | Dipped beam                                                                                  |      |
|    | High beam                                                                                    | 56   |
|    | Rear fog light                                                                               |      |
|    | (grey) Steered additional axle malfunction; see display message                              | 321  |
|    | (yellow) Steered additional axle malfunction; see display message                            | 322  |
|    | (red) Steered additional axle malfunction or power steering malfunction; see display message | 325  |
|    | Instruction to brake at high engine speeds                                                   |      |
| ③                                                                                   | Right-hand warning/indicator lamps                                                           |      |
|    | Differential locks                                                                           | 148  |
|    | Cab tilt lock                                                                                | 237  |
|    | Brake malfunction; see display message                                                       |      |
|    | Hill holder                                                                                  | 135  |
|   | HOLD function                                                                                | 136  |
|  | Continuous brake                                                                             | 137  |

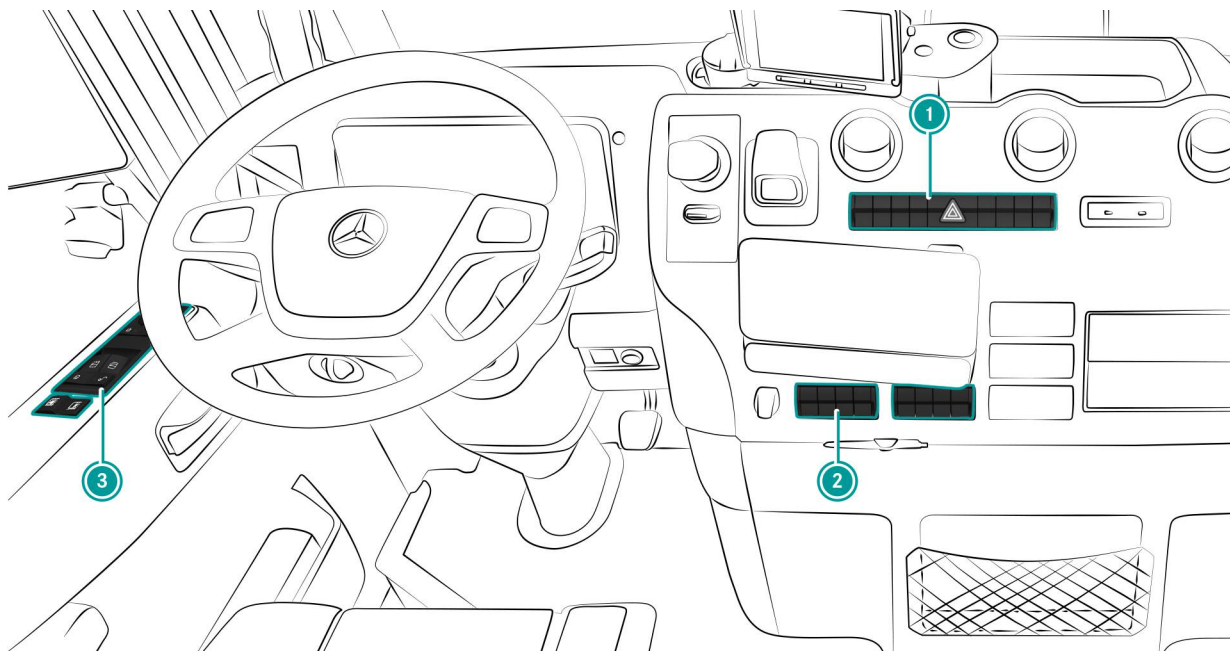
| Item                                                                                | Description                                          | Page |
|-------------------------------------------------------------------------------------|------------------------------------------------------|------|
|  | 24 V vehicle on-board electrical system fault        | 81   |
|  | Drive system ready for operation                     | 81   |
|  | High-voltage system malfunction                      | 82   |
|  | Active Brake Assist                                  | 165  |
|  | Automatic high voltage emergency shutoff malfunction | 28   |
| ④                                                                                   | Warning/indicator lamps in the status area           | 83   |

## At a glance

### Switch panels

#### Switch panels

#### Switch panels in the cockpit





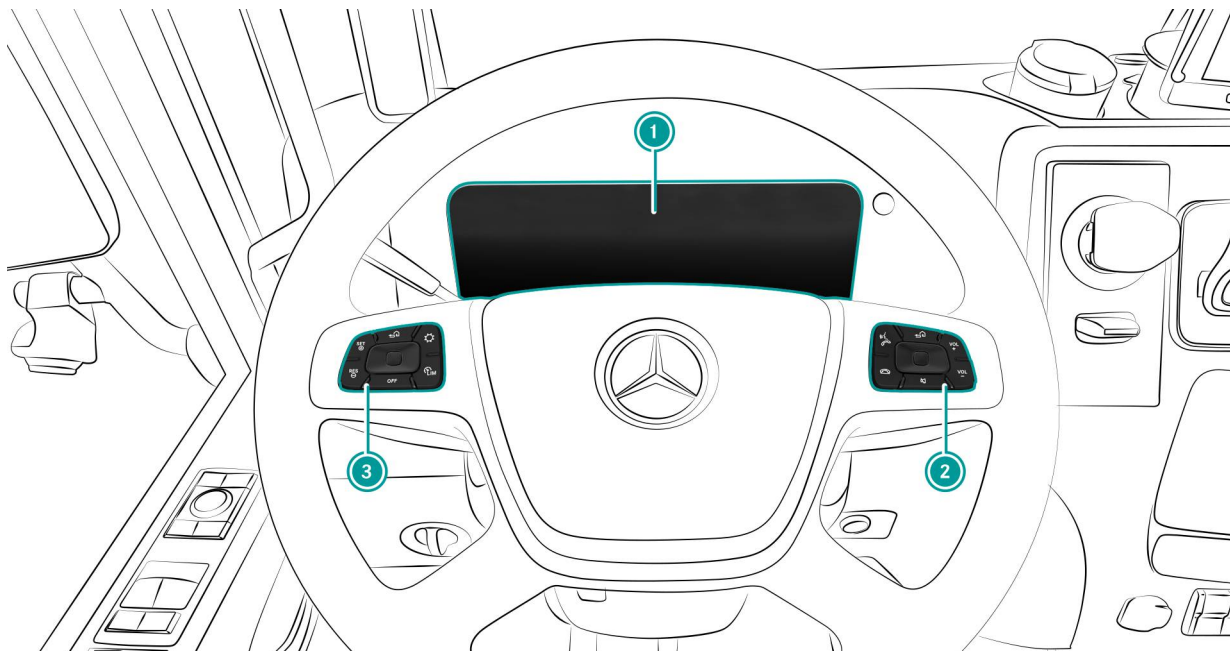
| Item | Description                                                                                                                     | Page |
|------|---------------------------------------------------------------------------------------------------------------------------------|------|
| ①    | Switch panel                                                                                                                    |      |
|      |  Activates and deactivates reversing assistant | 188  |
|      |  Rotating beacon                               | 57   |
|      |  Chassis-lowering feature on the front axle    | 196  |
|      | Automatic activation of the hazard warning light system in the event of power take-off                                          | 212  |
|      |  Hazard warning light system                   |      |
|      |  Work lamp (manoeuvring assistance)            | 57   |
|      | Work/surround lighting (is automatically switched off when the parking brake is released)                                       |      |
|      |  Differential lock                             | 148  |
|      |  Stability Control Assist                      | 147  |
| ②    | Switch panel                                                                                                                    |      |
|      |  Hill holder                                   | 135  |
|      |  Frequent-stop brake                          | 134  |
|      |  Opens/closes folding door                   | 40   |

| Item | Description                                                                                                                                          | Page |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------|------|
|      |  Activates/deactivates exterior button for folding door             | 40   |
|      |  Level control system: STOP                                         |      |
|      |  Work lamp (switched off automatically when the vehicle pulls away) |      |
| ③    | Driver's door switch panel                                                                                                                           |      |
|      | Adjusts the outside mirrors                                                                                                                          | 50   |
|      |  Switching the mirror heater on/off                                 | 51   |
|      |  Opens/closes the side windows                                      | 44   |
|      |  Lock vehicle                                                       | 38   |
|      |  Unlock vehicle                                                     | 38   |
|      |  Retracting the roller sunblind                                     | 45   |
|      |  Extending the roller sunblind                                      | 45   |

## At a glance

### Multifunction steering wheel

#### Multifunction steering wheel



Example: buttons on the multifunction steering wheel

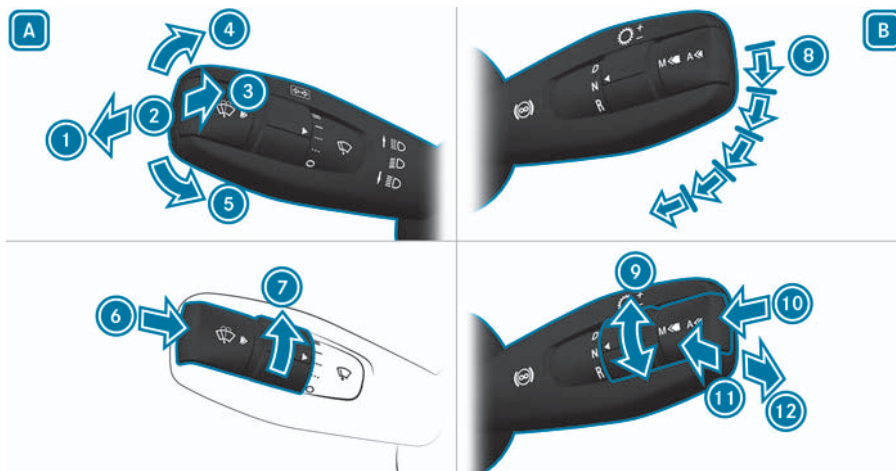
| Item | Description                                                                                                                                            | Page |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| ①    | Instrument cluster                                                                                                                                     |      |
| ②    | Right-hand switch panel on steering wheel                                                                                                              |      |
|      | Navigate/confirm/select in the multimedia system                                                                                                       | 87   |
|      |  Main menu and Back button (multimedia system)                        |      |
|      |  Increases the volume                                                 |      |
|      |  Decreases the volume                                                 |      |
|      |  Switches the mute function on/off                                    |      |
|      |  Ends/rejects a call                                                  |      |
|      |  Making/accepting a call                                              |      |
|      |  Activates the voice control system                                   |      |
| ③    | Left-hand switch panel on steering wheel                                                                                                               |      |
|      | Navigate/confirm/select in the on-board computer                                                                                                       | 79   |
|      |  Main menu and Back button (on-board computer)                        |      |
|      |  Settings                                                            |      |
|      |  Short button press: Select cruise control/Proximity Control Assist | 155  |

| Item | Description                                                                                                                                                                 | Page |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
|      |  Long button press: Selecting the limiter                                                  | 154  |
|      |  Switch off Limiter/Cruise control/Proximity Control Assist                                |      |
|      |  Switching on and setting the current speed/speed limit, adjust set speed/speed limit      |      |
|      |  Switching on and calling the stored speed/speed limit, reducing the set speed/speed limit |      |

## At a glance

### Combination switch and multifunction lever

#### Combination switch and multifunction lever



| Item     | Description                                                                                                                             | Page |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>A</b> | Combination switch                                                                                                                      |      |
| ①        | Main beam on (when dipped beam is switched on)                                                                                          | 56   |
| ②        | Main beam off (when dipped beam is switched on)                                                                                         | 56   |
| ③        | Headlamp flashing                                                                                                                       | 56   |
| ④        | Turn signal indicator, right                                                                                                            | 56   |
| ⑤        | Turn signal indicator, left                                                                                                             | 56   |
| ⑥        | Wiping with washer fluid/single wipe                                                                                                    | 65   |
| ⑦        | Windscreen wipers                                                                                                                       | 64   |
| <b>B</b> | Multifunction lever                                                                                                                     |      |
| ⑧        | Continuous brake                                                                                                                        | 137  |
| ⑨        | Automatic gearshift selection – direction of travel selection                                                                           | 143  |
| <b>D</b> | Drive/driving forwards                                                                                                                  |      |
| <b>N</b> | Neutral position                                                                                                                        |      |
| <b>R</b> | Reversing                                                                                                                               |      |
| ⑩        | Drive program selection<br><b>A/M</b> automatic drive program with the drive and vehicle-specific shift program or manual drive program | 140  |

| Item | Description                                                        | Page |
|------|--------------------------------------------------------------------|------|
| 11   | Automatic gearshift – manual upshifting, automatic drive program   | 144  |
|      | Manual drive program                                               | 144  |
| 12   | Automatic gearshift – manual downshifting, automatic drive program | 144  |
|      | Manual drive program                                               | 144  |

## Occupant safety

### Restraint system

#### Restraint system

#### Protection provided by the restraint system

In the event of an accident, the restraint system can reduce the forces to which the vehicle occupants are subjected.

A seat belt can only provide the best level of protection if it is worn correctly.

For the restraint system to provide its full protection, each occupant must observe the following:

- Fasten seat belts correctly.
- Be in an almost upright position with their back against the seat backrest.
- Sit with feet resting on the floor whenever possible.
- Always secure persons under 1.50 m in height in an additional restraint system suitable for Mercedes-Benz trucks.

Always observe the notes on adjusting the seats (→ Page 46) and adjusting the steering wheel (→ Page 49).

However, no system available today can completely prevent injuries and fatalities in every accident situation. In particular, the seat belt generally does not protect against objects penetrating the vehicle from the outside.

#### Limitations of the protection provided by the restraint system



**Danger**

#### Risk of injury or death due to modifications to the restraint system

Vehicle occupants may no longer be protected as intended if alterations are made to the restraint system.

- Never alter the parts of the restraint system.
- Never tamper with the wiring or any electronic component parts or their software.

If it is necessary to adjust the vehicle to accommodate a person with disabilities, contact a qualified specialist workshop.

Daimler Truck AG recommends contacting an authorised Mercedes-Benz Trucks Service Centre.

#### Operational readiness of the automatic high-voltage emergency shutoff

When the vehicle is switched on, the function of the automatic high-voltage emergency shutoff is checked during the self-test. If the restraint system/crash sensing warning lamp  does not light up in the instrument cluster, the components of the automatic high-voltage emergency shutoff are operational.

Overview of warning/indicator lamps (→ Page 83).

#### Automatic high-voltage emergency shutoff malfunction

If the restraint system/crash sensing warning lamp  lights up in the instrument cluster, a malfunction has occurred in the automatic high-voltage emergency shutoff.

The display message **High-voltage system malfunction Not possible to restart** also appears ( → Page 300).



### Warning

#### **Risk of fatal injury due to malfunctioning automatic high-voltage emergency shutoff**

In the event of an accident, the high-voltage on-board electrical system may not be deactivated as intended.

You may receive an electric shock if you touch the damaged components of the high-voltage on-board electrical system.

- ▶ Have the automatic high-voltage emergency shutoff checked and repaired immediately at a qualified specialist workshop.
- ▶ After an accident, switch off the vehicle immediately.

Mercedes-Benz recommends that you have the vehicle towed or transported to a qualified specialist workshop in the event of a malfunctioning automatic high-voltage emergency shutoff.

### Seat belts

#### **Protection provided by the seat belt**

Always fasten your seat belt correctly before starting a journey. A seat belt can only provide the best level of protection if it is worn correctly.



### Danger

#### **Risk of injury or death due to incorrectly fastened seat belt**

If the seat belt is not worn correctly, it cannot perform its intended protective function.

In addition, an incorrectly fastened seat belt can also cause injuries, for example, in the event of an accident or when braking or changing direction suddenly.

- ▶ Always ensure that all vehicle occupants have their seat belts fastened correctly and are sitting properly.

Always observe the notes on adjusting the seats ( → Page 46) and adjusting the steering wheel ( → Page 49).

In order for the correctly worn seat belt to provide the intended level of protection, each vehicle occupant must observe the following information:

- The seat belt must not be twisted and must fit tightly and snugly across the body.
- The seat belt must be routed across the centre of the shoulder and as low down across the hips as possible.
- The shoulder section of the seat belt should not touch your neck nor be routed under your arm or behind your back.
- Avoid wearing bulky clothing, e.g. a winter coat.
- Push the lap belt down as far as possible across your hips and pull tight with the shoulder section of the belt. Never route the lap belt across your abdomen.
- Pregnant women must also take particular care with this.
- Never route the seat belt across sharp, pointed, abrasive or fragile objects.

## Occupant safety

### Seat belts

- Only one person may use each seat belt at any given time.
- Never secure objects with a seat belt if the seat belt is being used by one of the vehicle's occupants.  
Also ensure that no objects, e.g. a cushion, are ever placed between a person and the seat.
- Your feet should always be placed on the ground. Do not put your feet on the cockpit, for example. Otherwise, when braking or in the event of an accident, you could slide underneath the seat belt.

Be sure to observe the instructions and safety instructions on "Children in the vehicle" when children are travelling in the vehicle (→ Page 32).

To secure objects, luggage or loads, always observe the instructions and safety instructions on stowage spaces and stowage compartments (→ Page 68).

#### Limitations of the protection provided by the seat belt



#### **Danger**

#### **Risk of injury or death due to incorrect seat position**

The seat belt does not offer the intended level of protection if you have not moved the seat backrest to an almost vertical position.

In particular, you may slide underneath the seat belt and injure yourself.

- ▶ Adjust the seat properly before commencing your journey.
- ▶ Always ensure that the seat backrest is in an almost vertical position and that the shoulder belt is routed across the centre of your shoulder.



#### **Danger**

#### **Risk of injury or death when additional restraint systems are not used for persons with a smaller build**

Persons under 1.50 m in height may not be able to fasten their seat belt correctly

without a suitable additional restraint system.

- ▶ Always secure persons under 1.50 m in height in a suitable additional restraint system.



#### **Danger**

#### **Risk of injury or death due to damaged or modified seat belts**

Seat belts cannot provide protection in the following situations:

- the seat belt is damaged, has been modified, is extremely dirty, bleached or dyed.
- the seat belt buckle is damaged or extremely dirty.
- modifications have been made to the seat belt anchorage or seat belt retractor.

Seat belts may sustain non-visible damage in an accident, e.g. due to glass splinters.

Modified or damaged seat belts could tear or fail in the event of an accident, for example.



- ▶ Never modify the seat belt system, for example the seat belt, seat belt buckle, seat belt anchorage and seat belt retractor.
- ▶ Make sure that the seat belts are not damaged, are not worn and are clean.
- ▶ Always have the seat belts checked immediately after an accident at a qualified specialist workshop.

Daimler Truck AG recommends that you use seat belts that have been specifically approved for the relevant type of vehicle by Daimler Truck AG. Otherwise, your vehicle's general operating permit could be invalidated.



### NOTE

#### Damage caused by trapping the seat belt

If an unused seat belt is not fully retracted, it may become trapped in the door or in the seat mechanism.

- ▶ Always ensure that an unused seat belt is fully retracted.

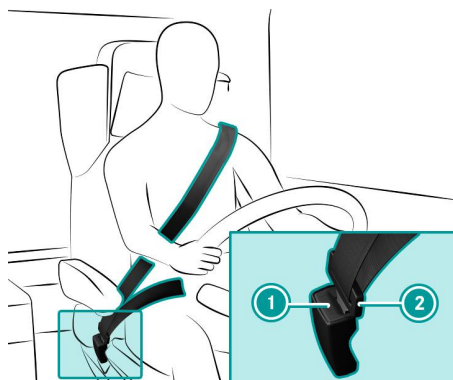
#### Fastening and adjusting the seat belts

If the seat belt is pulled out quickly or sharply, the seat belt retractor locks. The belt strap cannot be pulled out any further.



### Note

A seat belt can only provide the best level of protection if it is worn correctly. Observe the notes on fastening the seat belt (→ Page 29).



- ▶ Always engage the seat belt tongue ② of the seat belt into the seat belt buckle ① of the corresponding seat.

#### Releasing the seat belt

- ▶ Press the release button in the seat belt buckle and guide the seat belt back with the seat belt tongue.

#### Function of the seat belt warning for the driver and co-driver

The seat belt warning lamp in the instrument cluster is a reminder that all vehicle occupants must wear their seat belts correctly.

A warning tone may also sound.

Depending on the vehicle equipment, the seat belt warning lamp goes out and the warning tone is switched off:

- The driver's seat belt is fastened.
- The driver and co-driver have fastened their seat belts.

## Occupant safety

### Travelling safely with children in the vehicle

Overview of warning/indicator lamps  
(→ Page 83).

#### Travelling safely with children in the vehicle

#### Always observe when children are travelling in the vehicle



#### Note

Be sure to observe the safety instructions relevant to the situation. You will be able to identify possible risks and avoid dangers when children are travelling in the vehicle (→ Page 32).

#### Be diligent

Bear in mind that failing to take due care when securing the child in the child restraint system may have serious consequences. Always be diligent and secure a child carefully before each journey.

Infants and children must never travel sitting on the lap of a vehicle occupant.

In order to better protect children of under 12 years of age or under 1.50 m in height, Daimler Truck AG strongly recommends observing the following:

- Always secure the child in a child restraint system suitable for this Mercedes-Benz truck.
- The child restraint system must be appropriate to the age, weight and size of the child.
- The vehicle seat must be suitable for the child restraint system to be fitted.

#### The generic term child restraint system

The generic term child restraint system is used in these operating instructions. A child restraint system is, for example:

- A baby car seat
- A rearward-facing child seat
- A forward-facing child seat
- A child booster seat with a backrest and seat belt guides

The child restraint system must be appropriate to the age, weight and size of the child.

#### Observe laws and regulations

Always observe the legal requirements when using a child restraint system in the vehicle.

Ensure that the child restraint system is approved in accordance with the valid test specifications and guidelines. Further information can be obtained at a qualified workshop. Daimler Truck AG recommends that you use an authorised Mercedes-Benz Trucks Service Centre for this purpose.

#### Detecting risks, avoiding dangers

#### Securing systems for child restraint systems in the vehicle

Only use the following securing system for child restraint systems:

- the vehicle's seat belt system

### Advantages of a rearward-facing child restraint system

Babies or infants should preferably travel in a suitable rearward-facing child restraint system. In this case, the child is seated in the opposite direction to the direction of travel and is facing backwards.

Babies and infants have comparatively poorly developed neck muscles in relation to the size and weight of their head. A rearward-facing child restraint system can reduce the risk of injury to the cervical spine in an accident.

### Always secure child restraint systems correctly



**Danger**

### Risk of injury or death due to an incorrectly installed child restraint system

The child cannot then be protected or restrained as intended.

- ▶ Always comply with the manufacturer's installation instructions for

the child restraint system and its correct use.

- ▶ Make sure that entire base of the child restraint system always rests on the seat cushion of seat.
- ▶ Never place objects under or behind the child restraint system, e.g. cushions.
- ▶ Only use child restraint systems with its intended genuine cover.
- ▶ Always replace damaged covers with genuine covers.

Use child restraint systems recommended for Mercedes-Benz Trucks wherever possible.



**Danger**

### Risk of injury or death due to unsecured child restraint systems in the vehicle

If the child restraint system is incorrectly fitted or not secured, it can become detached.

The child restraint system may be thrown around and hit vehicle occupants.

- ▶ In addition, always be sure to correctly install any unused child restraint systems.
- ▶ Always comply with the manufacturer's installation instructions for the child restraint system.

- Be sure to observe the installation and operating instructions provided by the manufacturer of the child restraint system being used as well as the vehicle-specific notes at all times:
  - Securing the child restraint system with the seat belt on the front passenger's seat (→ Page 35).
- Observe the warning labels in the vehicle interior and on the child restraint system.

## Occupant safety

### Travelling safely with children in the vehicle

#### Do not make modifications to child restraint systems



#### **Danger**

#### **Risk of injury from modifications to the child restraint system**

The child restraint system can no longer function as intended. There will be an increased risk of injury.

- ▶ Never alter a child restraint system.
- ▶ Only install accessories that the manufacturer of the child restraint system has specifically approved for that child restraint system.

Daimler Truck AG recommends Daimler Truck AG care products for cleaning child restraint systems recommended for Mercedes-Benz trucks.

#### Only use child restraint systems which are not damaged



#### **Danger**

#### **Risk of injury or death due to the use of damaged child restraint systems**

Damaged child restraint systems, those subjected to stress during an accident or their securing systems may no longer provide protection as intended.

It may not be possible to restrain the child.

- ▶ Replace damaged child restraint systems or those subjected to stress during an accident immediately.
- ▶ Have the securing systems of child restraint systems checked at a qualified workshop immediately before a child restraint system is installed again.

#### Avoid direct sunlight



#### **Danger**

#### **Risk of burns when the child seat is exposed to direct sunlight**

If the child restraint system is exposed to direct sunlight, its parts may heat up considerably.

Children could suffer burns from these parts, particularly from the metal parts of the child restraint system.

- ▶ Always ensure that the child restraint system is not exposed to direct sunlight.
- ▶ Protect the child restraint system, e.g. with a blanket.
- ▶ If the child restraint system was exposed to direct sunlight, let it cool down before securing the child in it.
- ▶ Never leave children unattended in the vehicle.

### Observe the following when stopping or parking



**Danger**

#### Risk of accident and injury due to children left unattended in the vehicle

If you leave children unattended in the vehicle, some of the things that might happen include:

- They may open doors, thereby endangering other persons or road users.
- They may get out and be struck by oncoming traffic.
- They may operate vehicle equipment and become trapped.

In addition, the children could also set the vehicle in motion, for example by:

- Releasing the parking brake.
  - Changing the transmission position.
  - Starting the vehicle.
- Never leave children unattended in the vehicle.

- When leaving the vehicle, always take the key with you and lock the vehicle.
- Keep the key out of the reach of children.



**Danger**

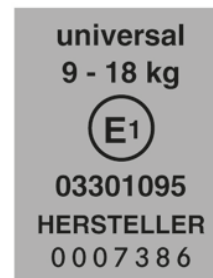
#### Danger to life due to exposure to extreme heat or cold in the vehicle

If persons particularly children are subjected to prolonged exposure to intense heat or cold, there is a risk of severe injury or even death.

- Never leave persons particularly children unattended in the vehicle.

### Securing the child restraint system with the seat belt on the co-driver's seat

Child restraint systems in the "Universal" category are marked with an orange approval label.



Example of an approval label

### When fitting a belt-secured child restraint system on the co-driver's seat, always observe the following:

|                                     |                                                                                                                                           |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> | When using a child restraint system in the "Universal" category, ensure that it is approved for the vehicle seat.                         |
| <input checked="" type="checkbox"/> | Observe the manufacturer's installation and operating instructions for the child restraint system used.                                   |
| <input checked="" type="checkbox"/> | The backrest of a forward-facing child restraint system must, as far as possible, rest flat against the backrest of the co driver's seat. |

## Occupant safety

### Notes on pets in the vehicle

|   |                                                                                                                                                                                           |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ☑ | The child restraint system must not be put under strain between the roof and the seat cushion and/or be fitted facing the wrong direction. Adjust the seat backrest angle as appropriate. |
| ☑ | The child restraint system must not be put under strain by the head restraint. Adjust the head restraints as appropriate.                                                                 |
| ☑ | Never place objects under or behind the child restraint system, e.g. cushions.                                                                                                            |

- ▶ Move the co-driver's seat as far back as possible.
- ▶ Fully retract the seat cushion length adjustment.
- ▶ Adjust the seat cushion angle in such a way that the front of the seat cushion is in the topmost position and the rear of the seat cushion in the lowest position.
- ▶ Adjust the seat backrest to an almost vertical position.
- ▶ Fit the child restraint system.
  - ◁ The entire base of the child restraint system must always rest

on the seat cushion of the co-driver's seat.

- ▶ Always make sure that the shoulder belt strap is correctly routed from the seat belt outlet of the vehicle to the shoulder belt guide on the child restraint system.
  - ◁ The shoulder belt strap must be routed forwards and downwards from the seat belt outlet.
- ▶ If necessary, adjust the seat belt outlet and the co-driver's seat as appropriate.

### Notes on pets in the vehicle



**Danger**

**– Risk of accident and injury due to animals left unsecured or unattended in the vehicle**

If you leave animals unattended or unsecured in the vehicle, they could press buttons or switches, for example.

An animal may, for example:

- Activate vehicle equipment and become trapped, for example.
- Switch systems on or off and endanger other road users.

Unsecured animals may also be thrown about the vehicle in the event of an accident or sudden steering and braking manoeuvres and injure vehicle occupants in the process.

- ▶ Never leave animals in the vehicle unattended.
- ▶ Always correctly secure animals while driving, e.g. using a suitable animal carrier.

### Electronic key

#### Overview of electronic keys

You require a key coded for the vehicle to use the following functions:

- Switching on the drive system
- Shifting gears

#### **i** Note

If a vehicle key is lost, obtaining a replacement is a time-consuming process.

This can only be done through an authorised Mercedes-Benz Trucks Service Centre.

Daimler Truck AG therefore recommends that you always keep an easily accessible replacement key with you for emergencies.

If you lose a key, have it disabled at an authorised Mercedes-Benz Trucks Service Centre.



Key with remote control and lamp check

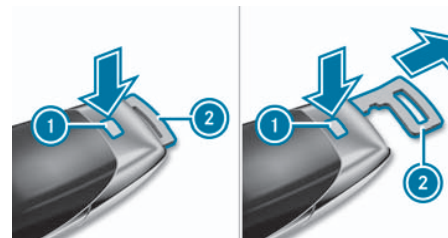
If the key's battery is low or almost discharged, the instrument cluster will show you a corresponding display message.

Change the battery of the key (→ Page 42).

Observe the information on the "Vehicle key radio operating permit" (→ Page 279).

#### Using the emergency key element

##### Removing



- ▶ Press release knob ①.
  - ◁ Emergency key ② is pushed out slightly.
- ▶ Pull out emergency key ② until it engages in the intermediate position.

#### **i** Note

You can use the intermediate position of emergency key ② to attach the key to a key ring.

- ▶ Press release button ① again and fully remove emergency key ②.

##### Inserting

- ▶ Press release knob ①.

## Opening and closing

### Unlocking and locking the central locking system

- ▶ Insert emergency key ② to the intermediate position or fully until it engages.

#### Deactivating the function of the key

##### Vehicles with keyless start

If you deactivate the function of the relevant key, it is no longer possible to authorise driving with a keyless start. Activate the function of the relevant key so that the function is available again as usual.

You can also deactivate the function of the key to reduce the energy consumption of the key if you do not use the vehicle or a key for an extended period of time.

- ▶ **To deactivate:** Press and hold the  button of the key.
- ▶ With  button pressed, immediately press the  button of the key twice in rapid succession.
  - ◁ The indicator lamp of the key lights up once briefly and once long.
- ▶ **To activate:** Press any button on the key.

or

- ▶ Hold the key against the contact point for emergency operation (→ Page 125).

#### Carry out lamp check with the key

The lamp check assists you in making your departure checks and helps you identify defective bulbs/LEDs.

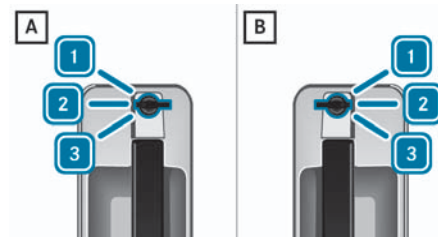
- ▶ Apply the parking brake.
- ▶ Press the  button on the key.
  - ◁ Low beam headlamps, marker lights, tail lights and licence plate lighting light up continuously. The turn signal, main-beam, brake light, daytime running lights, reversing light, fog light and the work lamp are switched on one after the other. This cycle is repeated three times.

If you want to cancel the lamp check, press the button  again or release the parking brake.

### Unlocking and locking the central locking system

#### External door lock

#### Unlocking/locking using the key



- ▶ Insert the key into the lock of left-hand hinged door **A** or right-hand hinged door **B** in position **2**.
- ▶ **To release:** Turn the key to the **3** position.
  - ◁ Both doors are unlocked.
- ▶ **To lock:** Turn the key to the **1** position.
  - ◁ Both doors are locked.

For vehicles with the folding door design, the folding door must be closed before it is locked.



#### Unlocking/locking using the remote control

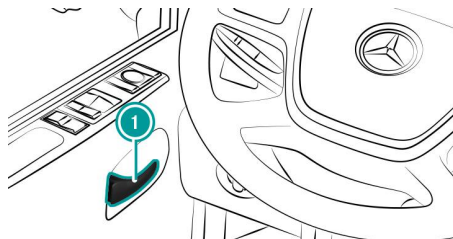
- ▶ **To release:** Press the  button on the key.
  - ◁ The doors are unlocked.
- ▶ **To lock:** Press the  button on the key.
  - ◁ Both doors are locked.

**For vehicles with a folding door:** When the vehicle is switched on, the folding door closes if you press the  button on the key.

- ▶ Press the  button within a minute of switching the vehicle off.
  - ◁ The folding door is closed and the vehicle is locked.

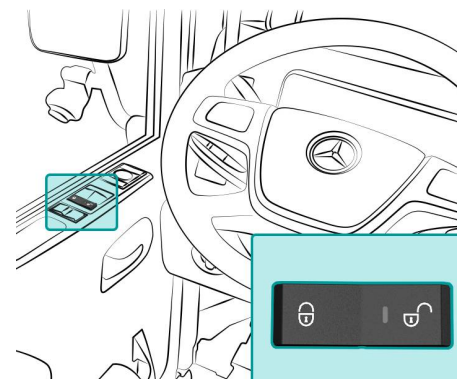
#### Internal door lock

#### Locking/unlocking using the door handle



- ▶ Close the doors.
- ▶ **To lock:** Push the door handle  against the door.
  - ◁ The respective door is locked.
- ▶ **To open:** Pull the door handle .

#### Locking/unlocking centrally using the buttons



- ▶ Close both doors. For vehicles with a folding door, the folding door must be closed.
- ▶ **Central locking:** Press the button .
- ▶ **Central unlocking:** Press the button .
  - ◁ The indicator lamp on the  button will flash.

## Opening and closing

### Instructions for the safety locking feature

#### Instructions for the safety locking feature

With the safety locking feature, you can electrically lock and unlock the cab doors.

The following requirements must be met in order to lock the doors:

- The driver's door is closed.
- Vehicles with co-driver's revolving door: the co-driver's door is closed.

#### **i** Note

Vehicles with folding door: if you lock the vehicle while the folding door is open, the folding door will automatically be closed.

When the vehicle has been locked, the turn signal indicators flash three times.

When the vehicle has been unlocked, the turn signal indicators flash once.

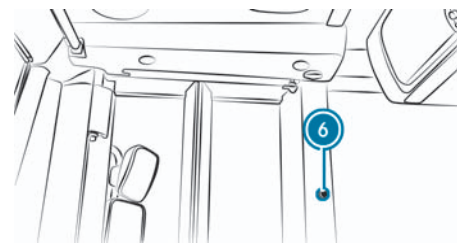
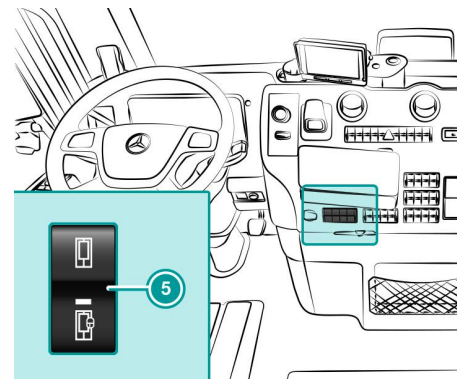
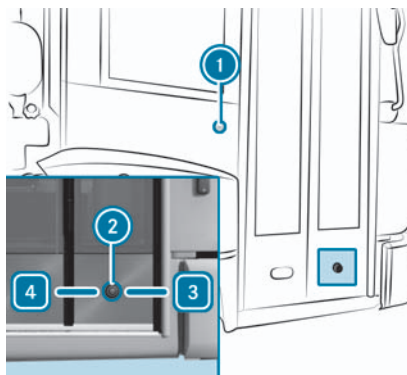
#### **i** Note

To ensure the safety locking feature, the following steps must be carried out after any interruption to the power supply:

- Switch the power supply off and on using the start/stop button.
- Lock and unlock the vehicle once.

#### Folding door

##### Overview of folding door



- ① Exterior button, opens/closes folding door
- ② Lock cylinder
- ③ Lock cylinder position, unlocked
- ④ Lock cylinder position, locked
- ⑤ Unlock/lock the cab doors

- ⑥ Vehicles with co-driver's revolving door: Locking the cab doors
- ⑥ Internal button, opens/closes folding door

### Unlocking and locking the folding door

On vehicles with lock cylinders ②, it is possible to unlock and lock the folding door from the outside.

- ▶ **To unlock:** Insert the key into lock cylinder ② and turn it to position ③.
- ▶ **To lock:** Insert the key into lock cylinder ② and turn it to position ④.

### Unlocking and locking the cab doors

You can use the electronic key or the ⑤ button to unlock and lock the cab doors electrically.

- ▶ **To unlock and lock:** Press ⑤ button below.
  - ◁ When the indicator lamp in the ⑤ button comes on, the cab doors are locked.

### Opening and closing the folding door

- ▶ Vehicles with lock cylinders: unlock the door using the door lock.
- ▶ Switch on the power supply.
- ▶ Press the ⑤ button above.

or

- ▶ Press internal button ⑥.

or

- ▶ Press exterior button ①.
  - ◁ The folding door opens or closes.

#### **i** Note

Vehicles with lock cylinders: If you have not unlocked the door lock and you press the button ⑤, a warning tone sounds.

#### **i** Note

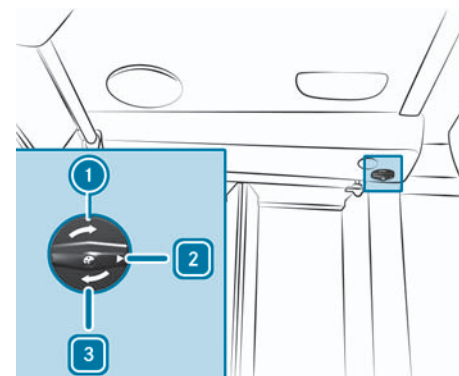
If you lock the vehicle electrically while the folding door is open, the folding door will be closed automatically.

You are only able to open or close the folding door at speeds below 3 km/h.

If you pull away with the folding door open, it will automatically close at a speed above 3 km/h.

If the blockage detection detects an obstacle while the folding door is being closed, the folding door opens again automatically.

### Opening the folding door in an emergency



- ▶ Vehicles with lock cylinders: unlock the external door lock of the folding door before driving off.
- ▶ **To open:** Turn control knob ① to the emergency position ③.

## Opening and closing

### Replacing the key battery

- ▶ Open the folding door manually.
- ▶ **To close:** Turn control knob ❶ to the drive position ❷.
  - ◁ The folding door closes  $\frac{2}{3}$  and is not locked.
- ▶ The folding doors close.

### Replacing the key battery



#### Warning

#### Risk of fatal injuries due to swallowed batteries

Batteries contain toxic and corrosive substances. Swallowing batteries may cause severe internal burns to occur within two hours.

There is a risk of fatal injury!

- ▶ Keep batteries out of the reach of children.
- ▶ If the cover and/or battery compartment does not close securely, do not use the key any longer and keep it out of the reach of children.
- ▶ If batteries are swallowed, seek medical attention immediately.



#### ENVIRONMENTAL NOTE

#### Environmental damage due to improper disposal of batteries



Batteries contain pollutants.

It is illegal to dispose of them with the household rubbish.



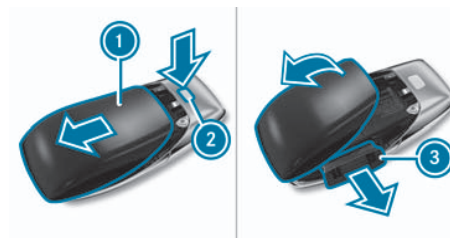
- ▶ Dispose of batteries of in an environmentally responsible manner. Take discharged batteries to a qualified specialist workshop or to a collection point for used batteries.

#### Requirements

- You will need a CR 2032 type 3 V button cell.

Daimler Truck AG recommends that you have the battery replaced at a qualified specialist workshop.

- ▶ Remove the emergency key.



- ▶ Press release button ❷ down fully and slide cover ❶ in the direction of the arrow.
- ▶ Fold out cover ❶ in the direction of the arrow and remove.
- ▶ Slide off battery compartment ❸ and take out the discharged battery.
- ▶ Insert the new battery into battery compartment ❸. Observe the positive pole marking in the battery compartment and on the battery when doing this.
- ▶ Slide in battery compartment ❸.
- ▶ Re-attach cover ❶ and push it until it engages.

#### Getting in and out



#### **Danger**

#### **Risk of accident and injury due to children left unattended in the vehicle**

If you leave children unattended in the vehicle, some of the things that might happen include:

- They may open doors, thereby endangering other persons or road users.
- They may get out and be struck by oncoming traffic.
- They may operate vehicle equipment and become trapped.

In addition, the children could also set the vehicle in motion, for example by:

- Releasing the parking brake.
  - Changing the transmission position.
  - Starting the vehicle.
- ▶ Never leave children unattended in the vehicle.

- ▶ When leaving the vehicle, always take the key with you and lock the vehicle.
- ▶ Keep the key out of the reach of children.



#### **Danger**

#### **Risk of injury due to excessive load on the grab handles**

If you apply your full body weight to the grab handle or pull it abruptly, the grab handle may be damaged or become loose from its anchorage.

- ▶ Use the grab handles only to stabilise the seating position or to assist in getting in and out.



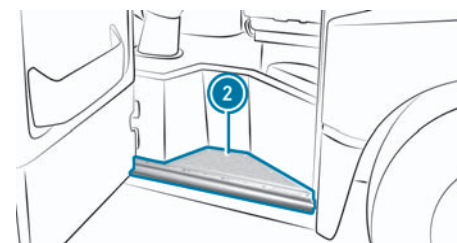
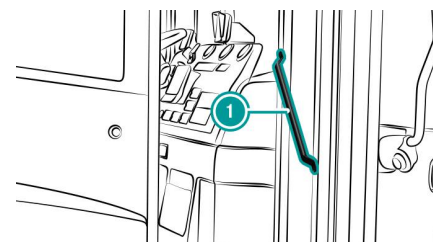
#### **Note**

When travelling with children in the vehicle, observe the supplemental notes (→ Page 32).

To get in and out safely, you should always use the steps and grab handles. They are the only elements designed to

bear such a load. Do not jump out of the cab.

Keep the steps, entrances, grab handles and your footwear free of dirt. This will make your footing safer.



- ▶ Use the easy entry and exit feature on the suspension seat (→ Page 48).
- ▶ Use grab handles ① and steps ② when getting in and out.

## Opening and closing

### Side windows

#### Side windows

#### Opening and closing the side windows



**Danger**

#### Risk of becoming trapped when opening a side window

When opening a side window, parts of the body could be drawn in or become trapped between the side window and window frame.

- ▶ When opening the side window, make sure that nobody is touching the window.
- ▶ If someone becomes trapped, release the button immediately or pull it in order to close the side window again.



**Danger**

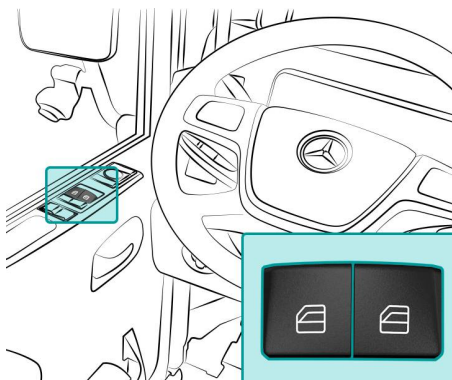
#### Risk of becoming trapped when closing the side window

Body parts may become trapped in the closing area when closing a side window.

- ▶ During the closing process, make sure that no body parts are in the closing area.
- ▶ If someone is trapped, release the button immediately or press the button in order to reopen the side window.

#### Requirements

- The ignition lock is in the radio position or the vehicle is switched on.



The image shows the version with two hinged doors

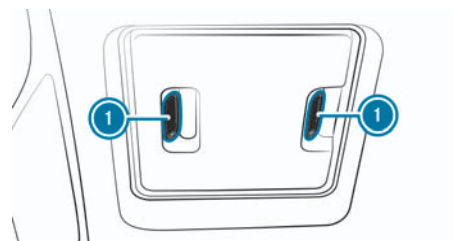
- ▶ **To open and close:** Push or pull the left- or right-hand button  until

the corresponding side window has reached the desired position.

- ▶ **To use convenience opening:** Briefly push the left- or right-hand button  past the point of resistance. The corresponding side window will open fully.

#### Roof

#### Opening and closing the roof hatch

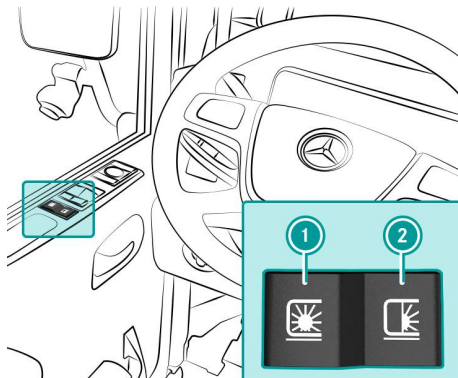


You can open the roof hatch at one end (front or back) or at both ends (front and back).

- ▶ Push the roof hatch upwards at the handles  or pull it down.

#### Retracting/extending the roller sunblind

##### Extending the roller sun blind



- ▶ Tap the ② button briefly.
  - ◁ The roller sun blind will extend as far as the end position.

or

- ▶ Press and hold the ② button until the roller sun blind has reached the desired position.

##### Retracting the roller sun blind

- ▶ Briefly press the ① button repeatedly until the roller sun blind has reached the desired position.

or

- ▶ Press and hold the ① button until automatic operation starts.
  - ◁ The roller sun blind will retract completely.
- ▶ **To interrupt automatic operation:**  
Briefly press one of the two buttons.

## Driver cockpit

### Notes on the seats

#### Notes on the seats



#### **Danger**

#### **Risk of injury if vehicle settings are adjusted while the vehicle is in motion**

You may lose control of the vehicle in particular in the following situations:

- If you adjust the driver's seat, steering wheel or mirror while the vehicle is in motion.
- if you fasten your seat belt while the vehicle is in motion.
- ▶ before starting the vehicle: Adjust the driver's seat, steering wheel or mirror and fasten the seat belt.



#### **Danger**

#### **Risk of becoming trapped when adjusting a seat**

When you are adjusting a seat, you or a vehicle occupant may become trapped on the seat's guide rail, for example.

- ▶ Make sure that no parts of the body are in the seat's movement area when adjusting the seat.



#### **Danger**

#### **Risk of injury or death due to incorrect seat position**

The seat belt does not offer the intended level of protection if you have not moved the seat backrest to an almost vertical position.

In particular, you may slide underneath the seat belt and injure yourself.

- ▶ Adjust the seat properly before commencing your journey.
- ▶ Always ensure that the seat backrest is in an almost vertical position and that the shoulder belt is routed across the centre of your shoulder.



#### **Danger**

#### **Risk of accident due to the driver's seat not being engaged**

The driver's seat may move unexpectedly while the vehicle is in motion.

This could cause you to lose control of the vehicle.

- ▶ Always ensure that the driver's seat is engaged before starting the vehicle.



#### **Danger**

#### **Risk of becoming trapped due to insufficient space for the suspension seat**

If there is insufficient space available, the suspension seat may trap parts of the body between the steering wheel and the suspension seat.

- ▶ Make sure that there is sufficient space for the suspension seat to move in.
- ▶ Fully lower the suspension seat before getting out of the seat.



**Danger****Risk of becoming trapped when pressing the bellows inwards**

If you press the bellows of the suspension seat inwards, your hand could become trapped.

- ▶ Do not press the bellows inwards.

**NOTE****Damage to the seats and the seat heating due to fluids, pointed objects or insulating materials**

To prevent damage to the seats and the seat heating, observe the following instructions:

- ▶ Do not spill any fluids onto the seats. If something does get spilt onto the seats, dry the seats as quickly as possible.
- ▶ Do not switch on the seat heating if the seat covers are wet or damp. Do not use the seat heating to dry the seats.

- ▶ Clean the seats as recommended; see the "Cleaning and care" section.
- ▶ Do not transport heavy loads on the seats. Do not place any sharp objects, such as knives, nails or tools, on the seats. Only use seats for people if possible.
- ▶ When using the seat heating, do not cover the seats with insulating materials, such as blankets, coats, bags, protective covers, child seats or booster seats.

**NOTE****Damage to the seats from objects in the cab**

Make sure that no objects block the seats in the cab.

Otherwise, the seats may be damaged.

- ▶ Keep the area for adjusting the seats in clear.

Your seat must be adjusted such that you can fasten your seat belt correctly. When doing so, observe the following:

- Place the seat backrest in a near-vertical position and sit almost upright. Do not drive with your seat backrest leaning back at a steep angle.
- Your arms should be slightly bent when you are holding the steering wheel.
- Avoid seat positions that prevent the seat belt from being routed correctly. The shoulder section of the belt must run across the centre of your shoulder and fit snugly across your chest. The lap belt must always fit snugly and as low as possible on your pelvis, i.e. over the hip joints.
- Sit at a suitable distance from the pedals to ensure that you can depress them properly.
- Make sure that you have the best possible view of the vehicle's immediate surroundings through the windscreen and the side windows, as well as the mirrors. Adjust the seat height accordingly, if necessary.

## Driver cockpit

### Operating the seat

Observe the safety instructions on "Children in the vehicle" (→ Page 32).

If your vehicle is fitted with a suspension seat, always use the easy entry and exit feature to get out.

To enable use of a suspension seat, your vehicle's compressed-air system must have a reserve pressure of at least 700 kPa (7 bar, 100 psi).

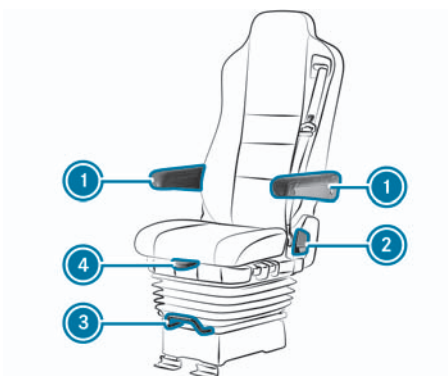
A seat with an integrated seat belt is a safety-critical component part and restraint system. Be sure to also read the safety instructions on "Occupant safety" (→ Page 28).

You can find information on cleaning seats under "Cleaning and care" (→ Page 215).

Always have any work on seats carried out at a qualified workshop.

### Operating the seat

#### Standard/comfort suspension seat



Suspension seat setting (example: comfort suspension seat)

Before adjusting the seats, read the "Notes on the seats" section. Depending on the seat model, certain adjustment options may not apply.

- ▶ **To adjust the armrests:** Fold up the armrests ①.
- ▶ Use the handwheel on the underside to adjust the angle of seat armrests ①.
- ▶ **To adjust the seat backrest:**

- Lift your weight off the seat backrest.
- Pull lever ② up and hold it in place.
- Move the seat backrest into the desired position by applying pressure to or lifting your weight off it.

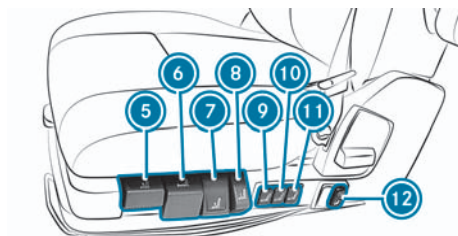
▶ **To adjust the seat fore-and-aft position:**

- Pull lever ③ up and hold it in place.
- Push the seat back/forwards until it is in the desired position.
- Release lever ③.
- Push the seat back/forwards until it engages audibly.

▶ **To adjust the seat cushion length:**

- Pull lever ④ up and hold it in place.
- Push the seat cushion back/forwards until it is in the desired position.

## Adjusting the multifunction steering wheel



Suspension seat setting (example: comfort suspension seat)

- ▶ **To adjust the seat cushion angle:**
  - Pull lever ⑥ up and hold it in place.
  - Move the seat cushion or backrest into the desired position by applying pressure to or lifting your weight off it.
- ▶ **To adjust the seat height:** pull lever ⑦ up or push it down by one increment.
  - ◁ The seat will be raised or lowered by one increment.
- ▶ **To adjust the vibration damper:** Adjust the vibration damper using lever ⑤.
- ▶ **To use the easy entry and exit feature:** Push lever ⑧ down.

- ◁ The seat will be lowered fully.
- ▶ Pull lever ⑧ up.
  - ◁ The seat will return to the last height set.

### **i** Note

You can adjust the contour of the backrest (lumbar support) and the side contours to provide support for your spine.

- ▶ **To adjust the seat backrest contour:** Press the upper or lower section of switch ⑨, ⑩ or ⑪.
  - ◁ This will make the respective seat backrest contour firmer or softer.
- ▶ **To switch seat heating on:** Press the button ⑫ at the top (first heating level) or bottom (second heating level).
- ▶ **To switch seat heating off:** Push switch ⑫ into the centre position.

### Adjusting the multifunction steering wheel



**Danger**  
Risk of injury if vehicle settings are adjusted while the vehicle is in motion

You may lose control of the vehicle in particular in the following situations:

- if you adjust the driver's seat, the head restraints, the steering wheel or the mirror while the vehicle is in motion.
- if you fasten your seat belt while the vehicle is in motion.
- ▶ before starting the vehicle: adjust the driver's seat, head restraints, steering wheel and mirror in particular and fasten your seat belt.

## Driver cockpit

### Outside mirrors



#### **Danger**

#### **Risk of accident due to unlocked steering wheel**

If the steering wheel remains unlocked while the vehicle is in motion, it may adjust unexpectedly.

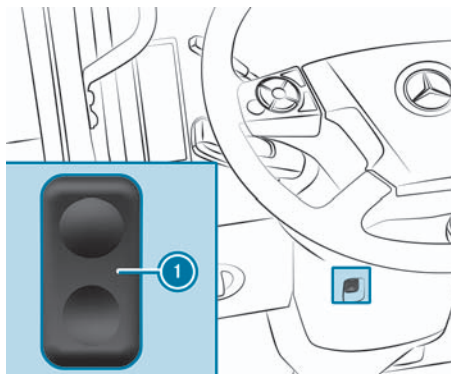
- ▶ Ensure that the steering wheel is locked before every journey.
- ▶ Do not unlock the steering wheel while the vehicle is in motion.

#### **Requirements**

- The vehicle is stationary.
- The parking brake is applied.

Observe the following when adjusting the steering wheel:

- First move the seat into the correct position (→ Page 46).
- Adjust the steering wheel angle to as low as possible.
- After adjusting the steering wheel, ensure that you can move freely and steer correctly.



- ▶ Press ❶ button below.
  - ◁ The steering column is unlocked.
- ▶ Adjust the multifunction steering wheel height and angle.
- ▶ Press the ❶ button above.
  - ◁ The steering column is locked.



#### **Note**

The steering wheel locks approximately ten seconds after unlocking.

### Outside mirrors

#### **Adjusts the outside mirrors**



#### **Danger**

#### **Risk of injury if vehicle settings are adjusted while the vehicle is in motion**

You may lose control of the vehicle in particular in the following situations:

- if you adjust the driver's seat, the head restraints, the steering wheel or the mirror while the vehicle is in motion.
- if you fasten your seat belt while the vehicle is in motion.
- ▶ before starting the vehicle: adjust the driver's seat, head restraints, steering wheel and mirror in particular and fasten your seat belt.

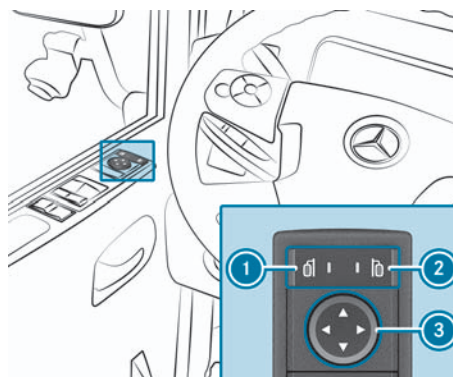


### Danger

#### Risk of accident if the distance is misjudged when using the outside mirror

The outside mirrors reduce the size of the image. The objects in view are in fact closer than they appear.

- You should therefore always ascertain the actual distance of other road users by looking over your shoulder.



Driver's door switch panel

- ①  Left outside mirror
- ②  Right outside mirror
- ③  Adjusts the outside mirrors

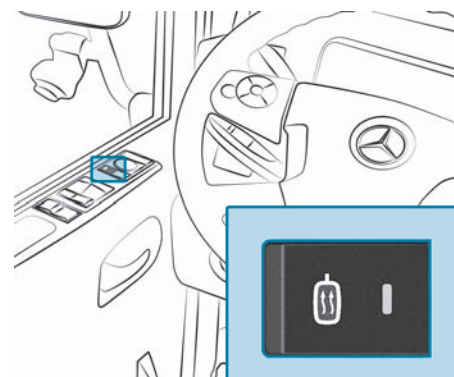
Adjust the front bumper-view mirror, kerb view mirror and wide-angle mirror manually.

- Switch on the vehicle.
- **To adjust the outside mirrors:**  
Press button  for the left outside mirror or button  for the right outside mirror.

◁ The indicator lamp on the button will light up.

- Push the  button forwards or backwards and on the right or left, until the outside mirror is in the correct position.

#### Activating/deactivating the mirror heater



- Switch on the vehicle.
- **To switch on/off:** Press the button .

## Driver cockpit

### Exterior lighting

- ◁ When the indicator lamp on the button lights up, the mirror heater is switched on.

#### Exterior lighting

##### Lamp check

The lamp check is only an aid. Responsibility for the correct vehicle lighting in accordance with the legal requirements always rests with the driver.

All the vehicle's exterior lighting is electronically monitored. A malfunction in the exterior lighting is signalled to the driver via the indicator lamp  on the instrument cluster.

Follow the instructions in the display messages on the instrument cluster.

Check the vehicle lighting with a second person.

Depending on the equipment, you may be able to check the lighting with the electronic vehicle key (→ Page 38).

#### Light switch overview



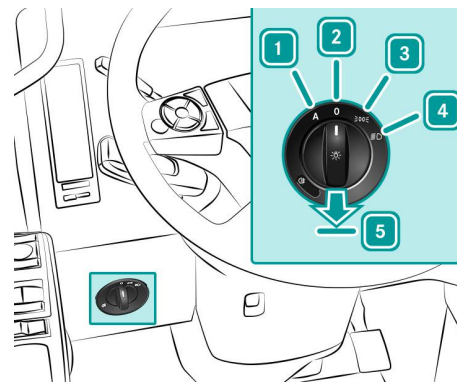
**Danger**

**Risk of accident if the exterior lighting is covered by the side board**

If you open the rear side board, the rear exterior lighting is covered.

This could delay other road users in recognising the vehicle as an obstacle.

- Secure the rear area of the vehicle according to the national requirements, e.g. using a warning triangle.



Light switch (example)

- |   |                                                                                     |                                                                                    |
|---|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1 | <b>A</b>                                                                            | Automatic driving lights and daytime running lights                                |
| 2 | <b>0</b>                                                                            | Lights off or daytime running lights                                               |
| 3 |  | Standing and tail lights; licence plate lighting; perimeter and side marker lights |
| 4 |  | Dipped beam/main beam                                                              |
| 5 |                                                                                     | Rear fog light                                                                     |

You can operate the main beam and perform headlamp flashing using the combination switch (→ Page 56).

A warning tone will sound in the following cases:

- You open the driver's door with the dipped beam switched on and the power supply switched on.
- You open the driver's door with the standing lights or dipped beam switched on and the ignition lock in the **0** position.

#### Activating/deactivating low beam

- ▶ Turn the light switch to .
  - ◁ The low beam and the standing light will be switched on. The  indicator lamp on the instrument cluster will light up.

#### Switching the daytime running lights on/off

- ▶ Start the vehicle.

- ▶ Turn the light switch to position **0**.
  - ◁ The daytime running lamps will be switched on.

or

- ▶ Turn the light switch to **A**.
  - ◁ If the low beam and standing lights are automatically switched off, the daytime running lights will go on.

If the light switch is turned to , the daytime running lights will automatically switch off.

#### Activating/deactivating the automatic driving lights



**Danger**

#### Risk of accident when the dipped beam is switched off in poor visibility

When the light switch is set to **A**, the dipped beam may not be switched on automatically if there is fog, snow or

other causes of poor visibility such as spray.

- ▶ In such cases, turn the light switch to .

The automatic driving lights are only an aid. The responsibility for vehicle lighting rests with the driver.

Turn the light switch from **A** to  immediately in the event of darkness or fog. Otherwise, the driving light may be temporarily interrupted.

- ▶ Switch on the vehicle.
  - ▶ Turn the light switch to **A**.
    - ◁ The low beam and standing lights will be switched on or off automatically depending on the light conditions. If the dipped beam is switched on, the  indicator lamp on the instrument cluster will switch on.
- Vehicles with daytime running lights: If the low beam and standing lights are automatically switched off, the daytime running lights will switch on.

## Driver cockpit

### Exterior lighting

#### Note

If the vehicle is switched off manually or automatically (→ Page 123), the vehicle lighting switched on by the automatic driving lights will also be switched off. If necessary, switch the vehicle lighting on manually.

#### Note

If the rain and light sensor is malfunctioning, the dipped beam and standing lights will be switched on automatically and stay on.

### Switching the rear fog light on/off

- ▶ Turn the light switch to  or .
- ▶ **Switch on rear fog light** Pull the light switch out to stage .
  - ◁ The rear fog light and indicator lamp  next to the light switch will go on.

### Setting the switch-off delay time

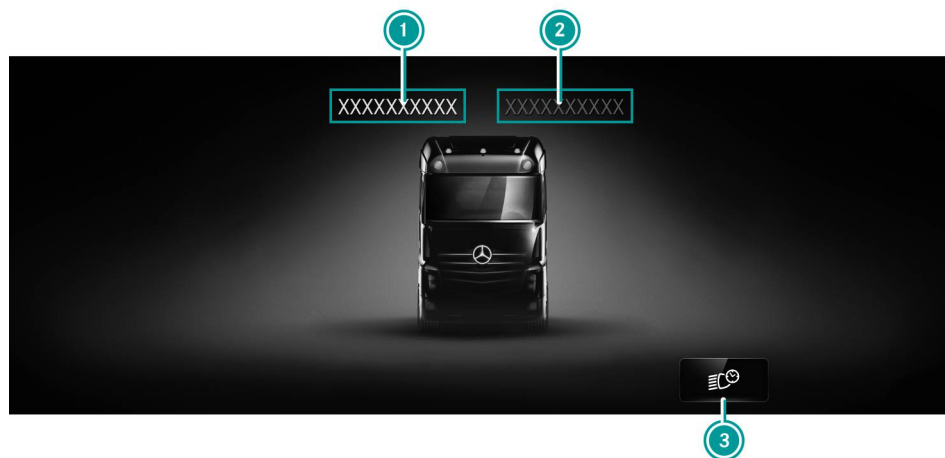
Multimedia system:

 Lights  Exterior lighting

### Overview of exterior lighting menu

- ▶ Press the  button below the multimedia display to access the lighting menu.





Exterior lighting menu (example)

- ① Menu **Exterior lighting**
- ② Menu **Interior lighting**
- ③  Switch-off delay time setting

► Select the **Exterior lighting** ① function.

◁ The exterior lighting menu is shown.

You have the following options for operating the menus:

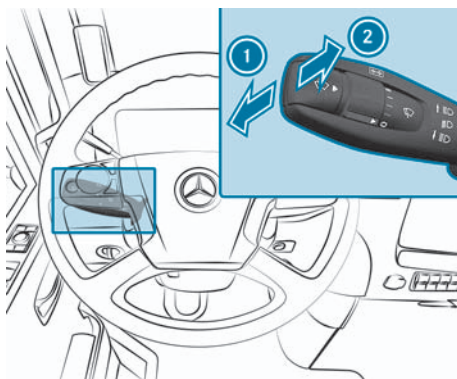
- via the multimedia system touch-screen
- with the right Touch Control on the multifunction steering wheel
- Press the  button in the menu.
  - ◁ The controller and the current setting of the switch-off delay time are displayed.
- Move the controller between OFF and 90 seconds to the desired time.

## Driver cockpit

### Exterior lighting

#### Combination switch

#### Using the main beam and headlamp flashing



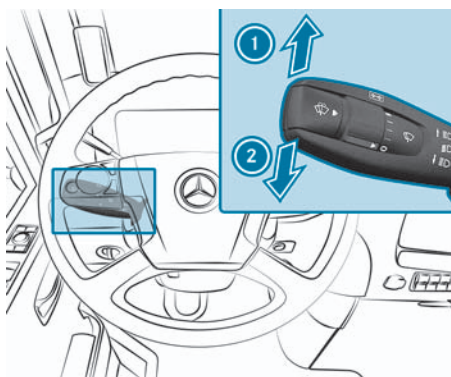
- Switch on the vehicle.

#### Switching on high beam

- Switch on dipped beam .
- Press the combination switch in the direction of arrow ① and latch into place.
  - ◁ The  indicator lamp on the instrument cluster will come on.

- **Headlamp flasher:** Briefly pull the combination switch in the direction of arrow ②.
  - ◁ The  indicator lamp on the instrument cluster and the high beam will light up briefly.

#### Using the turn signal indicators



- **To indicate:** Push the combination switch upwards to indicate right ① or downwards to indicate left ② until it engages.

- ◁ The corresponding turn signal lamps and the  or  indicator lamp on the instrument cluster will flash.

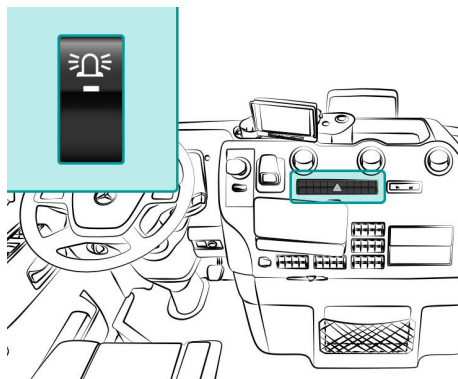
- **To flash:** When you are overtaking or changing lanes, briefly tap the combination switch in the desired direction to indicate right ① or indicate left ②.

- ◁ The corresponding turn signal lamps and the  or  indicator lamp on the instrument cluster will flash five times.

- **To cancel the flash:** Briefly tap the combination switch in the opposite direction.

### Operating the interior lighting

#### Switching the rotating beacon on/off



- ▶ **To switch on:** Press the  switch at top.
  - ◁ The indicator lamp on the button will light up.
- ▶ **Switching off:** Press the  switch at the bottom.
  - ◁ The indicator lamp on the button will go out.

When driving the vehicle with a rotating beacon, observe the legal requirements for the country in which you are currently driving. If the required field of vision is not

satisfied due to bodies or attachments, secure the vehicle using additional lights.

#### Note

Depending on the body of the vehicle, the rotating beacon can be switched on and off automatically when the power take-off is switched on and off (→ Page 212).

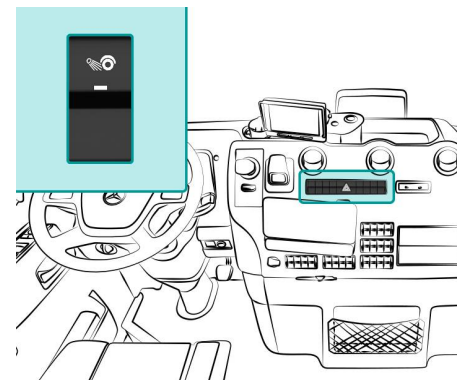
#### Activating/deactivating the work lamps (manoeuvring assistance)

#### Danger

#### Risk of accident due to glare from work lamps

Other road users may be dazzled by the work lamps.

- ▶ Only use the work lamps on public roads when on a work assignment.
- ▶ Ensure that no other road users are dazzled.



During manoeuvring, the work lamps help by illuminating the area in front of the rear wheels.

- ▶ Switch on the low beam.
- ▶ **To switch on/off:** Press the button .
  - ◁ If the indicator lamp on the button lights up, the work lamps are switched on.

#### Operating the interior lighting

Multimedia system:

 Lights  Interior lighting

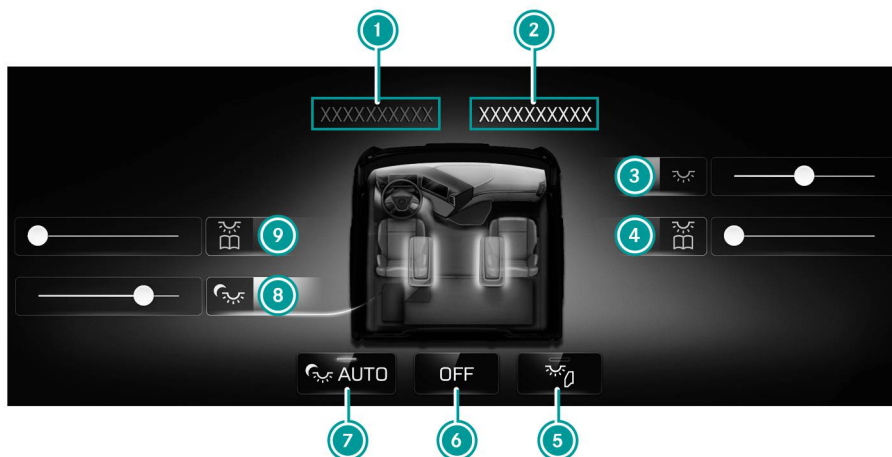
## Driver cockpit

### Operating the interior lighting

#### Overview of interior lighting menu

- Press the button  below the multimedia display to access the lighting menu.

## Operating the interior lighting



Interior lighting menu (example)

- ① Exterior lighting menu
  - ② Interior lighting menu
  - ③ Interior lamps
  - ④ Co-driver's reading lamp
  - ⑤ Automatic interior lighting control
  - ⑥ **OFF** Switches off all interior lighting
  - ⑦ **AUTO** Automatic night lighting control
  - ⑧ Night lighting
  - ⑨ Driver's reading lamp
- Select the **Interior lighting** ② function.

◁ The interior lighting menu is shown.

### Note

Depending on the vehicle's equipment, functions/buttons may not be present.

You have the following options for operating the menus:

- via the multimedia system touch-screen

## Driver cockpit

### Replacing bulbs

- with the right Touch Control on the multifunction steering wheel
- ▶ **Switching the function or driving system on/off:** Press the corresponding button in the menu.
  - ◁ The status is displayed in the car view. The function is activated when the indicator lamp in a button is lit.

#### Note

If you close the doors when the automatic interior lighting control is activated, the interior lamps will switch off after a delay.

- ▶ **To dim the lights:** move the corresponding controller next to the button.

#### Note

If you drive faster than 30 km/h, the brightness of the interior lamps will be automatically dimmed down.

### Replacing bulbs

#### Notes on changing a bulb

Daimler Truck AG recommends that you have the headlamp bulbs changed at a qualified workshop.



**Danger**

#### **Risk of burns from hot component parts whilst replacing a bulb**

Bulbs, lamps and plugs can become very hot during operation.

When replacing a bulb, you may burn yourself on these component parts.

- ▶ Allow the component parts to cool down before replacing the bulbs.

A bulb may explode in the following cases:

- You touch the bulb when it is hot.
- You drop the bulb.
- You scratch the bulb.

Do not use a bulb if it has been dropped or if its glass tube has been scratched.

Protect bulbs from moisture during operation and do not allow bulbs to come into contact with liquids.

Bulbs and lights are a major element in vehicle safety. Therefore, ensure that all bulbs are always working. The lamp check can help you when you are inspecting the bulbs (→ Page 52).

When you are changing a bulb, also note the following points:

- To prevent a short circuit, switch off the lighting system and the vehicle before changing the bulb.
- Wear safety glasses and clean gloves.
- Always replace faulty bulbs with certified new bulbs with the correct wattage and voltage.
- Touch new bulbs only with a clean, lint-free cloth or similar. Do not perform this work with wet or oily fingers.
- Check the contacts for corrosion and clean them if necessary.
- Ensure that seals are seated correctly and replace damaged seals.

- If the new bulb does not light up, consult a qualified workshop.

Have the LED lights changed at a qualified workshop. The rear LED lamp unit can only be replaced as a whole.

### Bulbs

#### Front bulbs

|                                            |              |
|--------------------------------------------|--------------|
| Low beam                                   | LED module   |
| High beam                                  | LED module   |
| Position lamp                              | LED module   |
| Turn signal light, front                   | PY21W 24 V   |
| Daytime running lights                     | LED module   |
| Turn signal light, side; side marker light | PY21/5W 24 V |
| Perimeter lamps                            | C5W 24 V     |

#### Interior lighting

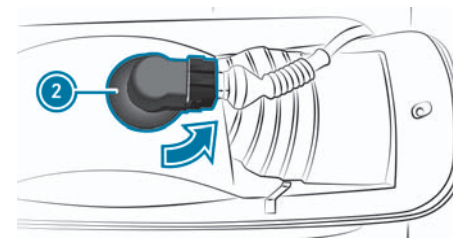
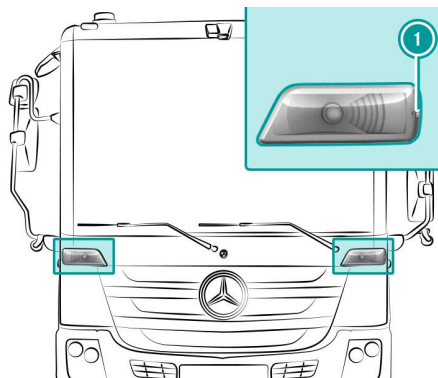
|                           |                  |
|---------------------------|------------------|
| Dome lamp: Interior light | P18W 24 V        |
| Dome lamp: Reading light  | R10W 24 V        |
| Dome lamp: Night lighting | EBS-R4 1.2W 24 V |

### Additional bulbs

|                    |            |
|--------------------|------------|
| Side marker lights | LED module |
| Exit lamp          | W5W 24 V   |

### Replacing front bulbs

#### Turn signal light, front

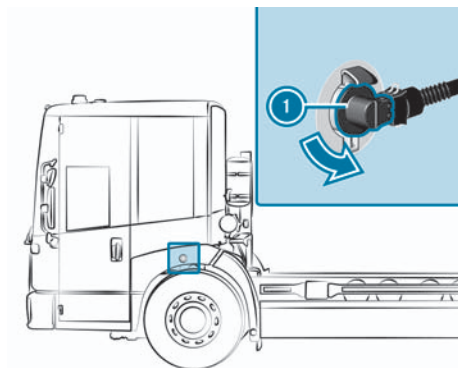


- ▶ Unscrew screw ① and remove the turn signal lamp.
- ▶ Gently turn bulb mount ② anti-clockwise and remove it.
- ▶ Gently turn the bulb anti-clockwise and remove it.
- ▶ Insert the new bulb and gently turn it clockwise.

## Driver cockpit

### Replacing bulbs

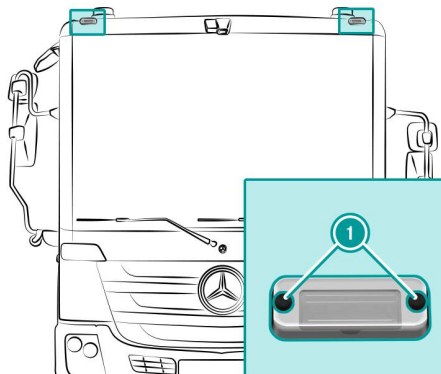
#### Turn signal light, side; side marker light



Turn signal light, side; side marker light (taking the left wheel arch as an example)

- ▶ Gently turn bulb mount ① anti-clockwise and remove it.
- ▶ Gently turn the bulb anti-clockwise and remove it.
- ▶ Insert the new bulb and gently turn it clockwise.

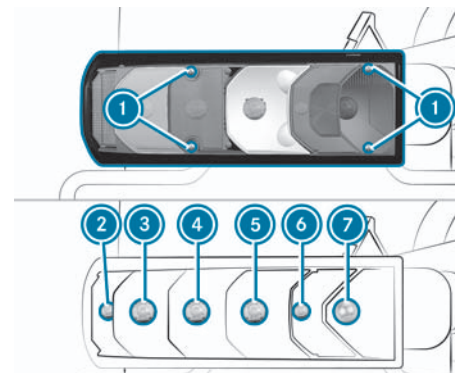
#### Perimeter lamps



- ▶ Unscrew screws ①.
- ▶ Remove the clearance lamp and pull out both cable connectors on its underside.
- ▶ Remove the lens from the rubber seal.
- ▶ Remove the bulb.
- ▶ Insert the new bulb.
- ▶ When installing, align the clearance lamp with the recess and fasten it in place.

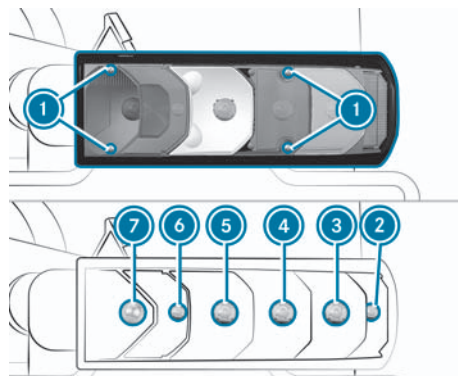
#### Replacing rear bulbs

##### Replacing bulbs



6-chamber lamp unit, rear left (example)





6-chamber lamp unit, rear right (example)

- ① Screws
- ② Clearance light/side marker light
- ③ Turn signal light
- ④ Brake light
- ⑤ Reversing light
- ⑥ Tail light
- ⑦ Rear fog light

The number plate lamp is behind the reflector unit.

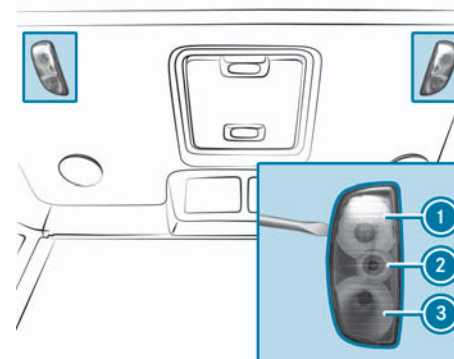
If there are LED modules in the tail light, do not change the light source for the turn signal light, standing lights, brake light or the clearance lights. If necessary, have

these light sources changed at a qualified specialist workshop.

- Observe the notes on changing a bulb (→ Page 60).
- Unscrew screws ①.
- Remove the lens.
- Gently turn the bulb anti-clockwise and remove it.
- Insert the new bulb and gently turn it clockwise.

### Replacing interior lighting bulbs

- Observe the notes on changing a bulb (→ Page 60).



- ① Interior lamp
- ② Night lighting
- ③ Reading lamp

- Prise off the lens with a screwdriver.
- Unclip the bulb mount.
- Gently turn the bulb anti-clockwise and remove it.
- Insert the new bulb and gently turn it clockwise.

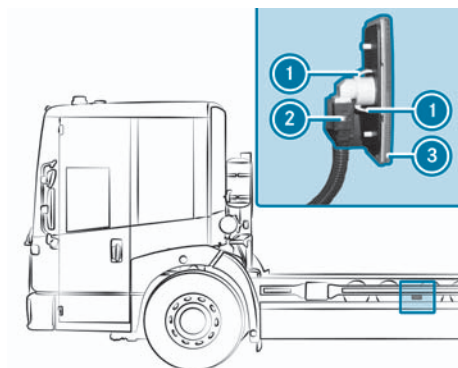
## Driver cockpit

### Good visibility

#### Changing additional bulbs

- Observe the notes on changing a bulb (→ Page 60).

#### Side marker lamps

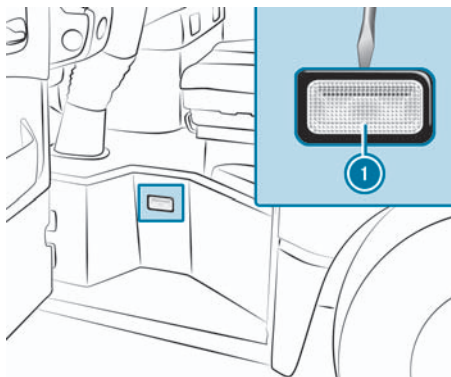


Side marker lamps (example)

- Press and firmly hold the retainer on cable connector ② with a screw-driver.
- Pull out cable connector ②.
- Press retainers ① on side marker lamp ③ together and hold them firmly.

- Replace side marker lamp ③.

#### Exit lamps on the step and in the door trim



Exit lamp, left (example)

- Prise off exit lamp ① with a screw-driver.
- Gently turn bulb mount anti-clockwise and remove it.
- Gently turn the bulb anti-clockwise and remove it.
- Insert the new bulb and gently turn it clockwise.

#### Good visibility

#### Using the windscreen wipers



#### NOTE

##### Damage to the windscreen on vehicles with a rain sensor

In dry weather, dirt or optical factors may cause the windscreen wipers to make unwanted wiping movements.

This may damage the wiper rubbers or scratch the windscreen.

- Switch off the windscreen wiper in dry weather.



#### NOTE

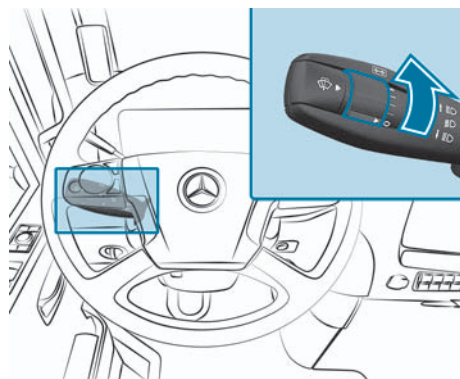
##### Damage to the windscreen caused by the windscreen wipers being activated when starting a new journey

If the windscreen wipers are not switched off before parking the vehicle, this may cause the windscreen wipers to make unwanted wiping movements when starting a new journey. This may damage the wiper blades or windscreen,

particularly if the windscreen is dirty or covered with ice.

- ▶ Switch off the windscreen wipers before parking the vehicle.

Worn or damaged wiper blades can cause smearing on the windscreen. This can cause malfunctions in vehicles with a rain sensor.



Wiper switch on the combination switch

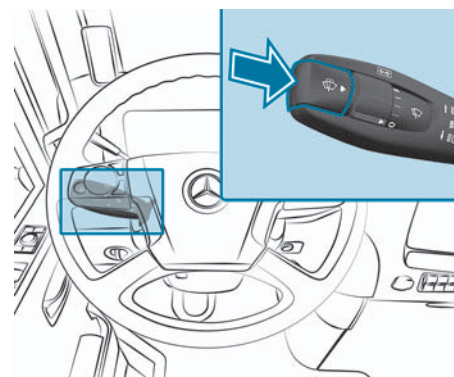
- 0** Windscreen wipers off
- ...** Slow intermittent wiping or wiping with the rain sensor
- ....** Fast intermittent wiping or wiping with the rain sensor

- Continuous wiping, slow
- ==** Continuous wiping, fast

- ▶ **To switch on:** Switch on the vehicle.
- ▶ Turn the wiper switch to the desired position depending on the intensity of the rain.

**Vehicles with rain sensor:** If the **...** or **....** setting is used, the appropriate wipe frequency will be set depending on the precipitation. In the position **....**, the rain sensor is more sensitive than in the position **...**, causing the windscreen wipers to wipe more frequently. If the rain sensor fails, the windscreen wiper will automatically switch to the wiping interval corresponding to the position of the switch.

### Using the windscreen washer system



Combination switch (example)

- ▶ **Single wipe:** Briefly press the  button.
- ▶ **Wiping with washer fluid** Press and hold the  button.

### Switching the windscreen heater on/off

The windscreen heater is ready for operation when the vehicle is started and

## Driver cockpit

### Power supply

switches off automatically after approx. 15 minutes.

- **To switch on/off:** In the multimedia system, under the menu **Climate control**, press the button .
- ◄ When the indicator lamp on the  button comes on, the windscreen heater function is switched on.

### Notes on winter operation

#### Headlamps



#### NOTE

#### Damage to the headlamps when removing ice

If you use an ice scraper to remove ice and snow from the headlamp, you may scratch the plastic cover of the headlamp.

- Use a de-icer spray suitable for plastic covers.

### Windscreen wipers

At temperatures of approximately 5°C to -5°C and during snowfall, direct the air to the windscreen ( or  position) using the air-distribution control (→ Page 75). You can also switch on the windscreen heater. With these settings, the wiper blades on the windscreen will be heated. You can thereby prevent smearing or the snow freezing on the wiper rubber.

### Power supply

#### Overview of sockets



Example: 24 V socket

Depending on the vehicle equipment, 12 V and 24 V sockets can be fitted to the cockpit.

#### 12 V sockets

You must not exceed the maximum permissible load of 180 W (15 A) for the 12 V socket.

## Battery temperature warning system



### Note

The 12 V socket is not labelled.

### 24 V sockets

If you are using multiple 24 V sockets at the same time, you must not exceed a maximum total load of 360 W.

### Battery temperature warning system

#### Function of the battery temperature warning system

The temperature of your vehicle's high-voltage battery is monitored.



### Danger

#### Risk of fire when the high-voltage battery is overheated

When the temperature of a high-voltage battery is too high, there is a risk of fire.

When the battery temperature warning system determines that the temperature of a high-voltage battery is too high, a continuous warning tone sounds via the alarm system in the cab.

The alarm system's LED in the cab also lights up red.

- ▶ Stop the vehicle immediately in accordance with the traffic conditions.
- ▶ If possible, park the vehicle outdoors and keep a safe distance from other vehicles and buildings.
- ▶ Do not continue driving.
- ▶ Ensure that all vehicle occupants exit the vehicle immediately.
- ▶ If heat or smoke develops, leave the danger zone, get as far away from the vehicle as possible, and call the fire service immediately.
- ▶ Consult a qualified specialist workshop even if there are no external signs of a fire.

When a high-voltage battery is overheated, a display message in the instrument cluster may also be shown, which is independent of the battery temperature warning system (→ Page 301).

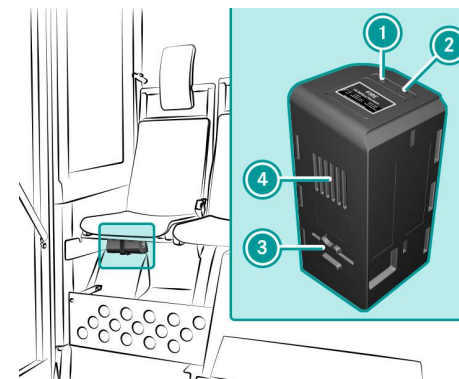
The system is supplied with voltage via the 24 V on-board electrical system of the vehicle. To ensure that the alarm system

also functions if the 24 V vehicle electrical system fails, the alarm system is also equipped with a 9 V battery.



### Note

The alarm system is functional even when the vehicle has been switched off.



Example: alarm system in the cab

The status of the system is signalled via LED ② on the alarm system:

- Green LED – the system functioning correctly.
- Red LED – fire alarm

## Driver cockpit

### Practical tips

A permanent warning tone will also sound.

- LED does not light up – fault in the alarm system's power supply.  
An intermittent warning tone also sounds.

If the LED ② lights up red and a continuous warning tone sounds:

- The high-voltage battery has overheated. There is a risk of fire!
- Stop the vehicle immediately in accordance with the traffic conditions.
- If possible, stop the vehicle in the open air and ensure that all vehicle occupants get out.
- If smoke is present, leave the danger zone and call the fire service immediately.
- Consult a qualified specialist workshop even if there are no external signs of a fire.

If LED ② does not light up and an intermittent warning tone sounds, have the alarm system checked at a qualified specialist workshop.

Never cover alarm system speakers ④ (e.g. with clothing).

**Testing the alarm system:** The function of the alarm system can be tested by pressing button ①. While button ① is pressed, an LED lights up red and a warning tone sounds. If you press and hold button ① and an alarm is not triggered, have the alarm system checked at a qualified specialist workshop.

**Replacing the battery:** The alarm system is also equipped with a 9 V battery in the battery compartment ③. Replace the 9 V battery annually and after a 24 V on-board electrical system battery failure. Test the alarm system after replacing the battery.

### Practical tips

#### Stowage spaces and stowage compartments

#### Notes on stowage spaces and compartments



**Danger**

#### **Risk of injury due to objects being stowed incorrectly**

If objects in the vehicle interior are stowed incorrectly, they may slide or be thrown around and hit vehicle occupants. In addition, cup holders, open stowage spaces and mobile phone receptacles will not always be able to retain all objects that they contain in the event of an accident.

There is a risk of injury, particularly in the event of braking manoeuvres or abrupt changes in direction.

- ▶ Always stow objects such that they cannot be thrown around in such situations.

- ▶ Always make sure that objects do not protrude from stowage spaces, luggage nets or stowage nets.
- ▶ Close the lockable stowage spaces before starting a journey.
- ▶ Stow and secure objects that are heavy, hard, pointed, sharp-edged, fragile or too large in the load compartment.

**Danger**

### **Risk of injury if the maximum permitted load is exceeded**

Objects may be flung out of the stowage compartment. There is a risk of injury for vehicle occupants, particularly in the event of sudden braking or abrupt changes in direction.

- ▶ Always comply with the maximum permitted load of the stowage compartment.

**Danger**

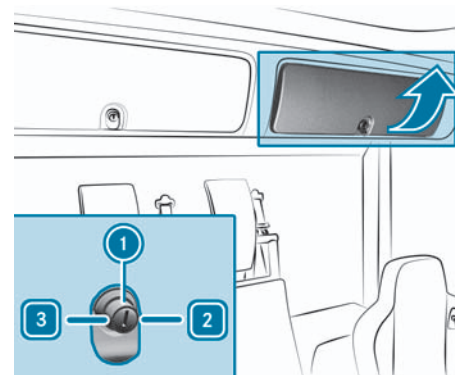
### **Risk of injury if the maximum permitted load is exceeded**

Objects may be flung onto the road surface.

- ▶ Always comply with the maximum permitted load of the stowage compartment.
- ▶ Always ensure that the stowage compartment is locked before every journey.

Do not place loads of more than 8 kg in the stowage compartment with cover.

### **Opening and closing stowage compartments**



Stowage compartment (example)

- ▶ Insert the key into the lock ① on the left-hand or right-hand stowage compartment flap.
- ▶ **To lock:** Turn the key to the ③ position.
  - ◁ The stowage compartment flap will then have been locked.
- ▶ **To release:** Turn the key to the ② position.

## Driver cockpit

### Communications

- ◁ The stowage compartment flap will then have been unlocked.
- ▶ **To open:** Remove the key in position **2**.
- ▶ Press the lock cylinder and swing the stowage compartment flap up.

#### Notes on cup holders / bottle holders



**Danger**

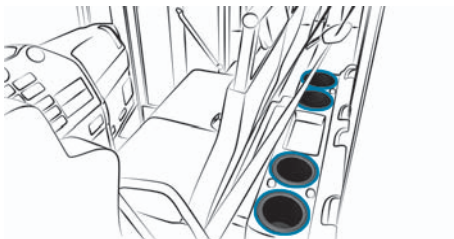
#### Risk of accident or injury when using the cup holder while driving

The cup and bottle holders cannot be used to secure containers while the vehicle is in motion.

If you use cup or bottle holders while the vehicle is in motion, containers may be thrown around and liquid may be spilled. The vehicle occupants may come into contact with the liquid and if it is hot, they could be scalded. You could be distracted from traffic conditions and you may lose control of the vehicle.

- ▶ Only use the cup and bottle holder when the vehicle is stationary.

- ▶ Only put suitable containers into the cup or bottle holders
- ▶ Close the container, particularly if the liquid is hot.



Cup holders (example)

#### Activating/deactivating the air horn/horn

- ▶ In the multimedia system, under the menu **Operation** item **Switch**, press the button .
- ◁ When the indicator lamp on the  button lights up, the horn has switched over to the air horn. If you sound your horn after that, it will be the air horn.

#### Communications

#### Telephone

#### Notes on the telephone



**Danger**

#### Risk of distraction from information systems and communications equipment

If you operate information systems and communications equipment integrated in the vehicle when driving, you could be



distracted from the traffic situation. You could also lose control of the vehicle.

- ▶ Only operate this equipment when the traffic situation permits.
- ▶ If you cannot be sure of this, stop the vehicle in accordance with the traffic conditions and operate the equipment while the vehicle is stationary.

Observe the legal requirements for the country you are currently in while operating the telephone or other communications equipment.

The vehicle may be equipped with an integrated Bluetooth® hands-free system. To charge the Bluetooth® mobile phone in the vehicle, you will need a suitable mobile phone charger bracket. These are available from retailers of Mercedes-Benz Trucks accessories.

You can operate the mobile phone using the  and  buttons on the multi-function steering wheel.

The Daimler Truck AG installation specifications must be observed if you subsequently install one of the following communication devices:

- Mobile phone
- Two-way radio
- Fax machine

#### Notes on climate control



#### NOTE

##### **Damage to the air-conditioning system due to rare use**

If the air-conditioning system is not used for an extended period of time, the refrigerant compressor may be damaged.

- ▶ Switch on the air-conditioning system for at least approx. ten minutes once a month.

The heating and air-conditioning system are operational only if the vehicle has been started.

The air-conditioning system regulates the temperature and air humidity in the cab.

The filter installed largely filters out dust particles, pollen and unpleasant odours from the outside air that is fed in. An added filter reduces the flow of air into the cab. Depending on the operating conditions and environmental influences, the interval for replacing the filter may be shorter than prescribed.

Using the heating and climate control functions has a direct negative effect on the vehicle's range.

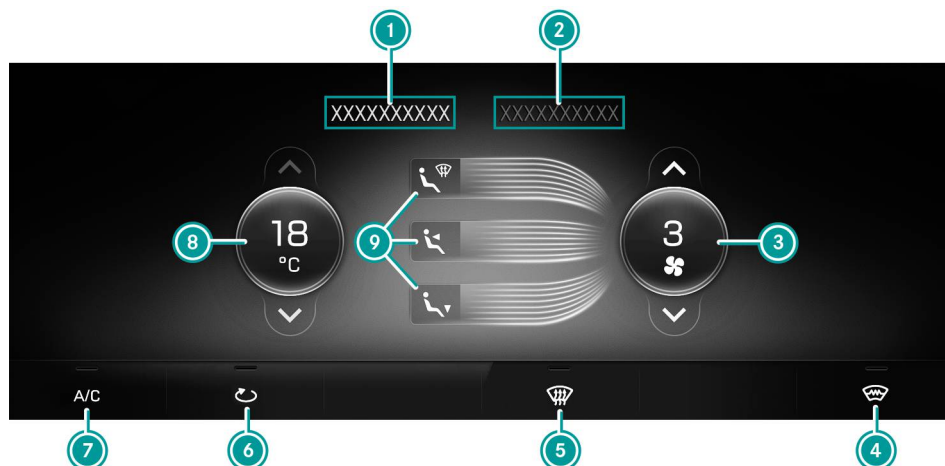
When the state of charge of the high-voltage battery is low, the performance of the heating and climate control functions is reduced.

Observe the following notes to minimise energy consumption:

- Select as low a blower setting as possible.
- Use the windscreen demisting function only briefly until the misted-up windscreen is clear again.
- Ventilate the vehicle briefly in warm weather. This will accelerate the cooling process and the desired temperature will be reached more quickly.

## Overview of the multimedia system climate control functions

### Overview of the multimedia system climate control functions



Climate control menu (example: moving vehicle)

- ① Climate control menu
- ② Departure time menu
- ③ Sets the airflow ( → Page 75)
- ④ Switching the windscreen heater on/off ( → Page 65)
- ⑤ Demists the windscreen ( → Page 75)
- ⑥ Switching air-recirculation mode on/off ( → Page 75)
- ⑦ Switching the air-conditioning system on/off ( → Page 74)
- ⑧ Sets the temperature ( → Page 74)
- ⑨ Sets the air distribution ( → Page 75)

Press the button below the multimedia display to access the climate control menu shown above.

### Note

The functions shown on the display after the climate menu opens depend on whether the vehicle is stationary or moving.

## Climate control

### Cooling with air dehumidification

#### Note

Depending on the vehicle's equipment, functions/buttons may not be present.

Some functions have an indicator lamp above the symbol. The function is switched on if the indicator lamp is lit.

#### Cooling with air dehumidification

##### Notes on cooling with air dehumidification

The air-conditioning system is only operational if the vehicle has been started.

In vehicles with an air-conditioning system, you can dehumidify and cool the interior air to a set temperature in warm weather.

You can switch between the  (cooling without air dehumidification) and  Dry (cooling with air dehumidification) functions by pressing and holding the button .

#### Note

Depending on the function selected, the display on the button will change.

The "Cooling without air dehumidification" function consumes less fuel than the "Cooling with air dehumidification" function.

#### Note

Condensation may drip from the underside of the vehicle when cooling mode is active.

#### Switching cooling with air dehumidification (air-conditioning system) on/off

##### Switching on cooling without air dehumidification

- ▶ In the multimedia system, in the **Climate Control** menu, briefly press the button .

##### Switching cooling with air dehumidification on

- ▶ In the multimedia system, in the **Climate Control** menu, press and hold the button .
  - ◁ The display for button  will switch from  to  Dry.
- ▶ Briefly press the Dry button .

##### Deactivating

- ▶ Briefly press the Dry button  or  again.

#### Setting the temperature

It is possible to set the temperature in the **Climate control** menu of the multimedia system or using the buttons under the multimedia system.

##### Setting via the multimedia system

- ▶ In the **Climate control** menu, increase or reduce the set value using the buttons  or  (→ Page 73).

##### Setting via buttons

## Setting the air distribution/airflow

- Press button  or  to increase or reduce the setting.

### Setting the air distribution/airflow

#### Air distribution

The symbols for air distribution have the following meanings:

|                                                                                   |                                                                                   |                                                                      |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|
|  |  | Directs additional air towards the windshield                        |
|  |                                                                                   | Directs additional air towards the windshield and throughout the cab |
|  |                                                                                   | Directs additional air into the footwell                             |
|  |                                                                                   | Directs air via the centre and side vents only                       |

- In the multimedia system, set the air distribution that you require by pressing the corresponding buttons under the [Climate control](#) menu.

#### Airflow

- In the multimedia system, in the menu [Climate control](#) increase/reduce the airflow  using the buttons  / .

### Demisting/defrosting the windscreen and windows

Select the following setting only for as long as it takes for the windscreen to be clear again.

- **Vehicles with windshield heater:**  
Switch on the windscreen heater ( → Page 65).
- In the multimedia system, under the menu [Air conditioning](#), press the button .
- Open the side air vents completely and turn them outwards to the side windows ( → Page 76).

### Switching air-recirculation mode on/off

#### Note

When air-recirculation mode is switched on, the windows may mist up more quickly, especially in low outside temperatures. Switch air-recirculation mode on only briefly, e.g. when dust or unpleasant odours are causing annoyance.

- Close the windows and roof hatch.
- In the multimedia system, under the menu [Air conditioning](#), press the button .

### Setting pre-entry climate control for departure time

#### Danger

**Danger to life due to exposure to extreme heat or cold in the vehicle**

If persons – particularly children – are subjected to prolonged exposure to

## Climate control

### Air vents

intense heat or cold, there is a risk of severe injury or even death.

- ▶ Never leave persons – particularly children – unattended in the vehicle.

Multimedia system:

  **Climate control**  **Departure time**

If the vehicle is connected to a charging station, you can set the pre-entry climate control for the cab to a set departure time.

- ▶ Set the departure time  
(→ Page 204).

### Air vents

#### Notes on air vents



**Danger**

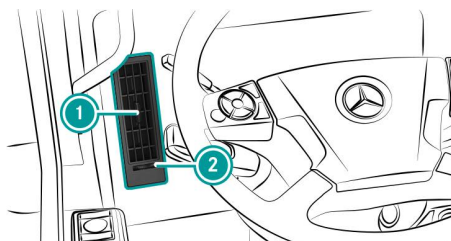
**Risk of burns or frostbite due to being too close to the air vents**

Very hot or very cold air can flow from the air vents.

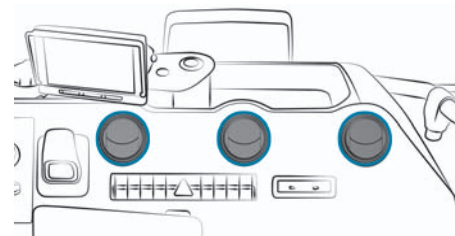
- ▶ Make sure that all vehicle occupants always maintain a sufficient distance from the air vents.
- ▶ If necessary, direct the airflow to another area of the vehicle interior.

Always keep all air vents and ventilation grilles in the cab free so that the air can flow into the cab unhindered.

### Adjusting the air vents



Side air vent (example on left-hand side)



Centre air vents

- ▶ **To open:** Turn adjustment wheel ② to position .
- ▶ **To close:** Turn adjustment wheel ② to position .
- ▶ **To adjust:** Use air vent on handle ① to slide into the required direction.

or

- ▶ Adjust the air distribution by opening and closing the flaps on the air vent.

### Instrument cluster

#### Notes on the instrument cluster



**Danger**

#### Risk of distraction from information systems and communications equipment

If you operate information systems and communications equipment integrated in the vehicle when driving, you could be distracted from the traffic situation. You could also lose control of the vehicle.

- ▶ Only operate this equipment when the traffic situation permits.
- ▶ If you cannot be sure of this, stop the vehicle in accordance with the traffic conditions and operate the equipment while the vehicle is stationary.



**Danger**

#### Risk of accident if the instrument cluster fails

If the instrument cluster fails or malfunctions, you will not detect any limitations to the functioning of safety-relevant systems. This may impair operating safety.

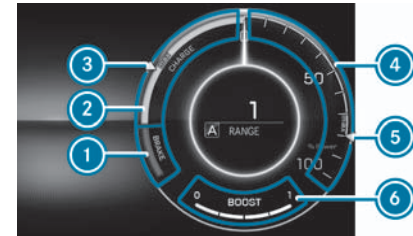
- ▶ Park the vehicle safely as soon as possible and notify a qualified specialist workshop.

Observe the legal requirements for the country you are currently in while operating the instrument cluster.

#### Function of the power availability display

The power availability display provides information during the journey about when electrical power is being delivered or recovered through recuperation.

The power availability display can help you to develop an energy-saving driving style.



Power availability display (example)

- ① Braking power (service brake)
- ② Recuperated energy (recuperation) ( → Page 138)
- ③ Maximum possible recuperation
- ④ Output
- ⑤ Maximum possible output
- ⑥ BOOST display

When you start the drive system and the indicator lamp **READY** is displayed in the instrument cluster, the vehicle is ready to drive. The needle moves to position **0**.

#### Output

The current power output is shown on the drive power scale ④.

The maximum mark ⑤ shows a restriction of the currently available drive power.

## Instrument cluster and on-board computer

### Instrument cluster

If you have selected the drive program **RANGE**, **MUNICIPAL** or **ECONOMY**, the maximum mark ⑤ is shown in green. In this drive programs, an automatic limitation of the available output takes place.

#### **i** Note

For more information on the drive programs, see the section "Drive programs and drive functions" (→ Page 141).

If the maximum mark ⑤ is shown in yellow, the following may be the cause:

- The high-voltage battery is not yet at operating temperature.
- The high-voltage battery's state of charge is too low.
- The drive system is malfunctioning.

#### Recuperation

The recuperation scale ② shows how much energy is recovered during recuperation.

The maximum mark ③ shows a restriction of the currently possible recuperation.

If the maximum mark ③ is displayed, the following may be the cause:

- The high-voltage battery is not yet at operating temperature.
- The high-voltage battery is fully charged.
- The drive system is malfunctioning.

#### Braking power

The braking power scale ① shows how much energy was expended during deceleration by the service brake.

Energy expended by the service brake cannot be recuperated and is lost.

#### BOOST display

In the event of a kickdown, increased output can be called up for a short time.

The BOOST display ⑥ shows how long this increased drive power is still available.

The BOOST display ⑥ fills up during normal driving operation and empties during high output of the drive system.

#### **i** Note

Even without kick-down, the BOOST display ⑥ may slowly empty during high power delivery.

#### Function of the charge level display



Charge level display (example)

Charge level display ① shows you the charge level of the high-voltage battery and the estimated range.

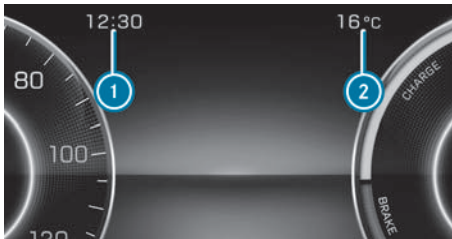
When the drive system is ready for operation and the yellow indicator lamp  on the charge level display lights up, the charge level of the high-voltage battery has reached the reserve range.

Charge the high-voltage battery from a charge level of less than 20%.



You can find information about charging the high-voltage battery in the section "Charging the high-voltage battery" (→ Page 201).

#### Overview of time and outside temperature



Example: instrument cluster

You should pay special attention to road conditions when temperatures are around freezing point.

Changes in the outside temperature will be displayed after a short delay.

Switch on the vehicle. The display on the speedometer will show the time ① and outside temperature ②.

#### Odometer overview



Trip distance/total distance instrument cluster menu

The **Kilometre reading** menu on the instrument cluster also shows the trip distance ① and total distance ②.

#### On-board computer

##### Notes on/overview of the on-board computer



**Danger**

##### **Risk of distraction by information systems and communications equipment**

If you operate information systems and communications equipment integrated in the vehicle when driving, you may be

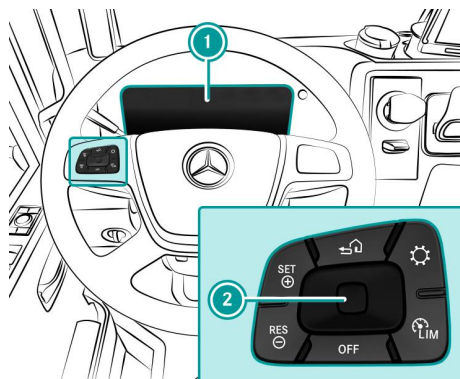
distracted from what is happening in traffic. You may also lose control of the vehicle.

- ▶ Only operate this equipment when the traffic situation permits.
- ▶ If you cannot be sure of this, stop the vehicle in accordance with the traffic conditions and operate the equipment while the vehicle is stationary.

You must observe the legal requirements for the country in which you are currently driving when operating the on-board computer.

## Instrument cluster and on-board computer

### On-board computer



On-board computer (example)

- ① Instrument cluster
- ② To navigate/confirm/select on the on-board computer

 Main menu and back button

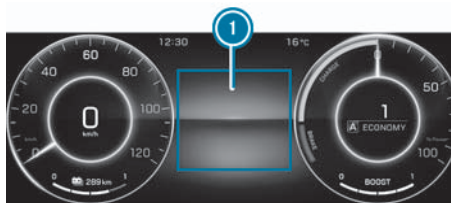
You can operate the on-board computer via the left-hand button group on the multifunction steering wheel.

The on-board computer will provide you with the following information when you are driving:

- Energy consumption
- Trip time
- Operating statuses

- Maintenance due dates
- Malfunctions
- Causes of malfunctions
- Actions to be taken

### Overview of menus on the on-board computer



Instrument cluster (example)

- ① Menu display
-  Supply pressure
-  Distance
-  Trip computer
-  Tachograph
-  Assistance
-  Range/energy consumption
-  Eco Support
-  Telephone

 Audio

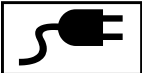
You can use the button group on the left of the multifunction steering wheel to navigate through the menus and to open/close individual menus (→ Page 79).

### Note

The number and order of the menus depends on your vehicle's equipment and the type of vehicle.

## Warning-/indicator lamps 24-V-on-board electrical system and high-voltage system

### Warning-/indicator lamps 24-V-on-board electrical system and high-voltage system

| Warning/indicator lamp                                                                                                                | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>24 V on-board electrical system warning lamp</p> | <p>The red 24 V on-board electrical system warning lamp is lit.</p> <p>There is a malfunction in the 24 V on-board electrical system.</p> <ul style="list-style-type: none"> <li>► Refer to the display message in the instrument cluster ( → Page 312).</li> <li>► Contact a qualified specialist workshop.</li> </ul>                                                                                                                                                                                                                                                                |
|  <p>Ready to drive indicator lamp</p>                | <p>The  indicator lamp lights up in the instrument cluster after the vehicle has been started.</p> <p>The drive system is ready to drive.</p>                                                                                                                                                                                                                                                                                                                                                         |
|  <p>Charging cable connected warning lamp</p>        | <p>The charging cable connected warning lamp is lit.</p> <p> (grey): The charging cable connector is inserted in the vehicle socket.</p> <p> (blue): The high-voltage battery is charging.</p> <p>Start-off protection is active. The drive system cannot be started.</p> <p>If you would like to start the vehicle:</p> <ul style="list-style-type: none"> <li>► End the charging process ( → Page 208).</li> </ul> |

## Instrument cluster and on-board computer

### Warning-/indicator lamps 24-V-on-board electrical system and high-voltage system

| Warning/indicator lamp                                                                                                                        | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Reserve high-voltage battery warning lamp</p>            | <p>The yellow reserve high-voltage battery warning lamp.</p> <p>The charge level of the high-voltage battery has dropped into the reserve range.</p> <p>► Charge the high-voltage battery ( → Page 201).</p>                                                                                                           |
|  <p>High-voltage battery warning lamp</p>                    | <p>The high-voltage battery warning lamp is lit while the vehicle is ready to drive <span>READY</span>.</p> <p>The high-voltage battery has a malfunction.</p> <p>► Observe the yellow ( → Page 299) or red display message ( → Page 300) in the instrument cluster.</p>                                               |
|  <p>High-voltage on-board electrical system warning lamp</p> | <p>The high-voltage on-board electrical system warning lamp is lit while the vehicle is ready to drive <span>READY</span>.</p> <p>There is a malfunction in the high-voltage on-board electrical system.</p> <p>► Observe the yellow ( → Page 302) or red display message ( → Page 303) in the instrument cluster.</p> |
|  <p>Electrical drive warning lamp</p>                        | <p>The electrical drive warning lamp is lit.</p> <p>There is a malfunction in the drive system.</p> <p>► Observe the yellow ( → Page 303) or red display message ( → Page 303) in the instrument cluster.</p>                                                                                                          |

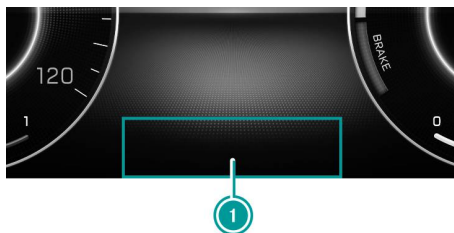
## Overview of warning/indicator lamps in the status area

### Overview of warning/indicator lamps in the status area

#### Safety notes

If you ignore warning/indicator lamps, you may not be able to acknowledge failures or malfunctions in components or systems. Driving/braking characteristics may be affected and the reliability and road safety of your vehicle may be impaired. Have the affected system checked and repaired by a qualified specialist workshop. Always observe warning/indicator lamps and take the corresponding action.

#### Overview



Status area on the instrument cluster

If there is a fault, warning, or operating status information, a warning/indicator lamp will come on in the status area ① of the instrument cluster. The warning/indicator lamp will light up in a different colour depending on the priority of the fault, warning, or operating status information. The warning/indicator lamp may also light up in addition to a display message.

|  |                                                                                                     |
|--|-----------------------------------------------------------------------------------------------------|
|  | Work lamps                                                                                          |
|  | Automatic high-voltage emergency shut-off malfunction/crash sensing ( → Page 28)                    |
|  | Seat belt warning ( → Page 31)                                                                      |
|  | Assistance systems faulty ( → Page 315)                                                             |
|  | Lane Keeping Assist switched off ( → Page 180)                                                      |
|  | Frontguard Assist malfunction ( → Page 162)                                                         |
|  | The audible warning from Sideguard Assist has been deactivated ( → Page 188).                       |
|  | Sideguard Assist only available to a limited extent (e.g. only available on one side) ( → Page 181) |

|  |                                                                                                |
|--|------------------------------------------------------------------------------------------------|
|  | Active Brake Assist unavailable ( → Page 181)                                                  |
|  | Coolant level too low ( → Page 225)                                                            |
|  | (Grey) Energy-saving mode active                                                               |
|  | (Red) 24 V on-board electrical system malfunction ( → Page 312)                                |
|  | (Yellow) High-voltage battery malfunction; see corresponding display message ( → Page 299)     |
|  | (Red) High-voltage battery malfunction; see corresponding display message ( → Page 300)        |
|  | (Red) High-voltage battery ageing, vehicle start not possible for safety reasons ( → Page 304) |
|  | (Yellow) Electric drive malfunction; see corresponding display message ( → Page 303)           |
|  | (Red) Electric drive malfunction; see corresponding display message ( → Page 303)              |
|  | Service due date; see corresponding display message                                            |
|  | Trailing axle ( → Page 197)                                                                    |
|  | Steered additional axle centred ( → Page 198)                                                  |
|  | Steering angle disparity in steered additional axle ( → Page 198)                              |

## Instrument cluster and on-board computer

### Overview of warning/indicator lamps in the status area

|                                                                                   |                                                                          |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------|
|  | Cargo liftgate (see separate operating instructions)                     |
|  | Electric power take-off system<br>( → Page 212)                          |
|  | Continuous brake malfunction<br>( → Page 137)                            |
|  | Tyre pressure monitor warning message, see corresponding display message |
|  | Tyre pressure monitoring system<br>( → Page 260)                         |
|  | Frequent-stop brake ( → Page 134)                                        |
|  | Reversing assistant ( → Page 188)                                        |
|  | Brake pad wear                                                           |

### Overview and operation

#### Overview of the multimedia system



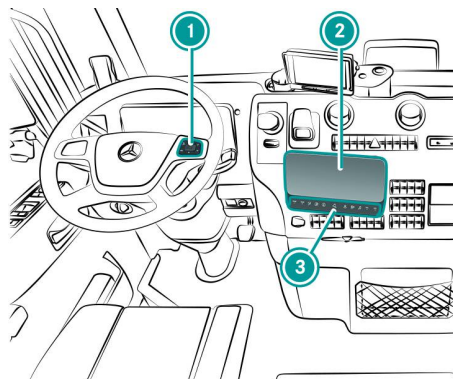
**Danger**

#### Risk of distraction from information systems and communications equipment

If you operate information systems and communications equipment integrated in the vehicle when driving, you could be distracted from the traffic situation. You could also lose control of the vehicle.

- ▶ Only operate this equipment when the traffic situation permits.
- ▶ If you cannot be sure of this, stop the vehicle in accordance with the traffic conditions and operate the equipment while the vehicle is stationary.

You must observe the legal requirements for the country in which you are currently driving when operating the multimedia system.



You can operate the multimedia system with the following central control elements:

- Touch Control ①  
Operation takes place using cursor control.
- Multimedia system display ②
- Buttons on the operating strip ③

#### Notes on the multimedia system display

Observe the notes on caring for the interior.

The visibility of the displays is affected by the following conditions:

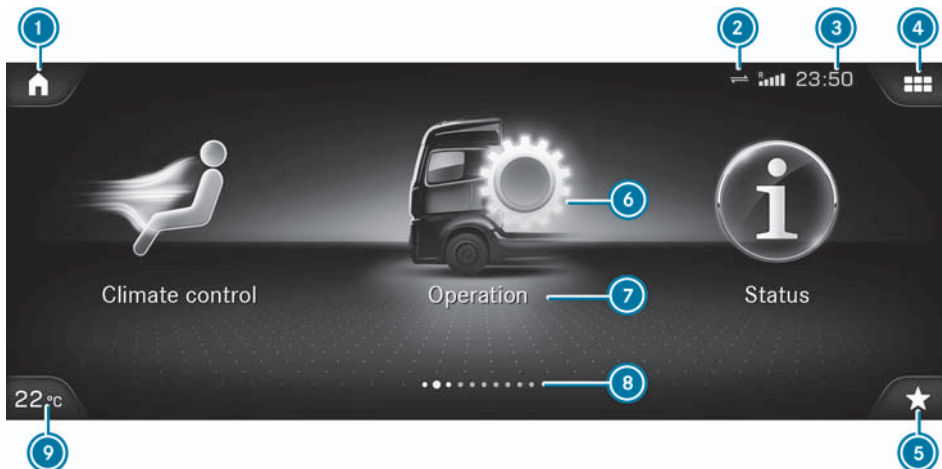
- Seat position
- Setting of the image position
- Light conditions
- Wet roadway
- Objects on the display cover
- Polarisation in sunglasses

**Automatic temperature switch-off:** if the temperature is too high, the brightness is subsequently reduced. The multimedia system display may then switch off completely for a while.

## Multimedia system

### Overview and operation

#### Home screen overview



- ① Calls up the home screen
- ② Connection status with the telematics server
- ③ Time
- ④ Calls up the virtual switch
- ⑤ Favourites
- ⑥ Calls up an application using the symbol
- ⑦ Name of the application, beneath the current selection or information
- ⑧ Number of applications and currently selected display area
- ⑨ Display of the temperature set for the climate control

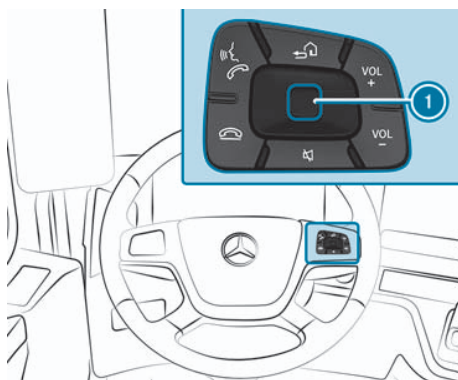
The following menus are available:

- Climate control
- Operation
- Status
- Lights
- Telephone
- Radio
- Media
- Connect



### Touch Control

### Operating Touch Control (multimedia system)



Right-hand button group on the multifunction steering wheel

- ① Navigation in the multimedia system: Swipe using Touch Control  
To confirm/select
- Home screen and Back button (multimedia system)
- Vol. Increases the volume
- Vol. Decreases the volume
- Mutes

- Ends/rejects a call
- Makes/takes a call
- Activates the voice control system

- ▶ **To select a menu item:** swipe up, down, left or right on Touch Control ①.
- ▶ Press Touch Control ①.
- ▶ **To open a menu or list:** Press Touch Control ①.
- ▶ **To close the menu or list:** Press the button .
- ▶ **Moving the digital map:** Swipe in all directions.

### Setting the sensitivity for the Touch Control

Multimedia system:

- ▶ Select **Fast**, **Moderate** or **Slow**.

### Switching acoustic operating feedback for the Touch Control on or off

Multimedia system:

The function is supported by the selection in a list.

- ▶ Select **Acoustic operating feedback**.
- ▶ Set to **Normal**, **Loud** or **Off**.
  - ◁ If the function is activated you will hear a clicking sound when scrolling in a list. When the beginning or end of the list is reached you will hear another clicking sound.

### Operating the touchscreen

#### Tapping

- ▶ **To select a menu item or entry:** Tap on a symbol or an input.
- ▶ **Character entry via keyboard:** Press briefly on a key.

## Multimedia system

### Overview and operation

#### Single-finger swipe

- ▶ **Navigating in menus:** Swipe up, down, left or right.
- ▶ **Moving the digital map:** Swipe in all directions.

#### Two-finger swipe

- ▶ **To increase or decrease the map scale:** Pull two fingers apart or together.
- ▶ **To turn the map:** Use two fingers to turn to the left or right.

#### Touching, holding and moving

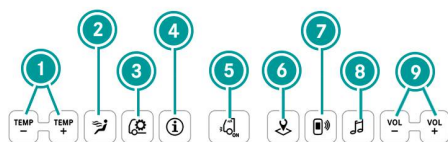
- ▶ **To move the map:** Touch the touchscreen and move your finger in any direction.
- ▶ **To set the volume on a scale:** Touch the touchscreen and move your finger to the left or right.

#### Touching and holding

- ▶ **To call up a global menu in applications:** Touch and hold the touchscreen until the **OPTIONS** menu appears.

#### Applications

##### Calling up applications using the operating strip



- ① Lowers/increases the temperature of the climate control
- ② Air conditioning
- ③ Operation
- ④ Status
- ⑤ Lights
- ⑥ Navigation
- ⑦ Telephone
- ⑧ Radio/media
- ⑨ Volume

- ▶ **To call up the application:** press the desired button.

##### Calling up applications using the home screen

The home screen is shown in the multimedia system display when you start the vehicle.

- ▶ Select the application by swiping and tapping.

or

- ▶ **In any display:** Press the button  on the Touch Control.

or

- ▶ Briefly press the symbol  in the multimedia system display (→ Page 86).
  - ◁ The applications are displayed.
- ▶ Select the application by swiping and tapping.

##### Changing the arrangement of applications on the home screen

- ▶ Call up the home screen.
- ▶ Touch and hold an application on the touchscreen until symbols appear.

- ▶ Tap on  or  and move the application in the menu.
- ▶ Tap on .

### Switching the sound on/off

#### Switching sound off/on at the multifunction steering wheel

- ▶ **Switching off:** Press the button .
  - ◁ The symbol  appears in the status line of the display.
- ▶ **To switch on:** Press the  button or change the media source.

### Note

Please note that you can switch the sound on or off in the menu currently active, e.g. radio, media.

### Adjusting the volume

#### On the multifunction steering wheel

- ▶ **To increase or decrease the volume:** Press the button Vol.  or Vol. .

- ◁ The volume of the current radio or media source is set. The volume of other audio sources can be adjusted separately.

Adjust the volume in the following situations:

- During a traffic announcement

### Favourites

#### Overview of favourites

Favourites offer you quick access to frequently used applications. It is possible to create 20 favourites in total.

You can select favourites from categories or you add favourites directly from an application.

#### Calling up favourites

- ▶ Tap on .
  - ◁ A maximum of five favourites are shown.

### Adding favourites

#### Selecting favourites from categories

- ▶ Tap on .
  - ◁ The favourites are displayed.
- ▶  Select **Create favourites**.
- ▶ Select the category.
- ▶ Select a favourite.
- ▶ **All positions are assigned in the favourites:** Confirm the displayed message with **OK**.
  - ◁ A list shows all the favourites.
- ▶ Select a favourite which should be overwritten.

#### Adding a favourite from an application

Examples for adding from an application are:

- Saving a contact.
- Storing radio stations.
- ▶ **To save a contact as a global favourite:** Select a contact.
  - ◁ The details are displayed.
- ▶ Press and hold a telephone number until the **OPTIONS** menu appears.
- ▶ Select **Save as a favourite**.

## Multimedia system

### Overview and operation

- ◁ The contact is added as a favourite.
- ▶ **To store a radio station as a global favourite:** Select a radio station.
- ▶ Press and hold the radio station until the **OPTIONS** menu appears.
- ▶ Select **Save as a favourite**.
  - ◁ The radio station is added as a favourite.

#### Renaming favourites

- ▶ Tap on .
- ◁ The favourites are displayed.
- ▶ Press and hold a favourite until the **OPTIONS** menu appears.
- ▶ Select **Rename**.
- ▶ Enter the name.
- ▶ Select **OK**.

#### Moving favourites

- ▶ Tap on .
- ◁ The favourites are displayed.

- ▶ Press and hold a favourite until the **OPTIONS** menu appears.
- ▶ Select **Move**.
- ▶ Move the favourite.
- ▶ Select **OK**.

#### Deleting favourites

- ▶ Tap on .
- ◁ The favourites are displayed.
- ▶ Press and hold a favourite until the **OPTIONS** menu appears.
- ▶ Select **Delete**.
- ▶ Select **Yes**.

#### Entering characters

##### Using the character input function



**Danger**

#### Risk of distraction from information systems and communications equipment

If you operate information systems and communications equipment integrated

in the vehicle when driving, you could be distracted from the traffic situation. You could also lose control of the vehicle.

- ▶ Only operate this equipment when the traffic situation permits.
- ▶ If you cannot be sure of this, stop the vehicle in accordance with the traffic conditions and operate the equipment while the vehicle is stationary.

You must observe the legal requirements for the country in which you are currently driving when operating the multimedia system.

Character entry can be carried out with these control elements:

- Touch Control
- Touchscreen (multimedia system display)

Character entry takes place by cursor control when using the Touch Control.

Character input can be started with a control element and resumed with another.

- ▶ When the keyboard is shown, enter the characters on the control element by swiping and pressing or by tapping (touchscreen).

You can use the character input function in the following situations, for example:

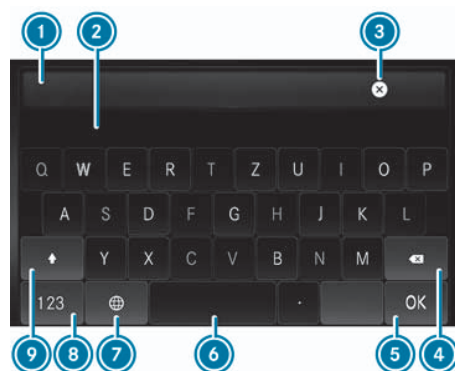
- Renaming a favourite.
- Entering a telephone number.

### Entering characters on the touchscreen

#### Requirements

- An online connection is required for some functions.

### Using the keyboard



- ① Input line
- ② Shows suggestions during entry (if available).
- ③ Deletes an entry.
- ④ Delete  
Pressing briefly deletes the last character entered.  
Pressing and holding deletes the entry.
- ⑤ Accepts an entry.
- ⑥ Enters a space.
- ⑦ Changes the keyboard language.

- ⑧ Switches to digits and special characters (level 2).  
ABC: switches to letters (level 1).
  - ⑨ Pressing briefly switches between upper-case and lower-case letters.  
Pressing and holding switches to upper-case letters permanently.
- ▶ Call up character entry, e.g. rename a favourite (→ Page 90).
    - ◁ The keyboard is shown.
  - ▶ Press briefly on a character key.
    - ◁ The character is entered in input line ①. Suggestions are shown in ②.
  - ▶ **Select suggestion:** Selecting one of the suggestions.
  - ▶ If available, display additional suggestions with or .
  - ▶ Resume character input.
  - ▶ **To enter alternative characters:**  
Press and hold a character.
  - ▶ Select the character.
  - ▶ **To end character entry:** Press the button .

## Multimedia system

### System settings

#### Setting the keyboard for character entry

Multimedia system:

→  >> Operation >> Settings >>  
Display and operation >> Controls >>  
Keyboards

- ▶ Confirm [Select keyboards](#).
  - ◁ Select the keyboard language in the list.

#### System settings

#### Display

#### Configuring display settings

#### Requirements

- The vehicle is stationary.

Multimedia system:

→  >> Operation >> Settings >>  
Display and operation

#### Display brightness

- ▶ Select [Display brightness](#).
- ▶ Select a brightness value.

#### Switching the display off/on

- ▶ **Off:** Select [Display off](#).
- ▶ **To switch on:** Select a menu on the operating strip.

#### Activating/deactivating the assistance view of Distance Assist DISTRONIC

- ▶ **Off:** Switch off the [Assistance view during distance control](#) .
- ▶ **On:** Switch on the [Assistance view during distance control](#) .

#### Time and date

#### Setting the time and date automatically

#### Requirements

- **If a tachograph or GPS device is available:** [Manual time adjustment](#) is not available as an option.

#### Note

If a tachograph or a GPS device is available in the vehicle, you can only change the time zones. The time and date are set automatically when your vehicle is equipped with a tachograph or GPS device.

Multimedia system:

→  >> Operation >> Settings >>  
Display and operation >> Time and date

- ▶ Switch off [Manual time adjustment](#) .
- ◁ The time and date are set automatically for the selected time zone and summer time option.

#### Note

The correct time is a requirement for the following functions:

- Route guidance with time-dependent traffic routing
- Calculation of the expected time of arrival

- Alarm function

### Setting the time zone

#### Requirements

- **For manual time zone setting:**  
The [automatic time zone](#) function is switched off.

Multimedia system:

→  → Operation → Settings →  
Display and operation → Time and date

#### Setting the time zone manually

- ▶ [Time zone:](#) must be selected.

The list of countries is displayed.

- ▶ Select a country and time zone if required.
  - ◁ The set time zone is displayed behind [Time zone:](#) .

#### Setting the time zone automatically

- ▶ Switch on [Automatic time zone](#).

### Setting the time and date format

Multimedia system:

→  → Operation → Settings →  
Display and operation → Time and date  
→ Setting a format

- ▶ Set the date and time format •.

### Setting the time and date manually

#### Requirements

- The [Manual time adjustment](#) function is activated.
- **For setting the date manually:** No GPS or tachograph is installed in the vehicle.

Multimedia system:

→  → Operation → Settings →  
Display and operation → Time and date

- ▶ **Setting the time:** Select [Set time](#).

#### Note

In vehicles with GPS, the time is set automatically according to the selected time zone.

- ▶ **Setting the date:** Select [Set date](#).

#### Note

In vehicles with GPS, the date cannot be set even if the time is adjusted manually. The date is set automatically according to the selected time zone.

### Setting the alarm clock

The [ALARM CLOCK](#) menu shows buttons for up to 15 alarms in a carousel. Only the alarm whose button was last pressed is active. If no repetition is set, the wake-up alarm is active once. When a new alarm clock is set up, it is active.

Multimedia system:

→  → Operation → Timer control  
→ [ALARM CLOCK](#)

## Multimedia system

### System settings

- ▶ Press the "Plus" button.
  - ◁ An alarm clock with the preset values is set.
- ▶ Open [Edit](#).
  - ◁ The corresponding button is shown in the display.

An alarm clock shows the following information:

- Wake-up alarm active or inactive
- the time set
- Days of the week for which a repetition has been set. If no day of the week is selected, the button does not display repeat information.
- a symbol for sound with two possible options: Radio and Standard Sound
- A symbol for the alarm light: when the alarm light is off, no symbol is displayed.

### Connectivity

#### Bluetooth®

#### Bluetooth® functions

Bluetooth® is a technology for wirelessly transferring data over short distances up to approx. 10 m.

You can use Bluetooth® to connect your mobile phone to the multimedia system and use the following functions, for example:

- Hands-free system with access to the following options:
  - Contacts ( → Page 103)
  - Call lists ( → Page 105)
- Listening to music via Bluetooth® audio

Bluetooth® is a registered trademark of Bluetooth Special Interest Group (SIG) Inc.

### Note

Bluetooth® is not available in all countries.

### Activating/deactivating Bluetooth®

Multimedia system:

 [Settings](#) [Connectivity](#)  
[Wi-Fi and Bluetooth](#)

- ▶ Activate or deactivate Bluetooth®.

### Calling up the device manager

#### Requirements

- The Bluetooth® connection on the Bluetooth® device is active.

Multimedia system:

 [Settings](#) [Connectivity](#)  
[Device manager](#)

If two Bluetooth® devices are connected with the multimedia system at the same time, these are shown in the device manager.



- ▶ Select **Device manager**.

### System language

#### Notes on the system language

This function allows you to determine the language for the menus. The selected language affects the characters available for entry.

#### Setting the system language

Multimedia system:

  
Language

- ▶ Set the language.

#### Setting the units

Multimedia system:

  
Units of measurement

- ▶ Select **Units of measurement**.

- ◀ Select the unit of measurement that you require from the list, e.g. **km** or **mi**.

### Resetting the multimedia system (reset function)

Multimedia system:

  
Resetting

Personal data is deleted, for example:

- Station presets
- Connected mobile phones

A prompt appears again asking whether you really wish to reset.

- ▶ Select **Yes**.

### Sound

#### Calling up the sound menu

Multimedia system:

  
Display and operation

The following functions are available:

- **Equaliser**
- **Balance and fader**
- **Auto. volume adjustment**

- ▶ Select a function.

### Adjusting treble, mid-range and bass settings

Multimedia system:

  
Display and operation

- ▶ Adjust **TREBLE**, **MID-RANGE** or **BASS** settings.

### Activating/deactivating automatic volume adjustment

Multimedia system:

  
Display and operation ▶ Auto. volume adjustment

## Multimedia system

### System settings

Automatic volume adjustment compensates for differing volumes when changing between audio sources.

- Switch [Auto. volume adjustment](#) on or off.

#### Adjusting the balance/fader

Multimedia system:



[Display and operation](#) ► [Balance and fader](#)

#### Adjusting the balance

- Move the volume distribution between the speakers on the right and left sides of the vehicle in the grid shown.

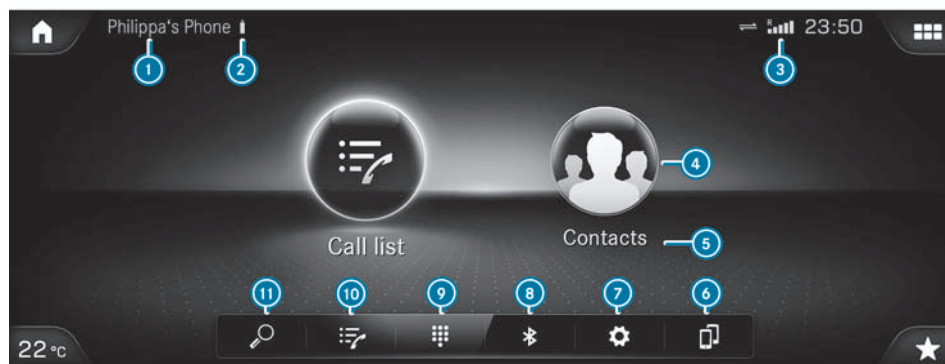
#### Setting the fader

- Move the volume distribution in the displayed grid between the front and rear sections of the vehicle.

### Telephony

### Telephony

### Telephone menu overview



- ① Bluetooth® device name of the connected mobile phone
- ② Battery status of the connected mobile phone
- ③ Signal strength of the mobile phone network
- ④ Calls up an application using the symbol
- ⑤ Name of the application, e.g. call list, contacts
- ⑥ Device manager
- ⑦ Options
- ⑧ Connects a device
- ⑨ Numerical pad
- ⑩ Call list
- ⑪ Contact search

The Bluetooth® interface is available for you to use on your phone. The mobile phone is connected directly via the multimedia system.

Symbols ① to ③ are not shown until a mobile phone has been connected to the multimedia system. The symbols depend on your mobile phone and your mobile phone network provider.

## Multimedia system

### Telephony

#### Bluetooth® profile overview

| Bluetooth® profile of the mobile phone | Function                                                      |
|----------------------------------------|---------------------------------------------------------------|
| PBAP (Phone Book Access Profile)       | Contacts are automatically displayed in the multimedia system |
| HFP (Hands Free Profile)               | Voice control is active                                       |
| BTA (BTAudio)                          |                                                               |
| AVRCP (Player)                         |                                                               |

#### Telephony operating modes overview

The following types of telephone mode are available:

- A mobile phone is connected to the multimedia system via Bluetooth® (→ Page 99).



#### Note

The Bluetooth® audio functionality can only be used with the primary mobile phone (→ Page 112).

#### Notes on telephony



#### Danger

##### **Risk of distraction from operating integrated communication equipment while the vehicle is in motion**

If you operate communications equipment integrated into the vehicle while driving, you may be distracted from traffic conditions. You could also lose control of the vehicle.

- ▶ Only operate this equipment when the traffic situation permits.
- ▶ If you cannot be sure of this, stop the vehicle in accordance with the traffic conditions and operate the equipment while the vehicle is stationary.



#### Danger

##### **Risk of accident from operating mobile communication equipment while the vehicle is in motion**

Mobile communication devices distract the driver from the traffic situation. This

can also cause the driver to lose control of the vehicle.

- ▶ As a driver, only operate mobile communication devices when the vehicle is stationary.
- ▶ As a vehicle occupant, only use mobile communication devices in the designated area, e.g. in the rear passenger compartment.

You must observe the legal requirements for the country in which you are currently driving when operating mobile communication equipment in the vehicle.

You can obtain further information from a Mercedes-Benz Trucks Service Centre.

#### Information on telephony

Call disconnection may occur when the vehicle is in motion in the following situations:

- An area may have insufficient network coverage.
- You move from one transmitter/receiver area (cell) into another and no voice channels are free.

- The SIM card used is not compatible with the network available.
- You are using a mobile phone with "Twincard" and the second SIM card is already logged in to the network.

The multimedia system supports calls with better speech quality in HD Voice®. This is conditional upon HD Voice® being supported by the participants' mobile phones and network providers.

Depending on the quality of the connection, the voice quality may fluctuate.

### Connecting a mobile phone

#### Requirements

- Bluetooth® is activated on the mobile phone (see the manufacturer's operating instructions).
- Bluetooth® is activated on the multimedia system (→ Page 94).

Multimedia system:



#### Searching for a mobile phone

- ▶ Select **Connect new device**.

#### Connecting a mobile phone (authorisation using Secure Simple Pairing)

- ▶ Select a mobile phone.
- ▶ A code is displayed in the multimedia system and on the mobile phone.
- ▶ **Codes match:** Confirm the code on the mobile phone.

#### Note

For older mobile phone models, enter a 1 to 16-digit number code on the mobile phone and on the multimedia system for authorisation.

#### Note

Up to 15 mobile phones can be authorised on the multimedia system.

#### Note

The connected primary mobile phone can also be used as a Bluetooth® audio device (→ Page 112).

#### Connecting a second mobile phone (two phone mode)

#### Requirements

- At least one mobile phone is already connected to the multimedia system via Bluetooth®.

Multimedia system:



- ▶ Select **Connect new device**.
- ▶ Select the mobile phone.
- ▶ Select **With <Device\_name>**.
  - ◀ The selected mobile phone is connected to the multimedia system.

#### Functions of the mobile phone in two phone mode

#### Overview of functions

| Functions of the mobile phone in the foreground | Functions of the mobile phone in the background |
|-------------------------------------------------|-------------------------------------------------|
| Full range of functions                         | Incoming calls                                  |

# Multimedia system

## Telephony

### Interchanging mobile phones (two phone mode)

Multimedia system:



- ▶ Select

After interchanging the mobile phones in one of the submenus, the mobile phone in the foreground is replaced by the mobile phone in the background.

### Changing the function of a mobile phone

Multimedia system:



#### Activating a function

- ▶ Select a grey symbol in the line of a mobile phone.
  - ◁ The corresponding function is activated.

#### Deactivating a function

- ▶ **A function is active:** select the coloured symbol in the line of a mobile phone.
  - ◁ The mobile phone is disconnected from the multimedia system.
- ▶ **Several functions are active:** select a coloured symbol in the line of a mobile phone.
  - ◁ The corresponding function is deactivated.

### Replacing mobile phones

Multimedia system:



- ▶ **No authorised mobile phone available:** Select [Connect new device](#).
- ▶ Select a mobile phone.
- ▶ **Newly authorised mobile phone:** Confirm the number code on the mobile phone.

#### Using in two phone mode

- ▶ Select [With <Device\\_name>](#).

- ◁ A newly authorised mobile phone is connected with the selected mobile phone in two phone mode.

If the mobile phone has already been authorised and connected in single telephone mode, it will be connected in future with the selected mobile phone in two phone mode.

If the mobile phone was previously connected with another mobile phone, this connection is cancelled.

### Disconnecting/de-authorising a mobile phone

Multimedia system:



- ▶ Select in the line of the mobile phone.
- ▶ **To disconnect:** Select [Disconnect](#).

- ◁ If applicable, the mobile phone will be automatically reconnected when the vehicle is next started.

- ▶ **To de-authorise:** Select **De-authorise**.

### Setting the reception and transmission volume

#### Requirements

- A mobile phone is connected (→ Page 99).

Multimedia system:



This function ensures optimal language quality.

- ▶ Select  in the line of the mobile phone.
- ▶ Select **Reception volume** and **Transmission volume**.
- ▶ Set the volume.

### Setting the ringtone

Multimedia system:



- ▶ Select **Ringtones**.
- ▶ Set the ringtone.

#### Note

If the mobile phone supports the transfer of the ringtone, you will hear the ringtone of the mobile phone instead of that of the vehicle.

### Starting/ending mobile phone voice recognition

#### Requirements

- The mobile phone in the foreground is connected with the multimedia system (→ Page 99).

#### Starting mobile phone voice recognition

- ▶ Press and hold the button  on the multifunction steering wheel for at least one second.

- ◁ You can now use the mobile phone voice recognition.

#### To end mobile phone voice recognition

- ▶ Press the button  or  on the multifunction steering wheel.

#### Note

If a mobile phone is connected via the smartphone integration, the voice recognition of this mobile phone is started or stopped.

### calls

#### Telephone operation

Multimedia system:



#### Making a call by entering the numbers

- ▶ Select .
- ▶ Enter a number.
- ▶ Select .

#### Accepting a call

- ▶ Select  **Accept**.

## Multimedia system

### Telephony

#### Rejecting/ending a call

- ▶ Select  **Reject/end**.

#### Putting a call on hold and rejecting a call

- ▶ Select **Call on hold**.
  - ◁ The call is put on hold or rejected.  
Depending on the mobile phone, behaviour when rejecting a call may vary (see manufacturer's operating instructions).
- ▶ If the call is put on hold: Select **Reject** .

#### Activating functions during a call

The following functions are available during a call:

|                                                                                    |                                                                            |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|   | <b>End call</b>                                                            |
|   | <b>microphone off</b>                                                      |
|   | <b>Numerical pad</b> (show to send DTMF tones)                             |
|  | <b>To phone</b> (call is changed from hands-free mode to on the telephone) |

- ▶ Select a function.

#### Putting an active call on hold

- ▶ Tap on the contact or the call.
  - ◁ The call is put on hold.

#### Note

The call on hold can be ended any time or re-activated by tapping again.

#### Conducting a call with several participants

##### Requirements

- There is an active call (→ Page 101).
- Another call connection has been established.

##### Switching between calls

- ▶ **To switch calls:**  Select **calls to be selected**.
- ▶ Select the contact.
  - ◁ The selected call is active. The other call is on hold.

##### Activating a call on hold

- ▶ Select the contact for the call on hold.

#### Ending the active call

- ▶  Select **end Call**.

#### Note

On some mobile phones, the call on hold is activated as soon as the active call is ended.

#### Accepting/rejecting a waiting call

##### Requirements

- There is an active call (→ Page 101).

If there is an incoming call while a call is being conducted, a notification is shown.

Depending on the mobile phone and the mobile phone network provider you will hear a knocking sound.

In addition, you will hear an acoustic signal in two phone mode, if the call is incoming on the other (still not active) mobile phone.

- ▶  Select **Accept**.
  - ◁ The incoming call is active.



The previous call is put on hold if only one mobile phone is connected with the multimedia system. If, during a call, you accept another call on the other phone when in two phone mode, the existing call will be ended.

The previous call is put on hold.

- ▶  Select **Reject**.

### Note

The function and behaviour depend on your mobile phone network provider and mobile phone (see the manufacturer's operating instructions).

## Contacts

### Information about the contacts menu

The contacts menu contains all contacts from existing data sources, e.g. mobile phone or memory card.

Depending on the data source you have the following number of contacts:

- Permanently stored contacts: 3000 entries
- Contacts loaded from the mobile phone: 5000 entries per mobile phone

You can perform the following tasks from the contacts menu:

- Operate the telephone, e.g. call a contact (→ Page 103)
- Navigate
- Additional options

If a mobile phone is connected to the multimedia system (→ Page 99) and automatic calling up (→ Page 103) is activated, the mobile phone's contacts will be displayed automatically.

Based on frequently used contacts as well as incoming and outgoing calls the multimedia system can show suggestions. These will be shown at the top of the contact list.

### Downloading mobile phone contacts

Multimedia system:



- ▶ Select the connected mobile phone.
- ▶ Select **Contacts & Call list**.

### Automatically

- ▶ Switch on **Automatically synchronise contacts**: .

### Manually

- ▶ Switch off **Automatically synchronise contacts**: .
- ▶ Select **Synchronise contacts**.

### Calling a contact

Multimedia system:



- ▶ Select  **Find contact**.
- ▶ Enter characters into the search field.
- ▶ Select the contact.
- ▶ Select the telephone number.
  - ◀ The number is dialled.

## Multimedia system

### Telephony

#### Editing the format of a contact's name

Multimedia system:



- ▶ Select **Name format**.

The following options are available:

- **Surname, first name**
- **Surname first name**
- ▶ Select an option.

#### Overview of importing contacts

##### Contacts from various sources

| Source                                                                                                  | Requirements                                                                                                                  |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
|  Bluetooth® connection | Bluetooth is activated in the multimedia system and on the respective device (see the manufacturer's operating instructions). |

#### Importing contacts into the contacts menu

Multimedia system:



- ▶ Select an option.

#### Calling a contact

Multimedia system:



- ▶ Select  **Find contact**.
- ▶ Enter characters into the search field.
- ▶ Select the contact.
- ▶ Select the telephone number.
  - ◀ The number is dialled.

#### Saving a contact as a favourite

Multimedia system:



- ▶ Press and hold on the desired contact in the phone book.
  - ◀ The contacts are saved as global favourites and appear on the home screen.

#### Deleting favourites in the telephone menu overview

Multimedia system:



- ▶ Tap on .
- ◀ The favourites are displayed.
- ▶ Press and hold a favourite until the **Options** menu appears.
- ▶ Select **Delete**.
- ▶ Select **Yes**.

#### Deleting all favourites

- ▶ Tap on  in the media display.
- ▶ Select **Telephone**.
- ▶ Select .
- ▶ Select the connected phone.
- ▶ Select **Delete all favourites**.

- ▶ Select **OK**.

## Call list

### Overview of the call list

Depending on whether your mobile phone supports the PBAP Bluetooth® profile or not, this can have different effects on the presentation and functions of the call list.

The PBAP Bluetooth® profile is supported:

- The call lists from the mobile phone are shown in the multimedia system.
- When connecting the mobile phone, you may have to confirm the connection for the PBAP Bluetooth® profile.

The PBAP Bluetooth® profile is not supported:

- The multimedia system generates a call list itself as soon as calls are made in the vehicle.
- This list is not synchronised with the call lists on the mobile phone.

### Making a call from the call list

Multimedia system:



- ▶ Select **Call list**.
- ▶ Select an entry.

### Calling up additional options in the call list

Multimedia system:



- ▶ Select **Call list**.
- ▶ For previously stored contacts: select in the line of an entry.
  - ◀ The search results are displayed.
- ▶ For contacts who have not been stored: select .

### Selecting options for suggestions in the call list

#### Requirements

- Call list has entries, such as incoming or missed calls, for example.

Multimedia system:



- ▶ Select **Call list**.
  - ◀ The following options are available:
    - **Save as a favourite**
    - **No longer suggest**
- ▶ Select an option.

### Deleting the call list

Multimedia system:



- ▶ Select the connected mobile phone.
- ▶ Select **Delete call list**.
- ▶ Select **Yes**.

## Multimedia system

### Radio



#### Note

This function is only available if your mobile does not support the PBAP Bluetooth® profile.

### Radio

#### Switching on the radio

Multimedia system:



- Alternatively: Press the button .

### Radio overview



- ① Active frequency band
- ② Station name or set frequency
- ③ Track, artist and radio text
- ④ Options
- ⑤ Full screen/DAB slide show
- ⑥ Mute
- ⑦ Station list
- ⑧ Search

### Setting the frequency band

Multimedia system:



- Select a frequency band.

### Selecting a radio station

Multimedia system:



- Swipe to the left or right on the control element.

## Multimedia system

### Radio

#### Selecting a radio station list

Multimedia system:



- ▶ Select .
- ▶ Select a station.

#### Searching for radio stations using station names or direct frequency entry

Multimedia system:



- ▶ Select .
- ▶ Select .
- ▶ Enter a station name or frequency.

#### Storing radio stations

Multimedia system:



- ▶ Select a radio station.

#### Memory slots are available in station presets

- ▶ Select .

or

- ▶ Press and hold on the radio station.

#### Replacing the entry in the station preset

- ▶ Press and hold on the entry in the station preset.
- ▶ Select **Yes**.

#### Editing station presets

Multimedia system:



#### Moving stations:

- ▶ Press and hold on a saved station.
- ▶ Select **Move**.

- ▶ Move the station to the new position.

#### Deleting a station:

- ▶ Press and hold on a station.
- ▶ Select **Delete**.

#### Replacing a station:

- ▶ Press and hold on a station.
- ▶ Select **Replace a radio station**.

#### Activating/deactivating the frequency fix function

Multimedia system:



- ▶ Select **Frequency fix** to switch it on or off.
  - ◁ If the function is activated, the set frequency is kept even if the reception is poor.

## Switching traffic announcements on or off

Multimedia system:



- ▶ Switch **Traffic announcements** on or off.

## Setting the traffic information service volume increase

Multimedia system:



- ▶ Select **More sound settings**.
- ▶ Select **Navigation and traffic announcements**.
- ▶ Select **Traffic information service volume increase**.
- ▶ Adjust the value.
- ▶ To accept the value: select .

## Display radio text

Multimedia system:



- ▶ Switch **Show radio text information** on or off.

## Media

## Information about media mode

## Information about supported formats and data storage media

**Danger****Risk of distraction when handling data storage medium**

If you handle a data storage medium while driving, you may be distracted from traffic conditions. You could also lose control of the vehicle.

- ▶ Only handle a data storage medium when the vehicle is stationary.



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**The multimedia system supports the following formats and data storage media:**

|                                |                                                                   |
|--------------------------------|-------------------------------------------------------------------|
| Permissible file systems       | FAT32, exFAT, NTFS                                                |
| Permissible data storage media | USB devices, iPod®/iPhone®, MTP devices, Bluetooth® audio devices |

## Multimedia system

### Media

|                   |                                              |
|-------------------|----------------------------------------------|
| Supported formats | <b>Audio:</b> MP3, WMA, AAC, WAV, FLAC, ALAC |
|-------------------|----------------------------------------------|

#### Note

Observe the following notes:

- Up to 50,000 supported files can be managed by the multimedia system.
- Data storage media up to 2 TB are supported (32-bit directory).
- Due to the large variety of available music files in terms of encoders, sampling frequency and data transfer rate, playback cannot be guaranteed.
- It cannot be guaranteed that all USB devices can be played back due to the wide variety of USB devices available.
- Copy-protected music files or DRM (Digital Rights Management) encrypted files cannot be played back.
- MP3 players must support Media Transfer Protocol (MTP).

#### Information on copyright protection and trademarks

Audio files that you create or reproduce yourself for playback are generally subject to copyright protection. In many countries, reproductions, even for private use, are not permitted without the prior consent of the copyright holder. Make sure that you know about the applicable copyright regulations and that you comply with these.



### Overview of the media menu



- ① Previous track or fast rewind
- ② Next track or fast forward
- ③ Album cover
- ④ Active analogue external media source
- ⑤ Track and artist
- ⑥ Current track number/track in playback list and active data storage medium
- ⑦ Repeat
- ⑧ Settings
- ⑨ Connects a device
- ⑩ Timeline
- ⑪ Controls playback
- ⑫ Categories
- ⑬ Music search
- ⑭ Random playback

## Multimedia system

### Media

#### Connecting data storage media with the multimedia system

#### Connecting USB devices

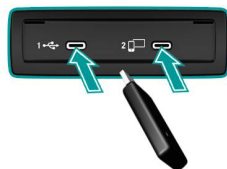


#### NOTE

#### Damage due to high temperatures

High temperatures can damage USB devices.

- ▶ Remove the USB device after use and take it out of the vehicle.



The multimedia connection unit is on the co-driver's side of the multimedia display and is equipped with two type C USB ports.

- ▶ Connect the USB device to the USB port.



#### Note

USB hubs are not supported by the multimedia system.



#### Note

When connecting multiple Apple® devices at the same time, the multimedia system will only connect to the device that is connected first. Additional Apple® devices connected at the same time will only be supplied with power.

#### Searching for and authorising the Bluetooth® audio equipment

#### Requirements

- Bluetooth® is activated on the multimedia system and audio equipment (→ Page 94).
- The audio equipment supports the Bluetooth® audio profiles A2DP and AVRCP.
- The audio equipment is "visible" for other devices.

Multimedia system:



#### Authorising a new Bluetooth® audio device

- ▶ Select [Connect new device](#).
- ▶ Select an audio device.
  - ◁ Authorisation starts. A code is displayed on the multimedia system and on the mobile phone.



#### Note

The selected BTA device is set as the Primary Phone.

- ▶ If the codes are identical, confirm on the audio equipment.
- ▶ Select [Only as a Bluetooth audio device](#).
  - ◁ The Bluetooth® audio device is connected to the multimedia system.

#### Connecting previously authorised Bluetooth® audio equipment

- ▶ Select a Bluetooth® audio device from the list.

### Starting media playback

#### Requirements

- A data storage medium is connected to the multimedia system.

Multimedia system:



- ▶ Select **USB** or **Bluetooth** as a media source.

### Controlling media playback

Multimedia system:



- ▶ **To pause playback:** must be selected.
- ▶ **To continue playback:** must be selected.
- ▶ **Repeat track:** must be selected.
  - ◁ For repeating there are the following options:

- **Select once:** The active playlist is repeated.
- **Select twice:** The current track is repeated.
- **Select three times:** Repeat is deactivated.

- ▶ **To play back tracks in a random order:** must be selected.
- ▶ **To move forward or backward within a track:** Tap at the relevant point of the timer bar.
- ▶ **To select the next track:** must be selected.
- ▶ **To select the previous track:** must be selected.
- ▶ **Fast track scrolling:** or press and hold or .
- ▶ **Show current track list:** Select the title image displayed.

### Additional options for setting media playback

Multimedia system:



### Calling up additional options

- ▶ Select .

The following options are available:

- [Play similar tracks](#)
  - [Surprise mix](#)
  - [Adding to favourite songs](#)
  - [Deleting from favourite songs](#)
- ▶ Select an option.

### Media search

#### Information about the search function in categories

Under , you can search through all available media files. There are several categories available for selection.

Available categories with Bluetooth® audio equipment:

- [Current tracklist](#)
- The folders and categories of the connected device are shown.

Available categories with audio files:

- [Current tracklist](#)

## Multimedia system

### Media

- Playlists
- Folders
- Albums
- Artists
- Track
- Music genres
- Year
- Composers
- Podcasts (Apple® devices)
- Audio books (Apple® devices)

#### Starting a search in categories

Multimedia system:



- ▶ Select a category.
- ▶  Select and enter a search term.

#### Sorting results shown or playing back all media found

- ▶  must be selected.
- ▶ **Play all search results in the category:** Select [Play all](#).
  - ◁ If, for example, the album category is active, all albums

found by the desired artist will be played back.

- ▶ **Sort results alphabetically:** Select [Sort from A-Z](#).
- ▶ **Sort results in reverse alphabetical order:** Select [Sort from Z-A](#).

#### Note

The available options depend on the selected category and the connected device.

#### Using the keyword search

Multimedia system:



You can look for content using the keyword search with free text input.

- ▶ Select .
- ◁ A keyboard for character entry appears.
- ▶ Enter the term searched for.

#### Note

The search begins with the first character entered. The more characters entered the more concrete the search results become.

- ▶ Select the desired entry from the result list.
  - ◁ If the selected result is an album, song or a playback list then playback is started. If the selected result is a new category then this is opened in the search.

#### Searching for a track according to mood

Multimedia system:



Using the My Music search, you can find music tracks suitable for a mood.

- ▶ Select [Mood](#).
  - ◁ A grid with the following moods appears:
    - [Positive](#)

- Calm
  - Sombre
  - Energetic
- Slide the control into the required position.

## Truck Data Centre

### Information about Truck Data Centre

#### Information about Truck Data Centre

The Truck Data Centre is the telematics control unit of Daimler Truck AG. It enables the transmission of all data generated by the vehicle and driver.

A valid framework agreement with a respective service booking is required for use of the Mercedes-Benz Trucks connectivity services. The Truck Data Centre is a prerequisite for using Mercedes-Benz Trucks connectivity services.

#### Connectivity services

##### Mercedes-Benz Uptime

You can obtain further information at [www.uptime-info.mercedes-benz.com](http://www.uptime-info.mercedes-benz.com) or at an authorised Mercedes-Benz Trucks Service Centre.

##### Remote Online

With Remote Online a connection is established with vehicle's own Wi-Fi network. It is the interface for using vehicle-related applications via mobile end

devices (BYOD). For access to the Wi-Fi network, the mobile end device must be authorised once via the multimedia system. In addition, a vehicle-related application such as the Remote Truck app, for example, is required. This can be obtained in the App Store (Apple®) or Play Store (Android). The range of the connection is approx. 25 m.

#### Mercedes-Benz Truck App Portal

You can obtain further information on the Mercedes-Benz Truck App Portal website: <https://truck-app-portal.mercedes-benz-trucks.com/#/customer>

#### Fleetboard

Further information can be found from <https://www.fleetboard.com>

## Notes on electric mode

**Warning**
**Risk of death and fire due to modified and/or damaged components of the high-voltage on-board electrical system**

The high-voltage on-board electrical system is under high voltage. If you modify component parts in the high-voltage on-board electrical system or touch damaged component parts, you may be electrocuted. In addition, modified and/or damaged components may cause a fire.

In the event of an accident or impact to the vehicle underbody, components of the high-voltage electrical system may become damaged even if the damage is invisible.

- ▶ Never make any modifications to the high-voltage on-board electrical system.
- ▶ Do not switch on or use the vehicle if its high-voltage on-board electrical system components have been modified or damaged.

- ▶ Never touch damaged components of the high-voltage on-board electrical system.
- ▶ After an accident, do not touch any components of the high-voltage on-board electrical system.
- ▶ After an accident, have the vehicle transported away.
- ▶ Have the components of the high-voltage on-board electrical system checked at a qualified specialist workshop and replaced if necessary.

**Danger**
**Risk of chemical burns and poisoning from damaged high-voltage battery**

If the housing of the high-voltage battery has been damaged, electrolyte and gases may escape.

- ▶ Avoid contact with the skin, eyes, or clothing.
- ▶ Immediately rinse electrolyte splashes off with water and seek medical attention straight away.

**Danger**
**Risk of fire and explosion from excessive internal pressure in the high-voltage battery**

In the event that the vehicle catches fire, flammable gas can escape and ignite.

- ▶ If you notice an unusual smell, smoke, or burn marks, immediately discontinue the charging process.
- ▶ Leave the danger zone immediately. Secure the danger zone at a suitable distance.
- ▶ Call the fire service.

The high-voltage emergency shutdown automatically switches off the high-voltage on-board electrical system irreversibly depending on the accident situation. The instrument cluster will display a grey display message in this case.

The operational readiness of the high-voltage emergency shutdown is checked during the self-test (→ Page 28). If the restraint system warning lamps  and  light up on the instrument cluster, this indicates a malfunction in the auto-

## Driving mode

### Manual disconnection of the high-voltage on-board electrical system

matic high-voltage emergency shutoff. In this case, observe the display message.

Observe the following notes on vehicle noise emissions and the acoustic vehicle alerting system:

- The vehicle is equipped with a purely electric drive system and develops considerably lower vehicle noise emissions than vehicles with a combustion engine.  
For this reason, the vehicle is equipped with a sound generator, a front loudspeaker, and a reverse warning device which acts as an acoustic vehicle warning system (AVAS).
- If the parking brake is released and the transmission is engaged in position **N** or **D**, the sound generator produces speed-appropriate driving noises up to speeds of approx. 25 km/h via the front speaker.  
When the transmission is in position **R**, the reverse warning device (→ Page 200) produces a warning

tone via the rear speaker when reversing.

This helps other road users – especially pedestrians and cyclists – to hear your vehicle better.

- From speeds of approx. 25 km/h, the audible vehicle alerting system (AVAS) switches off with decreasing volume.
- Despite the sound generator, the vehicle may still not be heard by other road users. Adapt your driving style accordingly.
- Deactivating the audible vehicle alerting system (AVAS) or covering up the speakers/reverse warning device with sound-reducing material is not permitted.
- Structural modifications to the audible vehicle alerting system (AVAS) are not permitted. Observe the information about attachments, add-on equipment, installations and conversions (→ Page 3).

The front speaker of the audible vehicle alerting system (AVAS) is on the side of the bumper.

### Manual disconnection of the high-voltage on-board electrical system

#### Requirements

- The vehicle is possibly damaged, and the high-voltage on-board electrical system is not irreversibly disconnected via the automatic high-voltage emergency shutoff.
- The vehicle is switched off and secured against rolling away.

#### Note

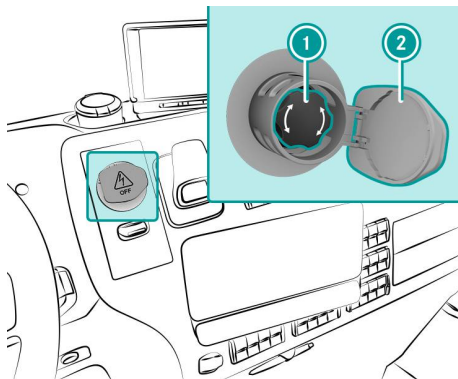
If the high-voltage emergency shutoff has been activated, the instrument cluster displays a corresponding message. If the high-voltage emergency shutoff has a malfunction, the restraint system/crash sensing warning lamp  will light up in the instrument cluster.



## Manual disconnection of the high-voltage on-board electrical system

### Using the high-voltage disconnect device in the driver cockpit

You can temporarily disconnect the high-voltage on-board electrical system for a visual check of the vehicle when it is parked. While the system is disconnected, the vehicle is not operational.



- ▶ **To disconnect the high-voltage on-board electrical system:** open the cap ② and press the switch ① until it engages.

◁ The high-voltage on-board electrical system is disconnected. The capacitors of the vehicle's high-

voltage on-board electrical system are completely discharged after approximately five seconds.

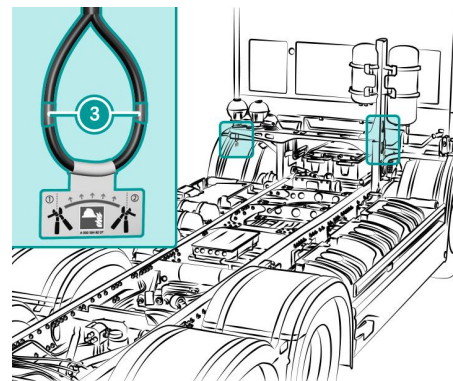
- ▶ Secure switch ① against reactivation and fold cap ② back over the switch.

All work on the drive system may only be carried out at a qualified specialist workshop, even if it has been deactivated manually.

- ▶ **To connect the high-voltage on-board electrical system:** if necessary, fold cap ② open and remove the safeguard against reactivation.
- ▶ Turn switch ① clockwise until it unlocks and springs back to its starting position.
- ▶ Fold cap ② back over the switch.

### Using the high-voltage rescue separation points

The high-voltage rescue separation points are located on both sides behind the cab and are only to be used by rescue teams in an emergency.



Example: high-voltage rescue separation points

- ▶ Pull the cable loop out slightly and, using an appropriate tool, cut it at positions ③ as shown on the sign and remove the severed piece of cable.
- ◁ The high-voltage on-board electrical system is disconnected. The capacitors of the vehicle's high-voltage on-board electrical system are completely discharged after approximately five seconds.

All work on the drive system may only be carried out at a qualified specialist

## Driving mode

### Driver assistance/driving safety systems and your responsibility

workshop, even if it has been deactivated manually.

#### Driver assistance/driving safety systems and your responsibility

Please bear in mind that the driver assistance/driving safety systems described here can only reduce the risk of accidents within system limits and the laws of physics.

Poor road and weather conditions as well as certain traffic situations can impair the function of driver assistance/driving safety systems.

Driver assistance/driving safety systems are only aids. They cannot replace your awareness of your surroundings and do not relieve you of your responsibility under road traffic laws. Pay attention to the traffic conditions at all times and intervene when necessary.

For vehicles operated in Ukraine: Please also observe the information on “Radio approval for the radar sensor system” (→ Page 282).

#### Driving

##### Notes on preparing for a journey

Please note that all electronic safety systems in the vehicle merely serve to assist you. They do not relieve you of the obligation to carry out a visual check of the vehicle and the trailer/semitrailer before starting a journey.



**Danger**

#### Risk of injury from unsecured items in the vehicle

If objects, luggage or loads are not secured or not secured sufficiently, they could slip, tip over or be thrown around and thereby hit vehicle occupants.

There is a risk of injury, particularly in the event of braking manoeuvres or abrupt changes in direction!

- ▶ Always stow objects in such a way that they cannot be thrown around.
- ▶ Before the journey, secure objects, luggage or loads against slipping or tipping over.



**Danger**

#### Risk of accident due to objects in the driver's footwell

Objects in the driver's footwell may impede pedal travel or block a depressed pedal.

This jeopardises the operating and road safety of the vehicle.

- ▶ Stow all objects in the vehicle securely so that they cannot get into the driver's footwell.
- ▶ Always ensure that there is sufficient clearance for the pedals.
- ▶ Always install the floor mats securely and as specified.
- ▶ Do not use loose floor mats and do not place floor mats on top of one another.



### Danger

#### Danger of accident due to uneven loading

The driving characteristics, as well as steering and braking characteristics, may be greatly impaired.

- ▶ Load the vehicle evenly.
- ▶ Secure the load against sliding.

#### Preparing for a journey



### Danger

#### Risk of chemical burns and poisoning from damaged high-voltage battery

If the housing of the high-voltage battery has been damaged, electrolyte and gases may escape.

- ▶ Avoid contact with the skin, eyes, or clothing.
- ▶ Immediately rinse electrolyte splashes off with water and seek medical attention straight away.



### Danger

#### Risk of accident or death due to damaged vehicle parts

Damaged vehicle parts may lead to malfunctions and system failures.

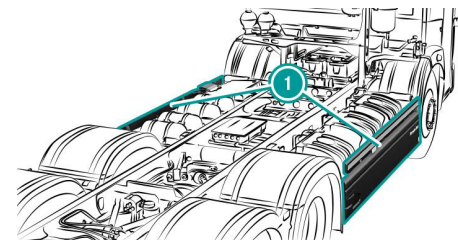
The operating safety of the vehicle may be at risk.

- ▶ Before each journey, check the vehicle for visible damage.
- ▶ Have damage to the vehicle checked and repaired immediately at a qualified specialist workshop.

#### Visual and function check on the vehicle exterior

Before commencing your journey, carry out the following checks on the vehicle:

- ▶ Observe the notes on operating safety (→ Page 4).



Platform truck (example)

Your vehicle is equipped with a side underride guard ①, which provides additional damage protection for the high-voltage batteries. In the event of a side impact, the high-voltage on-board electrical system may be switched off automatically by the side impact protection sensors fitted near the underride guard ①.

- ▶ Check the side underride guard ① for visible damage.
  - ◁ If the side underride guard ① or the high-voltage batteries show visible signs of damage, e.g. due to an accident, have the vehicle checked and repaired immediately by a qualified specialist workshop.
- ▶ Check the vehicle for leaks.

## Driving mode

### Driving

- ▶ Make sure that the licence plate number, vehicle lighting, turn signals, and brake lights are not dirty or damaged.

#### Note

The lamp check of the electronic key helps you check the vehicle lighting (→ Page 37).

- ▶ Check that the outline markings of all attachments and add-on equipment are in good condition.
- ▶ Check the wheels and tyres (→ Page 261). When doing so, observe the following:
  - General condition
  - Visible damage and cracks
  - Tyre pressure
  - Tyre tread depth
  - The wheels are fitted securely
- ▶ Make sure that the vehicle is loaded correctly.
- ▶ Make sure that the dropside and exterior flaps are locked securely and not damaged.

- ▶ In wintry conditions, remove accumulations of snow and ice from the vehicle (→ Page 210).

Have all faults and damage rectified. If necessary, have the causes determined and rectified at a qualified specialist workshop.

#### Check the emergency equipment/first-aid kit

#### Note

You will find an overview of emergency equipment and first-aid kits in the "Breakdown assistance" section (→ Page 235).

- ▶ Check the following emergency equipment to make sure that it is accessible, complete, and ready for use:
  - High visibility vest
  - Warning triangle
  - Warning beacon
  - First-aid kit
  - Fire extinguisher

- ▶ Check that the first-aid kit is usable at regular intervals. Note the use-by dates of the contents.
- ▶ Have the fire extinguisher checked every one to two years.
- ▶ The fire extinguisher must be refilled after each use.

You must observe the legal requirements for the country in which you are currently driving when carrying emergency equipment.

#### Checking the vehicle illumination system

- ▶ Switch on the vehicle.
    - ◁ If a bulb on the vehicle is faulty, the instrument cluster will show the display messages  and  for the towing vehicle.
  - ▶ Check the vehicle lighting, turn signals, and brake lights with the help of a second person.
- or
- ▶ Check the vehicle lighting using the vehicle key lamp check (→ Page 38).

- Replace the faulty bulbs  
(→ Page 60).

### Information on pre-installation for alcohol tester with immobiliser

The pre-installation is prescribed as part of the General Safety Regulation and provides for a standardised interface in the cab, which is intended for the connection of a breathalyser from third-party providers.

By retrofitting an alcohol tester, companies can prevent the vehicle from being put into operation without a successful breathalyser test being completed.

### Note

Country-specific regulations may prescribe or permit a bypass switch for the alcohol tester. This can be used in an emergency to start the vehicle without carrying out a breathalyser test.

### Note

Further information on retrofitting an alcohol tester with immobiliser can be obtained from an authorised Mercedes-Benz Trucks Service Centre.

### Note

Information on operating the alcohol tester has been specified in the vehicle's operating instructions.

### Notes on the ignition lock

### Danger

### Risk of accident and injury due to children left unattended in the vehicle

If you leave children unattended in the vehicle, some of the things that might happen include:

- They may open doors, thereby endangering other persons or road users.

- They may get out and be struck by oncoming traffic.
- They may operate vehicle equipment and become trapped.

In addition, the children could also set the vehicle in motion, for example by:

- Releasing the parking brake.
- Changing the transmission position.
- Starting the vehicle.
- Never leave children unattended in the vehicle.
- When leaving the vehicle, always take the key with you and lock the vehicle.
- Keep the key out of the reach of children.

### Danger

### Risk of accident when the vehicle is switched off while driving

If during an emergency you switch off the vehicle while you are driving, safety-relevant functions will be restricted or no longer available.

## Driving mode

### Driving

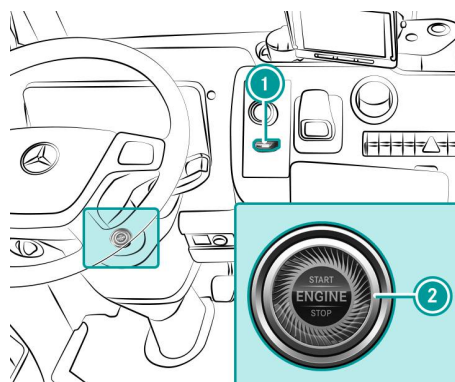
This applies, for example, to the compressed-air system, which will still be supplied from the compressed-air reservoirs but is no longer filled.

You will then have to stop immediately in accordance with the traffic regulations. If the reservoir pressure is too low, it will no longer be possible to brake the vehicle.

- For this reason alone you should never switch off the vehicle while you are driving.

For vehicles with an electronic ignition lock, also pay attention to the information about the radio operating permit (→ Page 280).

#### Switch positions of the electronic ignition lock



- ① Key in storage compartment
- ② Start/stop button

#### The ignition lock has the following switch positions:

|   |                                                                                                                 |
|---|-----------------------------------------------------------------------------------------------------------------|
| 0 | The vehicle is switched off.                                                                                    |
| 1 | Radio position – the power supply for some consumers is switched on.                                            |
| 2 | The vehicle is switched on.                                                                                     |
| 3 | Ready to drive – the vehicle has been started. The check lamp <b>READY</b> in the instrument cluster lights up. |

If you press start/stop button ② when the brake is not depressed, the ignition lock will switch to the next switch position. If you press the start/stop button in switch position ②, the ignition lock will switch to switch position 0.

If you press start/stop button ② when the vehicle is stationary and has been started, the vehicle will switch off. The indicator lamp **READY** in the instrument cluster goes out.

#### Switching on the radio position and the vehicle

##### Requirements

- The key is in the vehicle.
- The key battery is charged.

To switch the ignition lock to the radio position, press the button ② once. In the radio position, the power supply is switched on for some consumers.

The power supply is switched off again when one of the following conditions is met:

- You open the driver's or co-driver's door.
- Press the button  twice.

To switch on the vehicle, press the button  twice. When the vehicle is switched on, the indicator lamps in the instrument cluster light up.

The vehicle is switched off again if one of the following conditions is met:

- You do not start the vehicle within 30 minutes.
- Press the button  once.

### Note

If the vehicle is switched off automatically, vehicle lighting switched on by the automatic driving lights is also switched off. If necessary, switch the vehicle illumination system on manually ( → Page 52).

### Note

In an emergency, you can switch off the vehicle while driving. To do this, press the button  for about three seconds or press the button  three times within three seconds. Be sure to observe the safety instructions at the beginning of this chapter.

### Ignition run-on

The ignition run-on function allows you to leave the vehicle switched on and take the key out of the vehicle with you. This means you can leave and lock the cab when the vehicle is switched on, for example. The ignition run-on function is limited to a maximum duration of 30 minutes. The ignition lock then switches to position  and the vehicle switches off.

### Start the vehicle

#### Requirements

- The key is in the vehicle.
- The key battery is charged.
- The high-voltage battery is charged sufficiently.

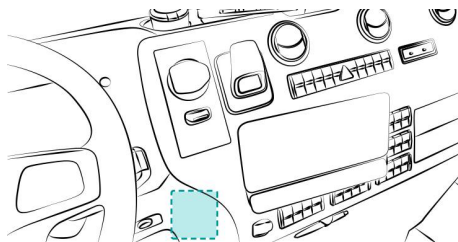
- ▶ Shift the transmission to .
- ▶ Switch off the cargo liftgate, see the separate operating instructions.
- ▶ Depress the brake pedal and keep it depressed.
- ▶ Press the start/stop button once.
  - ◀ The vehicle is started when the indicator lamp  on the instrument cluster lights up.

If the vehicle does not start and the display message [Hold key to contact point as shown in Owner's Manual](#) is displayed on the instrument cluster, the vehicle can be started in emergency operation mode.

## Driving mode

### Driving

#### Start the vehicle in emergency operation mode



Contact point for the vehicle key

- ▶ Hold the vehicle key against the contact point and press the start/stop button.
- ▶ If the vehicle starts up: Have the vehicle key checked at a qualified workshop.
- ▶ If the vehicle does not start: Consult a qualified workshop

#### Cycle the ignition

- ▶ Stop the vehicle in accordance with the traffic conditions
- ▶ Switch off the vehicle.
- ▶ Lock the vehicle using the remote control and wait until the red indicator

lamp on the parking brake lever goes out.

- ◁ The red indicator lamp on the parking brake lever goes out after two to three minutes.
- ▶ After a further waiting time of ten seconds, unlock the vehicle using the remote control.
- ▶ Start the vehicle.
  - ◁ The vehicle is started when the indicator lamp **READY** on the instrument cluster lights up. A function test of the vehicle systems is started.

If warning messages advising against continued driving are still displayed in the instrument cluster after a waiting time of approximately 60 seconds, do not pull away.

Consult a qualified workshop.

#### Safety inspections

##### Checking the supply pressure in the compressed-air brake system



**Danger**

#### Risk of accident due to leaking compressed-air brake system

If a compressed-air brake system is leaking or there is insufficient reserve pressure, it will not be possible to brake the vehicle.

- ▶ Only set the vehicle in motion if the necessary reserve pressures are reached.
- ▶ In the event of pressure loss while driving, immediately bring the vehicle to a halt in accordance with the traffic conditions.
- ▶ Secure the vehicle using the parking brake.
- ▶ Consult a qualified specialist workshop immediately.



- ▶ Start the vehicle and wait until the  warning lamp in the instrument cluster goes out.
- ▶ In the instrument cluster, call up the **Supply pressure** menu window and check the current supply pressure.
- ▶ Adjust the compressed-air delivery rate if necessary ( → Page 214).
- ▶ Observe the display messages and the indicator lamps in the status area of the instrument cluster for the supply pressure and the compressed-air brake system.

### Check the function of the compressed-air dryer



**Danger**

### Risk of accident due to excessively high condensation level

If the condensation level in the compressed-air reservoirs is too high, the braking effect may be reduced, or the compressed-air brake system may fail.

- ▶ Check the compressed-air system for condensation before starting a journey.
- ▶ If the compressed-air brake system has an excessively high condensation level, have it checked at a qualified workshop immediately.
- ▶ Start the vehicle.
- ▶ Check the reservoir pressure in the compressed-air brake system ( → Page 126).

If it is detected that there are too few regeneration phases, the instrument cluster will show the yellow display message  **Condensation in the compressed-air reservoir**.

If the condensation level in the compressed-air reservoirs is too high, the display message can no longer be acknowledged. In this case, have the compressed-air brake system checked immediately at a qualified workshop.

### Checking the vehicle height



**Danger**

### Risk of accident due to lowered or raised chassis

The braking and handling characteristics may be greatly impaired. The permitted vehicle height may be exceeded.

- ▶ Set the driving level before you set off.

If the yellow indicator lamp  lights up in the instrument cluster, this means the chassis is not at driving level.

- ▶ Set driving level ( → Page 193).
  - ◁ The indicator lamp  in the instrument cluster goes out.

## Driving mode

### Driving

#### Notes on pulling away



**Danger**

#### **Risk of accident due to the vehicle rolling unintentionally on uphill slopes**

If you reverse start with the vehicle on a steep uphill slope, the vehicle may briefly roll downhill.

Before reverse starting on a steep uphill slope:

- ▶ Activate the HOLD function.

or

- ▶ Apply the parking brake.



**NOTE**

#### **Damage to the power steering due to overload**

If, for example, the wheels are touching a kerb, holding the steering at the stop for too long or turning the steering wheel too forcefully could cause the steering to be damaged or malfunction.

- ▶ Do not hold the steering wheel in the fully turned position for longer than ten seconds.
- ▶ Do not steer too forcefully if the wheels are close to objects.

Perform a brake test before starting a journey. Pay attention to the traffic situation while doing this. If you notice a reduction in braking power when testing the brakes, stop the vehicle as soon as possible while paying attention to traffic conditions. Have the brake system checked and repaired at a qualified workshop.

#### Notes on stopping and switching off the engine



**Danger**

#### **Risk of accident when the vehicle is switched off while driving**

If during an emergency you switch off the vehicle while you are driving, safety-relevant functions will be restricted or no longer available.

This applies, for example, to the compressed-air system, which will still be supplied from the compressed-air reservoirs but is no longer filled.

You will then have to stop immediately in accordance with the traffic regulations. If the reservoir pressure is too low, it will no longer be possible to brake the vehicle.

- ▶ For this reason alone you should never switch off the vehicle while you are driving.



**Danger**

#### **Risk of accident and injury due to children left unattended in the vehicle**

If you leave children unattended in the vehicle, some of the things that might happen include:

- They may open doors, thereby endangering other persons or road users.
- They may get out and be struck by oncoming traffic.

- They may operate vehicle equipment and become trapped.

In addition, the children could also set the vehicle in motion, for example by:

- Releasing the parking brake.
- Changing the transmission position.
- Starting the vehicle.
- ▶ Never leave children unattended in the vehicle.
- ▶ When leaving the vehicle, always take the key with you and lock the vehicle.
- ▶ Keep the key out of the reach of children.

When travelling with children in the vehicle, observe the supplemental notes (→ Page 32).

### Stopping and switching off the engine

#### Stopping

- ▶ Stop the vehicle.
- ▶ Apply the parking brake.
- ▶ Shift the transmission to neutral.

### Switching off the vehicle

- ▶ Press and release the start/stop button.
- ▶ Secure the vehicle against rolling away; use chocks if necessary.

### Brakes

#### Notes on the brake system

Always observe the information in the yellow or red display messages (→ Page 306) shown in the instrument cluster when there is a malfunction in the brake system. In addition, an indicator lamp lights up in the status area of the instrument cluster and a warning tone may sound. Have the brake system checked and repaired at a qualified workshop.

The vehicle is equipped with an electronic braking system (EBS) when it leaves the factory and can include the following functions:

- ABS (anti-lock braking system)
- ASR (acceleration skid control system)

- ALB (automatic load-dependent brake)
- Hill holder
- BAS (Brake Assist System)

### Checking the compressed-air brake system for leaks



#### Danger

#### Risk of accident due to leaking compressed-air brake system

If a compressed-air brake system is leaking or there is insufficient reserve pressure, it will not be possible to brake the vehicle.

- ▶ Only set the vehicle in motion if the necessary reserve pressures are reached.
- ▶ In the event of pressure loss while driving, immediately bring the vehicle to a halt in accordance with the traffic conditions.
- ▶ Secure the vehicle using the parking brake.

## Driving mode

### Brakes

- ▶ Consult a qualified workshop immediately.

#### Note

Do not let anyone enter or exit the vehicle during the test. This avoids mistaking pressure loss from the pneumatic folding door, air-sprung cab suspension system, air-sprung seats or the level control system for a leak.

- ▶ Stop the vehicle on a level surface.
- ▶ Apply the parking brake.
- ▶ Use chocks to protect the vehicle against rolling away.
- ▶ Release the parking brake.
- ▶ Switch on the vehicle.
- ▶ In the instrument cluster, call up the **Supply pressure** menu window.
- ▶ Start the vehicle and wait until the system-dependent supply pressure is shown (→ Page 292).
- ▶ Switch off the vehicle.
  - ◁ The parking brake is automatically applied.

- ▶ Switch on the vehicle.
- ▶ Release the parking brake again.
- ▶ In the instrument cluster, call up the **Supply pressure** menu window again.
- ▶ Depress the brake pedal and keep it in this position.
- ▶ After a brief moment, read off the supply pressure.
- ▶ Read off the supply pressure once again after approximately one minute.

If no significant loss of pressure is detected in the **Reserve pressure** menu window after that minute, the compressed-air brake system is not leaking.

If a significant loss of pressure is detected, the compressed-air brake system is leaking.

- ▶ If the compressed-air brake system is leaking, have it checked and repaired at a qualified workshop.

### ABS (anti-lock braking system)

#### Function of ABS

ABS controls the brake pressure so that the wheels do not lock under braking. This means that the vehicle can still be steered while braking.

#### ABS display check



**Danger**

#### Risk of skidding when ABS malfunctions

If ABS malfunctions, the wheels may lock during braking.

- ▶ Drive on carefully.
- ▶ Have the ABS checked at a qualified specialist workshop immediately.

The anti-lock protection function is not guaranteed in the following situations:

- After switching on the vehicle, the  indicator lamp is not displayed in the status area of the instrument cluster.

- The vehicle has been switched on for three seconds and the indicator lamp is still lit.
- The indicator lamp lights up after the vehicle has pulled away.

### Braking with anti-lock protection

- ▶ **If ABS intervenes:** Continue to depress the brake pedal until the braking situation has passed.
  - ◁ When ABS is intervening, the continuous brake is switched off. The indicator lamp  in the instrument cluster remains on.
- ▶ **If emergency braking is engaged:** Vigorously depress the brake pedal.
  - ◁ Under certain conditions, the hazard warning light system switches on automatically.

### Electronic parking brake

#### Notes on the electronic parking brake



**Danger**

#### Risk of accident and injury due to children left unattended in the vehicle

If you leave children unattended in the vehicle, some of the things that might happen include:

- They may open doors, thereby endangering other persons or road users.
- They may get out and be struck by oncoming traffic.
- They may operate vehicle equipment and become trapped.

In addition, the children could also set the vehicle in motion, for example by:

- Releasing the parking brake.
- Changing the transmission position.
- Starting the vehicle.
- ▶ Never leave children unattended in the vehicle.

- ▶ When leaving the vehicle, always take the key with you and lock the vehicle.
- ▶ Keep the key out of the reach of children.



**Danger**

#### Risk of accident on uphill or downhill slopes

The parking brake may not be sufficient to secure the vehicle on uphill and downhill slopes.

A loaded vehicle or a vehicle with a trailer may roll away.

- ▶ In the check position, check whether the braking force of the parking brake is sufficient to hold the towing vehicle.
- ▶ If it does not hold the towing vehicle, secure the vehicle additionally, including the trailer, with chocks.
- ▶ Additionally secure the trailer using its parking brake.

## Driving mode

### Brakes



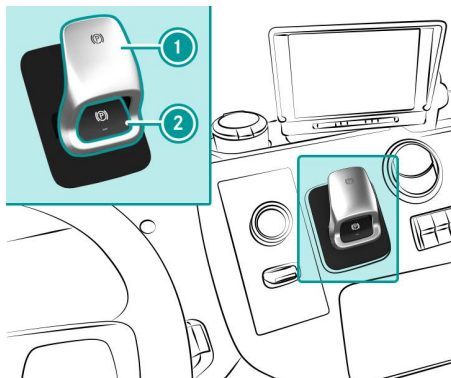
#### Danger

#### Risk of skidding and accident due to emergency braking with the parking brake

If you brake the vehicle using the parking brake, no ABS braking regulation takes place. The wheels could lock. There is an increased risk of skidding and/or accident.

- ▶ Do not pull the parking brake lever past the resistance point.
- ▶ If the wheels lock, immediately move the parking brake lever in the release direction to the point at which the wheels begin to turn again.

#### Using the electronic parking brake



Electronic parking brake (example)

- ① Lever for electronic parking brake
- ② Button for electronic parking brake

#### Applying the electronic parking brake manually

Requirements:

- The vehicle is stationary.

- ▶ Press the button .

or

- ▶ Pull the lever ① past the resistance point.
  - ◁ The parking brake is applied. The indicator lamp in the button and the indicator lamp on the instrument cluster light up red continuously.



#### Note

Only when the indicator lamp in the button and the indicator lamp on the instrument cluster come on does this indicate that the parking brake has been engaged correctly.

#### Releasing the electronic parking brake manually

Requirements:

- The vehicle is switched on.
  - The pressure in brake circuits 1 and 2 is sufficient.
- ▶ Depress the brake pedal.
  - ▶ Press the button .

- ◁ The parking brake is released.  
The indicator lamp in the button  and the indicator lamp  on the instrument cluster go off.

### Applying the electronic parking brake automatically

Requirements:

- The vehicle is stationary.
- The vehicle is switched on.
- Switch off the vehicle.

or

- Open the driver's door.
  - ◁ The parking brake is automatically applied.  
The indicator lamp in the button  and the indicator lamp  on the instrument cluster light up red continuously.

### Releasing of the electronic parking brake automatically

Requirements:

- The vehicle is switched on.

- The pressure in brake circuits 1 and 2 is sufficient.
- The doors are closed.
- Shift transmission from **N** to **D** or **R**.
  - ◁ The HOLD function of the electronic brake system is activated.  
The indicator lamp  is displayed in the instrument cluster.  
The vehicle is held by the service brake.  
The parking brake is released.
- Depress the accelerator pedal.
  - ◁ The HOLD function of the electronic brake system is deactivated.  
The vehicle starts moving.  
The indicator lamp  in the instrument cluster goes out.

### Test the parking brake (check position)

Requirements:

- The pressure in brake circuits 1 and 2 is sufficient.
- The parking brake is applied.

- The lever  is in the starting position.
- Pull the lever  past the resistance point and hold it there.
  - ◁ The check position of the parking brake is activated.  
The display message [Performing check position](#) appears in the instrument cluster.

#### Note

During the test, the vehicle is only held by the spring-loaded brake cylinder.

### Secondary braking with electronic parking brake

Requirements:

- The parking brake is released.
- Depending on the desired strength of the auxiliary braking, pull lever  no further than the resistance point.
  - ◁ The parking brake brakes the vehicle depending on the lever position.  
If the pressure in the spring-type actuator is less than 6,5 bar, the

## Driving mode

### Brakes

indicator lamp  lights up in the instrument cluster.

#### Activating the workshop mode of the electronic parking brake

Requirements:

- The vehicle is stationary.
- The parking brake is applied.
- The vehicle is switched off.

If you activate workshop mode of the electronic parking brake, all automatic parking brake functions are deactivated.

To avoid automatic application of the electronic parking brake when the vehicle is being towed, activate workshop mode and tow the vehicle with the front axle raised.

- ▶ Switch on the vehicle, but do not start it.
- ▶ Press and hold the button .
- ▶ Switch off the vehicle.
  - ◁ Workshop mode of the electronic parking brake is activated. The instrument cluster shows a corresponding display message

after the vehicle is switched on again.

If you drive faster than 30 km/h while workshop mode is activated, this mode will deactivate.

#### Frequent-stop brake

##### Notes on the frequent-stop brake



**Danger**

#### Risk of accident due to excessively low brake pressure of the frequent-stop brake

If you use the frequent-stop brake to secure the vehicle to prevent it from rolling away, the brake pressure may be too low.

This may cause the vehicle to roll away even though the frequent-stop brake is applied.

- ▶ Never exit the driver's seat when the frequent-stop brake is applied, and be ready to brake.

- ▶ If the vehicle rolls away, also apply the service brake.



**Danger**

#### Risk of accident due to the frequent-stop brake being applied in winter

If, in wintry road conditions, you brake with the frequent-stop brake applied, the wheels may lock just before the vehicle stops. Even if you remove your foot from the brake pedal, the wheels will remain locked.

This may cause the vehicle to skid or slip away, e.g. on uphill or downhill slopes.

- ▶ Never apply the frequent-stop brake in wintry road conditions.



**Danger**

#### Risk of accident due to the frequent-stop brake not being applied

If you do not stop the vehicle with the service brake, e.g. when coasting, the hill holder will not activate. The vehicle may roll away.



- In order to apply the frequent-stop brake, always use the service brake to brake the vehicle to a standstill.

The frequent-stop brake uses less compressed air than the parking brake. Use the frequent-stop brake if you frequently pull away and stop for short periods of time, e.g. in refuse collection operation. The frequent-stop brake does not replace the service brake or the parking brake. You will find further information on stopping in the "Pulling away and stopping" section (→ Page 129).

When the frequent-stop brake is released, the  indicator lamp in the status area goes out.

If you switch the vehicle off while the frequent-stop brake is applied, the frequent-stop brake will remain applied.

### Applying and releasing the frequent-stop brake

#### Activating



Example: frequent-stop brake switch

- Press the bottom section of the  /  switch.
  - ◄ The  indicator lamp in the instrument cluster lights up.

#### Deactivating

- Press the upper section of the  /  switch and move the switch to the centre position.
  - ◄ The indicator lamp  on the instrument cluster goes out.

### Using the hill holder



#### Danger

#### Risk of accident due to the hill holder being activated in winter

If you brake in wintry road conditions with the hill holder switched on, the wheels may lock shortly before stopping. Even if you remove your foot from the brake pedal, the wheels will remain locked. This may cause the vehicle to skid or slip away, e.g. on uphill or downhill slopes.

- Do not activate the hill holder in wintry road conditions.



#### Danger

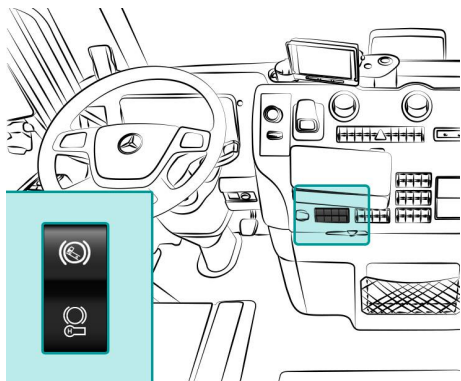
#### Risk of accident due to the hill holder not being activated

If you do not stop the vehicle with the service brake, e.g. when coasting, the hill holder will not activate. The vehicle may roll away.

## Driving mode

### Brakes

- In order activate the hill holder, always use the service brake to brake the vehicle to a standstill.



Example: hill holder switch

The hill holder assists you when pulling away on uphill or downhill slopes. The hill holder prevents the vehicle from rolling. This facilitates controlled pulling away.

- **To switch on:** Press the  /  switch on top.
- **Switching off:** Press the  /  switch below and move the switch to the centre position.

If you do not depress the brake pedal while the vehicle is stationary and the hill holder is active, a warning tone sounds briefly. The hill holder is deactivated and the  indicator lamp in the instrument cluster goes out.

If you depress the brake pedal until the vehicle is at a standstill while the hill holder is activated, the hill holder is engaged. The check lamp  in the instrument cluster lights up.

If you then apply the parking brake, the hill holder is deactivated and the  indicator lamp in the instrument cluster goes out. The vehicle is then held by the parking brake. After the parking brake is released, the hill holder is no longer active. If you depress the brake pedal, the hill holder remains active after the parking brake is released.

If you depress the accelerator pedal and the vehicle pulls away, the hill holder is released automatically. The indicator lamp  in the instrument cluster goes out.

The hill holder must be re-engaged after the vehicle has been switched on again.

### Using the HOLD function

The HOLD function holds the vehicle in place without requiring you to continue applying the brakes, for example, when pulling away on a hillside or during periods spent in stationary traffic.

Do not use the HOLD function when off-road, on steep downhill or uphill gradients with slippery or loose surfaces. Under these conditions, the HOLD function may not be able to hold the vehicle.

The HOLD function is only an aid. The driver is always responsible for safeguarding the vehicle against unintentional rolling away.



**Danger**

**Risk of accident due to the HOLD function being activated when leaving the vehicle**

If the vehicle is only held by the HOLD function, it may roll away in the following situations:

- If there is a malfunction in the system or power supply.
  - If the HOLD function is switched off due to the accelerator pedal being depressed, e.g. by a vehicle occupant.
- Use the parking brake to secure the vehicle against rolling away.

► **To activate the HOLD function:**

Depress the brake pedal and, after a brief moment, quickly continue to depress it.

◁ The **HOLD** display lights up in the instrument cluster.

- Release the brake pedal.

**i Note**

If, instead of the **HOLD** indicator lamp, the  indicator lamp of the hill holder is shown in the instrument cluster, refer to the notes under "Hill holder" (→ Page 135).

The HOLD function will deactivate and the **HOLD** indicator lamp will go out in the following circumstances:

- You depress the accelerator pedal with a gear engaged.
- The electronic parking brake is applied.
- The vehicle is switched off.
- A system error occurs.
- The power supply is too low.

### Continuous brake

#### Function of the continuous brake

Depending on the brake stage of the continuous brake selected, the electric motor is operated as an alternator when in overrun mode and during braking. In order to charge the high-voltage battery while driving, the vehicle's kinetic energy is converted to electrical energy (recuperation).

The higher the selected brake stage, the more sharply the vehicle is braked by means of recuperation and the more electrical energy is fed into the high-voltage battery.

When the continuous brake is activated and you release the accelerator pedal

while the vehicle is in motion, recuperation in overrun mode is initiated.

As the vehicle speed decreases, the braking effect of the continuous brake increases.

The braking effect of the continuous brake is influenced by the following:

- The state of charge of the high-voltage battery
- The vehicle speed
- Driving on uphill or downhill gradients
- The vehicle load condition

You can set the brake stages of the continuous brake using the multifunction lever (→ Page 138).

Also observe the information on one-pedal driving mode (→ Page 139).

### System limits

The braking effect of the continuous brake may be reduced or not available at all in the following situations:

- when the state of charge of the high-voltage battery is nearly 100%

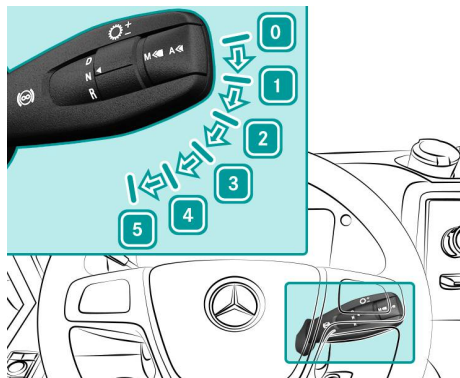
## Driving mode

### Brakes

- if the high-voltage battery is not yet at a normal operating temperature
- when the vehicle speed is less than 4 km/h
- in transmission position **N**
- during and after an ESP® intervention

#### Activating/deactivating the continuous brake

You can adjust the brake stages of the continuous brake (recuperation) manually using the multifunction lever in overrun mode.



Your vehicle is equipped with five brake stages (**1** – **5**).

Each brake stage increases the braking effect of the continuous brake and therefore the recuperation power by approximately 20%.

- ▶ **To switch on:** Pull the multifunction lever to the desired brake stage.
  - ◀ The indicator lamp  on the instrument cluster will come on.
- ▶ **Switching off:** Push the multifunction lever to position **0**.
  - ◀ The indicator lamp  on the instrument cluster goes out.

If the vehicle is switched on and the  indicator lamp flashes in the instrument cluster, the multifunction lever is not in position **0**.

When the  indicator lamp in the instrument cluster lights up, the continuous brake is malfunctioning. Have the continuous brake checked at a qualified specialist workshop.

#### Function of EcoRoll mode

EcoRoll mode is available in all drive-specific shift programs.

When EcoRoll mode is activated, the vehicle is not decelerated by the continuous brake when the accelerator pedal is released and no recuperation takes place. The vehicle rolls freely.

If EcoRoll mode is deactivated, the vehicle's movement energy is used to charge the high-voltage battery (recuperation) when the accelerator pedal is released.

In the multimedia system, you can activate and deactivate EcoRoll mode under the **Operation** menu, menu item **Settings, Assistance systems**.

#### Note

The last EcoRoll mode setting remains stored even after the vehicle is restarted.

### One-pedal driving mode

#### Function of one-pedal driving mode



**Danger**

#### Risk of accident due to insufficient deceleration in one-pedal driving mode

Irrespective of the position of the accelerator and the associated deceleration due to recuperation, the braking effect in one-pedal driving mode may not be sufficient depending on the situation.

- Always be ready to brake even when one-pedal driving mode is switched on.
- If necessary, also use the service brake to brake the vehicle.

The one-pedal driving mode function is a comfort function that allows the driver to decelerate and accelerate the vehicle using only the accelerator pedal.

When the one-pedal driving mode function is activated, the vehicle is driven or decelerated by means of recuperation, ac-

cording to the position of the accelerator pedal.

The position of the accelerator pedal has the following effects:

- 0% pedal travel – maximum deceleration via recuperation
- 1% - 29% pedal travel – recuperation, and thus deceleration, decreases as the pedal travel increases
- approx. 30% pedal travel – the vehicle rolls freely. There is neither deceleration nor drive.
- 31% - 100% pedal travel – the vehicle drive increases as the pedal travel increases. At 100% pedal travel, maximum output is available.

You can set the level of recuperation that occurs between 0% and 29% pedal travel using the brake stages of the continuous brake. Depending on the vehicle load, select a higher brake stage for a greater vehicle load ( → Page 138).

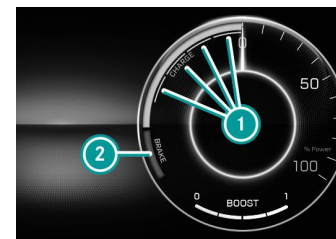
The current recuperation level or drive output is shown in the power availability display on the instrument cluster

( → Page 77). The indicator lamp  on the instrument cluster also lights up during recuperation.

If the vehicle speed is less than 20 km/h, the stored recuperation power is automatically reduced. At vehicle speeds below 4 km/h, the vehicle can no longer be decelerated using recuperation. Use the service brake to stop the vehicle.

#### Activating/deactivating one-pedal driving mode

##### Switching on



Power availability display (example)

The one-pedal driving mode function is always inactive when the engine is started.

- In the multimedia system, in the menu [Operation](#) menu item [Settings](#),

# Driving mode

## Transmission

Assist systems activate the One-pedal driving mode.

- Pull the multifunction lever to brake stage 1-5 (→ Page 138). Select a high brake level when the load condition is high.

- ◁ The number of bars 1 in the power availability display of the instrument cluster corresponds to the brake stage selected.

The braking power scale 2 shows how much energy was expended during deceleration by the service brake.

- **Switching off:** Push the multifunction lever to brake stage 0 (→ Page 138).

## Transmission

### Function of the multifunction lever and the gear display

When driving, the transmission control controls gear operation, e.g. in the following situations:

- when pulling away
- when manoeuvring
- When changing gear
- when stopping

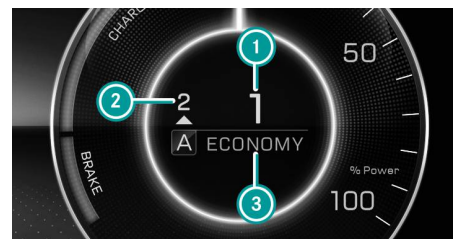
### Multifunction lever



- 1 Select direction of travel:
  - D** Driving forwards (→ Page 143)
  - N** Neutral position (→ Page 143)
  - R** Reversing
- 2 Select drive program (→ Page 142):
  - A** automatic drive program with a drive-specific shift program
  - M** Manual drive program
  - Manual downshift, in the automatic (→ Page 144)/manual (→ Page 144) drive program
- 3

- 4 **+** Manual upshift, in the automatic (→ Page 144)/manual (→ Page 144) drive program

### Gear display



- 1 Direction of travel and/or selected gear
- 2 Gear shifting recommendation (shift up ↑/down ↓) or preselected gear (flashing)
- 3 Drive program

| Possible displays | Meaning                 |
|-------------------|-------------------------|
| 1-2               | 1st and 2nd gear        |
| N                 | Neutral position        |
| R                 | Reverse gear            |
| A                 | Automatic drive program |

| Possible displays                              | Meaning                                                   |
|------------------------------------------------|-----------------------------------------------------------|
| A ECONOMY<br>A POWER<br>A RANGE<br>A MUNICIPAL | Automatic drive program with drive-specific shift program |
| M                                              | Manual drive program                                      |

### Drive programs and drive functions

#### Notes on drive programs

#### Automatic transmission with drive-specific shift program

The shift program is designed for the following functions:

- The performance-oriented, dynamic driving operation — **POWER**
- The fuel-efficient driving operation — **ECONOMY**
- The range extending driving operation — **RANGE**
- The refuse collection operation — **MUNICIPAL**

The **POWER** shift program has the following features:

- The maximum output is available.
- The adjustable speed in cruise control is 90 km/h.
- The maximum vehicle speed is 90 km/h.

The **ECONOMY** shift program has the following features:

- The available output is slightly reduced.
- The adjustable speed in cruise control is limited to 85 km/h.
- The maximum vehicle speed is 90 km/h.

The **RANGE** shift program has the following features:

- The available output is more significantly reduced.
- The heating and air-conditioning output is limited.
- The "kickdown" function is deactivated
- The adjustable speed in cruise control is limited to 80 km/h.

- The maximum vehicle speed is limited to 80 km/h.

The **MUNICIPAL** shift program is specially designed for use in refuse collection operation.

#### Note

The shift program **RANGE** is automatically activated if the state of charge is low. To increase the remaining range of your vehicle: You have the option of manually switching to the **ECONOMY** or **POWER** drive program.

#### Manual drive program

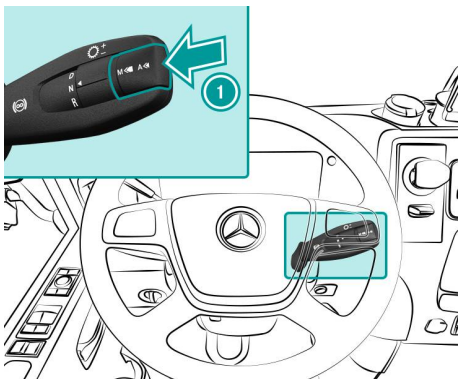
In the manual drive program, you change gears yourself. The gearshift recommendation is shown on the display of the instrument cluster.

In particularly difficult driving conditions, switch to the manual drive program. In this way, you can avoid the unwanted interruptions in tractive power that can occur during automatic gear changes.

## Driving mode

### Transmission

#### Selecting drive program



After the vehicle is started, the **ECONOMY** drive program is always switched on.

After the display check, the display shows e.g. **A ECONOMY**.

The drive program can be changed at any time.

- **To switch manual drive program on:** Press and hold the button **A/M** for ①.
  - ◁ The display will show the manual drive program **M**, the selected

gear and a gearshift recommendation.

- **To switch automatic on:** Briefly press the button ①.
  - ◁ The display will show the automatic drive program **A** and the selected and preselected gears.
- **Shift the automatic shift program:** Briefly press the button ①.
  - ◁ When the display shows **A**, the standard shift program is switched on. If, for example, the display shows **A ECONOMY** or **A POWER**, the drive-specific shift program is engaged.

#### Rocking free function

You can use the rocking-free drive function to rock the vehicle out of a hollow in the terrain in first gear or in reverse gear. After the rocking free drive function is switched on, the transmission control automatically switches to the manual drive program. If the rocking free function is switched on, the vehicle reacts more quickly to changes in the accelerator ped-

al position. The vehicle thereby can be set into a rocking motion.

After engaging the start-off gear, switch the rocking free drive function on or off via the multimedia system in the **Operation** menu, under the **Switch** menu item.

The rocking free function switches off automatically if you travel at a speed greater than approx. 8 km/h.

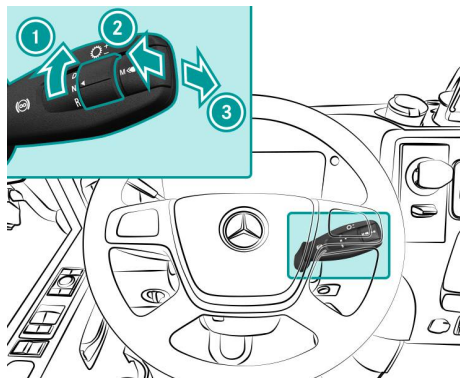
#### Activating/deactivating rocking free

- In the multimedia system, press the button  under the **Switch** menu item of the **Operation** menu.
  - ◁ When the indicator lamp in the  button lights up, the rocking free drive function is switched on.



### Pulling away and stopping the vehicle

#### Pulling away



- ▶ Start the vehicle.
- ▶ Depress the brake pedal.
- ▶ Turn the direction switch to the **D** position (drive/driving forwards) ①.
  - ◁ In all drive programs, the transmission control selects a suitable start-off gear, which can be changed manually, depending on the vehicle load.
- ▶ Release the brake pedal and depress the accelerator pedal.

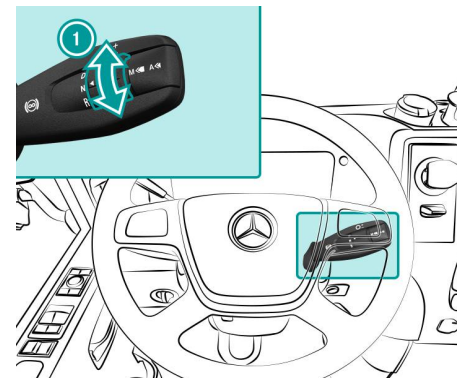
#### Changing the start-off gear

- ▶ Pull the multifunction lever up briefly (to shift up) ② or push it down briefly (to shift down) ③.
  - ◁ The transmission control shifts up or down one gear. When the selected gear appears in the display, the shift operation is completed.

#### Stopping

- ▶ Depress the brake pedal.
  - ◁ The transmission control shifts down according to the driving situation.

#### Shifting to neutral position



- ▶ Depress the brake pedal.
- ▶ Turn the direction switch to **N** position (neutral) ①.

## Driving mode

### Transmission

#### Driving in the automatic drive program

##### Using kick-down

###### **i** Note

The kick-down function is only available in the **ECONOMY** and **POWER** shift programs.

The kick-down serves to accelerate the vehicle to the maximum extent.

- ▶ Depress the accelerator pedal beyond the point of resistance as far as it will go.
- ▶ Ease off the accelerator pedal slightly once the desired speed is reached.

##### Decelerating

- ▶ Release the accelerator pedal.
- ▶ Depress the brake pedal.

or

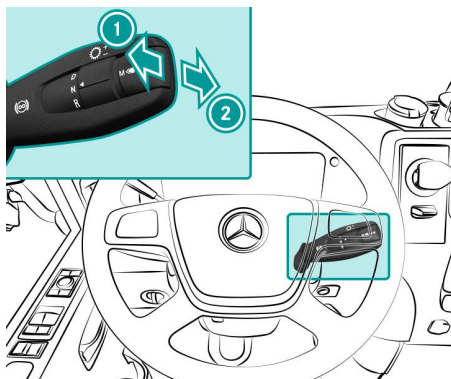
- ▶ Activate the continuous brake (recuperation).
  - ◁ The vehicle shifts down automatically according to the driving

situation or increases the recuperation.

###### **i** Note

If the one-pedal driving mode function is switched on, the vehicle can also be decelerated only by releasing the accelerator pedal (→ Page 139).

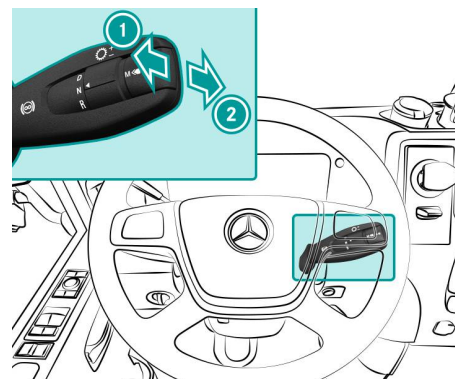
##### Selecting a gear manually



Another gear can also be selected manually. The automatic control functions are not changed by doing this.

- ▶ Pull the multifunction lever up briefly (to shift up) ① or push it down briefly (to shift down) ②.

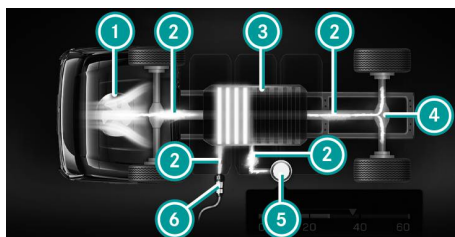
#### Driving using the manual drive program



- ▶ Pull the multifunction lever up briefly (to shift up) ① or push it down briefly (to shift down) ②.

### Operation

### Displaying the energy flow



Energy flow display (example)

- ① Cab climate control
- ② Energy flow
- ③ High-voltage battery
- ④ Electric motor
- ⑤ Electric power take-off system (power take-off)
- ⑥ Charging cable connector

► In the multimedia system , in menu **Status** choose the menu item **Energy flow**.

- ◁ The energy flow display shows the components of the high-voltage on-board electrical sys-

tem for climate control, the drive and the electric power take-off system. The energy flow between the components and the high-voltage battery is shown in colour.

The energy flow will be shown in different colours depending on the operating status:

- **White:** Constant energy flow, driving at constant speed or with moderate acceleration
- **Bronze:** High energy flow, e.g. strong acceleration (kick-down with boost effect)
- **Blue:** Energy flow when charging the high-voltage battery or through recuperation

The charging cable connector ⑥ and its energy flow are only displayed during a charging process.

Only when the electric power take-off system is switched on, the power take-off ⑤ is highlighted in colour in the display and the energy flow is shown. In addition, the

current power output and the possible, available power output are then shown in a bar graph.

### Notes on axle and wheel loads



#### NOTE

#### Damage to vehicle parts if the permissible gross mass is exceeded

Vehicle parts can be damaged in the following cases:

- The permissible gross mass is exceeded.
- The permissible axle load is exceeded.
- The permissible wheel load is exceeded.
- The permissible wheel load difference of 4% is exceeded.

The following vehicle parts could be damaged:

- Tyres
- Chassis
- Axle

## Driving mode

### Operation

Refer to the following permitted values:

- Maximum gross weight
- Axle and wheel load
- Wheel load difference of 4%

Observe the notes about attachments, add-ons, installations and conversions (→ Page 3).

While driving, regularly monitor the indicator and indicator lamps and the displays in the instrument cluster.

### Displaying axle loads and setting axle load indicator

#### Displaying axle loads

Vehicles with air suspension: The vehicle may be equipped with an axle load display. The axle load display is not calibrated, nor is it a system capable of being calibrated. The measured data provide only an approximate guide. The values cannot be used for official purposes. In order to avoid inaccuracies in the measurement, make sure that the vehicle is uniformly laden. When the chassis is at driving level, the axle load measuring sys-

tem determines the axle load from the pressure in the air suspension bellows.

- ▶ Park the vehicle on a level surface.
- ▶ Apply the parking brake.
- ▶ Set driving level (→ Page 193).
- ▶ Switch off the moving-off aid (→ Page 197).
- ▶ In the multimedia system , select in the **Status** menu the **Axles** menu item.
  - ◀ The axle load of the individual axles and the gross vehicle weight are displayed.

You can adjust the axle load display if there is a notable difference between the axle load display and the weighbridge results.

#### Setting the axle load display

- ▶ Drive the vehicle onto a single-axle load-measuring device.
- ▶ Displaying the axle loads.
- ▶ Press the Axle load calibration button .
  - ◀ The input window **Axle load calibration** is displayed.

- ▶ Select the desired axle.
- ▶ Correct the axle load according to the value shown by the single-axle load-measuring device.

### Function of ASR (anti-slip control)

If the drive wheels spin on both sides, ASR automatically reduces engine output.

If ASR intervenes, this has the following effects:

- The  indicator lamp flashes in the instrument cluster.
- Cruise control cannot be activated.

### Note

TC stands for Traction Control. This is the process whereby the power transmission between the tyres and the road surface is regulated by ASR.

If cruise control has already been activated, it will remain active. It is not possible, however, to accelerate or decelerate using cruise control.

When you switch on the vehicle, the  indicator lamp lights up in the instrument cluster and goes out again after approximately two seconds.

If the  indicator lamp does not go out, ASR has a malfunction. Have the fault rectified at a qualified specialist workshop.

#### Deactivating and activating ASR



**Danger**

#### Risk of skidding if ASR is deactivated

If you deactivate ASR, ASR cannot carry out vehicle stabilisation when pulling away and accelerating.

- ▶ ASR must only be deactivated in the following situations.



**Note**

Vehicles with Stability Control Assist lack the  button. Traction control (ASR function) is part of Stability Control Assist (→ Page 147). When Stability Control Assist is deactivated, traction control is also deactivated.

It may be best to deactivate ASR in the following situations:

- When using snow chains
- When driving on routes with a loose surface, e.g. gravel roads
- ▶ If the carriageway is slippery, engage the differential lock in an anticipatory fashion (→ Page 148).
- ▶ Observe the information on deactivating and activating Stability Control Assist (→ Page 147).

#### Function of Stability Control Assist

Stability Control Assist is operational at speeds above approximately 20 km/h, regardless of the operating status of the service brake or continuous brake. If Sta-

bility Control Assist intervenes, the indicator lamp  in the instrument cluster flashes.

If the indicator lamp  in the instrument cluster lights up continuously when the vehicle is started, Stability Control Assist has a fault. Have Stability Control Assist checked at a qualified specialist workshop.

Stability Control Assist stabilises the vehicle using the following automatic control interventions:

- Output reduction
- Targeted braking of individual wheels

#### Deactivating and activating Stability Control Assist



**Danger**

#### Risk of skidding and accident due to deactivating Stability Control Assist

If you deactivate Stability Control Assist, it cannot stabilise the vehicle.

## Driving mode

### Operation

- ▶ Only deactivate Stability Control Assist in the following situations.

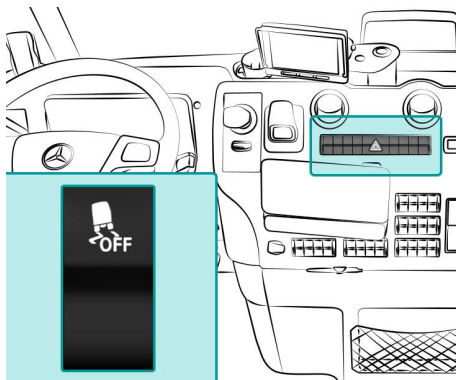
#### Note

When you start the vehicle, Stability Control Assist is automatically activated.

It may be better to deactivate Stability Control Assist in the following situations:

- when driving on a loose surface
- when driving with snow chains
- when operating a snow plough

ASR is also deactivated. Activate Stability Control Assist as soon as the situations previously described are no longer present.



Example: Stability Control Assist OFF button

- ▶ Press the  button.
  - ◀ If the  indicator lamp on the instrument cluster comes on, Stability Control Assist is deactivated.

### Using differential locks

#### Danger

##### **Risk of accident when the differential lock is engaged**

When driving off-road or driving with an engaged differential lock in the automatic drive program, the electronic management system may perform unwanted gear changes.

The vehicle may, for example, roll backwards on slopes.

- ▶ Always drive carefully and be ready to brake.
- ▶ Switch to the manual drive program.

#### Danger

##### **Risk of accident when differential lock is engaged on a firm surface**

If you engage differential locks on a firm and non-slip surface, you may lose control of the vehicle.

- ▶ Always disengage the differential locks immediately on a firm, high-traction surface.

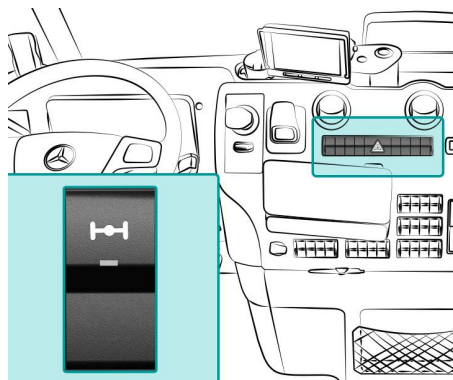


### NOTE

#### Damage to the differential as a result of improper use of the differential locks

Observe the following items to prevent damage:

- ▶ Avoid spinning drive wheels on one side, engage the differential locks in advance.
- ▶ Do not engage the differential locks when the accelerator or brake pedal is depressed.
- ▶ Only engage the differential lock when the vehicle is stationary or at walking pace.
- ▶ Pull away slowly after engaging the differential lock.
- ▶ Do not drive on high-grip roadways with engaged differential locks.
- ▶ Do not drive faster than 50 km/h with the differential locks engaged.



Differential lock switch (example)

The status of the differential lock is shown in the menu **Status** menu item **Axles** of the multimedia system.

When the differential lock is engaged, the display will show the associated status display  in yellow. When the differential lock is disengaged, the display will show the associated status display  in grey. If the status display flashes, the differential lock is not yet engaged or disengaged. The conditions for engaging or disengaging have not been fulfilled due to differing wheel speeds, for example. The

differential lock will be engaged automatically as soon as all engagement conditions are met.

Engage the differential locks, e.g. on slippery road surfaces or off-road, to prevent drive wheels from spinning on one side.

- ▶ In particularly challenging driving conditions, select the manual drive program to avoid unwanted gear shifting operations and interruptions to tractive power (→ Page 141).

### Applying

- ▶ Stop the vehicle.
- ▶ Press the  button above.
  - ◁ The indicator lamp on the  button will light up.
  - The indicator lamp  in the instrument cluster lights up.
  - The instrument cluster shows a display message with the current status of the differential lock.

### Deactivating

- ▶ Press  button below.
  - ◁ The indicator lamp in the  button goes out.

## Driving mode

### Driving systems

The instrument cluster shows a display message with the current status of the differential lock.

If the differential lock is engaged and you are driving faster than 50 km/h, the current setting of the differential lock is again shown in the display.

- ▶ Disengage the differential lock or drive slower than 50 km/h.

#### Driving systems

#### Reversing camera

#### Reversing camera display

The reversing camera is only an aid. It is not a substitute for your attention to your surroundings. The responsibility for safe manoeuvring and parking remains with you. When manoeuvring and parking, make sure that there are no persons, animals or objects in the manoeuvring area. You could otherwise endanger yourself and others.

The reversing camera cannot replace a safety officer or signaller. When manoeuv-

ring or parking, always seek assistance from a safety officer or signaller.

When reverse gear is engaged, the image from the reversing camera is automatically displayed. Depending on the equipment, the image from the reversing camera is shown on the multimedia system display or on the monitor of the blind spot camera system (→ Page 151).

It is not possible to change the display manually when the reverse gear is engaged.

If the display shows only a black, grey or blue image instead of the image from the reversing camera, the reversing camera is malfunctioning. In this case, have the reversing camera checked at a qualified specialist workshop.

Depending on the equipment, you can adjust the screen brightness in the multimedia system (→ Page 92) or on the display of the blind spot camera system (→ Page 151).

#### System limits

In the following situations, the reversing camera only has limited function, or does not function at all:

- The weather conditions are poor, e.g. heavy rain, snow, fog, storm or spray.
- The light conditions are poor, e.g. at night or with glare.
- if the surroundings are illuminated with fluorescent light, e.g. fluorescent lamps or LED lighting. The display may flicker.
- The camera lens is covered, dirty or fogged up. Observe the notes on cleaning the reversing camera (→ Page 219).
- The camera or the rear of the vehicle is damaged. In this case, have the camera, its position and setting checked at a qualified specialist workshop.

Do not use the reversing camera in these situations. You could injure others or collide with objects when parking.



**Note**

The contrast of the display may be impaired by sunlight or other light sources, e.g. when you are driving out of a garage. In these cases, be particularly attentive.

**Note**

If the display's usability is significantly restricted, e.g. in the event of pixel errors, have it repaired or replaced.

**Blind spot camera system****Notes on the blind spot camera system****Danger****Risk of distraction by information systems and communications equipment**

If you operate information systems and communications equipment integrated in the vehicle when driving, you may be distracted from what is happening in

traffic. You may also lose control of the vehicle.

- ▶ Only operate this equipment when the traffic situation permits.
- ▶ If you cannot be sure of this, stop the vehicle in accordance with the traffic conditions and operate the equipment while the vehicle is stationary.

Do not make any alterations to the blind spot camera system setting in the monitor. If you change the blind spot camera system's factory settings, the vehicle's registration for use on public roads can be revoked.

The blind spot camera system includes four cameras and one monitor. It replaces the front mirror and also displays images of the immediate surroundings of the vehicle. The blind spot camera system is only an aid. The cameras may show a distorted view of obstacles, show them incorrectly or not at all. The blind spot camera system is not a substitute for your attention to the immediate surroundings.

The responsibility for safe manoeuvring and parking remains with you. When manoeuvring and parking, make sure that there are no persons, animals or objects in the manoeuvring area.

You are always responsible for safety, and must always pay attention to your surroundings when parking and manoeuvring. This applies to the areas behind, in front of and beside the vehicle. You could otherwise endanger yourself and others.

The reversing camera cannot replace a safety officer or signaller. When manoeuvring or parking, always seek assistance from a safety officer or signaller.

Under the following circumstances, the cameras will not function, or will only function in a limited manner:

- if there is heavy rain, snow or fog
- if the area is lit by fluorescent light or LED lighting, for example
- if the camera lenses are dirty or covered

## Driving mode

### Driving systems

- if the vehicle components in or to which the cameras are fitted are damaged

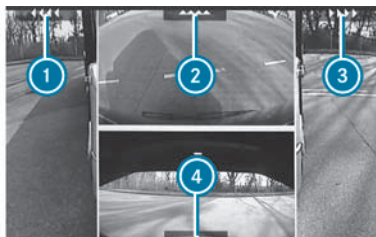
Be especially attentive in these situations. You could otherwise injure others during your manoeuvre or objects may damage the vehicle. If there is a malfunction, particularly if the front camera fails, have the camera system repaired at a qualified specialist workshop immediately.

You can make minor adjustments to the field of vision for the two blind spot cameras. Each camera's housing can be adjusted to a degree if you employ extra force. Mercedes-Benz recommends that you have this work carried out at a qualified specialist workshop.

An overview of the cameras can be found in the notes on cleaning (→ Page 219).

Observe the information on the monitor in the operating instructions provided by the manufacturer.

#### Monitor displays



Split-screen view including all camera images

- ① Symbol for left-side blind spot camera
- ② Symbol for front camera
- ③ Symbol for right-side blind spot camera
- ④ Symbol for reversing camera

Camera images are combined in a split-screen view or displayed on the monitor in full-screen mode. The ① to ④ symbols indicate the respective camera position.

Depending on your direction of travel, the monitor displays the following views:

- all four camera images in split-screen view when the vehicle is stationary

- the blind spot camera images as well as the reversing camera image as a split-screen view when reverse gear is engaged
- the front camera image when the vehicle is travelling forwards up to a speed of approx. 10 km/h
- when the vehicle is travelling forwards from a speed of approx. 10 km/h, depending on the selected setting
  - in automatic mode, the camera images relevant to your speed and traffic situation, such as when you make a turn or during manoeuvring. The front camera image will no longer be displayed from a speed of approx. 35 km/h
  - in manual mode, the selected camera image in full-screen mode

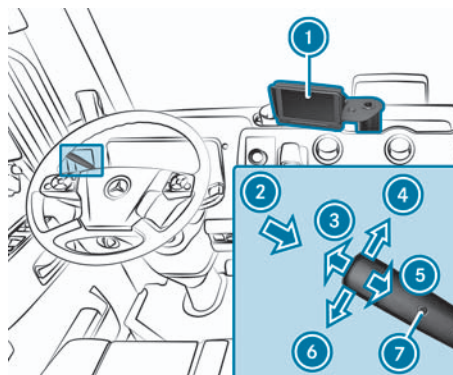
### Adjusting the blind spot camera system display

#### **i** Note

Manual selection of a camera image or switching to manual mode is only implemented by the system when the transmission is in idle mode or a forward gear.

When reverse gear is engaged, the image from the blind spot cameras and the reversing camera is always shown as a split-screen view.

### Selecting automatic mode



- ① Monitor (see the manufacturer's operating instructions)
  - ② To switch between automatic and manual mode
  - ③ To display the reversing camera image
  - ④ To display the right-side blind spot camera image
  - ⑤ To display the front camera image
  - ⑥ To display the left-side blind spot camera image
  - ⑦ Indicator lamp for manual mode
- Switch on the vehicle.

- Push the operating lever in the direction of arrow ②.
  - ◁ Indicator lamp ⑦ goes out and automatic mode is activated.
- Push the operating lever in the direction of arrow ③, ④, ⑤ or ⑥.
  - ◁ Monitor ① displays the corresponding camera image in full-screen mode for approximately five seconds.

### Selecting manual mode

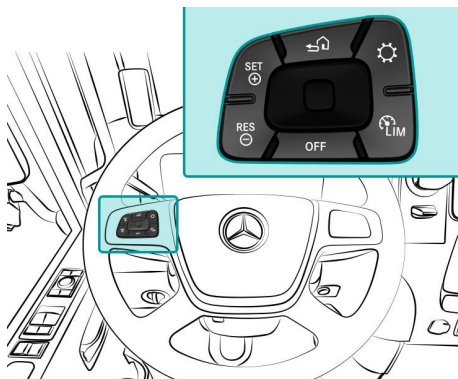
- Switch on the vehicle.
- Push the operating lever in the direction of arrow ②.
  - ◁ Indicator lamp ⑦ lights up and manual mode is activated.
- Push the operating lever in the direction of arrow ③, ④, ⑤ or ⑥.
  - ◁ Monitor ① displays the corresponding camera image in full-screen mode.

## Driving mode

### Driving systems

#### Limiters

##### Overview of the limiter



Short button press: Select cruise control (→ Page 155) or Proximity Control Assist (→ Page 158)

Long button press: Selecting the limiter



Activates and sets the current speed/limit speed, increases the set speed/limit speed



Activates and shows the stored speed/limit speed, decreases the set speed/limit speed



Deactivating the limiter

The **LIM** symbol in the instrument cluster shows the status of the limiter in colour:

- Grey symbol: The limiter is selected, but not engaged.
- White symbol: The limiter is engaged and restricts the vehicle speed to the limit speed you set.

#### Activating and deactivating the limiter

##### Selecting the limiter

The limiter restricts the vehicle speed to the set limit speed. You can accelerate the vehicle up to the limit speed using the accelerator pedal. In order to keep the set limit speed on downhill slopes, the limiter automatically brakes the vehicle with the continuous brake. If the set speed is exceeded, the **LIM** symbol will flash in the instrument cluster. If you cannot switch on the limiter, the instrument cluster shows the display message **--- km/h** in grey.

- ▶ Press and hold the button for

- ◁ The instrument cluster shows the **LIM** symbol in grey.

##### Activation while driving

- ▶ Select the limiter.
- ▶ Drive at the desired speed and briefly press the button.
  - ◁ The limiter is activated and the current vehicle speed is stored as the limit speed.

or

- ▶ Briefly press the button.
  - ◁ The limiter is activated and adopts the stored limit speed. The instrument cluster shows the **LIM** symbol and the set speed limit in white.

##### Increasing or decreasing the limit speed

You can change the settings of the limit speed while driving.

- ▶ Activate the limiter.
- ▶ **In 1 km/h increments:** Press the or button briefly until

the desired speed is displayed in the instrument cluster.

or

- ▶ Press and hold the  or  button until the desired speed is displayed in the instrument cluster.

### Driving

You can exceed the set limit speed, e.g. when overtaking.

- ▶ Briefly depress the accelerator pedal beyond the pressure point (kick-down).
  - ◁ The set limit speed is still shown and the  symbol flashes in the instrument cluster.
- ▶ When overtaking is complete, briefly release the accelerator pedal and depress it again.
  - ◁ The limiter again restricts the vehicle speed to the set limit speed.

### Deactivating

The limit speed remains stored if you deactivate the limiter.

- ▶ Press the  button.
  - ◁ The instrument cluster shows the  symbol in grey.

or

- ▶ Using the  button, select cruise control or the Proximity Control Assist.
  - ◁ The instrument cluster shows the  or  symbols and the set speed in grey.

### Cruise control

#### Overview of cruise control



**Danger**

#### Risk of accident due to unknown stored speed

If you call up the stored speed and it differs from the current speed, the vehicle will accelerate or brake.

If you do not know the stored speed, the vehicle may accelerate or brake unexpectedly.

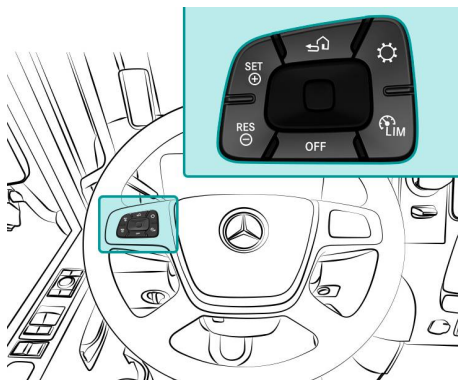
- ▶ Take the traffic situation into account before calling up the stored speed.
- ▶ If the stored speed is not known, store the desired speed again.

Do not use cruise control in the following situations:

- if the traffic conditions do not allow you to drive at a constant speed (e.g. in heavy traffic or on winding roads)
- if the drive wheels could lose traction on slippery roads, when braking or accelerating, for example
- if there is poor visibility, e.g. due to fog, heavy rain or snow

## Driving mode

### Driving systems



Short button press: Select cruise control/Proximity Control Assist  
Long button press: Selecting the limiter



Activate and set the current limit speed/increase the set limit speed



Activate and call up the set limit speed/decrease the set limit speed



Deactivates cruise control

The  symbol in the instrument cluster shows the status of cruise control by colour:

- Grey symbol: Cruise control has been selected, but not activated
- White symbol: Cruise control has been activated and maintains the set speed

#### Activating and deactivating cruise control

##### Selecting cruise control

Cruise control maintains the set vehicle speed on your behalf. If the set speed is exceeded on downhill slopes by more than the set speed tolerance, the continuous brake is applied automatically.

If you are driving at a speed slower than 15 km/h, you will not be able to activate cruise control.

If you cannot switch on the cruise control, the instrument cluster shows the display message --- km/h in grey.

Cruise control is automatically deactivated in the following situations:

- If you are driving at under 10 km/h
- you shift the transmission into neutral for more than five seconds

If cruise control is deactivated automatically, an acoustic signal sounds

- ▶ Press the  button repeatedly until the instrument cluster displays the  symbol in grey.

##### Activation while driving

- ▶ Select cruise control.
- ▶ Drive at the required speed and briefly press the  button.
  - ◁ Cruise control is activated and the current speed is set.

or

- ▶ Briefly press the  button.
  - ◁ Cruise control is activated and adopts the stored speed. The instrument cluster displays the  symbol and the set speed in white.
- ▶ Release the accelerator pedal.
  - ◁ In order to maintain the set speed, cruise control automatically brakes or accelerates the vehicle.

### Increasing and reducing the speed

When cruise control is active, you can change the speed setting while driving.

#### ► To adjust in 0.5 km/h increments:

Press  or  button repeatedly until the desired speed is shown in the instrument cluster.

or

#### ► To adjust in 5 km/h increments:

Press and hold the  or  button until the desired speed is displayed in the instrument cluster.

### Driving tips

In the **A ECONOMY** and **A RANGE** drive programs, cruise control maintains the set speed less forcefully. This explains why the vehicle speed and the set speed may differ slightly under certain circumstances. This results in lower energy consumption. In the **A RANGE** drive program, the speed can be set to a maximum of 80 km/h and in the **A ECONOMY** drive program, to a maximum of 85 km/h. In the **A ECONOMY** drive program, you can exceed the maximum speed of 85 km/h

by depressing the accelerator pedal, e.g. when overtaking.

You use the continuous brake to decelerate. Cruise control remains activated.

If you reset the continuous brake lever, but do not deactivate it, the vehicle will only accelerate on downhill slopes up to the set speed.

If the continuous brake is deactivated, the vehicle will accelerate to the last stored speed.

If cruise control is decelerating the vehicle using the continuous brake and you simultaneously depress the brake pedal, cruise control remains activated.

When the continuous brake is activated and you activate cruise control, the continuous brake regulates the set speed on downhill slopes.

### Overtaking

It is possible to exceed the set speed, e.g. when overtaking.

- Depress the accelerator pedal.

- Once the overtaking manoeuvre is finished, release the accelerator pedal again.

◁ Cruise control adjusts the vehicle's speed to the set speed.

### Deactivating

The speed remains stored if you deactivate cruise control.

- Press the  button.

or

- When cruise control accelerates the vehicle, depress the brake pedal.

◁ The instrument cluster shows the  symbol and the set speed in grey.

or

- Press and hold the  button to select the limiter.

◁ The instrument cluster shows the  symbol in grey.

or

- Select Proximity Control Assist with the  button.

## Driving mode

### Driving systems

- ◁ The instrument cluster displays the  symbol and the set speed in white. Proximity Control Assist is activated.

#### Proximity Control Assist

##### Overview of Proximity Control Assist



#### **Danger**

##### **Risk of accident due to the Proximity Control Assist's detection limitations**

Proximity Control Assist does not react:

- to pedestrians or animals.
- to stationary obstacles on the carriageway, e.g. stationary or broken-down vehicles.
- to oncoming vehicles or cross traffic

Proximity Control Assist is therefore unable to issue a warning or intervene in these situations.

- ▶ Always carefully observe the traffic conditions and be ready to brake at all times.

Proximity Control Assist reacts to stationary obstacles that the sensors have previously classified as moving, e.g. vehicles stopping at a traffic light.



#### **Danger**

##### **Risk of accident due to Proximity Control Assist's detection limitations**

Proximity Control Assist cannot always clearly identify other road users and complex traffic situations.

In these cases, Proximity Control Assist may unexpectedly accelerate or brake the vehicle.

- ▶ Drive on carefully and be ready to brake, particularly if Proximity Control Assist issues a warning.



#### **Danger**

##### **Risk of accident if Proximity Control Assist does not decelerate sufficiently**

Proximity Control Assist only brakes your vehicle with part of the maximum possible braking power. If this deceler-

ation is not sufficient, Proximity Control Assist will give you a visual warning.

- ▶ Apply the brakes yourself in these situations and try to take evasive action.



#### **Note**

Observe the notes on particular driving situations under "Active Brake Assist" (→ Page 171).

Proximity Control Assist has been developed for journeys on motorways and high-speed major roads. The system is not intended for use in urban traffic, for example.

Do not use Proximity Control Assist when towing or using the vehicle as a work machine.

The system may be impaired or may not function in the following situations:

- If there is poor visibility, e.g. due to insufficient road illumination or due to snow, rain, fog or heavy spray
- If there is glare, for example, from oncoming traffic, direct sunlight or



reflections (e.g. if the carriageway is wet)

- If the windscreen in the area of the camera is dirty, misted up or damaged
- If the windscreen in the area of the camera is obscured, e.g. due to a faulty windscreen wiper or a sticker
- if the carriageway is very narrow and winding
- If the radar sensor is impaired due to interference from other radar sources, such as strong radar reflections in car washes

Proximity Control Assist may not detect narrow vehicles driving in front, e.g. motorcycles or vehicles driving on a different line.

In particular, be aware of the following driving situations:

- Cornering, entering and exiting bends
- Your own vehicle driving on a different line or vehicles in front of you driving on a different line

- The loads of vehicles in front that protrude into the lane
- Other vehicles changing lane
- Vehicles turning off
- Overtaking
- Winding stretches of road
- Obstacles and stationary vehicles

Regularly clean the distance sensor cover and the windscreen in the camera's field of vision ( → Page 219).

If the distance sensor cover or the windscreen in the camera's field of vision is dirty or icy, its functionality may be impaired.

Do not use Proximity Control Assist in the following situations:

- On slippery roads. The drive wheels may lose their grip when braking or accelerating and the vehicle may skid.
- When there is low visibility, e.g. due to fog, heavy rain or snow.
- In cities and on country roads.

If Proximity Control Assist no longer detects a vehicle driving in front because it has turned off, for example, Proxim-

ity Control Assist may accelerate to the stored speed. On a filter lane or a slip road, this speed may be too high.

The  symbol in the instrument cluster shows the Proximity Control Assist status by colour:

- Grey symbol: Proximity Control Assist is selected, but not activated.
- White symbol: Proximity Control Assist is activated, but a vehicle in front has not been detected.
- White symbol and vehicle within the blue symbol: Proximity Control Assist is activated and a vehicle in front has been detected.

If a vehicle ahead has been detected, the instrument cluster additionally displays its driving speed and the distance to the detected vehicle.

## Driving mode

### Driving systems

#### Note

The **Assistance**  menu in the instrument cluster also shows the vehicle speed of the detected vehicle and the distance to the detected vehicle.

#### Activate and deactivate Proximity Control Assist

##### Functions and activation conditions

Proximity Control Assist controls the speed and supports you in automatically maintaining the distance from a vehicle detected in front. The Proximity Control Assist functions similarly to cruise control from a speed of 15 km/h.

If Proximity Control Assist detects a vehicle in front driving at a slower speed, your vehicle is slowed and the specified distance selected by you is maintained.

Proximity Control Assist brakes the vehicle with the continuous brake in the following situations:

- The vehicle exceeds the set speed, including the set speed tolerance, e.g. on a downhill gradient.

- A slower vehicle in front has been detected.

When the continuous brake slows the vehicle, the indicator lamp  in the instrument cluster lights up.

Proximity Control Assist may also brake the vehicle using the service brake in order to maintain the specified distance.

If the vehicle in front is no longer detected, e.g. if it changes lane, your vehicle will accelerate up to the set speed.

Proximity Control Assist cannot be activated or is deactivated automatically in the following situations:

- You shift the transmission into neutral for more than approximately five seconds.
- If you depress the brake pedal while Proximity Control Assist accelerates the vehicle.
- You select the reverse gear.
- There is a malfunction in the brake system or electronic management system.

- The distance sensor initialisation is not yet complete.

#### Note

At speeds of below 15 km/h, Proximity Control Assist cannot be switched on.

At vehicle speeds of below 10 km/h, Proximity Control Assist automatically switches off.

If Proximity Control Assist is deactivated automatically, an acoustic signal sounds.

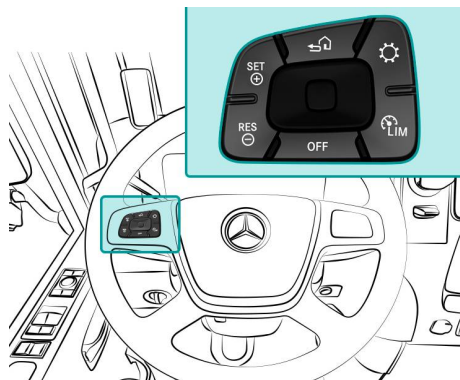
If you cannot switch on Proximity Control Assist, the instrument cluster shows the display message **-- .- km/h** in grey.

Proximity Control Assist remains activated in the following situations:

- You brake using the continuous brake.
- Proximity Control Assist decelerates the vehicle using the continuous brake or service brake and you simultaneously depress the brake pedal.

If your vehicle accelerates and you depress the brake pedal, Proximity Control Assist is deactivated automatically.

### Select Proximity Control Assist



Short button press: Select Proximity Control Assist / cruise control  
Long button press: Selecting the limiter



Activate and set the current limit speed/increase the set limit speed



Activate and call up the set limit speed/decrease the set limit speed



Deactivate Proximity Control Assist

- ▶ Press the  button repeatedly until the instrument cluster displays the  symbol in grey.

If you switch from cruise control to Proximity Control Assist and it was previously on, the instrument cluster displays the  symbol in white. Proximity Control Assist is activated. The vehicle adapts its speed to that of the vehicle in front, but only up to the desired and set speed.

### Activation while driving

- ▶ Drive faster than 15 km/h.
- ▶ Select Proximity Control Assist.
- ▶ Briefly press the  button.
  - ◁ Proximity Control Assist is activated and the current speed is set.

or

- ▶ Briefly press the  button.
  - ◁ Proximity Control Assist is activated and the last stored speed is set. The instrument cluster displays the  symbol and the set speed in white.

- ▶ Release the accelerator pedal.
  - ◁ The vehicle adapts its speed to that of the vehicle in front, but only up to the desired and set speed.

### Increasing or reducing the speed

You can only change the speed setting while driving.

- ▶ **To adjust in 0.5 km/h increments:**  
Press  or  button repeatedly until the desired speed is shown in the instrument cluster.
- or
- ▶ **To adjust in 5 km/h increments:**  
Press and hold button  or  until the desired speed is displayed on the instrument cluster.

### Setting a specified distance to the vehicle in front

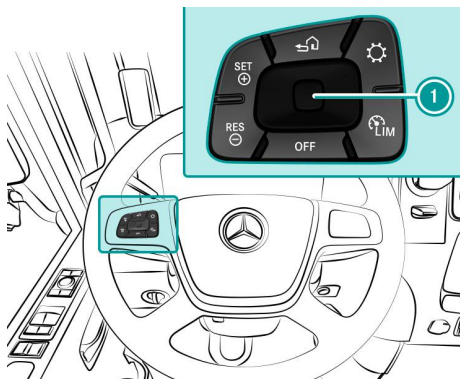
The specified distance for Proximity Control Assist can be set to five levels. After a vehicle start, the average target distance is automatically set.

Make sure that you maintain the minimum distance to the vehicle in front as required

## Driving mode

### Driving systems

by law. Adjust the specified distance to the vehicle in front if necessary.



- ▶ Press the  button to open the specified distance display in the instrument cluster.
- ▶ Swipe the  button to change the specified distance:
  - Swipe bottom to top to increase the specified distance.
  - Swipe top to bottom to reduce the specified distance.
- ◁ The bar display shows the specified distance you have selected.

### Overtaking

It is possible to exceed the set speed, e.g. when overtaking:

- ▶ Maintain a sufficient distance to the vehicle in front.
- ▶ Depress the accelerator pedal.
- ▶ Once the overtaking manoeuvre is finished, release the accelerator pedal again.
  - ◁ Proximity Control Assist adjusts the speed to the set speed.

### Deactivating

The speed remains stored if you deactivate Proximity Control Assist.

- ▶ Press the  button.
- or
- ▶ When Proximity Control Assist accelerates the vehicle, depress the brake pedal.
    - ◁ The instrument cluster shows the  symbol and the set speed in grey.

or

- ▶ Select the limiter with the  button.

◁ The instrument cluster shows the  symbol in grey.

or

- ▶ Using the  button, select the cruise control (→ Page 156).

◁ The instrument cluster displays the  symbol and the set speed in white. Cruise control is activated.

Observe the conditions that lead to automatic deactivation of Proximity Control Assist.

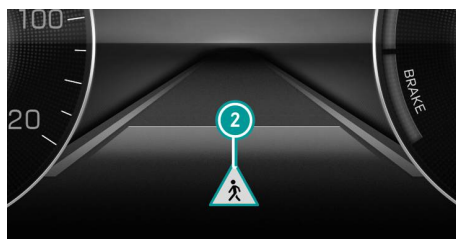
### Frontguard Assist

#### Notes on Frontguard Assist

Frontguard Assist can inform the driver about pedestrians or cyclists who are standing or moving in front of the vehicle and warn the driver.



① Monitoring range (example)



① Monitoring range (example)  
② Display in instrument cluster (example)

### Persons in the monitoring range – Info

If the system detects a pedestrian or cyclist within the sensors' monitoring range, it informs the driver by displaying the yellow

low  indicator lamp in the instrument cluster.

When the vehicle is switched on, the display is active from standstill up to a speed of 15 km/h (forward travel).

### Persons in the monitoring range – warning

If the system detects a pedestrian or cyclist within the sensors' monitoring range and classifies the situation as critical, it warns the driver by displaying the red  indicator lamp in the instrument cluster. An acoustic warning also sounds.

While the system issues an audible warning, the audio equipment installed at the factory and the hands-free system are muted.

The warning is active when driving forwards from vehicle start-up to a vehicle speed of 15 km/h.

If the driver depresses the accelerator pedal beyond the pressure point (kick-down), the colour of the indicator lamp in the instrument cluster changes from red

to yellow and the acoustic warning goes out.

The indicator lamp in the instrument cluster goes out in the following situations:

- The detected pedestrian or cyclist is no longer within the monitoring range of the sensors.
- The pedestrian or cyclist is no longer detected by the sensors.
- The distance to the detected pedestrian or cyclist is no longer evaluated as critical by the system in conjunction with the vehicle speed.

### System limits



#### Risk of accident due to limited detection capability of Frontguard Assist

Frontguard Assist cannot always clearly identify objects and complex traffic situations.

## Driving mode

### Driving systems

In such cases, Frontguard Assist may:

- inform or warn without justification
- may not inform or warn
- ▶ Always carefully observe the traffic conditions and be ready to brake at all times.

The system may be impaired or may not function in the following situations in particular:

- if the sensors are dirty, covered with ice, or obscured, for example, due to snow or heavy spray
- if the radar sensors are impaired due to interference, for example, from strong or inconvenient radar reflections caused by crash barriers or other infrastructure
- if the windscreen in the area of the camera is dirty, misted up or damaged
- if the windscreen in the area of the camera is obscured, e.g. due to a faulty windscreen wiper or a sticker

Frontguard Assist may not react correctly in the following situations in particular:

- If pedestrians or cyclists move into the sensors' monitoring range quickly
- on bends with tight radii
- to pedestrians or cyclists, who are obscured by other objects
- if the contour of a pedestrian does not stand out from the background
- if pedestrians or cyclists are no longer detected as such, e.g. due to special clothing or other objects
- pedestrians or cyclists in a tunnel

at very low vehicle speeds (e.g. when the vehicle crawls), the system may not issue a warning despite a critical situation.

#### Automatic deactivation of Frontguard Assist

If Frontguard Assist is malfunctioning or has been automatically deactivated due to an implement being used, the yellow  indicator lamp in the instrument cluster lights up. No information is then provided about objects in the monitoring range and no warning is issued by Frontguard Assist.

In the event of a malfunction, have the system checked at a qualified specialist workshop.

#### Activating/deactivating Frontguard Assist warning

When you start the vehicle, Frontguard Assist's visual and acoustic warning is automatically activated.



#### Note

The info display in the instrument cluster (yellow indicator lamp ) cannot be switched off.

- ▶ **Switching off:** Switch off in the multimedia system under [Operation](#) ►► [Settings](#) ►► [Assistance systems](#) the [Frontguard Assist](#) menu.
- ◁ Frontguard Assist's visual and acoustic warning is switched off.
- ▶ **To switch on:** Switch on in the multimedia system under [Operation](#) ►►

Settings ►► Assistance systems the Frontguard Assist menu.

- ◁ Frontguard Assist's visual and acoustic warning is switched on.

### Active Brake Assist 6

### Notes on Active Brake Assist 6

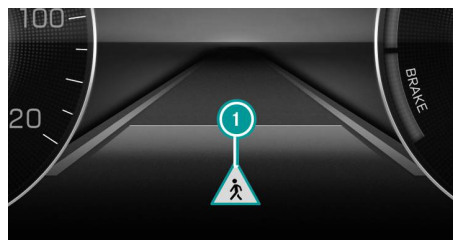
#### General notes

Active Brake Assist can have the following effects:

- Ideally avoid a collision with the vehicle in front.
- Reduce the danger of a collision with the vehicle in front or a stationary obstacle in the path of your vehicle.
- Reduce the consequences of a collision with the vehicle in front.
- Drivers are able to react to pedestrians and cyclists with partial braking and full-stop braking.

If Active Brake Assist detects the risk of a rear-end collision, a warning tone sounds and the instrument cluster shows the red

 indicator lamp for vehicles or  for pedestrians or cyclists.



① Display in the instrument cluster (example: pedestrian)

If you do not react to the visual and audible warning when a vehicle is detected and the danger persists, partial braking is initiated.

If a pedestrian/cyclist is detected, partial braking is initiated at the same time as the visual and acoustic warning.

If you do not react to the partial brake application and the danger persists, Active Brake Assist automatically initiates emergency braking.

### System limits

Full system performance is not yet available for a few seconds after switching on the vehicle or after driving off.

Active Brake Assist may be impaired or may not function in the following situations in particular:

- in snow, rain, fog or heavy spray
- if the carriageway is very narrow and winding
- If the radar sensors are impaired due to interference from other radar sources, such as strong radar reflections in car washes
- If there is glare, e.g. from oncoming traffic, direct sunlight, greatly varying light conditions or reflections (e.g. if the road surface is wet)
- If the windscreen in the area of the camera is dirty, obscured or damaged, for example, due to a faulty windscreen wiper or a sticker
- at very low vehicle speeds (e.g. when the vehicle is crawling)

## Driving mode

### Driving systems

#### Note

Following delivery of the vehicle, over the course of a certain distance, the Active Brake Assist sensors adjust automatically. During this teach-in process, Active Brake Assist is not available or available only to a limited extent.

In some circumstances, particularly those listed below, Active Brake Assist does not react correctly:

- to pedestrians or vehicles if they move quickly into the sensor detection range
- on bends with tight radii
- if the load of the vehicle in front protrudes into the lane in which you are driving.
- to pedestrians who are obscured by other objects
- to contours of a pedestrian, who does not stand out from the background
- to a pedestrian, who is no longer detected as such, for example, due to special clothing or other objects
- to pedestrians in a tunnel

Active Brake Assist can only react to pedestrians at speeds of up to 60 km/h.

#### Danger

##### **Risk of accident due to Active Brake Assist's limited detection capabilities**

Active Brake Assist cannot always clearly identify objects and complex traffic situations.

In such cases, Active Brake Assist may:

- issue an unnecessary warning and brake the vehicle
- possibly not issue a warning or intervene

Active Brake Assist is not a substitute for you paying attention to your surroundings and does not relieve you of your responsibility pertaining to road traffic law, in particular in relation to maintaining a safe distance, your speed and braking in good time.

- ▶ Always pay careful attention to the traffic situation.
- ▶ Be prepared to brake or swerve if necessary.

#### Danger

##### **Risk of accident due to Active Brake Assist's detection limitations**

Active Brake Assist does not react:

- to oncoming vehicles
- to animals

Active Brake Assist is therefore unable to issue a warning or intervene in these situations.

- ▶ Always carefully observe the traffic conditions and be ready to brake at all times.

#### Note

Active Brake Assist can also react to pedestrians or cyclists who are coming towards you or are moving at the edge of the road.

Active Brake Assist cannot override physical limits. You should always adapt your driving style to suit prevailing road and weather conditions.



The following vehicles in particular may not be detected by Active Brake Assist:

- narrow vehicles
- motorcycles
- vehicles not travelling in line with your vehicle

Please read the information on special driving situations that may lead to system limitations (→ Page 171).

Active Brake Assist may unexpectedly issue warnings or brake your vehicle in the following situations in particular:

- in car washes
- if there are stationary obstacles inside a tunnel
- on ferries
- in loading areas
- at toll gates
- in workshops

Critical driving situations are described as:

- situations with an imminent collision with the vehicle in front

- situations with imminent collision with a pedestrian or cyclist

The following may lead to Active Brake Assist having no reaction in critical driving situations:

- Active Brake Assist has not recognised the danger of the situation.
- Active Brake Assist is deactivated.
- Active Brake Assist has failed.
- Active Brake Assist is suppressed.

#### **Cancelling an Active Brake Assist brake application**

Due to technical and physical system limits, Active Brake Assist may warn the driver visually and acoustically, or may initiate a partial or emergency braking application although no critical driving situation is present.

You can suppress a warning tone and/or an emergency braking application initiated by Active Brake Assist as follows:

- By quickly depressing the accelerator pedal significantly beyond the current

position, or beyond the point of resistance (kick-down).

- By disengaging Active Brake Assist (depending on equipment) (→ Page 168)

You can only suppress an emergency braking application initiated by Active Brake Assist as follows:

- By depressing the accelerator pedal beyond the point of resistance to the kick-down position (depressing the accelerator pedal beyond the point of resistance without reaching the kick-down position is not sufficient, as opposed to the suppression of the warning tone or partial braking application described above).
- By disengaging Active Brake Assist (depending on equipment) (→ Page 168).

Please note that all suppression measures described above suppress warning tones and partial braking applications initiated by the system, also in critical driving situations. Depressing the accelerator pedal beyond the point of resistance

## Driving mode

### Driving systems

(kick-down) or deactivating Active Brake Assist (if installed) also suppresses an emergency braking application initiated by the system in critical driving situations. Only the visual warning remains in such cases.

If both the driving situation assessed as critical and the suppressive measure continue, a visual warning is issued and a warning tone sounds. The visual warning and brief warning tone are issued until the moment when the system would normally initiate the emergency braking application, which has, however, been suppressed due to the suppression measures. In this instance, Active Brake Assist does not initiate a partial or emergency braking application.

#### Notes on care

Regularly clean the distance sensor covers and the windscreen in the camera's field of vision (→ Page 219).

If the distance sensor cover or the windscreen in the camera's field of vision is

dirty or iced over, its functionality may be impaired.

#### Deactivating and activating Active Brake Assist 6

Depending on the vehicle equipment, it is possible to deactivate Active Brake Assist manually.

When you start the vehicle, Active Brake Assist is automatically activated.

If there is a malfunction in a system associated with Active Brake Assist, Active Brake Assist is automatically deactivated and the indicator lamp in the instrument cluster lights up. 

When the cause of the malfunction has been rectified, Active Brake Assist switches back on automatically and the indicator lamp  in the instrument cluster goes out.

- **Deactivation:** In the multimedia system under the menu **Operation** ►► **Settings**, menu item **Assistance sys-**

**tems** switch the **Active Brake Assist 6** off.

- ◁ If Active Brake Assist was deactivated, the indicator lamp  will light up in the instrument cluster.

#### Note

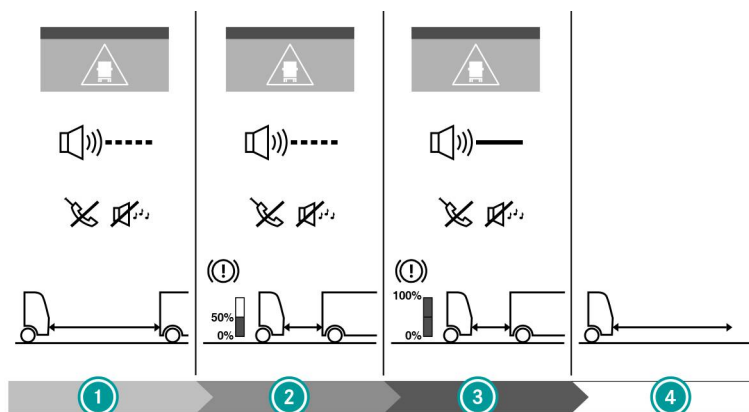
If the Active Brake Assist menu item is greyed out, Active Brake Assist cannot be deactivated in your vehicle.

#### Note

15 minutes after manual deactivation, Active Brake Assist is automatically re-activated and the indicator lamp  in the instrument cluster goes out.

- **Activation:** In the multimedia system under the menu **Operation** ►► **Settings**, menu item **Assistance systems** switch the **Active Brake Assist 6** on.
  - ◁ Once Active Brake Assist has been activated, the indicator lamp  in the instrument cluster will go out.

### Collision warning and emergency braking

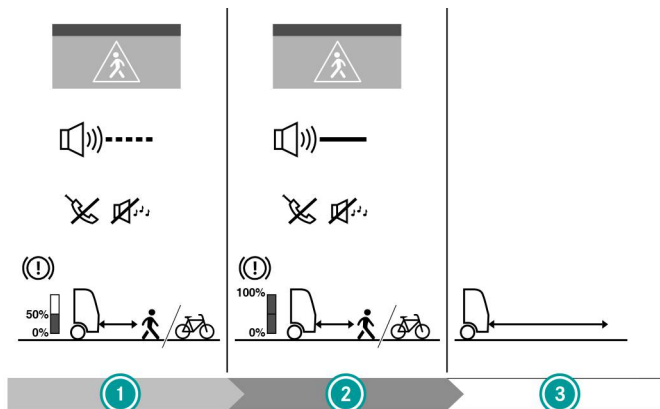


- ① Collision warning
- ② Partial braking
- ③ Emergency braking (maximum full-stop braking)
- ④ Emergency braking completed

Reaction to a vehicle (example)

## Driving mode

### Driving systems



Reaction to a pedestrian /cyclist (example)

- ① Collision warning with simultaneous partial braking
- ② Emergency braking (maximum full-stop braking)
- ③ Emergency braking completed

If there is a risk of a collision with the vehicle in front and Active Brake Assist issues a warning, the audio device and hands-free system installed at the factory are automatically muted.

#### Collision warning

The red indicator lamp lights up in  the instrument cluster when a vehicle, pedestrian or cyclist is  detected. An intermittent warning tone sounds.

#### Partial braking

The collision warning remains active. In addition, Active Brake Assist slows the vehicle with automatic partial braking.

#### Emergency braking

If you do not react to the collision warnings and partial brake application, Active Brake Assist automatically initiates emer-

gency braking (maximum full-stop braking) within the system limitations.

In addition, the red indicator lamps  or  light up in the instrument cluster and a continuous warning tone sounds.

At speeds greater than 50 km/h, during braking rapid flashing of the hazard warning lights (emergency braking flashing) is activated to warn following traffic.

After emergency braking, the instrument cluster shows the grey display message  **Emergency braking complete** and the hazard warning lights are switched on automatically at the normal flashing frequency.

After emergency braking to a standstill, the vehicle is secured by the HOLD function to prevent it from rolling away.

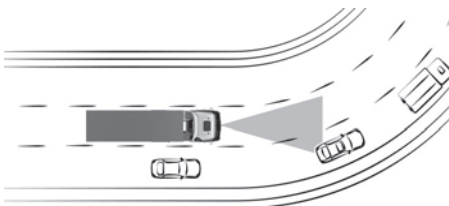
If an emergency braking manoeuvre has been performed:

- ▶ Remove the vehicle from the danger area as soon as possible while paying attention to the traffic situation.

- ▶ Switch off the vehicle and use the parking brake to secure it against rolling away.
- ▶ Make sure that the vehicle is in proper working order and that the load is secured properly.

### Information on specific driving situations

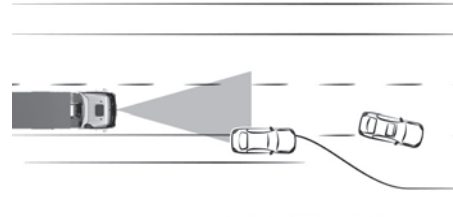
#### Cornering, entering and exiting bends



The ability of Active Brake Assist and the distance control assistant to detect vehicles on bends is limited:

- Active Brake Assist and the Proximity Control Assist may unexpectedly issue warnings or brake your vehicle.

### Vehicles not travelling in line with your vehicle and stationary vehicles



The ability of Active Brake Assist and the distance control assistant to detect vehicles not travelling in line with your vehicle or stationary vehicles is limited:

- Active Brake Assist and the Proximity Control Assist may unexpectedly issue warnings or brake your vehicle.
- Active Brake Assist and the Proximity Control Assist cannot issue warn-

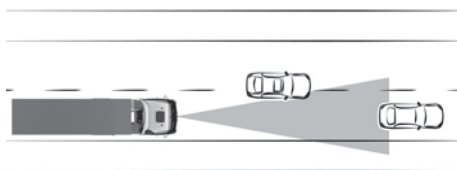
## Driving mode

### Driving systems

ings/brake although there is a critical driving situation.

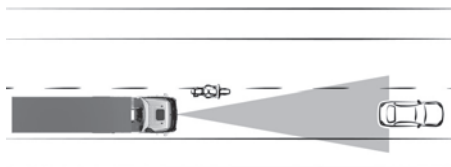
- The Proximity Control Assist may accelerate unexpectedly.

#### Other vehicles changing lane



The ability of Active Brake Assist and the Proximity Control Assist to detect vehicles pulling into your lane is limited.

- Active Brake Assist and the Proximity Control Assist may unexpectedly issue warnings or brake your vehicle.
- Active Brake Assist and the Proximity Control Assist cannot issue warnings/brake although there is a critical driving situation.
- The Proximity Control Assist may accelerate unexpectedly.

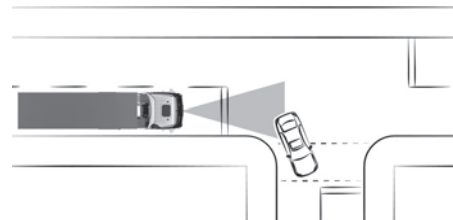


The ability of Active Brake Assist and the distance control assistant to detect vehicles pulling into your lane without maintaining a safe distance is limited. The system can only react to vehicles within the detection range of the sensors.

- Active Brake Assist and the Proximity Control Assist cannot issue warnings/brake although there is a critical driving situation.
- The Proximity Control Assist may accelerate unexpectedly.

Brake the vehicle. This will increase the distance to the vehicle in front.

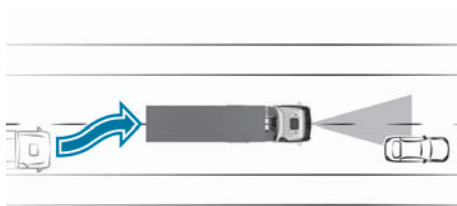
#### Vehicles turning off



The ability of Active Brake Assist and the Proximity Control Assist to detect vehicles turning off is limited:

- Active Brake Assist and the Proximity Control Assist may unexpectedly issue warnings or brake your vehicle.
- Active Brake Assist and the Proximity Control Assist cannot issue warnings/brake although there is a critical driving situation.
- The Proximity Control Assist may accelerate unexpectedly.

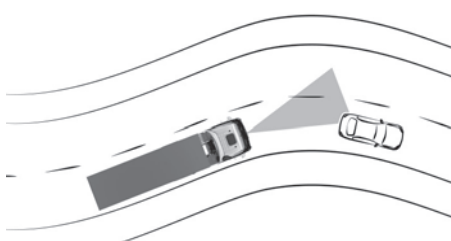
### Overtaking



When you are overtaking, Active Brake Assist and the Proximity Control Assist may in the following cases unexpectedly issue warnings or brake your vehicle if you:

- drive too close to the vehicle in front
- you are still in the lane of the vehicle in front

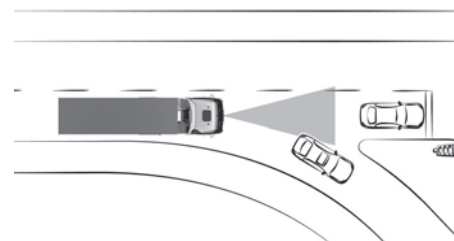
### Winding stretches of road



On winding stretches of road, Active Brake Assist and the Proximity Control Assist cannot always detect which lane the vehicle in front is driving in:

- Active Brake Assist and the Proximity Control Assist may unexpectedly issue warnings or brake your vehicle.
- Active Brake Assist and the Proximity Control Assist cannot issue warnings/brake although there is a critical driving situation.
- The Proximity Control Assist may accelerate unexpectedly.

### Obstacles and stationary vehicles



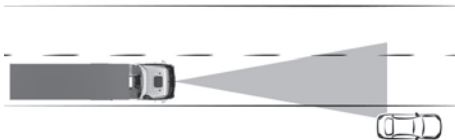
Active Brake Assist and the Proximity Control Assist may be too late detecting obstacles or stationary vehicles in front of the detected vehicle:

- Active Brake Assist and the Proximity Control Assist may unexpectedly issue warnings or brake your vehicle.
- Active Brake Assist and the Proximity Control Assist cannot issue warnings/brake although there is a critical driving situation.
- The Proximity Control Assist may accelerate unexpectedly.

## Driving mode

### Driving systems

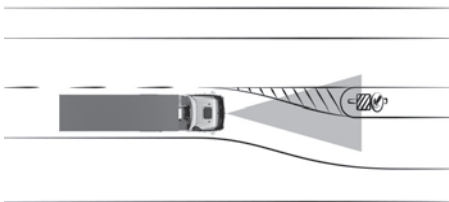
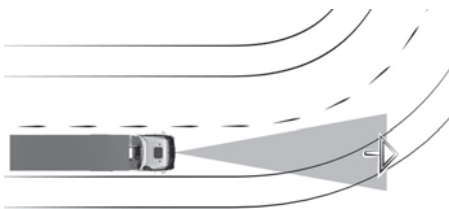
#### Vehicles parked at the roadside



Active Brake Assist and the Proximity Control Assist may react to vehicles parked at the roadside:

- Active Brake Assist and the Proximity Control Assist may unexpectedly issue warnings or brake your vehicle.
- The Proximity Control Assist may accelerate unexpectedly.

#### Stationary objects

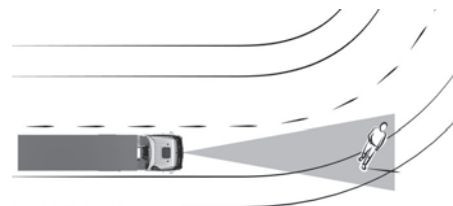


Active Brake Assist can also unexpectedly issue warnings and brake the vehicle if it detects stationary objects next to your lane. Examples of this are:

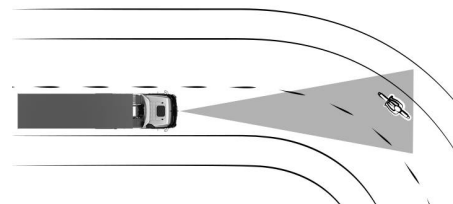
- Broken-down vehicles
- Road boundaries, e.g. marker posts
- Signs
- Bridges
- Traffic islands

- Parked vehicles
- People

#### Pedestrians and cyclists



Active Brake Assist may also unexpectedly issue a warning or brake for people moving along the roadside on a bend.



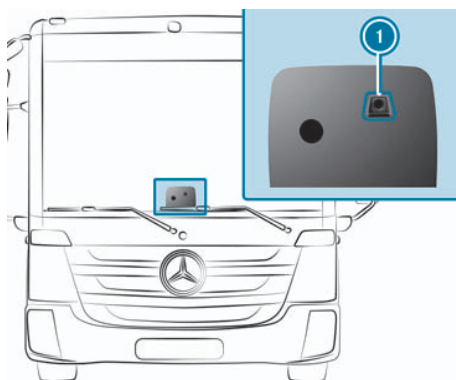
Active Brake Assist may also unexpectedly issue a warning or brake in the event of oncoming cyclists on a bend.



### Traffic Sign Assist

#### Function of Traffic Sign Assist

Traffic Sign Assist is only an aid. Always adjust your speed to suit the traffic situation. The driver is responsible for complying with traffic regulations at all times.



Traffic Sign Assist detects traffic signs with a multifunction camera (1). It assists you by displaying detected speed limits and overtaking restrictions and warning signs in the instrument cluster.

Traffic Sign Assist uses the map data stored and can, in the following situations, update the display without having detected a traffic sign:

- when the vehicle changes roads, e.g. motorway exit or slip road
- when the map indicates an updated maximum permissible speed

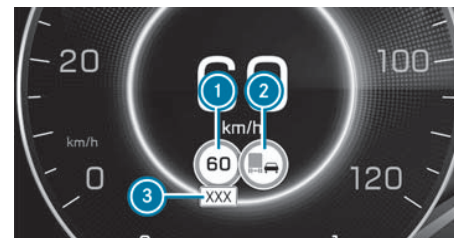
#### **i** Note

In countries without map data, the function is restricted.

Beside the information from the map, legally applicable speed limits in the respective country are also taken into account.

The camera also detects traffic signs with a restriction indicated by an additional sign (e.g. when wet). These will be taken into account in the display for the speed limit or displayed separately if they cannot be allocated accordingly.

### Display in instrument cluster



Display Traffic Sign Assistant (example)

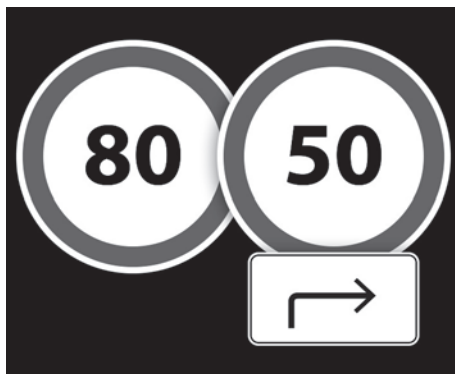
- ① Maximum permissible speed
- ② Alternative maximum permissible speed, overtaking restrictions and warnings
- ③ Additional sign with restriction

If more than two traffic signs are detected by the system, Traffic Sign Assist selects the two most relevant traffic signs for the display.

## Driving mode

### Driving systems

#### Speed limits on slip roads



Speed limits detected in the area of an exit are displayed in the instrument cluster with an arrow as an additional symbol. The additional symbol is displayed even if the actual traffic sign does not have an additional symbol.

At the same time, the permissible speed of the road being travelled is displayed in the instrument cluster to the left.

#### Warning when the maximum permissible speed is exceeded

The system can warn you if you unintentionally exceed the maximum permissible speed. To do this, the system issues a warning tone and the corresponding traffic sign in the instrument cluster flashes red.

In certain situations, a warning tone or a warning in the instrument cluster does not sound.

No warning tone is issued in the following cases:

- If the vehicle is slowed down by the driver
- If the vehicle speed is less than 20 km/h

In the following cases, no warning is issued in the instrument cluster:

- In the area of the tachograph control if there is no explicit speed limit
- If the warning function has been switched off (→ Page 177)

#### Warning when an overtaking manoeuvre is detected in a no-overtaking zone

The system can give you a visual warning if you start an overtaking manoeuvre in a no-overtaking zone. The corresponding traffic sign flashes red in the instrument cluster.

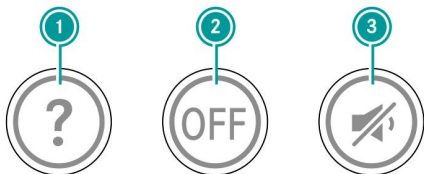
#### Acoustic information when the maximum permissible speed is changed

The system can inform you by means of an audible notification when a new maximum permissible speed is detected.

The function can be deactivated in the multimedia system (→ Page 177).

#### Display of system statuses

In addition to the display of traffic signs in the instrument cluster, the system also informs you of restrictions on individual functions.



- ① No applicable speed limit could be detected.
- ② The system has been switched off or is unavailable due to a malfunction.
- ③ The warning tone for when the maximum speed is exceeded is switched off.

### **i** Note

The display ③ is shown for ten seconds after the warning tone is switched off.

### System limits

The system may be impaired or may not function in the following situations:

- if the traffic signs are hard to detect, e.g. due to dirt or snow, or because

- they are covered, or due to insufficient lighting
- if the information in the digital road map is incorrect or not up to date
- when using a non-ISO compliant trailer
- if the signs are ambiguous, e.g. traffic signs at roadworks, in adjacent lanes or at intersections
- if there are dynamic signs, e.g. LED displays
- if there are signs that do not meet the official requirements
- if there is poor visibility, e.g. due to insufficient illumination of the road, highly variable shade conditions, or due to rain, snow, fog or heavy spray
- if there is glare, e.g. from oncoming traffic, direct sunlight or reflections
- if the windscreen in the area of the camera is dirty, misted up or damaged
- if the windscreen in the area of the camera is obscured, e.g. due to a faulty windscreen wiper or a sticker

- if there are no lane markings, or several unclear lane markings are present for one lane, e.g. around roadworks
- When the vehicle is switched on after a major change in the load: Restart the vehicle after a major change in the load.

Instruction labels for vehicles transporting hazardous goods are not taken into account by the system.

Make sure that the windscreen is always kept clean and unobstructed in the area of the camera (→ Page 219). Therefore, switch on the windscreen wiper to clean the windscreen, for example, or remove snow and ice from the windscreen.

### Activating/deactivating Traffic Sign Assist

- In the multimedia system, in the menu, [Operation](#) ► [Settings](#) ► [Assistance systems](#) select the menu item [Traffic Sign Assist](#).

## Driving mode

### Driving systems

- ◁ The following functions are available:
  - Activating/deactivating Traffic Sign Assist
  - Activating/deactivating Traffic Sign Assist warnings
  - Switching the acoustic information on or off when the maximum permissible speed changes
  - Display of the current map version

#### Note

If the system is unavailable due to a malfunction, the menu item is greyed out.

#### Note

When the vehicle is started, Traffic Sign Assist's warning and display functions are automatically activated. The acoustic information when the maximum permissible speed is changed remains in the last selected setting.

#### Note

Daimler Truck AG offers an annual map update to ensure that the system functions as efficiently as possible. The driver is responsible for keeping the map up to date.

### Lane Keeping Assist

#### Notes on Lane Keeping Assist

#### Danger

#### **Risk of accident despite Lane Keeping Assist**

Lane Keeping Assist cannot always clearly identify lane markings.

In such cases, Lane Keeping Assist may:

- Issue an unnecessary warning.
- Not issue a warning.
- Always pay particular attention to the traffic situation and keep within the lane, especially if Active Lane Keeping Assist alerts you.

The system may be impaired or may not function in the following situations:

- If there is poor visibility, e.g. due to insufficient road illumination or due to snow, rain, fog or heavy spray
- If there is glare, for example, from oncoming traffic, direct sunlight or reflections (e.g. if the carriageway is wet)
- If the windscreen in the area of the camera is dirty, misted up or damaged
- If the windscreen in the area of the camera is obscured, e.g. due to a faulty windscreen wiper or a sticker
- If there are no lane markings, or several unclear lane markings are present for one lane, for example, around roadworks
- If the lane markings are worn away, dark or covered up, for example, by dirt or snow
- If the distance to the vehicle in front is too small and the lane markings thus cannot, therefore, be detected

- If the lane markings change quickly, e.g. lanes branch off, cross one another or merge
- if the carriageway is very narrow and winding
- If there are highly variable shade conditions on the carriageway
- if attachments, such as a snow plough, restrict the camera's view of the lane markings
- After a major change in load with the vehicle switched on. Therefore, start the vehicle again after a significant change in load for Lane Keeping Assist to be available without any restrictions

Keep the windscreen around the camera free of dirt, snow or ice. Use the windscreen wipers or clean the windscreen by hand if required. Please note the information in "Notes on cleaning the vehicle exterior" (→ Page 217) and "Cleaning sensors" (→ Page 219).

### Overview

The lane markings are shown in the  Assistance menu in the instrument cluster.

The lane markings in the instrument cluster show the status of Lane Keeping Assist in colour:

- Grey lane marking: Lane Keeping Assist is either disengaged or engaged, but is not ready to issue warnings on the side of the vehicle in question.
- White lane marking: Lane Keeping Assist is engaged and ready to issue warnings on the side of the vehicle in question.
- Red lane marking: Lane Keeping Assist is engaged and is issuing a warning on the side of the vehicle in question.

### Functions and activation conditions

The Lane Keeping Assist monitors the area in front of your vehicle with a camera that is on the base of the windscreen. When Lane Keeping Assist is activated

and it detects lane markings on the carriageway, it warns you that you may be leaving your lane unintentionally. Lane Keeping Assist does not automatically keep the vehicle in its lane.

The following conditions must be fulfilled so that Lane Keeping Assist is ready to issue warnings:

- You are driving above approximately 60 km/h.
- The lane markings must be shown in white in the instrument cluster.

When you drive over the lane marking unintentionally, the following warnings are issued:

- The relevant lane marking is shown in red in the instrument cluster.
- The volume of the audio equipment or hands-free system is reduced and a warning tone sounds from the speaker on the corresponding side of the vehicle.

If the vehicle is not fully within the lane markings after a warning has been issued, no further warnings are possible.

## Driving mode

### Driving systems

Lane Keeping Assist does not issue a warning in the following situations:

- You have switched on the turn signal indicator.
- You clearly and actively steer, brake or accelerate.
- A driving safety system intervenes, e.g. Active Brake Assist or Stability Control Assist.

The warnings are then suppressed for a certain time period.

Lane Keeping Assist will warn you when changing lane if a turn signal indicator has been switched on for more than one minute.

#### Deactivating/activating Lane Keeping Assist

Depending on how the vehicle is equipped, it is possible to deactivate Lane Keeping Assist manually.

When you start the vehicle, Lane Keeping Assist is automatically activated.

► **To switch on and off:** In the multimedia system, under the menu **Operation**, menu item **Switch**, press the button .

- ◄ If the indicator lamp  in the instrument cluster and the indicator lamp of the button  light up, this means Lane Keeping Assist is deactivated.

#### Note

If there is no button  under the menu item **Switch** in the multimedia system, Lane Keeping Assist cannot be deactivated in your vehicle.

#### Attention Assist

##### Notes on Attention Assist

Attention Assist may be impaired or inoperative in the following situations:

- If there is poor visibility, e.g. due to insufficient road illumination or due to snow, rain, fog or heavy spray
- If the windscreen in the area of the camera is dirty, misted up or damaged
- If the windscreen in the area of the camera is obscured, e.g. due to a faulty windscreen wiper or a sticker
- If there are often no lane markings, or several unclear lane markings are present for one lane, for example, around roadworks
- If the lane markings are worn away, dark or covered up, for example, by dirt or snow
- If there is often glare, e.g. from oncoming traffic, direct sunlight or reflections (e.g. if the road surface is wet)
- on winding roads
- If the distance to the vehicle in front is too short and thus the lane markings often cannot be detected
- If attachments obstruct the camera's view of the lane markings
- When the vehicle is switched on after a major change in the load: Restart

the vehicle after a major change in the load.

Make sure that the windscreen is always kept clean and unobstructed in the area of the camera (→ Page 219). Therefore, switch on the windscreen wiper to clean the windscreen, for example, or remove snow and ice from the windscreen.

### Overview

Attention Assist helps you during long, monotonous journeys e.g. on motorways and trunk roads. Attention Assist is active at speeds above approximately 60 km/h.

If Attention Assist has detected typical indications of fatigue or increasing lapses in concentration on the part of the driver, it suggests taking a break. A warning is shown by Attention Assist irrespective of legally prescribed driving and rest periods or digital tachograph functions.

### Functions and activation conditions

Attention Assist assesses your level of fatigue or lapses in concentration by taking the following criteria into account:

- Personal driving style, for example, if you remain in the lane, your steering behaviour
- Journey-related parameters, such as the length of the journey

Attention Assist is restricted and a warning does not occur or is delayed in the following situations:

- If you are predominantly driving at under 60 km/h
- If lane markings are not present or difficult to distinguish
- on winding roads

The following interventions occur if Attention Assist detects typical indicators of fatigue or increasing lapses in concentration on the part of the driver:

- A warning tone is issued.
- The instrument cluster shows the display message  and **Attention Assist: take a break?**
- Lane Keeping Assist is reactivated automatically.

Take regular breaks in good time during long journeys. If you do not take a

break, Attention Assist can only warn you again after at least 15 minutes. When you switch off the vehicle or the vehicle is stationary for some time, Attention Assist resets the detection sequence.

### Deactivating/activating Attention Assist

After starting the vehicle, Attention Assist is always activated.

- ▶ In the multimedia system under the menu **Operation** ► **Settings**, menu item **Assistance systems** switch the **Attention Assist** on and off.

### Sideguard Assist 2

#### Function of Sideguard Assist 2

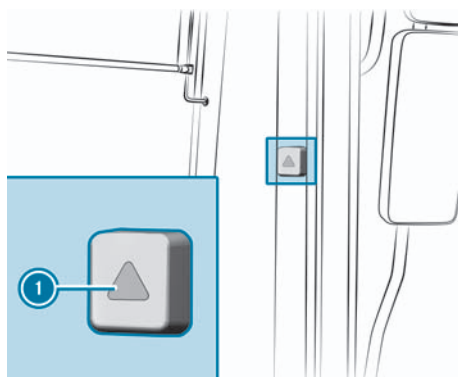
#### General notes

Sideguard Assist can warn of the risk of a side collision, especially when turning or changing lanes. It contains the functions of the legally prescribed turning assistant

## Driving mode

### Driving systems

(UN ECE R151, Blind Spot Information System).



Warning display on the A-pillar (example)

The system monitors the area next to the towing vehicle and the trailer or semitrailer on both sides with two radar sensors each. The radar sensors are integrated into the mudguards and the corners of the front bumper. Information from the front radar and the multifunction camera is also processed.

The system supports you when turning off and changing lanes with the warning display  in the A-pillar .

**Sideguard Assist Info:** The system detects a relevant object in the monitoring range. The warning display  on the corresponding side lights up yellow.

**Sideguard Assist Warning:** The system detects an increased risk of collision with a relevant object in the monitoring range. The warning display  on the corresponding side lights up red. A warning tone will also sound.

#### Note

Both display stages “Sideguard Assist Information” and “Sideguard Assist Warning” are referred to as warnings in the following, as they can transition directly into each other unless otherwise described.

The system is not active when reversing.

#### Note

Trailers with an interface ISO 11992-2: if trailer information such as length or number of axles is set incorrectly, the system could malfunction for the area surrounding the trailer or semitrailer. In such cases, trailer monitoring may issue unnecessary warnings or no warnings at all.

If you detect a trailer or semitrailer malfunction, contact the trailer manufacturer.

#### Note

Trailers without an interface according to ISO 11992-2: after coupling up a trailer or semitrailer, it is necessary to perform a number of cornering manoeuvres until the system has adjusted to the new trailer or semitrailer.



**i Note**

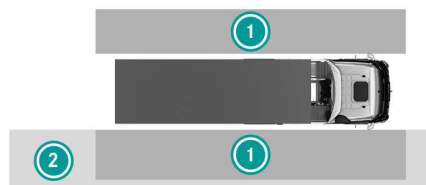
You can find information on whether your trailer or semitrailer includes an ISO 11992-2 interface in the operating instructions provided by the trailer or semitrailer manufacturer.

**Setting of the traffic direction**

The sensitivity of the Sideguard Assist warning output depends on the traffic direction set in the vehicle.

In countries in which traffic drives on the opposite side of the road to the country in which the vehicle is registered, it is necessary to change the setting in the multimedia system under the menu [Operation](#) **>> Settings >> Assistance systems >> TRAFFIC DIRECTION**.

Incorrect setting of the traffic direction can lead to unauthorised system reactions.

**Sideguard Assist monitoring range**

Overview (example using a left-hand drive vehicle)

The length of the warning area ① depends on the vehicle speed, but it always covers at least the vehicle length detected by the system, including any trailer /semitrailer.

The width of the warning range ① is equivalent to approx. one lane.

The extended information area ② results from the legal requirement in accordance with UN ECE R151 and is always located on the front passenger side. The length of this information-only area ② depends on the vehicle speed and is a maximum of 37 m.

**i Note**

Due to the size of the legal information area ②, a yellow warning display  may also appear for traffic following behind.



Proximity monitoring zone (example)

From a vehicle speed of approx. 40 km/h, warnings of vehicles rapidly approaching from behind are also given with the help of an approach monitoring zone ③.

**System limits**

Full system performance is not yet available for a few seconds after switching on the vehicle or after driving off.

Due to the system characteristics, warnings may be issued in error when driving

## Driving mode

### Driving systems

close to crash barriers or similar solid boundaries.

Due to the nature of the system, warnings may be interrupted when you are driving alongside particularly long vehicles, for example lorries, for a prolonged time.

Due to the nature of the system, the sensors may malfunction when driving next to uniform structures close to the road, e.g. tunnel walls for a prolonged period of time. A corresponding display message will be shown in the instrument cluster. (→ Page 316) If the fault is no longer present, the display message disappears and the system will be available again.

Depending on the situation and the trailer or semitrailer, the system may warn you too early or not at all.

Make sure that the covers for the radar sensors are free from dirt, ice and slush. (→ Page 219)

If Sideguard Assist is malfunctioning or has been automatically deactivated due to an implement being used, the yellow



or



indicator lamp lights up

in the instrument cluster. A warning is then not issued or not to the full extent about objects in the monitoring range. In the event of a malfunction, have the system checked at a qualified specialist workshop.

The system may be impaired or may not function in the following situations in particular:

- if the sensors are dirty, covered with ice, or obscured, for example, due to snow or heavy spray
- on very wide lanes
- if the carriageway is very narrow and winding
- if the radar sensors are impaired due to interference, for example, from strong or unfavourable radar reflections caused by crash barriers or other infrastructure
- If the windscreen in the area of the camera is dirty, misted up or damaged
- If the windscreen in the area of the camera is obscured, e.g. due to a faulty windscreen wiper or a sticker

- when manoeuvring: Sideguard Assist is not a manoeuvring aid.
- if the traffic direction is set incorrectly

The system may not fully recognise the trailer in the following situations:

- shortly after reversing
- shortly after coupling or hitching up

In this case, the system may mistakenly issue a warning, or it may not issue a warning at all.

In the following cases in particular, have the function of the radar sensors checked at a qualified specialist workshop:

- after a severe impact
- after damage to the sensors or their surroundings

Otherwise, it can no longer be guaranteed that Sideguard Assist will function correctly.

In some circumstances and in the following situations in particular, Sideguard Assist does not react correctly:

- to pedestrians or cyclists, who are obscured by other objects
- to pedestrians or cyclists, who suddenly change direction or speed
- to persons or vehicles moving very quickly within the detection range of the sensors
- to stationary objects or objects moving at the same speed as your own vehicle
- when turning into multiple lanes in the same general direction as vehicles in the adjacent lane
- during manoeuvres that involve pulling out



**Danger**

### Risk of accident due to limitations of Sideguard Assist with regard to the identification of objects and traffic conditions

The system cannot always clearly identify objects and complex traffic situations. In these cases, the system may:

- issue an unnecessary warning

- not issue a warning
  - ▶ Drive on carefully and be ready to brake, particularly if the system issues a warning.
  - ▶ Always pay careful attention to the traffic situation and maintain a safe distance at the side of the vehicle.

### Note

An object is classified as movable in the following cases:

- If it has been seen moving by a sensor at an earlier time and then stops
- If the multifunction camera detects a stationary pedestrian or cyclist

### Note

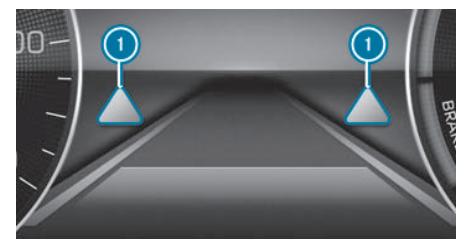
The system will only react to oncoming traffic if the system detects a pedestrian or cyclist.

A warning may not be issued despite the risk of collision for the following reasons in particular:

- The system has failed or there is a malfunction.
- The system has failed to detect the danger of the situation.
- If the traffic direction is set incorrectly
- The system's acoustic warning system is switched off. The visual warning and information display remains active in this case.

### Sideguard Assist displays in the instrument cluster

In addition to the display in the A-pillar, the warning display  is also shown in the instrument cluster menu [Assistance](#).

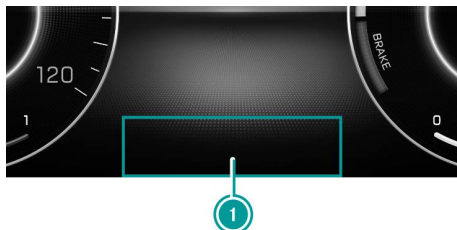


- ① Warning display in the menu [Assistance](#) (example)

## Driving mode

### Driving systems

The indicator lamps in the status area of the instrument cluster inform you of the current status of the system.

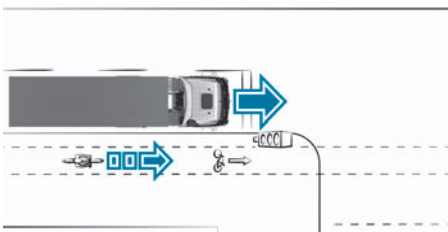


Status area in the instrument cluster (example)

#### The following indicator lamps can be displayed in the status area:

|                                                                                               |                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>(grey)   | The acoustic warning has been switched off ( → Page 188).                                                                                                                                 |
| <br>(yellow) | The system is only available to a limited extent (e.g. the system is only available on one side due to the failure of a sensor or automatic deactivation due to the use of an implement). |
| <br>(yellow) | The system is unavailable due to a malfunction or automatic deactivation due to the use of an implement.                                                                                  |

### Warning when turning

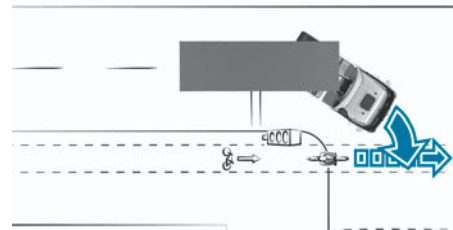


Example of turning right

The system warns of a moving object or an object classified as moving in the side monitoring zone up to a vehicle speed of approx. 40 km/h.

A moving object or an object classified as moving is located in the monitoring range. The yellow warning display  in the A-pillar lights up.

In addition, the corresponding warning indicator  in the instrument cluster **Assistance** menu lights up yellow.

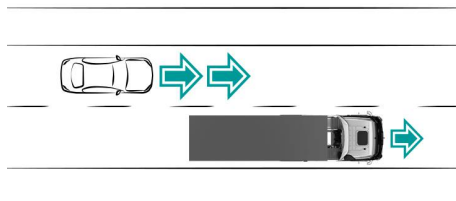


Example of turning right

A moving object or an object classified as moving is located in the monitoring range. If you steer or indicate in the direction of turning and the system detects an increased risk of collision, the red warning display  in the A-pillar flashes for approximately two seconds. A simultaneous warning tone sounds. The red warning light then lights up continuously while there is a risk of collision.

The warning light  in the instrument cluster menu **Assistance** also lights up red.

### Warning for vehicles approaching from behind



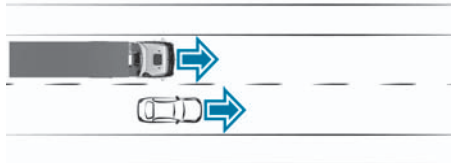
#### Example

As of a vehicle speed of approx. 40 km/h, the system warns of vehicles approaching quickly from behind in the neighbouring lane.

An object approaching quickly from behind is located in the approach monitoring zone. The yellow warning display  in the A-pillar lights up.

In addition, the corresponding warning indicator  in the instrument cluster **Assistance** menu lights up yellow.

### Warning during lane change

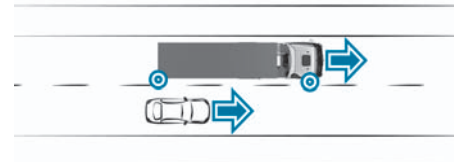


#### Example of overtaking manoeuvre

The system warns of moving objects or objects classified as moving in the neighbouring lane, regardless of the vehicle's own speed.

A moving object or an object classified as moving is located in the monitoring range. The yellow warning display  in the A-pillar lights up.

In addition, the corresponding warning indicator  in the instrument cluster **Assistance** menu lights up yellow.



#### Example of overtaking manoeuvre

There is a vehicle in the danger zone when changing lanes. If you indicate or steer in the direction of the object and the system detects an increased risk of collision (e.g. with a critical approach), the red warning display  in the A-pillar flashes for approximately two seconds. A simultaneous warning tone sounds. The red warning light then lights up continuously while there is a risk of collision.

The warning light  in the instrument cluster menu **Assistance** also lights up red.

## Driving mode

### Driving systems

#### Note

If a warning of the same object is issued several times while the light is flashing, only the visual system reaction is repeated. The acoustic warning tone is then suppressed.

#### Deactivating/activating warning tone of Sideguard Assist 2

When you start the vehicle, the system is automatically activated.

If Sideguard Assist is malfunctioning or has been automatically deactivated due to an implement being used, the yellow  or  indicator lamp lights up in the instrument cluster. A warning is then not issued or not to the full extent about objects in the monitoring range. In the event of a malfunction, have the system checked at a qualified specialist workshop.

- **Switching off:** Switch off in the multimedia system under [Operation](#) ►►

[Settings](#) ►► [Assistance systems](#) the [Sideguard Assist](#) menu.

- ◄ Sideguard Assist's warning tone is switched off.

The grey indicator lamp  in the instrument cluster lights up.

- **To switch on:** Switch on in the multimedia system under [Operation](#) ►► [Settings](#) ►► [Assistance systems](#) the [Sideguard Assist](#) menu.

- ◄ Sideguard Assist's warning tone is switched on.

The grey indicator lamp  in the instrument cluster goes out.

#### Pre-installation reversing assistant

#### Function and notes on reversing assistant

#### Notes on the pre-installation of the reversing assistant

Vehicles with pre-installation of the reversing assistant (code S9B): Your vehicle is equipped ex works with an interface for

the installation of a reversing assistant with rear area monitoring.

The pre-installation is compatible with sensor systems from various body manufacturers and accessory suppliers.

#### Note

For further information, refer to the Mercedes-Benz installation guidelines online at <https://bb-portal.mercedes-benz-trucks.com>

#### Note

You can have the reversing assistant installed by a qualified specialist workshop.

The capacity of this pre-installation is limited to the following:

- the ability to visually indicate to the driver via the instrument cluster the distance to an object when reversing
- the possibility to initiate braking at shunting speed

- the brake lamps switching on when braking is initiated
- limiting the vehicle speed to 7 km/h when reversing.
- a set up electrical wiring harness in the chassis
- a switch in the centre console
- a block on the accelerator pedal when braking is initiated by the reversing assistant
- a set up indicator lamp in the instrument cluster

### Function of the reversing assistant

The reversing assistant is only an aid. The responsibility for safe manoeuvring and parking remains with you. When manoeuvring and parking, make sure that there are no persons, animals or objects behind the vehicle.

The reversing assistant cannot replace a safety officer or signaller. When manoeuvring or parking, always seek assistance from a safety officer or signaller.

The reversing assistant supports you when reversing with the following functions:

- Display of the distance to an obstacle behind the vehicle in the instrument cluster:
  - at less than 2,50 m, the instrument cluster shows the display message **Ramp approach aid** and the distance to the obstacle.
  - at less than 1,50 m, the instrument cluster displays the rear area of the vehicle in red.
- sounds a warning tone when the distance to the obstacle is less than 1 m.
- automatically braking the vehicle to a standstill at a critical distance from the obstacle

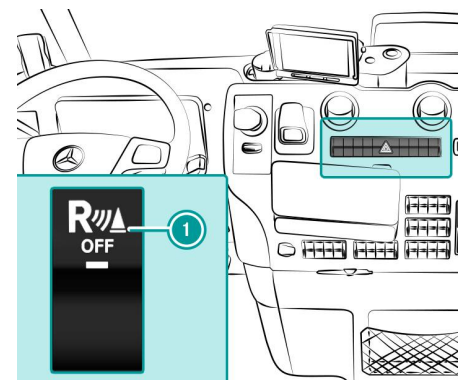
When the red  indicator lamp in the instrument cluster lights up, the reversing assistant is malfunctioning. Have the system checked at a qualified specialist workshop.

### Deactivating and activating the reversing assistant

If a reversing assistant has been retrofitted to your vehicle, it can be switched on and off using a switch in the centre console.

#### Note

After the vehicle is started, the reversing assistant is automatically activated.



- **To switch on and off:** Press the  button.

## Driving mode

### Level control system

- ◁ The red indicator lamp inside the button light ups when the reversing assistant is switched off.

#### Level control system

##### Notes on level control



**Danger**

#### **Risk of accident due to lowered or raised chassis**

The braking and handling characteristics may be greatly impaired if the chassis frame is modified. You may also exceed the permitted vehicle height.

- ▶ Set the driving level before you set off.



#### **NOTE**

#### **Damage to the vehicle due to driving in a lowered drive position**

Driving outside of driving levels 1 and 2 can cause damage to the vehicle's suspension.

- ▶ Only use driving levels 1 and 2 in manoeuvring mode.
- ▶ In driving mode, raise the chassis to driving level.

Observe the applicable legal requirements for each individual country for the permissible vehicle height.

You must raise or lower the chassis to pick up or set down swap bodies. If you continue a journey after having changed the chassis height, it is necessary to lower or raise the chassis to driving level.

If the yellow  indicator lamp in the instrument cluster lights up, the chassis is not at driving level or the level control is malfunctioning. Observe the additional information in the display message.

When loading or unloading the vehicle when it is switched off, refer to the information in the "Loading and unloading the vehicle when the vehicle is switched off" section (→ Page 193).

Level control can be used when the vehicle is stationary or while driving at up to approx. 30 km/h.

Depending on the vehicle equipment, there are the following options for operating the level control system:

- the operating unit on the driver's seat
- the buttons on the instrument panel
- the multimedia system (→ Page 192)

#### Activating and deactivating the level control operating unit

The operating unit is attached at the side between the driver's seat and the door.





Example: operating unit for fully air-sprung vehicles

- ① Preselection for raising or lowering the front axle
- ② Preselection for raising or lowering the entire vehicle
- ③ Preselection for setting the driving level
- ④ Preselection for raising or lowering the rear axle
- ⑤ Buttons
- ▶ To activate the operating unit, make a preselection for the front axle, rear axle, entire vehicle or driving level

- ◀ To activate the operating unit, make a preselection for the front axle, rear axle, entire vehicle or driving level
- ▼ To activate the operating unit, lower the chassis, set the driving level
- ▲ To activate the operating unit, raise the chassis, set the driving level
- ⊞ To activate the operating unit, end raising or lowering operation
- M1 M2 Press button briefly: to call up memory preset M1 or M2 for chassis height  
Press and hold button: to store memory preset M1 or M2 for chassis height
- ▶ Apply the parking brake.
- ▶ Switch on the vehicle.
  - ◀ The level control automatically adjusts the chassis frame to the previously stored height.
- ▶ If the reservoir pressure in the compressed-air system is too low, leave the vehicle switched on.

- ◀ The compressed-air system is charged.
- ▶ Take the operating unit out of the holder.

### Activating the operating unit

- ▶ Press the ▶, ◀, ▼, ▲ or ⊞ button briefly.

### Deactivating the operating unit

- ▶ Wait for approximately 60 seconds; do not press any buttons.
- or
- ▶ Drive at a speed of over approx. 30 km/h.
  - ◀ The operating unit is automatically deactivated.

## Driving mode

### Level control system

#### Raising or lowering the chassis with the operating unit



**Danger**

#### Risk of entrapment from vehicle lowering

When lowering the vehicle, other people may become trapped if their limbs are between the vehicle body and the tyres or underneath the vehicle.

- ▶ Make sure no one is underneath the vehicle or in the immediate vicinity of the wheel arches when you lower the vehicle.
- ▶ Press the or button to select front axle, entire vehicle or rear axle.
  - ◁ The LEDs for the selected preselection light up.
- ▶ Press the button to lower the chassis frame or the button to raise it.
  - ◁ If the chassis frame is not at driving level, the indicator lamp

in the instrument cluster will come on. In addition, the instrument cluster shows a yellow display message with or and **Set driving level**.

- ▶ Press the button to interrupt/end the raising or lowering operation.

#### Raising or lowering the chassis using the multimedia system



**Danger**

#### Risk of entrapment from vehicle lowering

When lowering the vehicle, other people may become trapped if their limbs are between the vehicle body and the tyres or underneath the vehicle.

- ▶ Make sure no one is underneath the vehicle or in the immediate vicinity of the wheel arches when you lower the vehicle.

When raising or lowering, depending on requirements, the front axle , overall

vehicle or rear axle can be selected.

- ▶ In the multimedia system, select the **Level control** menu item under the **Operation** menu.
- ▶ **Raising:** Press the button .
  - ◁ The chassis is raised as long as the button is pressed.
- ▶ **Lowering:** Press the button .
  - ◁ The chassis lowers as long as the button is pressed.

If the chassis frame is not at driving level, the indicator lamp in the instrument cluster will come on.

The instrument cluster also shows the display message or and **Set driving level**.

#### Storing/calling up the chassis height

- ▶ In the multimedia system, select the **Level control** menu item under the **Operation** menu.

### Storing

- ▶ Raise/lower the chassis to the desired height.
- ▶ In the multimedia system, press the button  for memory preset M1 or M2 for approximately two seconds.
  - ◁ The current chassis height is stored under the corresponding  button.

### Calling up

- ▶ Briefly press the  button for memory preset M1 or M2.
  - ◁ The chassis will be raised/lowered automatically to the stored height. The instrument cluster shows a display message with  or  and [Set the driving level](#).
- ▶ Press the button  – to interrupt or end the raising or lowering operation.

### Setting the driving level

#### Using the operating unit

- ▶ Press the button  or  to select driving level preselection .
  - ◁ The LEDs for driving level preselection  go on.
- ▶ Briefly press the  or  button.
  - ◁ The chassis will automatically be raised or lowered to the driving level. When the chassis frame is at driving level, the indicator lamp  in the instrument cluster will go out.
- ▶ Press the  button to interrupt/end the raising or lowering operation.

#### Using the multimedia system

- ▶ In the multimedia system, select the [Level control](#) menu item under the [Operation](#) menu.
- ▶ Press the   button.
  - ◁ The chassis will automatically be raised or lowered to the driving level. When the chassis frame is at driving level, the indicator lamp

 in the instrument cluster will go out.

- ▶ To interrupt or end the raising or lowering operation, press the   button again.

### Loading and unloading the vehicle when the vehicle is switched off



#### NOTE

#### Damage to shock absorbers due to rebounding of the chassis

If swap bodies are removed, this may result in damage to shock absorbers due to the sudden rebounding of the chassis.

- ▶ Fully lower the chassis before removing swap bodies.

Store a permanent chassis height for loading and unloading the vehicle.

- ▶ Vehicles with trailing axle: Apply load to the trailing axle ( → Page 197).
- ▶ If required, raise/lower the chassis to the desired height.

## Driving mode

### Level control system

- ▶ Leave the vehicle switched on until the pressure regulator cuts out.
- ▶ Press and hold the  button on the operating unit.

or

- ▶ In the multimedia system, press and hold the  /  button under the **Level control** menu item.
- ▶ Switch off the vehicle.
- ▶ Release the  button on the operating unit.

or

- ▶ In the multimedia system, release the  /  button under the **Level control** menu item.
  - ◁ If there is sufficient reservoir pressure in the compressed-air system, the height of the chassis is kept constant for approximately four to five hours.

### Special functions of the level control system

#### Notes



#### NOTE

#### Damage due to incorrect chassis height

There is a risk of damage to the vehicle in the following situations:

- if you start the work mode before the chassis has been lowered fully.
- if you raise the chassis to the driving level before the work mode is ended.
- ▶ Select the driving level as required and wait until the chassis has reached the level that you have selected.

#### Forced lowering

Depending on the function of the body and the auxiliary assembly, the electric power take-off system is automatically activated as needed. This allows the vehicle to be completely lowered on all

axles equipped with air suspension and increases the vehicle's tipping stability.

Observe the body manufacturer's operating instructions.

#### Residual air bellow pressure readjustment

Depending on the equipment, it is possible to readjust the residual air bellow pressure.

Vehicles with residual air bellow pressure readjustment: if residual air bellow pressure regulation is active, the air suspension pressure on each axle is constantly monitored and, if necessary, increased/decreased until it matches a parameterised value.

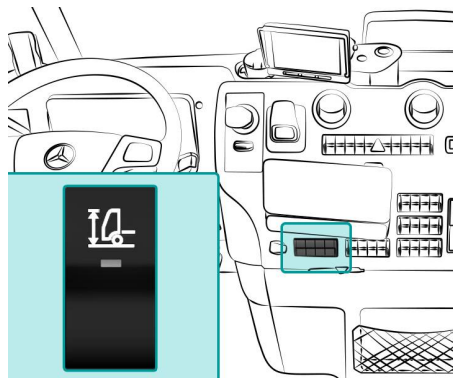
Vehicles without residual air bellow pressure readjustment: The air suspension pressure on the front axle is reduced until it matches a parameterised value. The air suspension valves are closed when the set value is reached. On the rear axle, the vehicle is lowered completely on all air-sprung axles. The air suspension valves on the rear axle subsequently remain open.

### Automatic deactivation of the special functions

Special functions are deactivated in the following cases:

- The vehicle is switched off and on again.
- The  button on the operating unit is pressed.
- In the multimedia system, the  /  button is pressed in the **Level control** menu item under the **Operation** menu.

### Using raised driving level (vehicles for transportation of bulk cargo)



Raise the chassis while driving for improved ride comfort.

- **Raising the chassis frame to raised driving level:** Press the  button above.

or

- In the multimedia system, select the **Level control** menu item under the **Operation**  menu.

- ◁ The indicator lamp in the  button and the  indicator lamp in the instrument cluster light up.

- **To lower the chassis frame to normal driving level:** Press the top  button again.

or

- In the multimedia system, under the **Operation** menu, select the **Level control**  menu item again.

- ◁ The indicator lamp in the  button goes out.  
When the chassis is lowered to the normal driving level, the  indicator lamp in the instrument cluster goes out.

### Notes on the chassis-lowering feature on the front axle

The chassis-lowering feature on the front axle makes entering and exiting the vehicle easier.

## Driving mode

### Additional axles

The chassis-lowering feature on the front axle is only available when the following conditions are met:

- The vehicle is stationary and has been started.
- There is sufficient reservoir pressure in the compressed-air system.

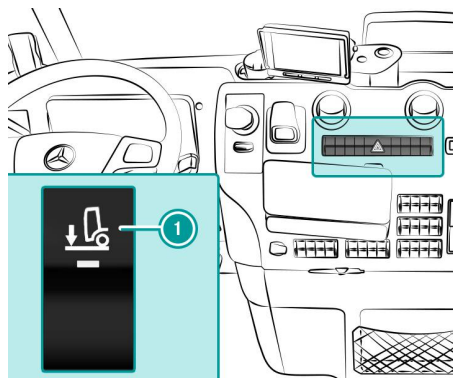
If you open the driver's or front passenger's door, the chassis is lowered on the front axle.

The chassis returns to the driving level in the following cases:

- The lowering process is cancelled.
- The driver's or front passenger's door is closed.
- The vehicle is set in motion.
- The vehicle is switched off.

#### Operating the chassis-lowering feature on the front axle

##### Activating or deactivating the chassis-lowering feature on the front axle



- ▶ **To activate:** Press the ❶ button above. The indicator lamp in the button ❶ comes on.
- ▶ **To deactivate:** Press ❶ button below. The indicator lamp on the button ❶ will go out.

##### Cancelling the chassis-lowering procedure on the front axle

- ▶ Press the  button on the operating unit.

or

- ▶ In the multimedia system, under the menu item **Level control** press the button  / **STOP**.
  - ◁ The chassis frame is raised to the previously set driving level.

#### Additional axles

##### Notes on starting-off aid



##### NOTE

##### Damage to tyres and axles due to prolonged use of the starting-off aid

The starting-off aid is only intended for short-term use if the vehicle's wheels are spinning on a slippery carriageway.

The axle load on the rear axle is increased when the starting-off aid is switched on.

Using the starting-off aid over an extended period of time may damage the rear axle and the wheels.

- ▶ Make sure that the starting-off aid is only switched on for a short time.

The starting-off aid can also be fitted with a limit speed or a time limit (reactivation lockout).

If the starting-off aid has no time limit, make sure that the starting-off aid is only switched on for a short time.

The starting-off aid with limit speed switches itself off automatically at a speed above 30 km/h. The starting-off aid can only be switched on again at a speed below 30 km/h.

The starting-off aid with reactivation lockout switches itself off automatically after 90 seconds. After 50 seconds, the starting-off aid with reactivation lockout can be switched on again.

The starting-off aid without reactivation lockout switches itself off automatically after 120 seconds and can be switched on again immediately.

### Switching starting-off aid on/off

- ▶ **Switching starting-off aid on:** In the multimedia system, from the **Operation** menu, select the item **Switch**.
- ▶ Press the  button in the multimedia system.
  - ◀ As long as the starting-off aid is switched on, the  indicator lamp in the status area of the instrument cluster lights up yellow.

### Note

If the starting-off aid has no time limit, stop the starting-off aid manually after a short time.

- ▶ **Ending the starting-off aid manually:** Press the  button again.
- or
- ▶ Press the  button on the operating unit of the level control (→ Page 190).
    - ◀ The  indicator lamp in the status area of the instrument cluster goes out.

### Loading and unloading the trailing axle

### Note

Observe the instructions in the "Axle and wheel loads" section (→ Page 145) and on the vehicle identification plate (→ Page 268).

- ▶ Switch on the vehicle.
- ▶ If the supply pressure in the compressed-air system is too low, leave the vehicle running.
- ▶ If the vehicle is empty or only partially laden, relieve the trailing axle before pulling away.
- ▶ Vehicles with a rear loading crane body: observe the information in the body manufacturer's operating instructions before relieving the trailing axle.
- ▶ When loading or unloading the vehicle, load the trailing axle.

## Driving mode

### Additional axles

- ▶ **Loading and unloading:** In the multimedia system, under the menu **Operation** item **Switch**, press the button .
  - ◁ When the trailing axle is relieved, the  indicator lamp in the status area lights up yellow.

#### Using steerable additional axle

##### To centre manually



#### NOTE

##### Damage to the tyres of vehicles with steerable additional axle

There is a risk of damaging the tyres if vehicles with steerable additional axles are manoeuvred on kerbs or pass through narrow passages.

- ▶ In such cases, centre the steering of the additional axle.
- ▶ Start the vehicle.
- ▶ In the multimedia system, press the button  under the **Switch** menu item of the **Operation** menu.

- ◁ The steering of the additional axle is centred. The wheels of the additional axle steer in a straight-ahead position.

When the indicator lamp  on the instrument cluster lights up, the additional axle is centred.

##### To release the steering function

- ▶ Press the  button again.
  - ◁ When the indicator lamp  on the instrument cluster goes out, the steering function of the additional axle is activated. The additional axle co-steers again.

If the steerable additional axle is disturbed or no longer steers, the instrument cluster displays a corresponding display message and the  indicator lamp in the status area of the instrument cluster lights up.

If the indicator lamp  lights up grey on the instrument cluster, the steering angle disparity between the wheels of the front axle on full lock and the steerable additional axle is too large. For example, the vehicle was parked with its steering

on full lock. The steerable additional axle does not co-steer.

- ▶ Turn the multifunction steering wheel to the left and right as far as it will go.
  - ◁ The steerable additional axle is picked up by turning the steering wheel. The additional axle co-steers again. The indicator lamp  on the instrument cluster goes out.

or

- ▶ Pull away slowly.
  - ◁ The additional axle co-steers again. The indicator lamp  on the instrument cluster goes out.



## Driving tips

## General driving tips

**Danger****Risk of accident when the vehicle is switched off while driving**

If during an emergency you switch off the vehicle while you are driving, safety-relevant functions will be restricted or no longer available.

This applies, for example, to the compressed-air system, which will still be supplied from the compressed-air reservoirs but is no longer filled.

You will then have to stop immediately in accordance with the traffic regulations. If the reservoir pressure is too low, it will no longer be possible to brake the vehicle.

- For this reason alone you should never switch off the vehicle while you are driving.

**Danger****Risk of accident on uphill or downhill slopes**

The parking brake may not be sufficient to secure the vehicle on uphill and downhill slopes.

A loaded vehicle or a vehicle with a trailer may roll away.

- In the check position, check whether the braking force of the parking brake is sufficient to hold the towing vehicle.
- If it does not hold the towing vehicle, secure the vehicle additionally, including the trailer, with chocks.
- Additionally secure the trailer using its parking brake.

## Reverse warning device

## Notes on the reverse warning device

**Danger****Risk of accidents due to persons or objects in the area in which you are manoeuvring**

Other road users may not hear or may ignore the warning tone of the reverse warning device. There is a risk of an accident occurring.

- Make sure that no persons or objects are present in the manoeuvring area when manoeuvring.
- If necessary, a second person must assist with manoeuvring.

The reverse warning device is a system designed to assist you in ensuring the safety of other road users.

When using the reverse warning device described here, observe the legal requirements for the country you are currently in.

## Driving mode

### Driving tips

When you switch on the vehicle and engage reverse gear, the reverse warning device is always set to loud.

#### Adjusting the reverse warning device

- **Adjusting the reversing warning device:** In the multimedia system, from the **Operation** menu, select the item **Switch**.

- ◄ The icons on the button  vary depending on the current system status.

Pressing this button changes the system status of the reverse warning device.

The different displays on the button have the following meaning:

-  Reverse warning device muted
-  Reverse warning device loud

If reverse gear has not been engaged, the muted setting of the reverse warning device remains active for around two minutes. After two minutes, the volume of the reverse warning device will be turned back up.

Vehicles with automatic activation of the hazard warning light system: The hazard warning lights are switched on when the reverse gear is engaged.

#### Using the reverse gear lock on waste collection vehicles

##### Waste collection vehicles only:

If the running boards in the rear area are subjected to a load, the vehicle speed is limited to a maximum of 30 km/h and the reverse gear lock is activated. The reverse gear lock prevents reverse gear being selected.

If the running boards in the rear area are loaded while reverse gear is engaged, a warning tone sounds and the vehicle switches off.

If the vehicle has been switched off by the reverse gear lock:

- Switch off the vehicle.
- Apply the parking brake.
- Switch on the vehicle again.

#### Notes on the warning tone

A warning tone will sound in the following cases:

- The driver's door is opened with the dipped beam switched on and the ignition lock in the radio position.
- The driver's door is opened with the standing lights switched on and the vehicle switched off.
- The immobiliser is activated.
- The brake pedal is not depressed when the hill holder is activated and the vehicle is stationary.
- Reverse gear is engaged.
- The vehicle is switched off when the frequent-stop brake is activated and the parking brake is released.
- When various driving systems issue warnings.

A warning tone sounds in addition to a display message in the instrument cluster in the following cases:

- The tyre pressure monitoring system displays a tyre pressure loss warning.

- The instrument cluster or the on-board computer is malfunctioning. Important operating information, maintenance information or warning and indicator lamps can no longer be displayed.
- A door is opened and the parking brake is not applied.

### Charging the high-voltage battery

#### Notes on charging the high-voltage battery

Information on the nominal voltage range and charging times can be found in the technical data ( → Page 292).



#### NOTE

#### Damage to the high-voltage battery due to long downtimes

Lithium-ion batteries will self-discharge on their own accord.

Downtimes lasting for several months may therefore cause deep discharge. This may damage the high-voltage battery.

- ▶ To prevent potential damage, please observe the following recommendations when handling the high-voltage battery.



#### NOTE

#### Accelerated ageing of the high-voltage battery due to not observing the following recommendations

As a result of its basic characteristics, the storage capacity of and the amount of energy available from the high-voltage battery decrease over the course of its life. As a result, the maximum electrical range that can be achieved by the vehicle is reduced and its maximum electrical output can be impaired.

The following factors could accelerate the ageing of the high-voltage battery:

- a high level of charge in the high-voltage battery, especially if the vehicle is idle for a lengthy period of time

### Charging the high-voltage battery

- leaving the vehicle idle for lengthy periods at high ambient temperatures
- ▶ To avoid accelerated ageing, please observe the following recommendations when handling the high-voltage battery.

Recommendations when handling the high-voltage battery:

- Charge the high-voltage battery to a charge level of 80% on average. This will prolong the service life of the high-voltage battery. Beyond a charge level of 90%, the charging capacity is significantly reduced in proportion to temperature.
- When leaving the vehicle idle for long periods, park up the vehicle with the high-voltage battery at a charge level between 30% and 50%.
- If leaving the vehicle idle for lengthy periods of time avoid, if possible, high and low outside temperatures.
- Check the high-voltage battery's charge level every four weeks ( → Page 78).

## Driving mode

### Charging the high-voltage battery

- If the charge level is below 20%, recharge the high-voltage battery.
- Do not disconnect the 24 V on-board electrical system battery, even for extended downtimes. Otherwise, the condition of the vehicle's high-voltage battery cannot be monitored.

You can contribute to reducing the vehicle's energy consumption in the following ways:

- an anticipatory driving style
- reduced use of electrical consumers
- having the vehicle regularly maintained

The charging time of the high-voltage battery may change over the course of its life.

#### Note

The on-board computer displays the charge level of the high-voltage battery in the instrument cluster (→ Page 78).



#### Warning

##### **Risk of fatal injuries when carrying out maintenance work during the charging process**

During the charging process, the vehicle's high-voltage on-board electrical system is under high voltage.

- ▶ Do not perform any maintenance work during the charging process.



#### Warning

##### **Risk of death from damaged components**

If you use a damaged component to connect the vehicle to the charging station, this may cause fires or an electric shock.

- ▶ For charging stations with a pre-installed charging cable:
  - Visually inspect the charging station for obvious defects, such as severe damage to the housing or the charging cable.
- ▶ For charging stations without a pre-installed charging cable:

- For safety reasons, only use charging cables that have been tested and approved by the manufacturer for charging the high-voltage battery in an electric vehicle.
- Never use damaged charging cables.
- Do not extend the charging cable.
- Do not use adapters.
- ▶ You must always follow the safety notes on the charging station.

The high-voltage battery can be charged as follows:

- charging through recuperation while the vehicle is in motion
- Stationary DC charging at a fast-charging station (mode 4) if one of the two following conditions is met:
  - the charging cable to the vehicle is not longer than 30 m in total.
  - the vehicle is charged completely alone on the charging station connector without a sub-distribution.

## Charging the high-voltage battery

### **i** Note

AC charging using a mains socket (mode 2), wallbox or charging station (mode 3) is not possible.

Observe any possible different local grid requirements of your current location when charging. Only use charging cables which conform to the grid requirements. Consult a qualified electrician or your local grid operator if you have any questions.

When charging at a public charging station, legal stipulations state that the charging cable to the vehicle must not be longer than 30 m in total. Be aware that parts of the charging cable may be routed underground. If in doubt, ask the charging station operator if this is the case before charging the high-voltage battery. When the vehicle is being charged at a private charging station, no other vehicles can be on the charging station connector being used. This is to prevent the interference of signals being received by radio communication devices in the vehicle or in close proximity to the charging station.

Most charging stations must be activated before the charging process, e.g. using an RFID card. Observe the on-site operator's instructions for the charging station.

The amount of energy dispensed for the charging process, shown by the charging station, may be higher than the amount of energy actually absorbed by the high-voltage battery. This is the result of different levels of charging losses and is referred to as recharge efficiency. Charging losses occur, for example, due to heat that builds up when charging or from auxiliary consumers that are switched on. Further information on recharge efficiency can be obtained at a qualified specialist workshop.

### System limits

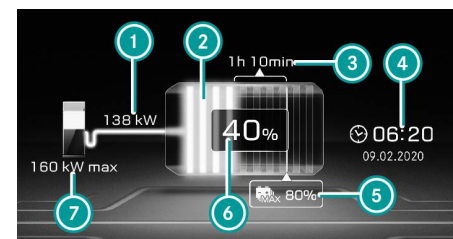
The charging time of the high-voltage battery may be increased in particular by the following:

- high or low outside temperatures
- the vehicle being non-operational for long periods without charge

- the maximum available charge current of the charging station
- the settings for the charging process in the on-board computer (→ Page 204)

### Function of the charging prediction

When the vehicle is switched off and is connected to the mains supply, the instrument cluster automatically shows the charge level display with a charging prediction. The display automatically switches off after approximately two minutes. When the vehicle is switched on, the multimedia system display continuously shows the charge level display.



① Current charging capacity [kW]

## Driving mode

### Charging the high-voltage battery

- ② Current charge level and energy flow
- ③ Remaining charging time [h/min]
- ④ Set departure time or projected end of charge cycle
- ⑤ Set charging limit [%]
- ⑥ Current charge level [%]
- ⑦ Available charging capacity [kW]

Depending on the state of charge, battery charge level ②, energy flow and charging station are displayed in colour as follows:

- Blue: active energy flow, charging process
- White: interruption in charging
- Bronze: malfunction, charging process interrupted

When the charging limit has been reached, the vehicle can continue to charge to maintain its charge level. If the vehicle is switched off and no departure time is set or instant charging is set, the charge is not maintained and the charging process is ended by the vehicle after 30 minutes and the high-voltage on-board electrical system is deactivated. If the

set departure time is less than 48 hours away, charge maintenance remains active until the departure time is reached.

#### Note

When the charging process has ended, the actual state of charge could differ from the set charging limit.

#### Note

The charging process can be restarted by removing and reinserting the charging cable connector.

In the event of a fault, the charge level display also shows the message **Fault Charging process interrupted**. If the battery charge level is displayed in bronze, a malfunction by the vehicle has occurred. If the charging station is displayed in bronze, a malfunction by the power supply has occurred.

### Setting the charging process

Multimedia system:

 **Operation** **Settings**  
Charging the battery

#### Setting the departure time

- ▶ Select **Departure time**.
- ▶ **To activate/deactivate:** Select one or more buttons with existing settings.
  - ◁ If the indicator lamp in the button is lit up, the setting is active. The charging capacity is automatically adjusted so that the set **Charging limit** is reached for the active departure time in a battery-saving manner. During the charging process, the drive system is preconditioned for optimal range and driving performance. You can also adjust the temperature in the cab prior to the active departure time.

## Charging the high-voltage battery

### Note

When a departure time is active, the yellow indicator lamp lights up in the  button under the multimedia system.

If the vehicle has been switched off and pre-entry climate control of the cab is active, the red indicator lamp lights up in the  button under the multimedia system.

### Note

When the departure time is reached, preconditioning and charge maintenance remain active for a further 30 minutes.

### Note

If the set departure time is more than 48 hours away, no preconditioning before the departure time takes place.

- ▶ **To change:**  Select setting below button.

or

- ▶ **To create as new:** Select the button .

- ▶ Set the **Time** and day(s) of the week (**Repeat**).
- ▶ Switch on **Air-condition cab**  or switch it off .
  - ◁ If climate control is activated, set the **Temperature**.  
The temperature in the cab will be adjusted prior to the departure time.
- ▶ **To delete:**  Select setting below button.
- ▶  Select **Delete**.
  - ◁ All button settings and the button are deleted.

### Setting the charging limit

- ▶ Select **Charging limit**.
- ▶ Set the charging limit as a percent.
  - ◁ The high-voltage battery is charged up to the set percentage as a maximum. The setting applies to all subsequent charging processes.

### Note

A charging limit of 80% is recommended. Continually setting higher charging limits could accelerate the ageing of the high-voltage battery (→ Page 201).

### Note

When a charging process is started, the display shows the charging prediction in addition to the settings (→ Page 203).

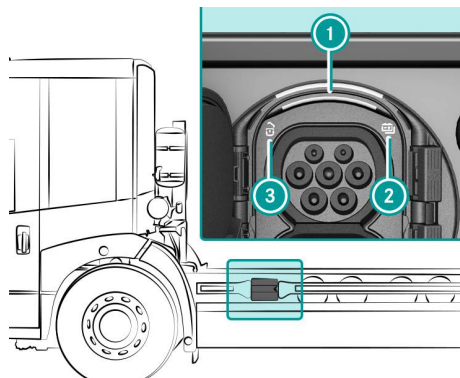
### Activating/deactivating instant charging

- ▶ Switch on **Instant charging**  or switch it off .
  - ◁ After inserting the charging cable connector, the charging process begins immediately at maximum charging capacity. The charging prediction shows a corresponding message.  
The drive system is not pre-conditioned. The instant charging function is deactivated after the vehicle is started.

## Driving mode

### Charging the high-voltage battery

#### Functions of the indicator lamps on the vehicle socket



- ① Socket lamp
- ② Charging process indicator lamp
- ③ Locking status indicator lamp

The upper arch of socket lamp ① is used for lighting and flashes or lights up as with indicator lamp ③. The lower arch is used for the status display and flashes or lights up as with indicator lamp ②.

#### Overview of locking status

| Locking status ③ | Display         | Meaning                                                  |
|------------------|-----------------|----------------------------------------------------------|
|                  | Lights up white | Vehicle socket unlocked, insert or remove charging cable |
|                  | Flashes white   | Malfunction during locking or unlocking                  |

#### Overview of the charging process status

| Status of charging process ② | Display          | Meaning                         |
|------------------------------|------------------|---------------------------------|
|                              | Pulses orange    | Connection is being established |
|                              | Pulses green     | Active energy flow              |
|                              | Lights up orange | Interruption in charging        |
|                              | Lights up green  | Charging process completed      |

| Status of charging process ② | Display     | Meaning                   |
|------------------------------|-------------|---------------------------|
|                              | Flashes red | Malfunction when charging |

#### Starting the charging process



#### Warning

#### Risk of death when charging at a damaged socket

The charging process uses high voltage.

If the charging cable, the vehicle socket or the mains socket are damaged, you could receive an electric shock.

- ▶ Only use an undamaged charging cable.
- ▶ Avoid mechanical damage such as crushing, abrading or driving over the cable.
- ▶ Have a damaged vehicle socket replaced at a qualified specialist workshop immediately.



## Charging the high-voltage battery

- ▶ Never insert the charging cable into a damaged vehicle socket.



### NOTE

#### Damage to or contamination of the vehicle socket with the socket flap open

- ▶ Always keep the socket cover and the socket flap closed if no charging cable is connected. These protect the vehicle socket from dirt and damage.
- ▶ Before closing the socket flap, ensure that the socket cover is closed correctly. Otherwise, there may be damage and it may no longer be possible to open the socket flap.



### NOTE

#### Damage to the vehicle socket or the charging cable connector due to incorrect handling

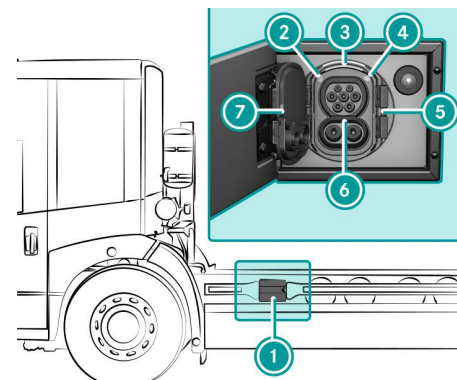
Do not use excessive force to insert the charging cable connector into the vehicle socket as far as it will go (max. 300 N).

Do not pull a locked charging cable connector out of the vehicle socket with excessive force. You may otherwise damage the vehicle socket, the charging cable connector or their contacts.

- ▶ If you feel there is increased resistance, pull the charging cable connector out of the socket and reinsert it.
- ▶ If the charging cable connector cannot be removed, release the charging cable connector at the end of the charging process using the emergency release.

### Requirements

- The electronic parking brake is applied.
- The charging cable is not under tension.



- ▶ Fold out socket flap ① to the left.
- ▶ Unlock the vehicle.
  - ◁ The indicator lamp ② and upper socket lamp ③ light up white.
- ▶ Press both catches ⑤ to the right.
  - ◁ Socket covers ⑦ are open.
- ▶ Remove foreign bodies from the vehicle socket.
- ▶ Insert the charging cable connector into vehicle socket ⑥ to the stop.

## Driving mode

### Charging the high-voltage battery

- ◁ Make sure that the inserted charging cable is not put under tension.

The indicator lamp  ② goes out. The indicator lamp  ④ and lower status display ③ pulse orange and, as soon as the high-voltage battery is charged, green.

When the charging cable is connected to the vehicle, the vehicle cannot be started or moved.

When the charging process is started, the estimated charging time is displayed in the instrument cluster. The charging prediction shows the predicted charge status at the set departure time or the time at which the high-voltage battery will be fully charged.

Observe any information which may be displayed in the instrument cluster:

- Charging prediction (→ Page 203)
- Display messages (→ Page 297)

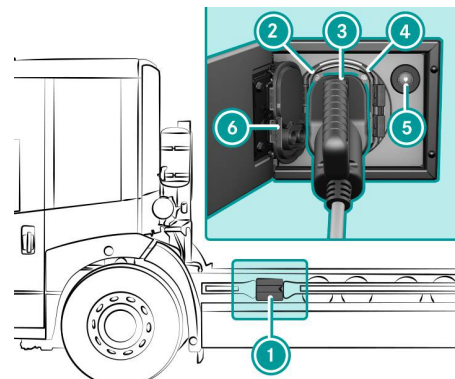
#### Note

Depending on the temperature, the fan and battery cooling system may audibly switch on during the charging process.

#### Note

If the vehicle is idle for lengthy periods and connected for the purpose of charging, the high-voltage battery will be charged automatically as needed or when electrical consumers are activated (e.g. climate control).

### Ending the charging process



- ▶ Press charging interruption button ⑤ and, if applicable, observe the additional notes on the charging station.
  - ◁ The charging process is ended after a short time. The indicator lamp  ④ lights up green.

## **i** Note

The charging process is automatically ended when the set charging limit or departure time is reached and also for prolonged trickle charging.

- ▶ Unlock the vehicle.
  - ◁ If the indicator lamp  ② then lights up white, the vehicle socket is unlocked.
- ▶ Remove charging cable connector ③ from the vehicle socket within 30 seconds, taking care not to use excessive force.
  - ◁ After the 30 seconds have elapsed, the vehicle socket will be locked again.

## **i** Note

If you cannot remove the charging cable connector, repeat the unlocking procedure. If the charging cable connector remains locked, unlock the charging cable connector with the emergency release (→ Page 209).

- ▶ Close socket cover ⑥ and socket flap ①.

## **i** Note

After charging cable connector has been disconnected, the left indicator lamp  ② on the vehicle socket remains lit for some time before going out.

## Unlocking the charging cable connector with the emergency release

### NOTE

#### Damage to the vehicle socket due to operation of the emergency release

If the emergency release is used to unlock the charging cable connector, parts of the vehicle socket could be damaged.

After the emergency release, it is no longer possible to charge the vehicle because the charging cable connector can no longer be locked.

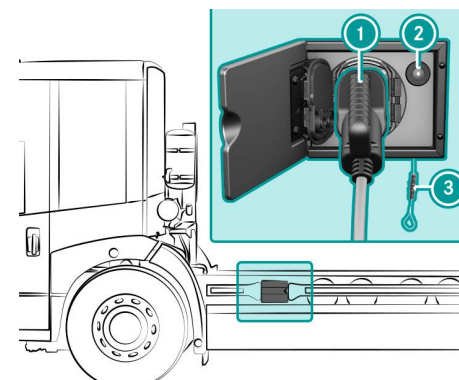
- ▶ Unlock the charging cable connector with the emergency release in an emergency only.

## Charging the high-voltage battery

- ▶ After an emergency release of the charging cable connector, have the vehicle checked and repaired immediately at a qualified specialist workshop.

## Requirements

- The charging cable connector cannot be disconnected.



- ▶ Press charging interruption button ②.
- ▶ Unclip cable ③ from the vehicle carrier and guide it vertically downwards.

## Driving mode

### Winter operation

- ▶ Pull cable ③ downwards and remove charging cable connector ① from the vehicle socket.
- ▶ Close the socket covers and the socket flap of the vehicle socket.

#### Winter operation

#### Winter driving

#### Preparing for a journey (winter operation)

#### Note

Remove snow and accumulations of ice on the vehicle before every journey. You can find information on cleaning the sensors under "Cleaning and care" (→ Page 219).

Before the start of the cold season make sure:

- The coolant contains sufficient antifreeze protection (→ Page 290).

- The windscreen washer system and headlamp cleaning system contains sufficient antifreeze (→ Page 226).
- Suitable winter tyres are fitted.
- Snow chains are carried in the vehicle.

#### Note

In wintry conditions, the law may require that winter tyres be fitted on the wheels of the drive axle. Find out which winter tyres are suitable for the intended use. Observe the legal requirements for the country you are currently in.

#### Notes on driving in winter

Please observe the following instructions on driving in winter:

- in snow, slush and on icy roads, fit snow chains to the drive wheels in good time.
- adapt your driving style to the wintry road conditions.
- If traction problems occur when driving with snow chains, disengage ASR

(→ Page 146) or Stability Control Assist (→ Page 147).

#### Snow chains

#### Notes on snow chains



#### Danger

#### Risk of accident due to tearing snow chains

If you drive too quickly with snow chains, they may tear and injure other persons and damage the vehicle.

- ▶ Comply with the maximum permissible speed for operating the vehicle with snow chains.



#### NOTE

#### Damage to the vehicle from snow chains

- ▶ Only use snow chains that are approved and recommended for Mercedes-Benz trucks. This prevents damage to the vehicle.

Contact a qualified workshop if you have any questions.

**NOTE****Damage to vehicle parts due to snow chains on the front axle**

Vehicle parts may be damaged by snow chains on the front axle.

- ▶ Do not install snow chains on the front axle.

The law may require that snow chains be removed again as soon as possible once the roadway is clear of snow. The vehicle's driving and braking characteristics will be adversely affected if you drive on roadways that are clear of snow with snow chains fitted to the vehicle.

The maximum permissible speed with snow chains fitted to the vehicle is 50 km/h.

When using the snow chains described here, observe the legal requirements for the country you are currently in.

Do not use twin chains on vehicles with roll control. Only fit snow chains on the exterior wheels.

**Vehicles with ASR/Stability Control**

**Assist:** If, when driving, traction problems arise while snow chains are being used, disengage ASR (→ Page 146) or Stability Control Assist (→ Page 147).

## Working mode

### Electric power take-off system

#### Electric power take-off system

#### Function of the electric power take-off system



**Danger**

#### Risk of accident on uphill or downhill gradients

The parking brake may not be sufficient to ensure that the laden vehicle does not roll away on uphill and downhill gradients.

- ▶ In the check position, check whether the parking brake will hold the laden vehicle.
- ▶ If the vehicle is not held, secure it using other methods, e.g. wheel chocks.

With the electric power take-off system (power take-off), you can drive auxiliary assemblies, for example, cooling systems.

The available power of the power take-off depends on the state of charge of the high-voltage battery.

You can use the power take-off in working or driving mode and during the charging process (→ Page 213).

If you switch on the vehicle, the power take-off is operational and the yellow  symbol appears on the instrument cluster. If the yellow symbol  does not appear on the instrument cluster, switch off the vehicle and restart it.

If the yellow symbol  still fails to appear on the instrument cluster, have the electric power take-off system checked at a qualified specialist workshop.

Depending on the function of the body and the ancillary components, the power take-off is activated automatically as needed. Depending on the equipment, special functions of the level control system can be activated automatically. Please observe the notes on the "Special functions of the level control system" (→ Page 194).

If the automatic activation of the hazard warning light system is switched on, the rotating beacons and/or hazard warning light system, for example, are

switched on and off automatically depending on the power take-off.

When the power take-off is active, the energy flow display shows the current status of the power take-off (→ Page 145).

You can find further information on the function of the power take-off in the operating instructions issued by the body manufacturer.

#### Switching electric power take-off system on and off

If you switch the vehicle off, the electric power take-off system (power take-off) is always operational. The activation conditions of the power take-off depend on the function of the body.

- ▶ Observe the body manufacturer's operating instructions.

#### Deactivating

- ▶ In the multimedia system, press the button  under the **Switch** menu item of the **Operation** menu.

- ◁ The power take-off is deactivated. The red indicator lamp in the  button and the yellow symbol  in the instrument cluster go out.

### Activating

- ▶ In the multimedia system, press the  button again under the **Switch** menu item in the **Operation** menu.
- or
- ▶ Switch the vehicle off and on again.

### Using the electric power take-off system during the charging process



#### Warning

#### Risk of death from damaged components

If you use a damaged component to connect the vehicle to the charging station, this may cause fires or an electric shock.

- ▶ For charging stations with a pre-installed charging cable:

- Visually inspect the charging station for obvious defects, such as severe damage to the housing or the charging cable.
- ▶ For charging stations without a pre-installed charging cable:
  - For safety reasons, only use charging cables that have been tested and approved by the manufacturer for charging the high-voltage battery in an electric vehicle.
  - Never use damaged charging cables.
  - Do not extend the charging cable.
  - Do not use adapters.
- ▶ You must always follow the safety notes on the charging station.



#### Warning

#### Risk of death when charging at a damaged socket

The charging process uses high voltage.

If the charging cable, the vehicle socket or the mains socket are damaged, you could receive an electric shock.

- ▶ Only use an undamaged charging cable.
- ▶ Avoid mechanical damage such as crushing, abrading or driving over the cable.
- ▶ Have a damaged vehicle socket replaced at a qualified specialist workshop immediately.
- ▶ Never insert the charging cable into a damaged vehicle socket.

If you want to use the electric power take-off system (power take-off) during the charging process at a charging station, observe the operating instructions issued by the body manufacturer.

- ▶ Observe the notes on charging the high-voltage battery ( → Page 201).
- ▶ Switch on the vehicle without a charging cable connector plugged in.

## Working mode

### Compressed air supply

- ▶ Plug the charging cable connector into the vehicle socket and start the charging process (→ Page 206).
  - ◁ If the power take-off is operational, the yellow symbol  appears in the instrument cluster.

If the vehicle is not switched on during the charging process, proceed as follows:

- ▶ End the charging process (→ Page 208).
- ▶ Remove the charging cable connector from the vehicle socket.
- ▶ Switch on the vehicle.
- ▶ Plug the charging cable connector into the vehicle socket and start the charging process again (→ Page 206).
  - ◁ If the power take-off is operational, the yellow symbol  appears in the instrument cluster.

### Compressed air supply

#### Adjusting the compressed-air delivery rate

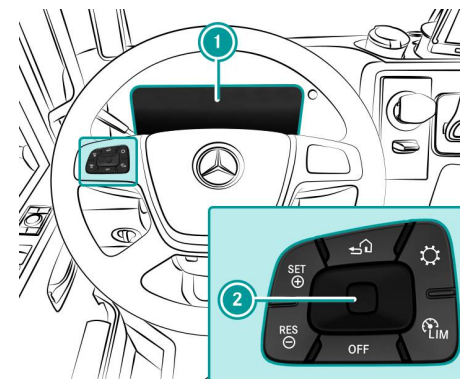
Requirements:

- The vehicle is stationary.
- The parking brake is applied.
- The transmission is in neutral position.
- The vehicle has been started.

Depending on how the vehicle is used, the compressed-air consumption of your vehicle may be increased, e.g. when operating with swap bodies. You can adapt the compressed air delivery rate of the air compressor and the associated operating noises to the operating conditions.

The following settings are available:

- min: Low compressed-air consumption (quiet)
- auto: Normal compressed-air consumption
- max.: Increased compressed-air consumption



- ▶ Press the  button on the multi-function steering wheel repeatedly until the input window **AIR COMPRESSOR** is displayed on the instrument cluster ①.
- ▶ Use Touch Control ② to set the desired compressed-air delivery rate.

#### Note

The last compressed-air delivery rate setting remains stored even after the vehicle is restarted.



#### Notes on cleaning the interior



**Danger**

#### Risk of injury or fatal injuries from bleached seat belts

Bleaching or dyeing seat belts can severely weaken them.

This can, for example, cause seat belts to tear or fail in an accident.

- ▶ Never bleach or dye seat belts.



#### ENVIRONMENTAL NOTE

#### Environmental pollution caused by environmentally irresponsible disposal

- ▶ Dispose of the cleaning materials in an environmentally responsible manner.



#### Note

Daimler Truck AG recommends that you only use care products that have been approved for Mercedes-Benz trucks.

These care products are available from an authorised Mercedes-Benz Trucks Service Centre.

Observe the notes on cleaning and care to avoid damage to the vehicle.

#### Seat belts

- Clean with lukewarm and soapy water.
- Do not use chemical cleaning agents.
- Do not dry the seat belt by heating it above 80 °C or exposing it to direct sunlight.

#### Displays

- Switch off the display and let it cool down.
- Clean the surface carefully with a cotton or microfibre cloth and a display care product (TFT-LCD) recommended for Mercedes-Benz trucks.
- Do not use any other cleaning products.

#### Cleaning the plastic trims

- Clean with a damp microfibre cloth.
- For heavy soiling: Use a cleaning agent recommended for Mercedes-Benz trucks.
- Do not use solvent-based care products and cleaning agents.
- Do not attach stickers, films or similar materials.
- Do not allow cosmetics, insect repellent or sun cream to come in contact with the plastic trim.

#### Trim elements

- Clean with a microfibre cloth.
- For heavy soiling: Use a cleaning agent recommended for Mercedes-Benz trucks.
- Do not use solvent-based cleaning agents, polishes or waxes.

#### Imitation leather seat covers

- Clean with a damp cloth and 1% soapy water.
- Do not use the microfibre cloth.

## Cleaning and care

### Notes on the use of a high-pressure cleaner

- Do not clean the seat cushions and backrests using steam cleaners or cleaning fluids.

#### Cloth seat covers

- Clean with a damp microfibre cloth and 1% soap solution and allow to dry.
- Do not clean the seat cushions and backrests using steam cleaners or cleaning fluids.

#### Steering wheel

- Clean with a damp cloth and mild soapy water.
- Wipe over again with a dry cloth.
- Do not use solvent-based cleaning and care products.
- Do not allow fluids with sticky ingredients to come into contact with the steering wheel or the steering-wheel buttons.

#### Notes on the use of a high-pressure cleaner



**Danger**

##### Risk of accident when using high-pressure cleaners with round-spray nozzles

The water jet cannot cause visible signs of damage.

Damaged components may fail unexpectedly.

- ▶ Do not use a high-pressure cleaner with round-spray nozzles.
- ▶ Have damaged tyres or suspension components replaced immediately.



**NOTE**

##### Damage due to the use of a high-pressure cleaner in the vehicle interior

The volume of water generated under pressure by the high-pressure cleaner and the associated spray may cause significant damage to the vehicle.

- ▶ Do not use a high-pressure cleaner in the vehicle interior under any circumstances.



**NOTE**

##### Component damage due to the use of a high-pressure cleaner

The following vehicle components can be damaged through direct application of a high-pressure cleaner:

- Door gaps
  - Air suspension bellows
  - Brake hoses
  - Balance weights
  - Electrical components
  - Electrical connectors
  - Seals
- ▶ Do not point the water jet at these components.

### Notes on the use of automatic car washes



#### NOTE

#### Malfunctions and damage to component parts of the high-voltage on-board electrical system due to the use of a high-pressure cleaner

- ▶ Do not direct the water jet at the high-voltage batteries or other component parts of the high-voltage on-board electrical system.

Keep the water jet moving constantly while cleaning.

#### Notes on the use of automatic car washes



#### NOTE

#### Damage when using an automatic car wash

If, when using an automatic car wash, the outside mirrors are folded out or the windscreen wipers are activated, they may be damaged.

- ▶ Before using an automatic car wash:

- Deactivate the windscreen wipers.
- Fold in the outside mirrors.

Make sure that the outside mirrors are fully folded out again when you leave the automatic car wash.

If the vehicle is very dirty, wash off excess dirt before cleaning the vehicle in an automatic car wash.



#### Note

After using an automatic car wash, wipe off the wax from the windscreen and the wiper rubbers. This will prevent smearing and reduce wiper noises caused by residue on the windscreen.

#### Notes on cleaning the exterior



#### Danger

#### Risk of an accident due to reduced braking effect after washing the vehicle

The braking effect of the brakes is reduced after washing the vehicle.

- ▶ After the vehicle has been washed, brake carefully while paying attention to the traffic conditions until the braking effect has been fully restored.



#### ENVIRONMENTAL NOTE

#### Environmental pollution caused by environmentally irresponsible disposal

- ▶ Dispose of the cleaning materials in an environmentally responsible manner.



#### ENVIRONMENTAL NOTE

#### Environmental damage due to improper cleaning of the vehicle exterior

If you clean the vehicle exterior in a washing bay that is not intended for this purpose, this could release polluting operating fluids and cleaning agents into the environment.

- ▶ Find a suitable washing bay for cleaning the vehicle exterior.

## Cleaning and care

### Notes on cleaning the exterior



#### Note

Daimler Truck AG recommends that you only use care products that have been approved for Mercedes-Benz trucks.

These care products are available from an authorised Mercedes-Benz Trucks Service Centre.

If you are cleaning higher parts of the vehicle, follow the instructions for the steps and grab handles (→ Page 219).

If you use a high-pressure cleaner to clean the vehicle, also observe the "Notes on using a high-pressure cleaner" (→ Page 216).

### Paintwork



#### NOTE

#### Paintwork damage caused by very stubborn or aggressive dirt

For example, the following types of dirt may cause lasting damage to the paintwork:

- Insect residues

- Bird excrement
  - Flash rust
  - Tree resins
  - Oils and greases
  - Fuels
  - Tar splashes
  - Road salt residues
- These types of dirt must be removed immediately.

### Radiator surface



#### NOTE

#### Damage to the drive system due to insufficient cooling

If the radiator fins are deformed or dirty, this may result in reduced cooling output.

The drive system cannot be sufficiently cooled.

- Check the radiator surface regularly and clean it if necessary.

Only direct the compressed air, steam or water jet perpendicular to the radiator

surface. Ensure that the radiator fins are not damaged during cleaning.

Have loss of coolant or damage checked at a qualified workshop.

### Light-alloy wheels



#### NOTE

#### Corrosion of the wheel nuts due to the use of incorrect cleaning agents

Using acidic or alkaline cleaning agents to clean light-alloy wheels may cause corrosion to the wheel nuts or the safety springs of the balance weights.

- Use cleaning agents that have been recommended for Mercedes-Benz trucks.

Clean the light-alloy wheels regularly.

### Exterior lighting



#### NOTE

#### Damage to the exterior lighting due to improper cleaning

Using unsuitable cleaning agents and cleaning cloths may scratch or damage the plastic covers of the exterior lighting.

- ▶ Clean the plastic covers with a wet sponge and a mild cleaning agent.

### Using steps



#### Danger

#### Risk of injury if unsuitable climbing aids are used

If you use openings in the bodywork or attachments as steps, there is a risk that you could:

- slip and/or fall.
- damage the vehicle and thus slip and fall.

- ▶ Always use non-slip and stable climbing aids, e.g. a suitable ladder.
- ▶ Keep steps and grab handles free of the following forms of dirt, including:
  - Mud
  - Soil
  - Snow
  - Ice
- ◁ This will improve the safety of your footing.

### Cleaning the sensors

If you need to clean areas of the vehicle which are high up:

- ▶ Use the steps and grab handles on the vehicle (→ Page 219) or no-slip and stable climbing aids.

### Clean the windscreen in the area of the rain/light sensor and multifunction camera



#### Danger

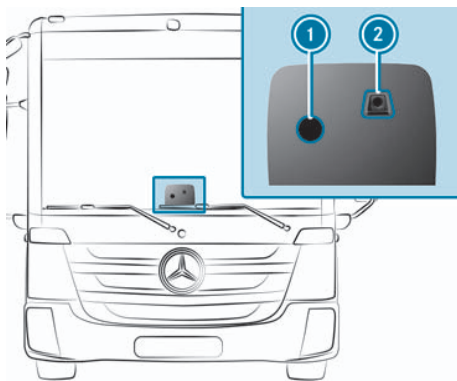
#### Risk of becoming trapped if the windscreen is being cleaned while the windscreen wipers are activated

If the windscreen wipers begin to move while you are cleaning the windscreen or wiper blades, you could become trapped.

- ▶ Always switch off the windscreen wipers and the vehicle before cleaning the windscreen or wiper blades.

## Cleaning and care

### Cleaning the sensors



- Regularly clean the windshield in the area of rain and light sensor ① and camera ② to prevent malfunctions.

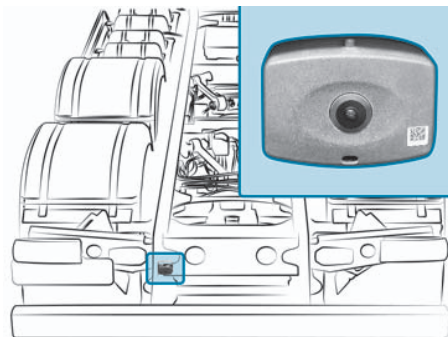
#### **i** Note

If the windshield is damaged in the area of the rain and light sensor and the camera, the functionality of various driving assistance/driving safety systems and the rain and light sensor may be impaired. Have the windshield replaced at a qualified workshop.

#### **i** Note

Note that it is only permissible to replace the windshield with a windshield approved by Daimler Truck AG.

### Cleaning the reversing camera



Reversing camera (example)

#### **i** Note

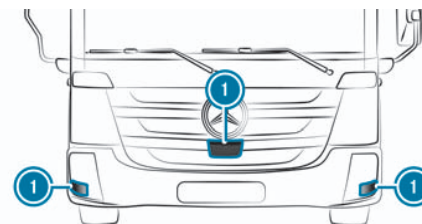
The position of the reversing camera may vary depending on the body and chassis frame.

Keep the camera lenses clean and free of snow or ice. Wet stubborn dirt on the

camera lenses with water. Then clean the camera lenses using commercially available glass cleaner and a damp cloth. Do not clean the camera lenses using a dry cloth.

- To prevent malfunctions, regularly clean the reversing camera.

### Cleaning the distance sensors

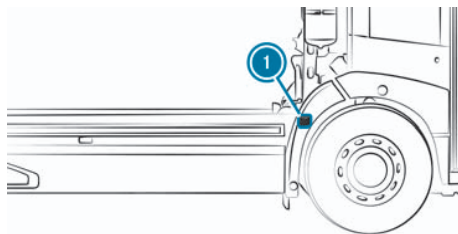


Distance sensors (example)

- To prevent malfunctions, regularly clean the distance sensors ①.

If the distance sensors are dirty, the display  Assistance systems malfunction Distance sensor viewing area blocked message appears in the instrument cluster.

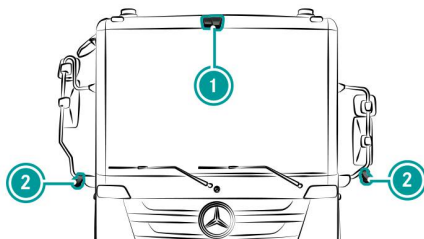
#### Cleaning the Sideguard Assist sensor



Sideguard Assist sensor (example)

- Clean the sensor ① regularly with water, shampoo and a soft cloth to prevent malfunctions.

#### Cleaning the Blind Spot Assist cameras



The camera system is made up of the front camera ①, two blind spot cameras ② and the reversing camera. You can find further information on the installation position of the reversing camera at the

rear of the vehicle in the body manufacturer's operating instructions.

Use clear water and a soft cloth to clean the camera lenses. Be careful not to apply wax to the camera lenses when you wax the vehicle.

- If necessary, remove the wax with water, shampoo and a soft cloth.

#### Mercedes-Benz Trucks Service

##### Information on service/maintenance work



**Danger**

##### **Risk of accident due to malfunctions or system failures**

If you do not have the prescribed service/maintenance work or any required repairs carried out, this could result in malfunctions or system failures.

- Always have the prescribed service and maintenance work, as well any required repairs, carried out at a qualified specialist workshop.

Always refer to the service and maintenance display of the maintenance system as well as the notes on the maintenance system (→ Page 223).

Have the specified service and maintenance work, as well as additional maintenance work, carried out within the remaining time period or the remaining mileage.

Additional maintenance work that is not load-dependent and that is not calculated

by the maintenance system is carried out at set time intervals.

Always have the prescribed service and maintenance work, as well any additional maintenance work, carried out and documented in a timely manner and according to the specifications of Daimler Truck AG. Otherwise, it can lead to loss of warranty entitlements after a report has been presented by the manufacturer.

Work relevant to safety or on safety-relevant systems must be carried out at a qualified specialist workshop.

Daimler Truck AG is continually working to improve and further develop its vehicles and therefore reserves the right to make modifications to the scope of service and maintenance work.



**Note**

You can always obtain up-to-date information concerning the maintenance of your vehicle from a qualified specialist workshop, e.g. a Mercedes-Benz Trucks Service Centre. This could include, for example, an overview of the service or

maintenance work or possible additional work.

##### **Notes on special service requirements**

For certain components, special service requirements may be necessary, e.g. after a running-in period.

Service and maintenance work will need to be performed more often or in addition if the vehicle is operated under arduous operating conditions or increased loads.



**Note**

You can obtain up-to-date information concerning the particular service requirements and service intervals of your vehicle at any time from a qualified specialist workshop. Daimler Truck AG recommends that you use a Mercedes-Benz Trucks Service Centre for this purpose.



## Notes on the oil change intervals for the major assemblies

Daimler Truck AG recommends using oil of the same quality grade as that used at the factory.

In particular, observe the notes on operating fluids (→ Page 289).

## Information on the digital service report

### Digital Service Booklet (DSB)

Service and maintenance work that has been performed is confirmed in our central database. You receive a printout of the service report which you must keep until the next service. In addition to the current service report, your vehicle's complete service history can be viewed at any time.

Your vehicle's service data is thus thoroughly documented and can no longer be lost. Every authorised Service Partner can provide you with information on your vehicle's service data. This means that

you can be assisted as quickly as possible, regardless of your location.

## Notes on the maintenance system



### Warning

#### Risk of fatal injuries when carrying out maintenance work during the charging process

During the charging process, the vehicle's high-voltage on-board electrical system is under high voltage.

- ▶ Do not perform any maintenance work during the charging process.



### Danger

#### Risk of accident due to malfunctions or system failures

If you do not have the prescribed service/maintenance work or any required repairs carried out, this may result in malfunctions or system failures.

- ▶ Always have the prescribed service and maintenance work, as well as any

## Notes on the maintenance system

required repairs, carried out at a qualified specialist workshop.



### ENVIRONMENTAL NOTE

#### Environmental pollution due to disposal in a environmentally irresponsible manner

If you have to do some maintenance work yourself for operating reasons, follow environmental protection requirements while you do so. You must comply with legal requirements when disposing of operating fluids, e.g. coolant. This also applies to all parts that have been in contact with operating fluids.

- ▶ Dispose of empty containers, cleaning cloths and care products in an environmentally responsible manner.
- ▶ Observe the instructions for care products.

You can, for example, set the operating fluids for the vehicle in the multimedia system, under the [Status](#) menu, [Maintenance](#).

## Maintenance

### Opening/closing the maintenance flap

The maintenance system calculates maintenance due dates for the vehicle based on the vehicle's operating conditions.

You must secure the vehicle on jack stands of sufficient load capacity if work is being carried out underneath the vehicle.

When working on the vehicle, comply with all safety regulations, such as the operating instructions, regulations concerning hazardous materials, environmental protection measures, work safety and accident prevention regulations.

You can call up the maintenance due dates calculated for the vehicle and its major assemblies in the on-board computer.

The display message for the service appointments appears in the instrument cluster for the first time 14 days before they are due.

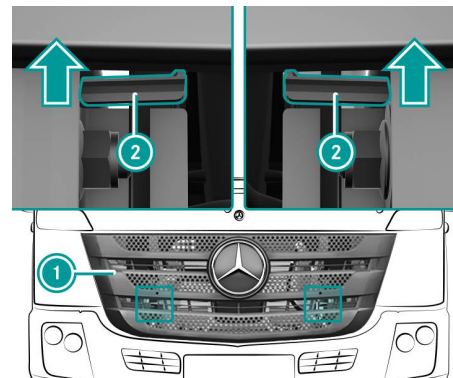
#### Note

A qualified workshop can program the first message to appear between 0 to 30 days before the due date.

If you switch on the vehicle and a service appointment is imminent or due, the service appointment will appear in a grey display message (→ Page 305).

If you miss a service appointment, the service appointment will appear in a yellow display message (→ Page 305).

### Opening/closing the maintenance flap



- ▶ **To open:** Push the release levers ② towards the rear and swing the maintenance flap ① upwards.
- ▶ **To close:** Swing maintenance flap ① downwards until it engages audibly.

### Check the coolant level



**Danger**

#### Risk of scalding from hot coolant

You may scald yourself if you open the cap when the drive system is at normal operating temperature.

- ▶ Let the vehicle cool down before opening the cap.
- ▶ When opening the cap, wear protective gloves and safety goggles.
- ▶ Open the cap slowly to release pressure.



#### NOTE

#### Damage to the major assemblies and the high-voltage battery due to low coolant level

If the vehicle is operated when the coolant level is too low, sufficient cooling output is not guaranteed. This can damage the major assemblies and the high-voltage battery.

- ▶ Check the coolant level regularly and top up the coolant if necessary.

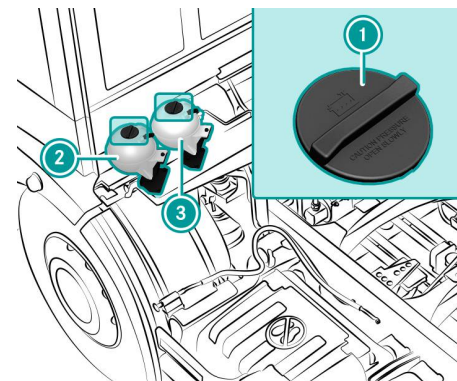
### Requirements

- The vehicle is parked on a level surface.
- The parking brake is applied.
- The vehicle is switched off.

Your vehicle is equipped with two coolant expansion reservoirs for the following coolant circuits:

- The drive system
- The high-voltage battery

If the drive cooling system or high-voltage battery cooling system has to be refilled, contact a qualified specialist workshop.



Coolant expansion reservoir (example)

- ① Lid
- ② Drive system
- ③ High-voltage battery

If the coolant level in the coolant expansion reservoir is too low, the following display messages may be shown:

- High-voltage battery: Yellow  (→ Page 303)
- Drive system: Yellow  (→ Page 304)

Only open the coolant expansion reservoir if the coolant temperature is below 50 °C. It is only possible to check whether

## Maintenance

### Refilling the windscreen washer system/headlamp cleaning system

the coolant level is correct if the coolant temperature is between 0 °C and 25 °C.

- ▶ Observe the notes on operating fluids ( → Page 289) and coolant ( → Page 290).
- ▶ Slowly turn the cap anticlockwise and allow the overpressure to escape.
- ▶ Unscrew and remove the cap.
- ▶ Check the coolant level.
  - ◁ The coolant expansion reservoir contains enough coolant if the coolant reaches the maximum mark.
- ▶ If necessary, top coolant up to the maximum mark.
- ▶ Put on the cap and tighten it as far as it will go.

#### Refilling the windscreen washer system/headlamp cleaning system



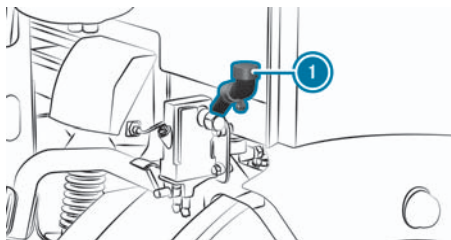
**Danger**

#### Risk of fire and injury from windscreen washer concentrate

Windscreen washer concentrate is highly flammable.

- ▶ Avoid fire, naked flames, smoking and the creation of sparks when using windscreen washer concentrate.

#### Topping up the washer fluid



The washer fluid reservoir for the windscreen washer system has a capacity of approximately 16 l.

If the washer fluid level in the washer fluid reservoir is too low, the instrument cluster

will display a grey message with the  symbol.

Add a concentrated windscreen washer concentrate to the washing water all year round in accordance with Mercedes-Benz Trucks Specifications for Operating Fluids. Adapt the mixing ratio to the outside temperature.

At temperatures above freezing, add a windscreen washer concentrate for the summer to prevent smearing. If there is a risk of frost, use a windscreen washer concentrate for winter to prevent the washer fluid from freezing on the windscreen.

- ▶ Observe the notes on operating fluids ( → Page 289).
- ▶ Pre-mix the washer fluid to the correct mixing ratio in a container.
- ▶ Unscrew and remove cap ① from the washer fluid reservoir.
- ▶ Top up the washer fluid reservoir .
- ▶ Replace cap ① and screw it on.

## Replacing the wiper blades

### Replacing the wiper blades



**Danger**

#### Risk of injury due to unsuitable climbing aid

When you use the steps and grab handles at the front of the vehicle, you may slip and/or fall.

- ▶ Always use nonslip and stable climbing aids, e.g. a suitable ladder.



**Danger**

#### Risk of becoming trapped if the windscreen wipers are switched on while wiper blades are being replaced

If the windscreen wipers begin to move while you are changing the wiper blades, you can be trapped by the wiper arm.

- ▶ Always switch off the windscreen wipers and the vehicle before replacing the wiper blades.

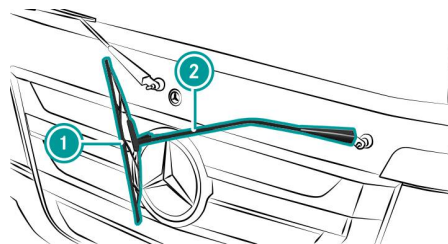


### NOTE

#### Damage to the maintenance flap or windscreen by the wiper arm

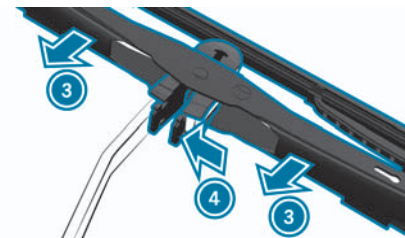
- ▶ If a wiper arm is folded forwards, do not open the maintenance flap.
- ▶ Do not fold the wiper arms against the windscreen without the wiper blades attached.

### Removing the wiper blades



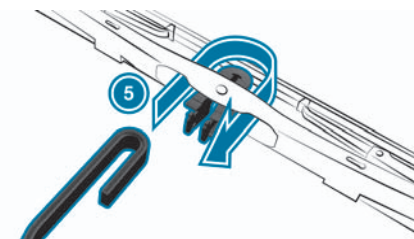
Wiper arm with wiper blade (example on the left-hand side of the vehicle)

- ▶ Fold wiper arm ② away from the windscreen.
- ▶ Place wiper blade ① in a perpendicular position.



- ▶ Push the safety spring in the direction of arrow ④ and slide wiper blade ① out of the bracket of wiper arm ② in the direction of arrow ③.
- ▶ Remove wiper blade ①.

### Fitting the wiper blades



- ▶ Slide the joint piece of the wiper blade ① into the bracket of wiper arm ② in the direction of arrow ⑤.

## Maintenance

### Notes on the major assemblies

- ▶ Press the wiper blade into the bracket of wiper arm ② until the safety spring engages audibly.
- ▶ Rotate wiper blade ① parallel to wiper arm ②.
- ▶ Fold wiper arm ② back onto the windshield.

Check the major assemblies for leak tightness regularly. If you note fluid loss, for example, coolant on the parking space, have the cause rectified immediately at a qualified specialist workshop.

and electrical contact points (earth contacts).

- ▶ Have any damage to the factory-fitted, anti-corrosion protection repaired at a qualified specialist workshop.

### Notes on the major assemblies



#### ENVIRONMENTAL NOTE

##### Pollution of the environment due to irresponsible disposal of operating fluids

Incorrectly disposing of operating fluids can cause considerable damage to the environment.

- ▶ Dispose of operating fluids in an environmentally responsible manner.

Operating fluids include:

- lubricants
- Coolant
- climate control system refrigerants

### Checking the anti-corrosion protection



#### NOTE

##### Damage to the anti-corrosion protection due to salt residue

Road salt is aggressive. You should wash the vehicle more often in the winter in order to remove road salt residue. Otherwise, this salt residue may damage the anti-corrosion protection.

All Mercedes-Benz Trucks cabs come treated with a cavity preservation.

- ▶ Check the vehicle regularly for corrosion damage, particularly the compressed-air lines, hydraulic lines

### Battery

#### Notes on the high-voltage battery



#### Danger

##### Risk of fire and explosion from excessive internal pressure in the high-voltage battery

In the event that the vehicle catches fire, flammable gas can escape and ignite.

- ▶ If you notice an unusual smell, smoke, or burn marks, immediately discontinue the charging process.
- ▶ Leave the danger zone immediately. Secure the danger zone at a suitable distance.
- ▶ Call the fire service.

Observe the notes on charging the high-voltage battery ( → Page 201).

|                                                                                   |                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Risk of explosion                                                                                                                                                                                                                                                               |
|  | Fire, naked flames and smoking are prohibited when handling the battery. Avoid creating sparks.                                                                                                                                                                                 |
|  | Electrolyte or battery acid is corrosive. Avoid contact with the skin, eyes and clothing. Wear suitable protective clothing, in particular gloves, an apron and a safety mask. Immediately rinse electrolyte acid splashes off with clean water. Consult a doctor if necessary. |
|  | Wear eye protection.                                                                                                                                                                                                                                                            |
|  | Keep out of the reach of children.                                                                                                                                                                                                                                              |
|  | Observe these Operating Instructions.                                                                                                                                                                                                                                           |

### Notes on the 24 V on-board electrical system battery



#### **Danger of chemical burns from battery acid**

Battery acid is caustic.

- ▶ Avoid contact with the skin, eyes, or clothing.
- ▶ Do not lean over the battery.
- ▶ Do not inhale battery gases.
- ▶ Keep children away from the battery.
- ▶ Immediately rinse battery acid off thoroughly with plenty of clean water and seek medical attention immediately.



#### **NOTE**

#### **Damage to the on-board electrical system batteries caused by receiving current directly**

Connecting consumers directly to the battery poles causes the battery sensors to make an incorrect determination.

This causes uncontrolled battery discharge, and both on-board electrical system batteries may be discharged.

- ▶ Do not connect consumers directly to the battery terminals.



#### **ENVIRONMENTAL NOTE**

#### **Environmental damage due to improper disposal of batteries**



Batteries contain pollutants.

It is illegal to dispose of them with the household rubbish.



Dispose of batteries of in an environmentally responsible manner. Take discharged batteries to a qualified workshop or to a collection point for used batteries.

## Maintenance

### Battery

**Observe the safety instructions and protective measures when handling the battery:**

|                                                                                   |                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Risk of explosion. Explosive oxyhydrogen gas is produced when batteries are being charged. Only charge batteries in a well-ventilated area.                                                              |
|  | Risk of explosion. Avoid creating sparks! Avoid fire, open flames and do not smoke when handling the battery.                                                                                            |
|  | Battery acid is caustic. Wear acid-resistant protective gloves! Splashes of acid on skin or clothing should be neutralised immediately using soapy water or acid neutraliser and then rinsed with water. |
|  | Wear eye protection. When mixing water and acid, the liquid may splash into your eyes. Rinse out any acid that splashes into eyes immediately using clean water and seek medical attention at once!      |
|  | Keep out of the reach of children. Children are not able to evaluate the risk involved in handling batteries and acid.                                                                                   |



Always observe the safety instructions, protective measures and procedures specified in these operating instructions when handling the on-board electrical system battery.

For safety reasons, Daimler Truck AG recommends that you only use batteries that have been approved for your vehicle by Daimler Truck AG.

#### General notes

Your vehicle is equipped with two 12 V on-board electrical system batteries connected in series. This results in an on-board electrical system voltage of 24 V.

In this section you will find notes and information about the on-board electrical system batteries. They will help you to ensure that the on-board electrical system batteries are charged and ready for use.

Observe the following points regarding battery capacity and performance:

- Battery capacity is limited and the operating time depends on the number of electrical consumers that are switched on and the duration used.

- The battery capacity depends on the following factors:
  - the age of the on-board electrical system battery
  - the outside temperature
- The on-board electrical system batteries are supplied with voltage exclusively from the high-voltage batteries and are therefore dependent on their charge.
- Intensive use of the on-board electrical system batteries has a direct effect on the vehicle's range.
- To be sure that you are always able to start your vehicle, observe the notes and instructions in the display messages for the on-board electrical system battery ( → Page 313).
- If the display message [Energy-saving mode active](#) appears in the instrument cluster, certain consumers, such as the audio equipment, are switched off ( → Page 311).



**Note**

Completely disconnect electrical consumers from the on-board electrical system when they are not needed.

If you park up your vehicle for longer than three weeks, observe the notes on parking up the vehicle (→ Page 234).

When storing batteries, comply with all safety regulations, such as operating instructions, regulations concerning hazardous materials, environmental protection measures, and work safety and accident prevention regulations.

Observe the instructions on care and maintenance of the on-board electrical system batteries (→ Page 231).

### Information on maintenance and care of the 24 V on-board electrical system battery

#### Parking the vehicle for an extended time period and storage

**NOTE**

#### Damage to the 24 V on-board electrical system battery due to deep discharge during long downtimes

- At least once a month, switch on the vehicle (high-voltage on-board electrical system) for approx. two hours to prevent a deep discharge of the 24 V on-board electrical system battery.

If you park up your vehicle for longer than three weeks, observe the notes on parking up the vehicle (→ Page 234).

When storing batteries, comply with all safety regulations, such as operating instructions, regulations concerning hazardous materials, environmental protection measures, and work safety and accident prevention regulations.

### Battery replacement and operating life

Have the on-board electrical system batteries replaced every three years at the latest.

Only have the on-board electrical system batteries replaced at a qualified specialist workshop, e.g. at an authorised Mercedes-Benz Trucks Service Centre.

Avoid fully discharging the on-board electrical system batteries. This can significantly reduce the operating life of the battery.

The on-board electrical system batteries can achieve a long service life if you keep the batteries adequately charged.

Observe the following notes for vehicles with AGM batteries (fleece battery):

- The on-board electrical system batteries are maintenance-free.
- They do not need topping up with distilled water.

## Maintenance

### Battery

#### Battery care



##### NOTE

#### Battery discharge due to leakage currents

Dirty battery terminals and battery surfaces cause leakage currents. This can result in the on-board electrical system battery discharging.

- ▶ Keep the battery terminals and battery surfaces clean and dry.



##### NOTE

#### Damage to the battery housing due to improper cleaning

If you use cleaning agents that contain fuel, they will attack the battery housing.

- ▶ Do not use cleaning agents that contain fuel.

Only use commercially-available cleaning agents to clean the battery housing.

#### Activating transport mode

##### Activating/deactivating

Transport mode includes a deactivation function for various electrical consumers when the vehicle is switched off, which protects the batteries from discharging when vehicles are parked up for longer periods of time.

In transport mode, all functions which are also deactivated in energy-saving mode are deactivated.

Additionally, the following functions are deactivated in transport mode:

- the work lamps
- the cargo liftgate
- charging management for recuperation
- the dipped beam (when the vehicle has not been started)
- the vehicle (after 15 minutes if no diagnostic device is connected)
- ▶ Press the hazard warning lights switch, the headlamp flasher and turn signal indicator simultaneously for

at least three seconds within three seconds of switching the vehicle on.

- ▶ **To activate:** Move the turn signal indicator down.
- ▶ **To deactivate:** Move the turn signal indicator up.

Active transport mode will be shown as a display message when the vehicle is switched on again.



##### Note

The use of transport mode is stored in the vehicle along with the kilometre reading and time of activation/deactivation.

#### Charging the 24 V on-board electrical system battery

Recharge the on-board electrical system battery regularly if the vehicle is parked up for longer periods of time. This ensures that the vehicle can be started.

- ▶ Observe the notes and information on the display messages for the

on-board electrical system battery (→ Page 313).

- If necessary, charge the on-board electrical system battery using the external charger connection point.



**Danger**

### Risk of explosion due to an explosive mixture of gas

An explosive mixture of gas may escape from the battery during the charging process and when providing starting assistance.

- Avoid fire, naked flames, sparks and smoking.
- Ensure sufficient ventilation.
- Do not lean over a battery.



**Danger**

### Risk of explosion when the battery is frozen

A discharged battery can freeze at temperatures near the freezing point.

If you then attempt to charge the battery, battery gas can escape.

- Always thaw a frozen battery before charging it.

If the warning and indicator lamps do not light up in the instrument cluster at temperatures around or below freezing, it is highly probable that the discharged on-board electrical system battery has frozen. In this case, do not charge the on-board electrical system battery.

The service life of an on-board electrical system battery that has been thawed may be reduced drastically.

Have a thawed on-board electrical system battery checked at a qualified specialist workshop.

Use a commercially available charger to charge the vehicle electrical system batteries. Make sure that the charging voltage is correct. Do not charge new on-board electrical system batteries with rapid charging.

For new on-board electrical system batteries, the charging current must not exceed

10% of the battery capacity. A higher charging current can damage the on-board electrical system batteries.

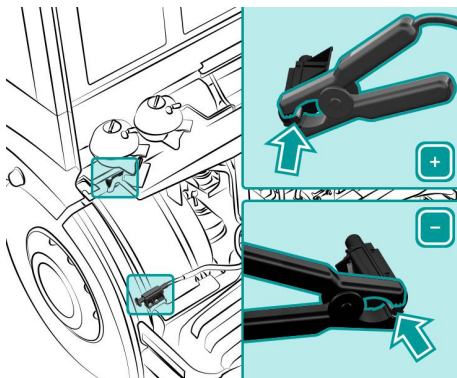


Briefly exceeding the maximum charging current of 30 A (24 V), e.g. when bridging deeply discharged batteries, is permitted for a maximum of two minutes.

Charge the on-board electrical system batteries at an ambient temperature of above 0 °C. The optimum ambient temperature is between 10 °C and 25 °C.

## Maintenance

### Battery



External charging connection point on the front left wheel arch

- ▶ Set the charging voltage to 24V. See the charger's operating instructions.
- ▶ Gently turn the black protective cap of the external charging connection point clockwise with the negative terminal clamp of the charger, slide it back and connect the negative terminal clamp to the  $\ominus$  negative terminal.
- ▶ Gently turn the red protective cap of the external charging connection point clockwise with the positive terminal clamp of the charger, slide it

back and connect the positive terminal clamp to the  $\oplus$  positive terminal.

- ▶ Switch on the battery charger. See the charger's operating instructions.

When the vehicle electrical system batteries are charged:

- ▶ Switch off the charger. See the charger's operating instructions.
- ▶ Disconnect the negative terminal clamp and the positive terminal clamp of the charger from the  $\ominus$  negative terminal and the  $\oplus$  positive terminal of the external charger connection point.
  - ◁ The cover caps of the external charger connection point spring back to their original position.

#### **i** Note

After charging/replacing the on-board electrical system batteries, the battery charge level calculated may not be correct. It takes approximately three days while the vehicle is in operation for the teach-in process to be completed. Display messages, such as, for example,

**Energy-saving mode active**, are only displayed correctly once the teach-in process has been completed.

#### Notes on parking up the vehicle

When parking up the vehicle, special measures according to the Mercedes-Benz Trucks Specifications for Operating Fluids need to be taken.

You can obtain further information in the notes on operating fluids (→ Page 289) or from an authorised Mercedes-Benz Trucks Service Centre.

### Function of telediagnosics

With telediagnosics, the Mercedes-Benz Trucks Service24h responder can call up detailed technical information for your vehicle. This way, the fault can be quickly identified and, depending on the situation, you can continue your journey. The requirement is an active telematics unit from Daimler Truck AG.

The following data can be called up by Mercedes-Benz Trucks Service24h:

- data necessary for identification of the vehicle (e.g. vehicle identification number/VIN)
- control unit diagnostic data (e.g. fault memory)
- current vehicle location data
- current total distance

No data related to the driver, the route or the load are transmitted.

### Vehicle tool kits

#### Overview of vehicle tool kits and emergency equipment

When working on the vehicle, comply with all safety regulations, such as the operating instructions, regulations concerning hazardous materials, environmental protection measures, work safety and accident prevention regulations.

Observe the notes on "Changing a wheel in the event of a flat tyre" (→ Page 240).

The vehicle tool kit and the emergency equipment are stowed in the vehicle as follows:

| Vehicle tool kit or emergency equipment | Storage location                                 |
|-----------------------------------------|--------------------------------------------------|
| Support rod                             | Stowage compartment behind the co-driver's seats |
| Drive-on wedge                          | Stowage space under the outer co-driver's seat   |

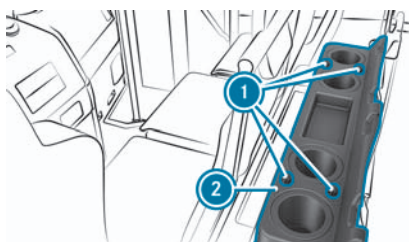
| Vehicle tool kit or emergency equipment              | Storage location                                 |
|------------------------------------------------------|--------------------------------------------------|
| Vehicle tool bags                                    | Stowage compartment behind the co-driver's seats |
| Hand wiper                                           | Stowage space behind the co-driver's seats       |
| Hand lamp                                            | Stowage compartment above the co-driver's seats  |
| Pump lever                                           | Stowage compartment behind the co-driver's seats |
| Tyre inflator hose                                   | Stowage compartment above the co-driver's seats  |
| Support block                                        | Stowage space under the outer co-driver's seat   |
| First-aid kit (hard case)/first-aid kit (soft sided) | Under the centre co-driver's seat                |
| Jack                                                 | On the chassis                                   |
| Vehicle document wallet                              | Driver's door stowage space                      |

## Breakdown assistance

### Vehicle tool kits

| Vehicle tool kit or emergency equipment | Storage location                                |
|-----------------------------------------|-------------------------------------------------|
| Warning triangle                        | Stowage space under the outer co-driver's seat  |
| Warning beacon                          | Stowage compartment above the co-driver's seats |
| High visibility vest                    | Stowage compartment above the co-driver's seats |
| Fire extinguisher                       | Under the outer co-driver's seat                |

#### Tool kit compartment behind the co-driver's seats



To access the vehicle tool kit or the emergency equipment behind the front passen-

ger's seats, unscrew and remove screws ① and remove cover ② from above.

#### Using the jack, pump lever and chock

##### Jack

##### **i** Note

Depending on the equipment, the jack has a maximum weight of 14,3 kg.

The maximum load capacity of the jack can be found on the sticker affixed to the jack.

If there is a malfunction, please contact a qualified workshop.

##### Maintaining the jack

- **After use:** Clean all moving parts and grease them again.
- **Every six months:** Fully extend and retract the reciprocating piston.

#### Pump lever (two-part)

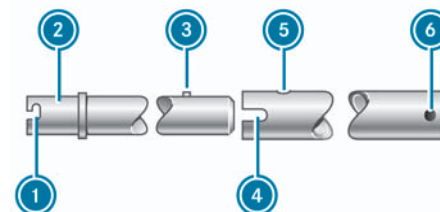


**Danger**

#### Risk of injury due to incorrect assembly of pump lever

During pumping, the lever can slide out of the guide.

- Make sure that the safety pin of the pump lever is engaged in the specified bore.



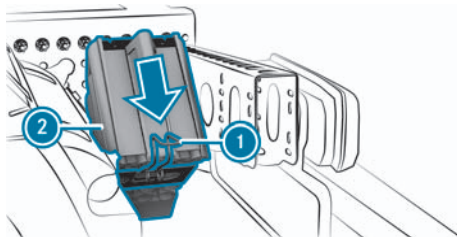
- ① Recess (jack)
- ② Jacking point (wheel spanner/cab tilting pump)
- ③ Locking pin
- ④ Recess (for operating the spare wheel winch and jack)
- ⑤ Hole for locking pin

- ⑥ Clamping sleeve for inserting the hand crank (spare wheel winch)

Use the pump lever to operate the following devices:

- Jack
- Wheel spanner
- Spare wheel winch
- Cab tilting pump
- ▶ **To assemble the pump lever:** Align and insert locking pin ③ into hole ⑤ of both pump lever parts.
  - ◀ Locking pin ③ will engage in hole ⑤.
- ▶ **To disassemble the pump lever:** Push locking pin ③ and pull apart the pump lever.

### Removing the chock



The storage location of the chocks may vary depending on the vehicle version and equipment.

- ▶ Press retainer ① in the direction of the arrow and hold it there.
- ▶ Pull out wheel chock ②.
- ▶ Release retainer ①.

### Inserting the chock

- ▶ Press retainer ① in the direction of the arrow and hold it there.
- ▶ Insert wheel chock ② into the bracket.
- ▶ Release retainer ①.

### Cab

### Tilting the cab

#### Before tilting the cab



**Warning**  
Risk of fatal injuries when carrying out maintenance work during the charging process

During the charging process, the vehicle's high-voltage on-board electrical system is under high voltage.

- ▶ Do not perform any maintenance work during the charging process.



### Risk of injury when tilting the cab

The cab may abruptly fall into the end position.

There is a risk of injury to persons in the area where the cab is tilted.

- ▶ Only tilt the cab if there is nobody in the tilting area.
- ▶ Do not go under the cab area when the cab is being tilted.



### Risk of injury due to moving parts

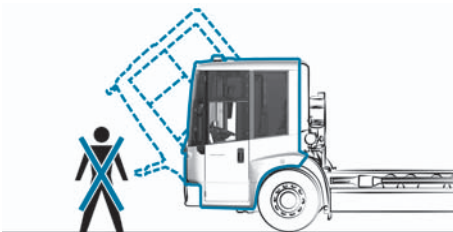
Components in the engine compartment can continue to run or start unexpectedly even when the vehicle is switched off.

## Breakdown assistance

### Cab

Observe the following before performing tasks in the engine compartment:

- ▶ Switch off the vehicle.
- ▶ Never reach into the danger zones surrounding moving components, e.g. the rotation area of the fan.
- ▶ Remove jewellery and watches.
- ▶ Keep items of clothing and hair away from moving parts.

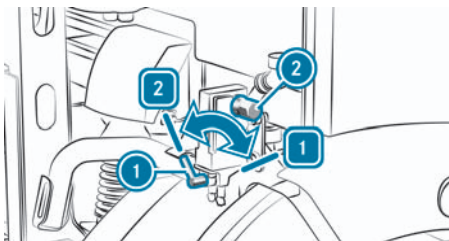


Perform the following before tilting:

- ▶ For safety reasons, keep the area in front of the cab clear.
- ▶ Apply the parking brake.
- ▶ Shift the transmission to neutral.
- ▶ Switch off the vehicle.
- ▶ Remove loose objects from the cab.
- ▶ Close the doors.

- ▶ Use chocks to protect the vehicle against rolling away.

#### Tilting the cab forwards (mechanical-hydraulic cab tilt system)



Cab tilting pump on right (example)

- 1 Tilting position
- 2 Drive position
- 1 Valve lever
- 2 Hexagon nut

Do not step on major assemblies or electrical components when the cab is tilted.

- ▶ Read the notes before tilting the cab.
- ▶ Open the maintenance flap (→ Page 224).
- ▶ Swing the valve lever 1 on the tilting pump into the tilting position 1.
- ▶ Fit the pump lever to the hexagon nut 2 on the tilting pump using the wheel spanner (vehicle tool kit).
- ▶ Move the pump lever up and down on the tilting pump until the cab tilts to the forward end position.
  - ◁ The cab will be unlocked automatically.
- ▶ If there is noticeable resistance when you operate the pump lever, check that the tilting position 1 has been set correctly on valve lever 1 on the tilting pump.
- ▶ When the cab is tilted to the end position, secure the cab with the support rod.
- ▶ If there is no noticeable resistance when you operate the pump lever, have the tilting hydraulics checked at a qualified workshop.



### Tilting the cab back into the drive position (mechanical-hydraulic cab tilt system)



**Danger**

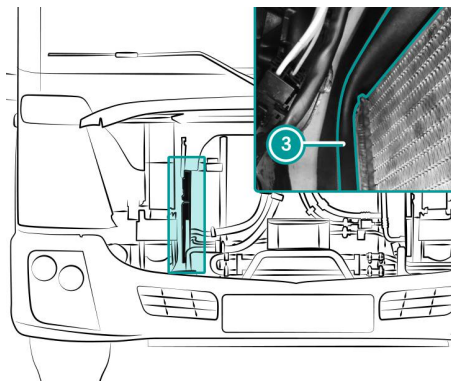
**Risk of accident and injury if the cab is not locked while the vehicle is in motion**

When the vehicle decelerates, the following risks, in particular, may arise:

- The cab may tilt forward.
- You may lose control of the vehicle.
- Persons in the cab could be thrown forwards.
- Persons or objects in the range of movement could be hit.

Ensure the following before every journey:

- The cab is locked.
- The cab is engaged in the drive position and the valve lever is in the drive position.
- The indicator lamp goes out when the vehicle is started.



Front right sealing lip (example)

Before tilting back the cab, ensure that the sealing lip ③ has been reinserted correctly. The sealing lip ③ must be flush and must not be twisted.

- ▶ Remove the support rod.
- ▶ Swing the valve lever ① on the tilting pump into the drive position ②.
- ▶ Fit the pump lever to the hexagon nut ② on the tilting pump using the wheel spanner.
- ▶ Move the pump lever up and down on the tilting pump until the cab is tilted back into the rear end position.

- ◁ The catch will engage audibly and the cab will lock automatically.
- ▶ Close the maintenance flap.
- ▶ Check whether the indicator lamp  on the instrument cluster goes out after you start the vehicle.
- ◁ If the indicator lamp  does not go out, repeat the process and tilt the cab back again.

### Problems with tilting the cab

#### Mechanical-hydraulic cab tilt system

#### The cab cannot be tilted.

Possible causes:

- The valve lever of the mechanical-hydraulic cab tilting pump is in the "Drive position" valve lever position.
- The tilting hydraulics are leaking or have failed.
- ▶ Turn the valve lever on the mechanical-hydraulic cab tilting pump to the "Tilting" valve lever position (→ Page 237).

## Breakdown assistance

### Flat tyre

- Have the tilting hydraulics repaired at a qualified workshop.

#### Flat tyre

##### Preparing for a wheel change



#### **Danger**

##### **Risk of injury through incorrect positioning of the jack**

If you do not position the jack correctly at the appropriate jacking point of the vehicle, the jack could tip with the vehicle raised.

- Only position the jack at the appropriate jacking point on the vehicle. The base of the jack must be positioned vertically under the jacking point of the vehicle.



#### **Danger**

##### **Risk of injury due to the vehicle tipping over**

On uphill and downhill slopes, the jack may tip over with the vehicle raised.

- Never change wheels on uphill or downhill slopes.
- Speak with a qualified workshop.



#### **Danger**

##### **Risk of injury due to the jack tipping over**

If you park a vehicle with air suspension, the air suspension can remain active for up to an hour, even if the vehicle is switched off. If you then raise the vehicle using the jack, the air suspension will attempt to even out the vehicle level.

The jack can tip over.

- Before raising the vehicle, press the Service button on the remote control for the air suspension.
  - ◄ This prevents the vehicle level from being automatically re-adjusted or manually raised or lowered.



#### **Danger**

##### **Risk of injury due to the use of a mount when changing a wheel**

If, when changing the outer wheel on twin tyres, you use a mount for driving with the inner wheel, the inner wheel will be under load. It will then not be possible to correctly press and tighten the wheel against its contact surface.

As a result, you can lose a wheel while driving.

- Always use the jack when changing a wheel.



#### **Danger**

##### **Risk of accident due to losing a wheel**

Oiled, greased or damaged wheel nut/wheel bolt threads or spherical spring washers can cause wheel nuts to come loose.

- Never oil or grease the threads or conical spring washers.

- ▶ If a thread or conical spring washer is damaged, consult a qualified workshop immediately.
- ▶ Have damaged wheel nuts, wheel bolts and conical spring washers replaced.
- ▶ Do not continue driving.



**Danger**

### Risk of injury when loosening a wheel

If you loosen a wheel which sits on the wheel bolts under tension, it may drop down or tip over.

- ▶ Only remove the last three wheel nuts after making sure that the wheel sits on the wheel bolts without tension.

Observe the following when changing a wheel on the vehicle:

- Only use wheel nuts that are approved for the vehicle.
- The wheel nuts for steel and light-alloy wheels are different.

- The wheel nuts for light-alloy wheels on the front and rear axles are different.

Observe the following notes when raising the vehicle:

- Make sure that the maximum load capacity for the jack is not exceeded. The maximum load capacity of the jack is stated on the adhesive label affixed to the jack.
- The jack is designed only to raise the vehicle for a short time, e.g. while a wheel is being changed. It is not suitable for raising the vehicle in order to perform work under the vehicle.
- Only position the jack at the appropriate jacking point on the vehicle. Make sure that the jack is correctly positioned on the jacking point before raising the vehicle.
- Secure the vehicle before raising it to prevent it from rolling away, for example, by applying the parking brake and/or using chocks. Do not release the parking brake while the vehicle is raised.

- The surface on which the jack is standing must be firm and level. The jack must be placed on an underlay if the ground surface is not firm.
- Ensure that the distance between the underside of the vehicle and the ground is no more than 30 mm. The vehicle could otherwise slip off the jack or tip over.
- Do not change a wheel on an uphill or downhill slope. The vehicle could otherwise slip off the jack.
- Do not place your hands or feet under the raised vehicle.
- Do not lie under the raised vehicle.
- Make sure that nobody is in the vehicle when it is raised.
- Do not start the vehicle when it is raised, and avoid otherwise jolting or shaking it. The vehicle could otherwise slip off the jack.
- ▶ Park the vehicle on a firm and level surface.
- ▶ Apply the parking brake.
- ▶ Use chocks to protect the vehicle against rolling away.

## Breakdown assistance

### Flat tyre

#### Note

For further information on tyre pressure, see the "Wheels and tyres" section (→ Page 264).

#### Spare wheel

You can find information on the location of the spare wheel and notes on removing the spare wheel in the body manufacturer's operating instructions.

#### Changing a wheel

##### Positioning the jack

- **Flat tyre, left:** Turn the steering wheel to the left as far it will go.

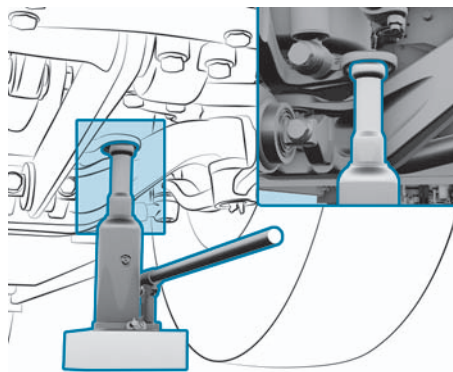
or

- **Flat tyre, right:** Turn the steering wheel to the right as far it will go.

#### Note

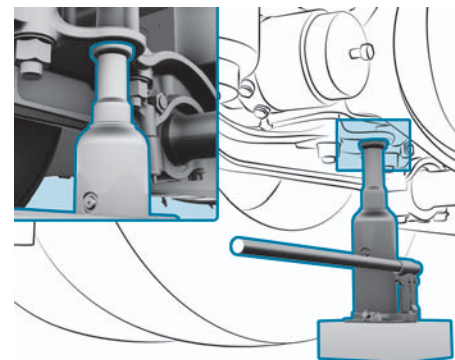
Only use the support block from the vehicle tool kit as a jack support.

##### Air-sprung front axle



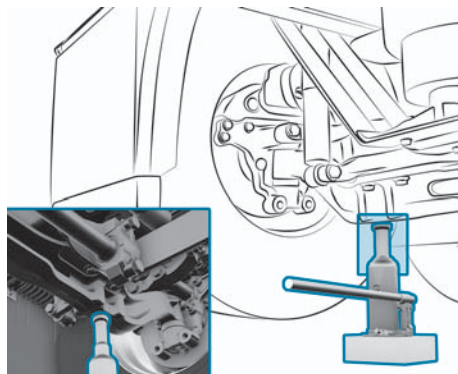
- Position the jack under the jacking point.

##### Air-sprung rear axle



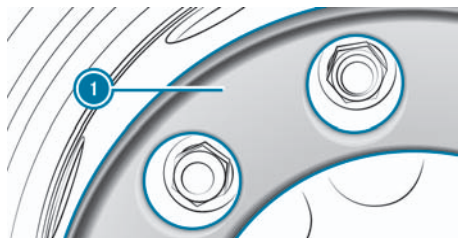
- Position the jack under the jacking point.

### Steered trailing axle



- Position the jack so that the jack plunger is centrally located under the axle carrier.

### Removing a wheel



- Unscrew the wheel nuts that secure wheel nut cover ①.
- Remove wheel nut cover ①.
- Remove the remaining wheel nuts.
- Remove the wheel.
- Vehicles with twin tyres: Unscrew the wheel nuts of the inner wheel on the connecting flange and remove the connecting flange.
- Vehicles with twin tyres: Remove inner wheel.

### Fitting a wheel

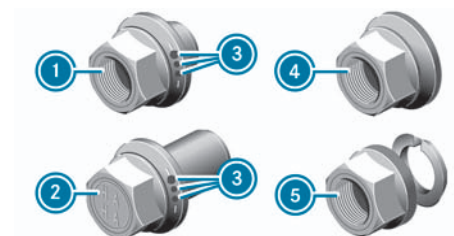


#### NOTE

#### Damage to the wheel nuts or wheel bolts due to the use of an impact wrench

If you use an impact wrench to tighten the wheel nuts or wheel bolts, you may damage them.

- Do not use an impact wrench to tighten the wheel nuts or wheel bolts.



Wheel nuts

- ① For single tyres with light-alloy wheels
- ② For twin tyres with light-alloy wheels
- ③ Wheel nut identification for light-alloy wheels
- ④ With pressure plate (hub centring) for steel wheels
- ⑤ With conical spring washer for steel wheels

### Before fitting a wheel

- Remove any rust and dirt from the contact areas of the wheel hub, disc wheel and wheel nuts.
- Lightly oil the friction contact surfaces between the pressure plate and the wheel nut.

### Flat tyre

#### Fitting a steel wheel

- ▶ Vehicles with single tyres: Fit the wheel in place and screw on two to three wheel nuts.
- ▶ Vehicles with single tyres: Screw on the remaining wheel nuts with the wheel nut cover.
- ▶ Vehicles with twin tyres: Fit both the wheels and screw on all wheel nuts.
- ▶ Tighten the wheel nuts in a crosswise pattern, observing the tightening torque while doing so (→ Page 294).
- ▶ Check the tyre pressure (→ Page 264).
- ▶ Retighten the wheel nuts after driving 50 km.



Assembly sleeve for twin tyres (vehicles with light-alloy wheels)

#### Fitting a light-alloy wheel

- ▶ Vehicles with single tyres: Fit the wheel in place and screw on two to three wheel nuts.
- ▶ Vehicles with single tyres: Screw on the remaining wheel nuts with the wheel nut cover.
- ▶ Vehicles with twin tyres: Place the assembly sleeve (vehicle tool kit) on the wheel bolt before fitting the inner wheel.
- ▶ Vehicles with twin tyres: Fit both the wheels and screw on two or three wheel nuts.
- ▶ Vehicles with twin tyres: Pull off the mounting sleeve.
- ▶ Vehicles with twin tyres: Screw on the remaining wheel nuts.
- ▶ Tighten the wheel nuts in a crosswise pattern, observing the tightening torque while doing so (→ Page 294).
- ▶ Check the tyre pressure (→ Page 264).
- ▶ Retighten the wheel nuts after driving 50 km.

#### Retightening the wheel nuts



**Danger**

#### **Risk of accident due to incorrect tightening torque**

The wheels may come loose if the wheel bolts or wheel nuts are not tightened to the prescribed torque.

- ▶ Ensure that the wheel bolts or wheel nuts are tightened to the prescribed tightening torque.
- ▶ If necessary, consult a qualified workshop and do not move the vehicle.
- ▶ Observe the instructions and safety instructions on "Changing a wheel in the event of a flat tyre" (→ Page 240).
- ▶ Observe the wheel nut tightening torques (→ Page 294).
- ▶ Have all of the wheels retightened after driving 50 km.
- ▶ For new or newly painted wheel rims, have the tightening torque of the wheel nuts checked after driving approx. 1000 km to 5000 km.

- Retighten the wheel nuts in a cross-wise pattern.

On light-alloy wheels, the wheel nuts are not flush with the wheel studs when tightened.

### Electrical fuses

#### Remove the main fuse carrier cover

#### Notes



**Danger**

#### Risk of accident and injury caused by overloaded lines

If you manipulate or bridge a faulty fuse or if you replace it with a fuse with a higher amperage, the electric lines could be overloaded.

This could result in a fire.

- Always replace faulty fuses with the specified new fuses having the correct amperage.

The fuse box is located in the electrical equipment compartment behind the driver's seat.

The individual electric circuits are protected by safety fuses or automatic circuit breakers.

Blown fuses or defective automatic circuit breakers must be replaced with equivalent fuses with the fuse ratings recommended in the fuse assignment diagram. Fuses with the same fuse rating are the same colour.

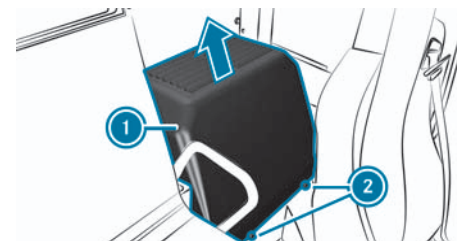
You can obtain further information from an authorised Mercedes-Benz Trucks Service Centre.

The fuse assignment diagram is on the inside of the main fuse carrier cover.

If the newly inserted fuse also blows, have the cause traced and rectified at a qualified specialist workshop.

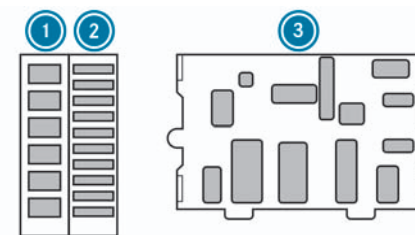
- If an electric circuit fails, switch off the consumer and the vehicle.

#### Remove the electrical equipment compartment cover



- Loosen screws ②.
- Pull the electrical equipment compartment cover ① up and out.

#### Fuse assignment diagram



Main fuse carrier (example)

- ① Fuses in module FA1
- ② Relays in module A31

## Breakdown assistance

### Electrical fuses

- ③ Fuses, relays and diodes in the PDM cabin module

#### Fuses in module FA1

| No. | Symbol                                                                            | Consumer                                             | Fuse rating |
|-----|-----------------------------------------------------------------------------------|------------------------------------------------------|-------------|
| F1  |                                                                                   | 12 V multimedia system                               | 10 A        |
| F2  |  | 12 V multimedia box                                  | 10 A        |
| F3  |  | Body manufacturer, 12 V terminal 30c                 | 5 A         |
| F4  |  | Top windscreen heater                                | 20 A        |
| F5  |  | Bottom windscreen heater                             | 20 A        |
| F6  |  | Drive control (eCPC 12 V)                            | 15 A        |
| F7  |  | ACH inverter 1 and 2/terminal 30c                    | 5 A         |
| F8  |                                                                                   | High-voltage on-board electrical system deactivation | 5 A         |

| No. | Symbol | Consumer           | Fuse rating |
|-----|--------|--------------------|-------------|
| F9  |        | AB12 crash sensing | 5 A         |
| F10 |        | Unassigned         |             |
| F11 |        | Unassigned         |             |
| F12 |        | Unassigned         |             |
| F13 |        | Unassigned         |             |
| F14 |        | Unassigned         |             |

#### Relays in module A31

| No. | Symbol                                                                             | Consumer                               |
|-----|------------------------------------------------------------------------------------|----------------------------------------|
| K1  |   | Rotating beacon                        |
| K2  |   | Top windscreen heater                  |
| K3  |   | Bottom windscreen heater               |
| K4  |   | Reversing assistant (ESA-R)            |
| K5  |   | Surround lighting                      |
| K6  |                                                                                    | 12 V terminal 15 for eCPC (24 V relay) |
| K7  |   | Unit enable (12 V relay)               |
| K8  |  | Folding door                           |
| K9  |                                                                                    | Cab water pump (12 V relay)            |

| No. | Symbol | Consumer                        |
|-----|--------|---------------------------------|
| K10 |        | AB12 crash sensing (12 V relay) |

#### Fuses in the PDM cabin module

| No. | Symbol                                                                               | Consumer                                      | Fuse rating |
|-----|--------------------------------------------------------------------------------------|-----------------------------------------------|-------------|
| F1  |   | Electric power-steering pump 2/3, terminal 15 | 10 A        |
| F2  |                                                                                      | Unassigned                                    |             |
| F3  |   | Body manufacturer, terminal 15                | 25 A        |
| F4  |                                                                                      | Unassigned                                    |             |
| F5  |   | Transmission control (eTCM), terminal 15      | 5 A         |
| F6  |   | Reversing assistant (ESA-R), terminal 15      | 10 A        |
|     |                                                                                      | Test box, terminal 15                         | 10 A        |
|     |  | Drive control (eCPC), terminal 15             | 10 A        |



| No. | Symbol                                                                            | Consumer                                                                    | Fuse rating |
|-----|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------|
| F7  |  | Acoustic vehicle alerting system (AVAS), terminal 15                        | 10 A        |
|     |  | Electric power-steering pump 4, terminal 15                                 | 10 A        |
| F8  |  | Fleet management system (FMS), terminal 15                                  | 10 A        |
|     |  | Brake system (EBS), terminal 15                                             | 10 A        |
|     |  | Toll Collect, terminal 15                                                   | 10 A        |
|     |  | Pre-installation performance-related heavy vehicle fee (LS-VA), terminal 15 | 10 A        |
|     |                                                                                   | TCO, terminal 15                                                            | 10 A        |
|     |                                                                                   | DC/DC converter, terminal 15                                                | 10 A        |
| F9  |                                                                                   | Unassigned                                                                  |             |

| No. | Symbol                                                                            | Consumer                                 | Fuse rating |
|-----|-----------------------------------------------------------------------------------|------------------------------------------|-------------|
| F10 |  | Mirror heater, terminal D+               | 15 A        |
| F11 |  | Work/surround lighting, terminal D+      | 15 A        |
| F12 |                                                                                   | Unassigned                               |             |
| F13 |                                                                                   | Unassigned                               |             |
| F14 |  | Voltage converter (T3)                   | 20 A        |
| F15 |  | Transmission control (eTCM), terminal 30 | 20 A        |
| F16 |  | Rotating beacon, terminal 30             | 15 A        |
| F17 |                                                                                   | Unassigned                               |             |
| F18 |  | Radar sensor (VRDU), terminal 30         | 10 A        |
|     |  | Reversing assistant (ESA-R), terminal 30 | 10 A        |
| F19 |                                                                                   | Unassigned                               |             |

| No. | Symbol                                                                              | Consumer                                                                    | Fuse rating |
|-----|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------|
| F20 |                                                                                     | Cab water pumps, terminal 30                                                | 25 A        |
|     |  | Body manufacturer (ABH), terminal 30                                        | 25 A        |
| F21 |  | Pre-installation performance-related heavy vehicle fee (LS-VA), terminal 30 | 10 A        |
| F22 |  | Electronic air processing unit (EAPU), terminal 30                          | 20 A        |
| F23 |                                                                                     | Unassigned                                                                  |             |
| F24 |  | Voltage converter (T2), terminal 30                                         | 20 A        |
| F25 |  | Voltage converter (T1), terminal 30                                         | 20 A        |
| F26 |                                                                                     | Test box, terminal 30                                                       | 10 A        |
| F27 |                                                                                     | Unassigned                                                                  |             |
| F28 |                                                                                     | Unassigned                                                                  |             |

## Breakdown assistance

### Electrical fuses

| No. | Symbol                                                                            | Consumer                                                                    | Fuse rating |
|-----|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------|
| F29 |  | Brake system (EBS), terminal 30                                             | 20 A        |
| F30 |  | Toll Collect, terminal 30                                                   | 10 A        |
|     |  | Pre-installation performance-related heavy vehicle fee (LS-VA), terminal 30 | 10 A        |
|     |  | Tachograph, terminal 30                                                     | 10 A        |
|     |  | Multi-purpose camera (MPC), terminal 30                                     | 10 A        |
| F31 |  | Common telematics platform (CTP), terminal 30                               | 10 A        |
|     |  | Fleet management system (FMS), terminal 30                                  | 10 A        |

| No. | Symbol                                                                            | Consumer                                                | Fuse rating |
|-----|-----------------------------------------------------------------------------------|---------------------------------------------------------|-------------|
|     |  | Video radar decision unit (VRDU), terminal 30           | 10 A        |
| F32 |                                                                                   | Unassigned                                              |             |
| F33 |  | Battery temperature warning system (BTWS), terminal 30  | 5 A         |
| F34 |  | Diagnostics connection, terminal 30                     | 10 A        |
| F35 |                                                                                   | Heating/air conditioning (HVAC)                         | 10 A        |
| F36 |  | Steering wheel buttons, terminal 30                     | 10 A        |
|     |  | Parking brake lever, terminal 30                        | 10 A        |
|     |                                                                                   | Breathalyser, terminal 30                               | 10 A        |
| F37 |                                                                                   | Heating/air conditioning (HVAC), black box, terminal 30 | 10 A        |

| No. | Symbol                                                                                | Consumer                                               | Fuse rating |
|-----|---------------------------------------------------------------------------------------|--------------------------------------------------------|-------------|
| F38 |    | Electrical immobiliser control unit (EIS), terminal 30 | 15 A        |
|     |    | Transmission control (eTCM), terminal 30               | 15 A        |
| F39 |                                                                                       | Level/roll control (CLCS), terminal 30                 | 10 A        |
| F40 |                                                                                       | Unassigned                                             |             |
| F41 |                                                                                       | Unassigned                                             |             |
| F42 |    | Driver's door module, terminal 30                      | 15 A        |
| F43 |    | Co-driver's door module, terminal 30                   | 15 A        |
|     |                                                                                       | TKS lever, terminal 30                                 | 15 A        |
| F44 |    | Cigarette lighter, terminal 30                         | 25 A        |
|     |  | Socket, terminal 30                                    | 25 A        |
| F45 |                                                                                       | Unassigned                                             |             |

| No. | Symbol                                                                            | Consumer                                       | Fuse rating |
|-----|-----------------------------------------------------------------------------------|------------------------------------------------|-------------|
| F46 |  | Seat heating, terminal 15                      | 15 A        |
| F47 |                                                                                   | Unassigned                                     |             |
| F48 |  | Central gateway, terminal 30                   | 5 A         |
|     |  | Steering angle sensor, terminal 30             | 5 A         |
| F49 |                                                                                   | Unassigned                                     |             |
| F50 |  | Brake system (EBS), terminal 30                | 10 A        |
| F51 |                                                                                   | Unassigned                                     |             |
| F52 |                                                                                   | Remote pin voltage converter (T1), terminal 15 | 10 A        |
|     |  | Blind spot camera system, terminal 15          | 10 A        |
|     |  | Body manufacturer (ABH), terminal 15           | 10 A        |

| No. | Symbol                                                                            | Consumer                                           | Fuse rating |
|-----|-----------------------------------------------------------------------------------|----------------------------------------------------|-------------|
| F53 |  | Video radar decision unit (VRDU), terminal 15      | 10 A        |
| F54 |                                                                                   | Unassigned                                         |             |
| F55 |  | Extension module cabin (XMC), terminal 30          | 20 A        |
| F56 |  | Blower, terminal 30                                | 20 A        |
| F57 |                                                                                   | Unassigned                                         |             |
| F58 |                                                                                   | Unassigned                                         |             |
| F59 |  | Signal acquisition module (ASAM) (A7), terminal 30 | 50 A        |
| F60 |  | Signal acquisition module (ASAM) (A7), terminal 30 | 60 A        |
| E1  |                                                                                   | Unassigned                                         |             |
| E2  |                                                                                   | Unassigned                                         |             |
| E3  |                                                                                   | Unassigned                                         |             |
| E4  |                                                                                   | Unassigned                                         |             |
| E5  |                                                                                   | Unassigned                                         |             |

| No. | Symbol | Consumer   | Fuse rating |
|-----|--------|------------|-------------|
| E6  |        | Unassigned |             |
| E7  |        | Unassigned |             |

## Relays in the PDM cabin module

| No. | Symbol                                                                              | Consumer          |
|-----|-------------------------------------------------------------------------------------|-------------------|
| K1  |  | Terminal 15 relay |
| K2  |  | Terminal D+ relay |
| K3  |  | Unassigned        |
| K4  |                                                                                     | Unassigned        |
| K5  |                                                                                     | Bridge            |
| K6  |  | Terminal 15 relay |
| K7  |  | Terminal 15 relay |
| K8  |                                                                                     | Unassigned        |

## Diodes in the PDM cabin module

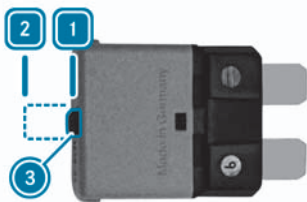
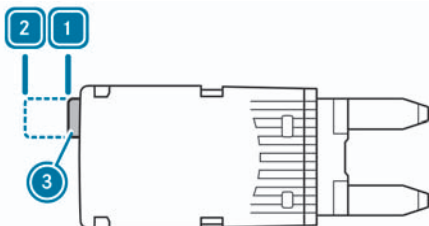
| No. | Symbol                                                                                | Consumer                    |
|-----|---------------------------------------------------------------------------------------|-----------------------------|
| D1  |    | Protection for horn         |
| D2  |                                                                                       | Unassigned                  |
| D3  |    | 12 V relay restraint system |
| D4  |  | 12 V relay restraint system |

## Breakdown assistance

### Information on the compressed-air system

#### Check automatic circuit breakers

#### Check and switch on an automatic circuit breaker



If an automatic circuit breaker is tripped, pin ③ moves to the OFF position ②.

- Pull the automatic circuit-breaker out of the module.
- Press pin ③ to ON position ①.
- Insert the automatic circuit breaker again.

- Switch the consumers on and check that they are functioning correctly.

If the automatic circuit-breaker is tripped again, have the electrical system checked at a qualified specialist workshop.

#### Information on the compressed-air system



#### NOTE

#### Damage to the compressed-air system due to contaminated and humid air

If the compressed-air system is filled via connection 28 on the electronic air processing unit, the compressed air does not pass through the compressed-air dryer.

If the reservoir pressure of the external compressed-air source is below 11 bar, it cannot be guaranteed that all the pressure circuits will be filled.

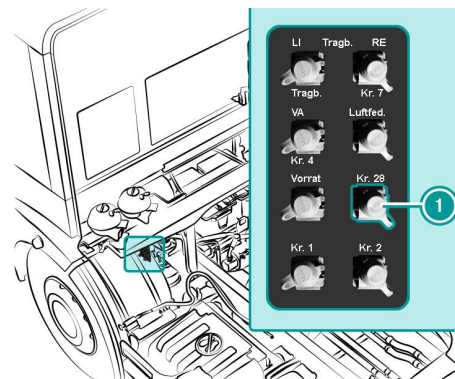
Fill pressure circuits with max. 12.5 bar.

Supply pressure that is too high, or contaminated and humid air from the ex-

ternal compressed-air source, can damage components of the compressed-air system.

- Fill the compressed-air system using a pressure between 11 bar and 12.5 bar.
- Fill the compressed-air system with clean and dry air.

System-dependent supply pressures can be found in the "Technical data" section (→ Page 292).



Connection 28 ① is connected to the electronic air processing unit. If the

compressed-air system is filled via connection 28 , the pressure circuit of the air suspension is also filled.

### Note

The electronic air processing unit contains the following components in a single unit:

- pressure regulator
- Compressed-air dryer
- multicircuit protection valve
- Pressure limiting valve
- Control unit
- Electronic parking brake

Check the following points before towing away:

- The spring brake actuators of the parking brake are released (→ Page 257).  
If the compressed-air supply is insufficient, release the spring-loaded brake cylinders manually (→ Page 257).
- Check whether workshop mode of the electronic parking brake is activated (→ Page 132).

### External charging, tow-starting and towing away

#### Carrying out external charging



**Danger**

#### **Risk of explosion when the battery is frozen**

A discharged battery can freeze at temperatures near the freezing point.

If you then attempt to charge the battery, battery gas can escape.

- ▶ Always thaw a frozen battery before charging it.

If the warning and indicator lamps do not light up in the instrument cluster at temperatures around or below freezing, it is highly probable that the discharged on-board electrical system battery has frozen. In this case, do not connect a second battery or charge the on-board electrical system battery.

The service life of an on-board electrical system battery that has been thawed may be reduced drastically.

After external charging, have the on-board electrical system battery checked at a qualified specialist workshop.



**Danger**

#### **Danger of chemical burns from battery acid**

Battery acid is caustic.

- ▶ Avoid contact with the skin, eyes, or clothing.
- ▶ Do not inhale battery gases.
- ▶ When carrying out maintenance work on the battery, wear acid-resistant protective clothing, particularly safety glasses, protective gloves and aprons.
- ▶ Do not lean over the battery.
- ▶ Keep children away from the battery.

If you come into contact with battery acid, observe the following:

- ▶ Immediately rinse the battery acid off your skin thoroughly with plenty of clean water and seek medical attention immediately.

## Breakdown assistance

### External charging, tow-starting and towing away

- ▶ If you get battery acid in your eyes, immediately rinse them thoroughly with plenty of clean water. Seek medical attention immediately.



**Danger**

#### **Risk of explosion due to an explosive mixture of gas**

An explosive mixture of gas may escape from the battery during the charging process and when providing starting assistance.

- ▶ Avoid fire, naked flames, sparks and smoking.
- ▶ Ensure sufficient ventilation.
- ▶ Do not lean over a battery.



**NOTE**

#### **Damage to the on-board electrical system battery or electronic components due to improper external charging**

- ▶ Only charge the on-board electrical system battery with a voltage of 24

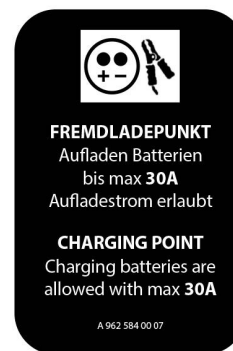
V via the external charger terminal point.

- ▶ When using a portable charger (batteries with a mains adapter), remove the mains plug before starting.
- ▶ Do not give other vehicles starting assistance.
- ▶ Use external charging cables protected from reverse polarity (jump leads) with a cable cross section of 35–50 mm<sup>2</sup> and isolated pole terminals.
- ▶ If the outside temperature has fallen below -10 °C, a discharged on-board electrical system battery may freeze. In this case, do not charge the vehicle. First, thaw out the battery.
- ▶ Do not connect the negative terminal clamp of the external charging cable (jump lead) to the chassis. Vehicle parts may otherwise be damaged.

#### **Requirements**

- Make sure that the vehicles are not touching.
- The parking brake is applied.
- all electrical consumers are switched off.
- The vehicle is switched off.

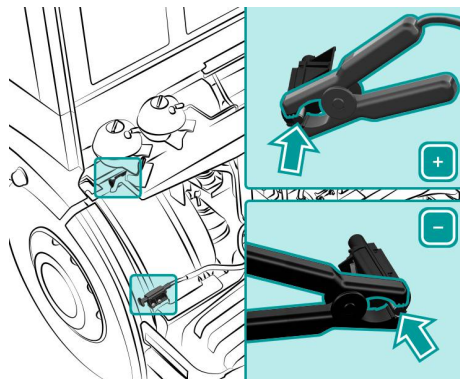
Observe the safety notes and protective measures when handling the on-board electrical system battery (→ Page 229).



Briefly exceeding the maximum charging current of 30 A (24 V), e.g. when bridging deeply discharged batteries, is permitted for a maximum of two minutes.

### External charging, tow-starting and towing away

#### Connecting the external charging cable



External charging connection point on the front left wheel arch

#### Connecting the external charging cable

- ▶ First, connect the positive terminal clamp of the external charging cable to the positive terminal of the other vehicle's battery.
- ▶ Gently turn the red protective cap of the external charger terminal point clockwise with the other positive terminal clamp of the external charging cable, slide it back and connect the positive terminal clamp to the + positive terminal.

- ▶ First, connect the negative terminal clamp of the external charging cable to the negative terminal of the other vehicle's battery.
- ▶ Gently turn the black protective cap of the external charger terminal point clockwise with the other negative terminal clamp of the external charging cable, slide it back and connect the negative terminal clamp to the - negative terminal.
- ▶ Keep the external charging cable connected until the on-board electrical system batteries are partly charged.
- ▶ Start the vehicle.

#### Disconnecting the external charging cable

- ▶ First, disconnect the negative terminal clamp of the external charging cable from the negative terminals.
  - ◁ The black cover cap of the external charger connection point springs back into its original position.

- ▶ Disconnect the positive terminal clamp of the external charging cable from the positive terminals.
  - ◁ The red cover cap of the external charger connection point springs back into its original position.

#### Notes on tow-starting and towing away



**Danger**  
**Risk of accident if the Active Brake Assist is activated when towed away or when the vehicle is used as a utility vehicle**

If, for example, the Active Brake Assist is activated while you are using your vehicle as a utility vehicle or you are towing, the vehicle may brake automatically.

The wheels may lose their grip.

- ▶ Always deactivate the Active Brake Assist in this or similar situations if possible.

## Breakdown assistance

### External charging, tow-starting and towing away



#### **Danger**

##### **Risk of accident due to lack of compressed-air supply**

If the compressed-air supply is not ensured, the spring-loaded parking brake can activate and the vehicle may then brake uncontrollably.

This could cause you to lose control of the vehicle.

- ▶ Always use a tow bar.
- ▶ Release the spring-loaded brake cylinder manually.
- ▶ After towing, have the spring-loaded brake cylinder reset to operational condition at a qualified specialist workshop as soon as possible.



#### **Danger**

##### **Risk of accident when towing away with the vehicle switched on**

When towing away with the front axle raised and the vehicle switched on, the ASR may brake the wheels of the rear axle in an uncontrolled manner.

The vehicle may break free.

- ▶ Switch off the vehicle before towing with the front axle raised.



#### **Danger**

##### **Risk of accident due to limited safety-related functions during the towing process**

Safety-related functions are limited or no longer available in the following situations:

- The vehicle is switched off.
- the brake system or power steering system is malfunctioning.
- the energy supply or the on-board electrical system is malfunctioning.

If your vehicle is being towed, considerably more power may be required for steering and braking.

- ▶ Use a tow bar.
- ▶ Make sure that the steering wheel can move freely, before towing the vehicle away.



#### **NOTE**

##### **Damage to the transmission when towing**

If the vehicle is towed with drive shafts installed, the transmission may be damaged.

Only in exceptional cases and briefly, e.g. when leaving danger zones, should the vehicle be towed with the drive shafts installed.

- ▶ Tow the vehicle on the shortest route possible to a suitable point to remove the drive shafts or load the vehicle onto a transporter.
- ▶ Tow the vehicle at no faster than walking pace.



#### **NOTE**

##### **Damage to the brake and air suspension when towing**

If the compressed-air supply of the vehicle is not guaranteed during towing, the following may occur:



### External charging, tow-starting and towing away

- the brake can overheat or be damaged
- the air suspension can be damaged
- ▶ Charge the compressed-air system or release the spring-loaded parking brake manually.
- ▶ While towing the vehicle away, monitor the driving level.



#### NOTE

#### Damage to attachments and bodies when towing away

Attachments and bodies can affect the vehicle height and width.

- ▶ Do not exceed the permitted vehicle height.
- ▶ Fold in the wind deflectors when towing away with the rear axle raised.
- ▶ Observe the maximum headroom clearance of underpasses.



#### NOTE

#### Damage to the drive system due to tow-starting

The vehicle cannot be started by tow-starting.

- ▶ Do not attempt to tow-start the vehicle.

Specialist knowledge beyond the scope of these operating instructions is required for towing the vehicle away. Only have your vehicle towed away by a professional towing and recovery company.

Before towing away, ensure that the high-voltage on-board electrical system is switched off and secured against being switched back on ( → Page 118).

Before towing the vehicle away, check whether workshop mode of the parking brake is activated ( → Page 132).

The following towing methods are permitted:

- raised at the rear and the steering wheel fixed in the centre position

- raised at the front and drive shafts removed
- at the coupling jaw with all axles on the ground and drive shafts removed

Do not exceed a maximum permissible towing speed of 60 km/h.

Only in exceptional cases, e.g. when leaving danger zones, should the vehicle be towed with the drive shafts installed.

If the drive shafts are not removed, observe the following:

- Ensure that the transmission is in neutral.
- Ensure that the oil pump of the drive axle is operational; to do so, switch on the vehicle.
- Tow the vehicle on the shortest route possible to a suitable point to remove the drive shafts or load the vehicle onto a transporter.
- Tow the vehicle at no faster than walking pace.

## Breakdown assistance

### External charging, tow-starting and towing away

#### **i** Note

Information on towing away and recovering can be found on the Internet at: <https://www.mercedes-benz-trucks.com>.

You can obtain further information from an authorised Mercedes-Benz Trucks Service Centre.

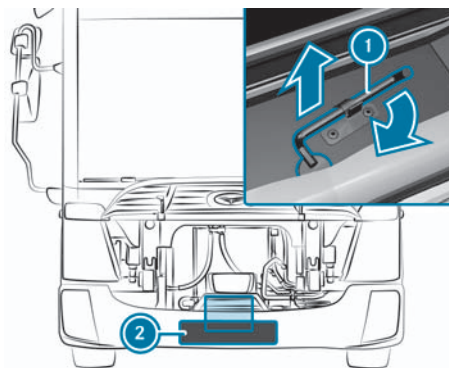
#### Manoeuvring and tow-starting/towing away

If the vehicle is being transported on a low loader, the permitted vehicle height could be exceeded. Pay attention to the headroom of structures, e.g. bridges.

- ▶ Observe the notes on towing to and away (→ Page 253).
- ▶ Before towing, activate the workshop mode of the parking brake (→ Page 132).
- ▶ If necessary, charge the compressed-air system using an external source of compressed air (→ Page 250).

- ▶ If there is insufficient reservoir pressure to release the parking brake, release the spring-loaded brake cylinders manually (→ Page 257).
- ▶ Before towing away, agree on a clear signal with the towing vehicle driver. Adapt your driving style to the more difficult conditions.

#### Coupling jaw



Coupling jaw

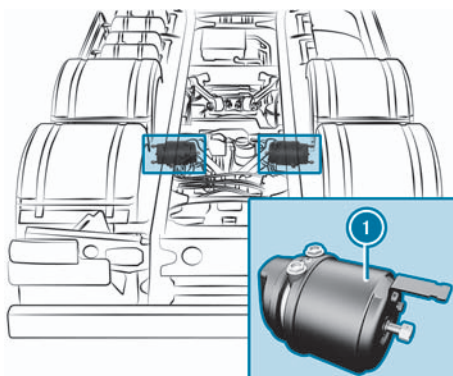
Use the front coupling jaw for manoeuvring and towing away.

- ▶ Open the maintenance flap (→ Page 224).

- ▶ Push the licence plate holder (2) down slightly and swivel it forwards.
- ▶ Turn the coupling pin (1) approximately 90° clockwise and pull it out from above.
- ▶ Attach the tow bar.
- ▶ Push the coupling pin (1) down through the eyelet of the tow bar.
- ▶ Turn the coupling pin (1) approximately 90° anti-clockwise and engage it in the locking mechanism.

### Spring-loaded brake

#### Overview of the arrangement of the spring-loaded brake cylinders



① Spring-loaded brake cylinder (example)

The arrangement of spring-loaded brake cylinders on the rear axle(s) depends on the axle version.

| Axes                       | 1 | 2              |
|----------------------------|---|----------------|
| 6x2/4                      | x | x <sup>1</sup> |
| <sup>1</sup> Trailing axle |   |                |

#### Manually releasing and moving the spring-loaded brake cylinder to the drive position

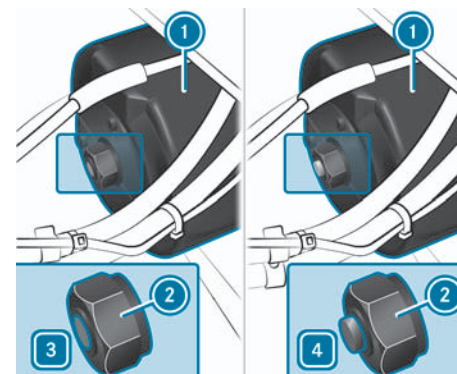


#### NOTE

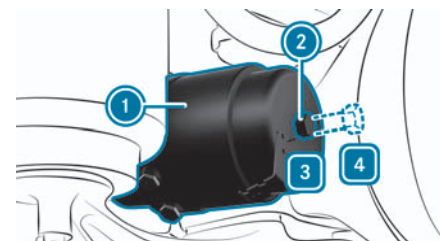
#### Damage to the spring-loaded brake cylinder

Using excessive torque when loosening or tightening the release screw can damage the spring-loaded brake cylinder.

- ▶ Loosen the release screw using a maximum of 70 Nm.
- ▶ Tighten the release screw using a maximum of 35 Nm.
- ▶ Do not use an impact wrench to loosen or tighten the release screw.



Spring-loaded brake cylinder with release screw and release indicator



Spring-loaded brake cylinder with release screw

- ① Spring-loaded brake cylinder
- ② Release screw
- ③ Drive position

## Breakdown assistance

### External charging, tow-starting and towing away

#### 4 Released position

If there is insufficient supply pressure to release the parking brake, release all spring-loaded brake cylinders manually in the event of an emergency.

The vehicles may be fitted with different types of spring-loaded brake cylinders depending on the axle, e.g. spring-loaded brake cylinders with a release screw or with a release indicator.

- ▶ Use chocks to protect the vehicle against rolling away.
- ▶ **To release the spring-loaded brake cylinder:** Turn release screw ② of the spring-loaded brake cylinder ① to the left until it stops in release position 4.

#### Moving the spring-loaded brake cylinder to the drive position

Make sure to move all spring-loaded brake cylinders back into the drive position.

- ▶ Charge the brake system until the shut-off pressure is reached ( → Page 292).

- ▶ Use chocks to protect the vehicle against rolling away.
- ▶ Release the parking brake manually ( → Page 132).
- ▶ Turn release screw ② of spring-loaded brake cylinder ① clockwise to the drive position 3.
- ▶ Tighten the release screw ②.

**Operating and road safety****General notes on wheels and tyres**

Tyres are of particular importance to the operating and road safety of the vehicle. Check vehicle tyres regularly (→ Page 261).

**Tyre load-bearing capacity, tyre speed rating and tyre types****Danger****Risk of injury through exceeding the specified tyre load-bearing capacity or the permissible speed rating**

Exceeding the specified tyre load-bearing capacity may lead to tyre damage and the tyres bursting.

- ▶ Therefore, only use tyre types and sizes approved for your vehicle model.
- ▶ Observe the tyre load-bearing capacity rating and speed rating required for your vehicle.

A tyre dealer, a qualified workshop or an authorised Mercedes-Benz Trucks Service Centre will be able to provide you with further information on the following:

- Tyre load-bearing capacity (LI, Load Index)
- Tyre speed rating
- Tyre age
- Causes and consequences of tyre wear
- Measures to be taken in the event of tyre damage
- Types of tyre for specific regions, areas of operation or conditions of vehicle use
- Interchangeability of tyres

In particular, observe the permissible tyre specifications for the country you are currently in. These requirements may stipulate a specific tyre type for your vehicle.

In addition, the use of specific tyre types may be advisable for certain regions and areas of operation.

**Notes on replacing tyres and retreaded tyres****Replacing tyres**

If you are replacing the standard tyres of your vehicle, use only the tyre and wheel rim sizes approved for your vehicle model.

A tyre dealer, a qualified workshop or an authorised Mercedes-Benz Trucks Service Centre will be able to provide you with further information. After replacing your tyres, carry with you the vehicle's type approval for the new tyre and wheel disk size as well as the manufacturer's certificate showing that the tyres may be used on the vehicle. Observe the legal requirements for the country you are currently in. An authorised Mercedes-Benz Trucks Service Centre can provide you with information on obtaining a manufacturer's certificate.

After replacing your tyres, it may be necessary for adjustments to be made to the control units. If necessary, have these adjustments carried out at a qualified workshop.

## wheels and tyres

### Operating and road safety

#### Retreaded tyres

Daimler Truck AG recommends that you only use wheels and tyres that have been approved for Mercedes-Benz trucks.

#### Balancing tyres

Daimler Truck AG recommends that you only use clip-on and adhesive wheel balance weights which have been approved for Mercedes-Benz trucks to balance the tyres. Adding balancing granulate, balancing beads or balancing gel to the tyres can damage the inliner of the tyre.

#### Notes on wheels with tyre pressure sensors

##### Tyre pressure monitoring system in the multimedia system

You can have the current tyre pressures, tyre temperatures and the tyre pressure sensor battery charge level displayed in the multimedia system.

To do this, in the multimedia system in the menu **Status** select the menu item **Tyres**.

Then press the  button, for example, in order to display the current tyre pressures.

#### Note

Pressing and holding the  button in the multimedia system displays the specified pressure set for the tyres.

The tyre pressure monitoring system is a convenience system that offers the following benefits:

- Assistance with regular tyre pressure monitoring
- Automatic warning in the event of tyre pressure loss

The tyre pressure monitoring system can provide reliable warnings only if the correct specified pressure for the tyres has been set on the on-board computer.

The tyre pressure monitoring system will detect an incorrect value if, for example:

- The vehicle is overloaded or incorrectly loaded.
- The vehicle is fitted with wheels that have a different tyre size and the

specified pressure on the on-board computer has not been corrected.

- An incorrect specified pressure has been set on the on-board computer.

You are responsible for setting the correct tyre pressure. Always have the correct specified pressure set on the on-board computer at a qualified workshop.

If radio transmitting equipment (e.g. cordless headphones, two-way radios) is operated inside the vehicle or in the vicinity of the vehicle, this may interfere with the operation of the tyre pressure monitoring system.

The multimedia system will display "—" instead of the tyre pressure in the following cases:

- A tyre pressure sensor is temporarily malfunctioning, e.g. due to radio transmitting equipment
- Values from the tyre pressure sensor are not yet available

Wait a few minutes and/or move the vehicle out of the operating range of the radio transmitting equipment.

## Notes on sound changes or undesired handling characteristics

After tyre pressure sensors have been replaced, the vehicle should remain stationary for 15 minutes.

Then drive the vehicle at over 30 km/h for at least ten minutes.

The tyre pressure values on the on-board computer may differ from those measured with a tyre pressure gauge. The tyre pressures shown by the on-board computer refer to those measured at sea level. At high altitudes, the tyre pressure values indicated by a tyre pressure gauge will be higher than those shown by the on-board computer. In this case, do not reduce the tyre pressure.

When the tyre temperature differs, different specified pressures may be shown for the tyres because of exposure to sunlight, for example.

### Tyre pressure loss warning system

If the tyre pressure in at least one tyre drops, the on-board computer will display the warning  in a display message. A warning tone will also sound. Each tyre pressure value for which a loss of pressure has been determined will be high-

lighted in red or yellow. When there is a notable loss of tyre pressure, the tyre pressure value will be shown in red.

### Mounting

You can recognise wheels with tyre pressure sensors by a red ring on the tyre valve. Observe the mounting instructions when changing a tyre.

A tyre dealer, a qualified workshop or an authorised Mercedes-Benz Trucks Service Centre will be able to provide you with further information.

### Storage

The battery of a tyre pressure sensor for the tyre pressure monitoring is activated automatically when it is mounted. The tyre pressure sensor will then begin to transmit data periodically. If you store wheels with tyre pressure sensors mounted, the service life of the tyre pressure sensor battery during road use is shortened. Therefore, tyre pressure sensors should only be mounted a short time before the wheel is used.

### Notes on sound changes or undesired handling characteristics

While driving, pay attention to vibrations, noises and unusual handling characteristics, e.g. pulling to one side. This may indicate that the wheels or tyres are damaged. If you suspect that a tyre is defective, reduce your speed immediately. Stop the vehicle as soon as possible to check the wheels and tyres for damage. Hidden tyre damage could also be causing the unusual handling characteristics. If you find no signs of damage, have the tyres and wheels checked at a qualified specialist workshop.

### Notes on regularly inspecting wheels and tyres

Tyres are of particular importance to the operating and road safety of the vehicle.

Therefore, regularly check the following points:

- Tyre condition
- Tyre age
- Tyre tread

## wheels and tyres

### Notes on regularly inspecting wheels and tyres

- Tyre pressure

#### Tyre condition



**Danger**

#### Risk of injury through damaged tyres

Damaged tyres can cause tyre pressure loss.

- ▶ Check the tyres regularly for signs of damage and replace any damaged tyres immediately.

Before starting your journey, check the tyre condition for the following points:

- External damage
- Foreign objects in the tyre tread
- Foreign objects between the tyres (on vehicles with twin tyres)
- Cracks or bulges
- Uneven tread wear or excessive wear on one side

Tyre damage can be caused by the following:

- The operating conditions of the vehicle

- Tyre ageing
- Kerbs
- Foreign objects
- Insufficient or excessive tyre pressure
- Weather conditions and environmental factors
- Contact with oil, grease, fuel

#### Tyre age



Tyres age, even if they are used infrequently or not at all. Operating and road safety diminish with the age of the tyres. For this reason, tyres more than six years old must be checked and, if necessary, replaced at a qualified workshop. This also applies to the spare wheel. The date of manufacture ① provides information about the age of a tyre. The first and second digits refer to the week of man-

ufacture, starting with "01" for the first calendar week of the year. The third and fourth digits refer to the year of manufacture. For example, a tyre that is marked "3818" was manufactured in week 38 in 2018.

#### Tyre tread



**Danger**

#### Risk of aquaplaning due to insufficient tyre tread

Insufficient tyre tread will result in reduced tyre grip.

If the road surface is wet, the risk of aquaplaning will increase, in particular where speed is not adapted to suit the conditions.

- ▶ You must regularly check the tyre tread depth and the condition of the tyre tread over the entire width on all tyres.

Minimum tread depth at

- Summer tyres: 3 mm
- M+S tyres: 4 mm



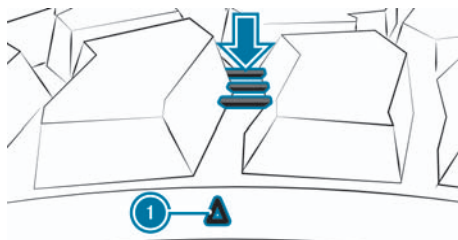
## Notes on regularly inspecting wheels and tyres

- For safety reasons, replace your tyres before reaching the legally prescribed limit for the minimum tread depth.

**Danger****Risk of aquaplaning due to insufficient tyre tread**

Depending on the depth of the water on the roadway, the vehicle may aquaplane.

- Avoid ruts and brake carefully.
- Change tyres if the tyre tread is insufficient.



① Example: Tyre tread wear indicator

A specified minimum tread depth is a legal requirement for all tyres. Observe the legal requirements for the country

you are currently in. The less tyre tread depth remaining, the poorer are the road adhesion and handling characteristics of the vehicle, particularly on wet or snow-covered roads. A tyre has reached the minimum tread depth when the tread wear indicator (arrow) is flush with the tyre tread.

**Tyre pressure****Danger****Risk of accident due to insufficient or excessive tyre pressure**

Tyres with insufficient or excessive tyre pressure pose the following hazards, in particular:

- The tyres may burst.
  - The tyres may wear excessively and/or unevenly.
  - The handling as well as steering and braking characteristics may be severely impaired.
- Observe the recommended tyre pressures and check the tyre pres-

sure of all the tyres including the spare wheel:

- Monthly
  - When the load changes
  - Before starting a long journey
  - If operating conditions change, e.g. off-road driving
- Adjust the tyre pressure as necessary.

Tyre pressure that is too low can have the following consequences:

- Severe heating
- Increased tyre wear
- Changed driving stability
- Increased energy consumption

Tyre pressure that is too high can have the following consequences:

- Increased braking distance
- Poorer tyre traction
- Increased tyre wear

## wheels and tyres

### Determining tyre pressures



**Danger**

#### **Risk of accident due to repeated pressure drop in the tyres**

The wheels, valves or tyres could be damaged.

Too low a tyre pressure can lead to the tyres bursting.

- ▶ Examine the tyres for foreign objects.
- ▶ Check whether the tyre has a puncture or the valve has a leak.
- ▶ If you are unable to rectify the damage, contact a qualified workshop.



**Danger**

#### **Risk of injury from lowering the tyre pressure when the tyres are warm**

The temperature and pressure of the tyre increase when the vehicle is in motion.

Lowering the tyre pressure on warm tyres can cause the tyres to burst.

- ▶ Do not lower the tyre pressure on warm tyres.

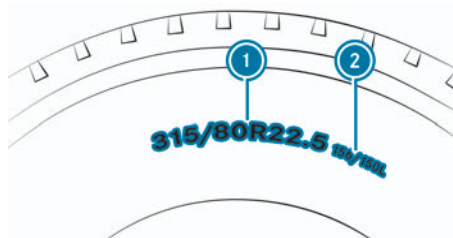
- ▶ Make sure you keep to the specified tyre pressure.

Before starting your journey, determine the specified tyre pressure and check the tyre pressures while the tyres are cold (→ Page 264).

### Determining tyre pressures

- ▶ Observe the notes in the "Tyre pressure" section of "Regular checking of the wheels and tyres" (→ Page 261).

Use the correct tyre pressures for each axle on the vehicle.



Example: Tyre size and load-bearing index

- ▶ Read off tyre size ① and load-bearing index ② on the tyre.

If the load-bearing index ② comprises two numbers, the first number before the "/" is for single tyres and the second number, after the "/", is for twin tyres.

- ▶ Determine the permissible axle load by checking the vehicle identification plate (→ Page 268).
- ▶ Select the tyre pressure table which corresponds to the tyre type.
  - Single tyres (→ Page 265)
  - Twin tyres (→ Page 266)
- ▶ Search tyre size ① in the tyre pressure table.
- ▶ Search load-bearing index ② which corresponds to the tyre type next to the tyre size in the tyre pressure table.
- ▶ Find the maximum permissible axle load in the tyre pressure table and read off the tyre pressure.

The tyre pressure changes by approx. 30 to 40 kPa (0.3 to 0.4 bar/4.4 to 5.8 psi) for every 10°C in air temperature. Bear this temperature-related change in tyre pressure in mind when checking tyre pressures indoors, where the temperature

## Tyre pressure table for single tyres

may be higher than the outside temperature.

**i Note**

For specific tasks, e.g. for winter services, a maximum vehicle axle load rating for fitting approved attachments may be increased. In such cases, the speed of the vehicle is limited. The tyre pressure will then deviate from the information in the following tyre pressure table. It needs to be increased in accordance with the E.T.R.T.O. standard and the information provided by the tyre manufacturer.

## Tyre pressure table for single tyres

## Tyres 295/80 R 22.5

Load-bearing index 150

- Axle load 6300 kg: 750 kPa (7.5 bar, 109 psi)
- Axle load 6700 kg: 800 kPa (8.0 bar, 116 psi)

Load-bearing index 152

- Axle load 6300 kg: 750 kPa (7.5 bar, 109 psi)
- Axle load 6700 kg: 800 kPa (8.0 bar, 116 psi)
- Axle load 7100 kg: 850 kPa (8.5 bar, 123 psi)

## Tyres 315/60 R 22.5

Load-bearing index 152

- Axle load 6300 kg: 775 kPa (7.75 bar, 112 psi)
- Axle load 6700 kg: 850 kPa (8.5 bar, 123 psi)
- Axle load 7100 kg: 900 kPa (9.0 bar, 131 psi)

Load-bearing index 154

- Axle load 6300 kg: 725 kPa (7.25 bar, 105 psi)
- Axle load 6700 kg: 800 kPa (8.0 bar, 116 psi)
- Axle load 7100 kg: 850 kPa (8.5 bar, 123 psi)
- Axle load 7500 kg: 900 kPa (9.0 bar, 131 psi)

## Tyres 315/70 R 22.5

Load-bearing index 154

- Axle load 6300 kg: 725 kPa (7.25 bar, 105 psi)
- Axle load 6700 kg: 800 kPa (8.0 bar, 116 psi)
- Axle load 7100 kg: 850 kPa (8.5 bar, 123 psi)
- Axle load 7500 kg: 900 kPa (9.0 bar, 131 psi)

Load-bearing index 156

- Axle load 6300 kg: 675 kPa (6.75 bar, 98 psi)
- Axle load 6700 kg: 725 kPa (7.25 bar, 105 psi)
- Axle load 7100 kg: 775 kPa (7.75 bar, 112 psi)
- Axle load 7500 kg: 850 kPa (8.5 bar, 123 psi)
- Axle load 8000 kg: 900 kPa (9.0 bar, 131 psi)

### Tyre pressure table for twin tyres

#### Tyres 315/80 R 22.5

Load-bearing index 154

- Axle load 6300 kg: 675 kPa (6.75 bar, 98 psi)
- Axle load 6700 kg: 725 kPa (7.25 bar, 105 psi)
- Axle load 7100 kg: 775 kPa (7.75 bar, 112 psi)
- Axle load 7500 kg: 825 kPa (8.25 bar, 120 psi)

Load-bearing index 156

- Axle load 6300 kg: 650 kPa (6.5 bar, 94 psi)
- Axle load 6700 kg: 700 kPa (7.0 bar, 102 psi)
- Axle load 7100 kg: 750 kPa (7.5 bar, 109 psi)
- Axle load 7500 kg: 800 kPa (8.0 bar, 116 psi)
- Axle load 8000 kg: 850 kPa (8.5 bar, 123 psi)

#### Tyre pressure table for twin tyres

#### Tyres 295/80 R 22.5

Load-bearing index 145

- Axle load 9500 kg: 625 kPa (6.25 bar, 91 psi)
- Axle load 10000 kg: 675 kPa (6.75 bar, 98 psi)
- Axle load 10500 kg: 725 kPa (7.25 bar, 105 psi)
- Axle load 11500 kg: 800 kPa (8.0 bar, 116 psi)

Load-bearing index 148

- Axle load 9500 kg: 600 kPa (6.0 bar, 87 psi)
- Axle load 10000 kg: 650 kPa (6.5 bar, 94 psi)
- Axle load 10500 kg: 700 kPa (7.0 bar, 102 psi)
- Axle load 11500 kg: 775 kPa (7.75 bar, 112 psi)
- Axle load 12000 kg: 800 kPa (8.0 bar, 116 psi)
- Axle load 12300 kg: 825 kPa (8.25 bar, 120 psi)

- Axle load 12600 kg: 850 kPa (8.5 bar, 123 psi)

#### Tyres 315/60 R 22.5

Load-bearing index 148

- Axle load 9500 kg: 650 kPa (6.5 bar, 94 psi)
- Axle load 10000 kg: 675 kPa (6.75 bar, 98 psi)
- Axle load 10500 kg: 725 kPa (7.25 bar, 105 psi)
- Axle load 11500 kg: 825 kPa (8.25 bar, 120 psi)
- Axle load 12000 kg: 850 kPa (8.5 bar, 123 psi)
- Axle load 12300 kg: 875 kPa (8.75 bar, 127 psi)
- Axle load 12600 kg: 900 kPa (9.0 bar, 131 psi)

#### Tyres 315/70 R 22.5

Load-bearing index 148

- Axle load 9500 kg: 600 kPa (6.0 bar, 87 psi)

- Axle load 10000 kg: 650 kPa (6.5 bar, 94 psi)
- Axle load 10500 kg: 700 kPa (7.0 bar, 102 psi)
- Axle load 11500 kg: 775 kPa (7.75 bar, 112 psi)
- Axle load 12000 kg: 800 kPa (8.0 bar, 116 psi)
- Axle load 12300 kg: 825 kPa (8.25 bar, 120 psi)
- Axle load 12600 kg: 850 kPa (8.5 bar, 123 psi)

## Load-bearing index 150

- Axle load 9500 kg: 600 kPa (6.0 bar, 87 psi)
- Axle load 10000 kg: 625 kPa (6.25 bar, 91 psi)
- Axle load 10500 kg: 675 kPa (6.75 bar, 98 psi)
- Axle load 11500 kg: 750 kPa (7.5 bar, 109 psi)
- Axle load 12000 kg: 800 kPa (8.0 bar, 116 psi)
- Axle load 12300 kg: 825 kPa (8.25 bar, 120 psi)

- Axle load 12600 kg: 850 kPa (8.5 bar, 123 psi)
- Axle load 13000 kg: 875 kPa (8.75 bar, 127 psi)

**Tyres 315/80 R 22.5**

## Load-bearing index 150

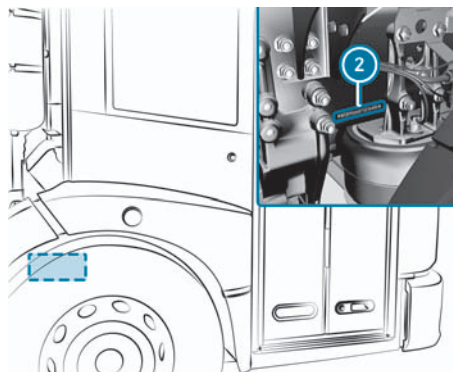
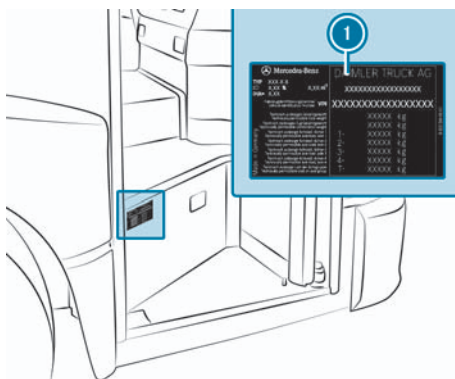
- Axle load 9500 kg: 575 kPa (5.75 bar, 83 psi)
- Axle load 10000 kg: 600 kPa (6.0 bar, 87 psi)
- Axle load 10500 kg: 650 kPa (6.5 bar, 94 psi)
- Axle load 11500 kg: 725 kPa (7.25 bar, 105 psi)
- Axle load 12000 kg: 750 kPa (7.5 bar, 109 psi)
- Axle load 12300 kg: 775 kPa (7.75 bar, 112 psi)
- Axle load 12600 kg: 800 kPa (8.0 bar, 116 psi)
- Axle load 13000 kg: 825 kPa (8.25 bar, 120 psi)

## Technical data

### Vehicle identification plate/axle loads

#### Vehicle identification plate/axle loads

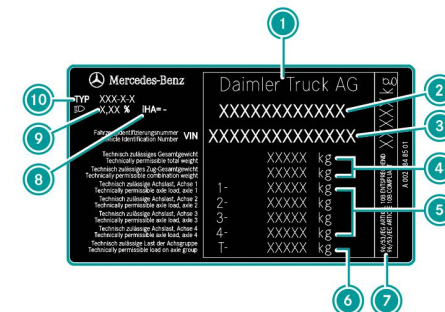
#### Vehicle identification plate and Vehicle Identification Number (VIN)



Vehicle Identification Number and vehicle identification plate (example: right-hand side of vehicle)

The Vehicle Identification Plate (1) is located on the door sill on the co-driver's side. The Vehicle Identification Number (VIN) (2) is stamped on the chassis frame in the right wheel well.

#### Information on the vehicle identification plate



Example: Vehicle identification plate

- ① Vehicle manufacturer
- ② EU general operating permit number (only in certain countries)
- ③ Vehicle Identification Number (VIN)
- ④ Maximum permissible gross vehicle weight (kg)
- ⑤ Permissible gross vehicle combination weight (kg)
- ⑥ Permissible axle loads of axles 1 to 4 (kg)

- ⑥ Permissible load of axle group T (kg)
- ⑦ Additional weight due to alternative drive (kg)
- ⑧ Rear axle ratio
- ⑨ Basic headlamp setting
- ⑩ Vehicle model or vehicle model designation

You will find the technically permissible gross vehicle combination weight on the vehicle identification plate or in the COC documents.

If you require further assistance, consult an authorised Mercedes-Benz Trucks Service Centre.

### Radio regulations

#### Vehicle installed radio components

**Only for EU and EFTA countries and countries that recognise the EU manufacturer's declaration of conformity:**



The following note applies to all wireless components of the vehicle and the information systems and communications equipment integrated in the vehicle:

The wireless components of the vehicle that receive and/or transmit radio waves are compliant with the basic requirements and all other relevant regulations stipulated by Directive 2014/53/EU. You can obtain further information from a Mercedes-Benz Trucks Service Centre.

#### For Israel only:

It is prohibited to make any change to a vehicle-installed radio component that could affect the wireless specifications of the device, including software changes, replacement of the original antenna, or adding the option to connect the device to an external antenna, without first obtaining approval from the Communications Ministry, because of concern about wireless interference.

#### For the United Kingdom only:

**UK  
CA**

The following note applies to all wireless components of the vehicle and the information systems and communications equipment integrated in the vehicle:

The manufacturers of the radio components declare that all radio equipment installed in the vehicle is in compliance with the relevant statutory requirements. The full texts of the declarations of conformity are available at the following web address:

<https://www.mercedes-benz-trucks.com/conformity>

#### Importer

Mercedes-Benz Trucks UK Limited

Delaware Drive, Tongwell

Milton Keynes, MK15 8BA

England

#### Importer of Mercedes-Benz spare parts

Mercedes-Benz Parts Logistics

## Technical data

### Radio regulations

Delaware Drive, Tongwell  
Milton Keynes, MK15 8BA  
England

#### For Brazil only:



Note on two-way radio systems in the vehicle:

These systems are not protected against harmful interference and must not cause interference in properly approved systems.

#### For Jamaica only:

All vehicle-installed radio components have been Type Approved by SMA.

#### For Russia only:



The manufacturers of the wireless components installed in the vehicle hereby declare that all wireless components installed in the vehicle comply with the technical regulations for two-way radios. You can obtain further information from an authorised Mercedes-Benz Trucks Service Centre.

#### For Thailand only:

Telecommunications equipment in this vehicle is in compliance with the requirements of the National Broadcasting and Telecommunications Commission (NBTC).

Radar sensor ARS4-A:

1) The telecommunications equipment and this device comply with the standard

or requirements of the National Broadcasting and Telecommunications Commission (NBTC).

2) When in use, this two-way radio system emits electromagnetic radiation that meets the health and safety standards of the National Telecommunications Commission (NTC).



**Type of wireless application and specification in accordance with 2014/53/EU****Note**

Not all two-way radio systems may be available in your vehicle. This depends on the equipment chosen or the national version.

## Technical data

### Radio regulations

Type of wireless application and specification in accordance with 2014/53/EU

| Component                     | Service                | Transmission frequency band                                                                                                                                                                             | Transmission output    |
|-------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Telematics platform<br>(CTP3) | 4G<br>LTE              | Band 1<br>Band 3<br>Band 4<br>Band 5<br>Band 7<br>Band 8<br>Band 12<br>Band 14<br>Band 17<br>Band 18<br>Band 19<br>Band 20<br>Band 25<br>Band 26<br>Band 28<br>Band 39<br>Band 40<br>Band 41<br>Band 66 | 0.25 W (Power Class 3) |
|                               | 3G<br>UMTS/HSDPA/HSUPA | Band 1<br>Band 5<br>Band 8                                                                                                                                                                              | 0.3 W (Power Class 3)  |
|                               | 2G GSM/GPRS/EDGE       | GSM 850 MHz<br>E-GSM 900 MHz                                                                                                                                                                            | 2 W (Power Class 4)    |

| Component                        | Service                    | Transmission frequency band                                                      | Transmission output    |
|----------------------------------|----------------------------|----------------------------------------------------------------------------------|------------------------|
| Telematics platform<br>(CTP3)    | 2G GSM/GPRS/EDGE           | DCS 1800 MHz<br>PCS 1900 MHz                                                     | 1 W (Power Class 1)    |
|                                  |                            | EDGE GSM 850 MHz<br>EDGE E-GSM 900 MHz<br>EDGE DCS 1800 MHz<br>EDGE PCS 1900 MHz | 0.5 W (Power Class 3)  |
|                                  | WLAN (IEEE802.11b)         | 2.4 GHz band                                                                     | 50 mW                  |
|                                  | WLAN (IEEE802.11g/n/ax)    | 2.4 GHz band                                                                     | 25 mW                  |
|                                  | WLAN (IEEE802.11a/n/ac/ax) | 5 GHz band                                                                       | 15 mW                  |
|                                  | Bluetooth®                 | 2.4 GHz band                                                                     | 5 mW                   |
| Telematics platform<br>(CTP2019) | 4G<br>LTE                  | Band I 2100 MHz<br>Band III 1800 MHz<br>Band V 850 MHz<br>Band VII 2600 MHz      | 0.25 W (Power Class 3) |
|                                  | 3G<br>UMTS/HSDPA/HSUPA     | Band I 2100 MHz<br>Band II 1900 MHz<br>Band V 850 MHz<br>Band VIII 900 MHz       | 0.25 W (Power Class 3) |
|                                  | 2G GSM/GPRS/EDGE           | GSM 850 MHz<br>E-GSM 900 MHz                                                     | 2 W (Power Class 4)    |
|                                  |                            | DCS 1800 MHz<br>PCS 1900 MHz                                                     | 1 W (Power Class 1)    |
|                                  | WLAN (IEEE802.11b)         | 2.4 GHz band                                                                     | 50 mW                  |

## Technical data

### Radio regulations

| Component                                       | Service                | Transmission frequency band                                                                                                                     | Transmission output    |
|-------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Telematics platform<br>(CTP2019)                | Wi-Fi (IEEE802.11g/n)  | 2.4 GHz band                                                                                                                                    | 25 mW                  |
|                                                 | Bluetooth®             | 2.4 GHz band                                                                                                                                    | 2.5 mW                 |
| Telematics platform<br>(CTP2019 from 10/2022)   | 4G<br>LTE              | Band I 2100 MHz<br>Band III 1800 MHz<br>Band VII 2600 MHz<br>Band VIII 900 MHz<br>Band XX 800 MHz<br>Band XXVII 700 MHz<br>Band XXXVII 2600 MHz | 0.25 W (Power Class 3) |
|                                                 | 3G<br>UMTS/HSDPA/HSUPA | Band I 2100 MHz<br>Band VIII 900 MHz                                                                                                            | 0.25 W (Power Class 3) |
|                                                 | 2G GSM/GPRS/EDGE       | E-GSM 900 MHz<br>DCS 1800 MHz                                                                                                                   | 2 W (Power Class 4)    |
|                                                 | WLAN (IEEE802.11b)     | 2.4 GHz band                                                                                                                                    | 50 mW                  |
|                                                 | Wi-Fi (IEEE802.11g/n)  | 2.4 GHz band                                                                                                                                    | 25 mW                  |
|                                                 | Bluetooth®             | 2.4 GHz band                                                                                                                                    | 2.5 mW                 |
| Radar sensor system<br>(ARS3-A, ARS4-B, ARS4-A) | Vehicle-interval radar | 76–77 GHz band                                                                                                                                  | 3.16 W                 |
| Radar sensor system<br>(SRR2, model L2C0055TR)  | Lateral radar          | 76–77 GHz band                                                                                                                                  | 1 W                    |
| Radar sensor system<br>(CR5TPCC)                | Corner radar           | 76–77 GHz band                                                                                                                                  | 0.11 – 0.92 W          |

| Component                                              | Service                           | Transmission frequency band | Transmission output |
|--------------------------------------------------------|-----------------------------------|-----------------------------|---------------------|
| Radio<br>(Daimler TCC Touch*,<br>1-DIN TCC MID*)       | Bluetooth®                        | 2402–2480 MHz               | ≤ 2.5 mW (4 dBm)    |
| Electronic ignition lock<br>(EIS, transmitter key)     | Short-range wireless technologies | 433 MHz band                | 10 mW               |
| Tyre pressure monitoring system<br>(TPM)               | Short-range wireless technologies | 433 MHz band                | 10 mW               |
| Android Black Box<br>(STG AP)                          | Bluetooth®                        | 1925–1975 MHz               | 25.6 dBm            |
|                                                        |                                   | 2402–2480 MHz               | 5.5 dBm             |
|                                                        | WiFi                              | 1715–1780 MHz               | 25.6 dBm            |
|                                                        |                                   | 2412–2484 MHz               | 18.5 dBm            |
|                                                        |                                   | 5150–5350 MHz               | 17.5 dBm            |
|                                                        |                                   | 5470–5725 MHz               | 17.5 dBm            |
|                                                        | GPRS 900                          | 880–914 MHz                 | 33 dBm              |
|                                                        |                                   | 885–910 MHz                 | 25.5 dBm            |
|                                                        | GPRS 1800                         | 837–857 MHz                 | 25.5 dBm            |
|                                                        |                                   | 1710–1785 MHz               | 30 dBm              |
| Instrument cluster<br>(Connect5)                       | WLAN 5 GHz                        | 5.725–5.875 GHz band        | < 25 mW             |
|                                                        |                                   | 5.150–5.350 GHz band        | < 200 mW            |
|                                                        | WLAN 2.4 GHz                      | 2.400–2.4835 GHz band       | < 20 mW             |
|                                                        | Bluetooth®                        | 2.400–2.4835 GHz band       | < 10 mW             |
| *1-DIN TCC LOW does not have Bluetooth® functionality. |                                   |                             |                     |

## Technical data

### Radio regulations

#### Manufacturers' declarations:

The complete texts of the declarations of conformity are available at the following internet address:

<https://www.mercedes-benz-trucks.com/conformity>

#### Radio-equipment approval of the tyre pressure monitoring system

##### Radio approval numbers

| Country   | Radio approval number                                                                                                      |
|-----------|----------------------------------------------------------------------------------------------------------------------------|
| Argentina | <br>TG6WU:<br>Numero de registro: H-25768 |

| Country | Radio approval number                                                                                                                                                                                                                                                                                                                 |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brazil  | <br><br>UL-BR 16.0721<br>Este equipamento opera em caráter secundário, isto é, não tem direito a proteção contra interferência prejudicial, mesmo de estações do mesmo tipo, e não pode causar interferência a sistemas operando em caráter primário |

| Country        | Radio approval number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| European Union | Hereby, Schrader Electronics Ltd declares that the radio equipment type TG6WU, TG6IECU, TG6ERX is in compliance with Directive 2014/53/EU.<br>The full text of the EU declaration of conformity is available at the following internet address: <a href="http://www.tpmseuroshop.com/documents/declaration_conformities">http://www.tpmseuroshop.com/documents/declaration_conformities</a><br>TG6WU: 433.92MHz<br>Maximum effective radiated power 10.691uW<br>TG6ERX: Category 2<br>Schrader Electronics Ltd. 11 Technology Park, Belfast Road, Antrim BT41 1QS, Northern Ireland United Kingdom |

## Technical data

### Radio regulations

| Country      | Radio approval number                                                                                                                                                                                                       |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Morocco      | TG6WU:<br>AGREE PAR L'ANRT MAROC<br>Numéro d'agrément: MR 12227<br>ANRT 2016<br>Date d'agrément: 2016-07-19<br>TG6ERX:<br>AGREE PAR L'ANRT MAROC<br>Numéro d'agrément: MR 12545<br>ANRT 2016<br>Date d'agrément: 2016-09-22 |
| Russia       |                                                                                                                                            |
| Serbia       |                                                                                                                                            |
| Singapore    | Complies with IMDA Standards<br>NO524-13<br>DA 105282                                                                                                                                                                       |
| South Africa | <br>TA-2020/6851<br>Approved                                                                                                               |
| Ukraine      | <br>109                                                                                                                                   |

| Country        | Radio approval number                                                             |
|----------------|-----------------------------------------------------------------------------------|
| United Kingdom |  |

### Radio operating permit for aerial amplifiers

#### Radio equipment approval numbers – RKE213E1

| Country or region | Radio approval number                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------|
| Argentina         | <br>Número de Inscripción: H-26206 |

| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brazil            | <br>3691-15-5298<br>Este equipamento não tem direito á proteção contra interferencia prejudicial a não pode causar interferencia em sistemas devidamente autorizados.<br>Para maiores informações, consulte o site da ANATEL:<br><a href="https://www.gov.br/anatel/pt-br">https://www.gov.br/anatel/pt-br</a> |

## Technical data

### Radio regulations

| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indonesia         | 76621/SDPPI/2021<br>7163<br><br>Dilarang melakukan perubahan spesifikasi yang dapat menimbulkan gangguan fisik dan/atau elektromagnetik terhadap lingkungan sekitarnya<br> |
| Israel            | 55-14267<br>Colmobil Israel<br>20 Ha - Amal Street<br>Afek Industrial Park<br>Rosh Ha - Ayin 48092                                                                                                                                                                                                                                           |
| Jordan            | TRC/LPD/2015/236                                                                                                                                                                                                                                                                                                                             |
| Malaysia          | <br>S00269                                                                                                                                                                                                                                                  |

| Country or region | Radio approval number                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------------|
| Morocco           | AGREE PAR L'ANRT MAROC<br>Numéro d'agrément: MR 10631<br>ANTR 2015<br>Date d'agrément: 16/07/2015           |
| Mexico            | IFETEL: RLVKARK15-1741                                                                                      |
| Oman              | TRA<br>R/2715/15<br>D090258                                                                                 |
| Zambia            | <br>ZM/ZICTA/TA /2019/3/11 |
| Serbia            | <br>W005 21                |
| Singapore         | Complies with IMDA Standards<br>Dealer's Licence No.: N2225-15                                              |
| South Africa      | <br>TA- 2015/1438          |

| Country or region          | Radio approval number                                                                                                                                                                                                                                                                                                                                             |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| South Korea                | B급 기기 (가정용 방송통신기자재) 이 기기는 가정용(B급) 전자파적합기기로서 주로 가정에서 사용하는 것을 목적으로 하며, 모든 지역에서 사용할 수 있습니다.This equipment is suitable for electromagnetic equipments for home (Class B) and it can be used in all areas.<br>해당 무선 설비는 운용 중 전파혼신 가능성이 있음 This radio equipment has possibility of radio interference during operation.<br>송신기 주파수: 433.92 MHz<br>Frequency: 433.92 MHz |
| Ukraine                    |                                                                                                                                                                                                                                                                                |
| United Arab Emirates (UAE) | TRA<br>Registered No: ER64693/18<br>Dealer: KATHREIN Automotive GmbH                                                                                                                                                                                                                                                                                              |
| United Kingdom             |                                                                                                                                                                                                                                                                              |



### Vehicle key radio operating permit

#### Radio operating permit numbers MS2

| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brazil            |  <p>00616-17-02930</p> <p>Este equipamento não tem direito á proteção contra interferencia prejudicial a não pode causar interferencia em sistemas devidamente autorizados.</p> <p>Para maiores informações, consulte o site da :<br/> <a href="https://www.gov.br/anatel/pt-br">https://www.gov.br/anatel/pt-br</a></p> |

| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indonesia         | <p>67372/SDPPI/2020<br/>7163</p>  <p>Dilarang melakukan perubahan spesifikasi yang dapat menimbulkan gangguan fisik dan/atau elektromagnetik terhadap lingkungan sekitarnya</p>  |
| Malaysia          |  <p>CIDF19000029</p>                                                                                                                                                                                                                                              |
| Morocco           | <p>AGREE PAR L'ANRT MAROC</p> <p>Numéro d'agrément: MR 13300</p> <p>ANRT 2017</p> <p>Date d'agrément: 2017-02-15</p>                                                                                                                                                                                                                               |
| Oman              | <p>TRA</p> <p>R/4136/17</p> <p>D080134</p>                                                                                                                                                                                                                                                                                                         |

| Country or region | Radio approval number                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------|
| Russia            |                                 |
| Zambia            |  <p>ZMB/ZICTA/TA/2018 /9/30</p> |
| Serbia            |                                 |
| Singapore         | <p>Complies with IMDA Standards</p> <p>Dealer's Licence No.: D103787</p>                                           |
| South Africa      |  <p>TA-2016/3314</p>            |

## Technical data

### Radio regulations

| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| South Korea       | B급 기기 (가정용 방송통신기자재) 이 기기는 가정용(B급) 전자파적합기기로서 주로 가정에서 사용하는 것을 목적으로 하며, 모든 지역에서 사용할 수 있습니다.This equipment is suitable for electromagnetic equipments for home (Class B) and it can be used in all areas.<br>해당 무선 설비는 운용 중 전파혼신 가능성이 있음 This radio equipment has possibility of radio interference during operation.<br>송신기 주파수: 433.92 MHz<br>Frequency: 433.92 MHz |
| Syria             | SyTRA<br>Registered No: FR00138-18<br>Dealer No: 1410086-16                                                                                                                                                                                                                                                                                                       |
| Ukraine           |                                                                                                                                                                                                                                                                                  |

| Country or region          | Radio approval number                                                             |
|----------------------------|-----------------------------------------------------------------------------------|
| United Arab Emirates (UAE) | TRA<br>Registered No: ER52668/17<br>Dealer No: 0018994/09                         |
| United Kingdom             |  |
| Belarus                    |  |

### Electronic ignition lock radio operating permit

#### Radio equipment approval numbers DAG16

| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brazil            | <br><br>01024-19-02930<br>Este equipamento não tem direito á proteção contra interferencia prejudicial a não pode causar interferencia em sistemas devidamente autorizados.<br>Para maiores informações, consulte o site da ANATEL:<br><a href="https://www.gov.br/anatel">https://www.gov.br/anatel</a> |

| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indonesia         | 82346/SDPPI/2022<br>9338<br><br>Dilarang melakukan perubahan spesifikasi yang dapat menimbulkan gangguan fisik dan/atau elektromagnetik terhadap lingkungan sekitarnya<br> |
| Jordan            | TRC/LPD/2018/252                                                                                                                                                                                                                                                                                                                             |
| Malaysia          | <br>CIDF19000029                                                                                                                                                                                                                                            |
| Morocco           | AGREE PAR L'ANRT MAROC<br>Numéro d'agrément: MR 16862<br>ANTR 2018<br>Date d'agrément: 18/06/2018                                                                                                                                                                                                                                            |
| Oman              | TRA<br>R/5785/18<br>D172249                                                                                                                                                                                                                                                                                                                  |

| Country or region | Radio approval number                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------|
| Zambia            | <br>ZM/ZICTA/TA/2018/8/19 |
| Singapore         | Complies with IMDA Standards<br>Dealer's Licence No.: D103787                                              |
| South Africa      | <br>TA- 2018/1781         |

| Country or region          | Radio approval number                                                                                                                                                                                                                                                                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| South Korea                | B급 기기 (가정용 방송통신기자재) 이 기기는 가정용(B급) 전자파적합기기로서 주로 가정에서 사용하는 것을 목적으로 하며, 모든 지역에서 사용할 수 있습니다.This equipment is suitable for electromagnetic equipments for home (Class B) and it can be used in all areas.<br>해당 무선 설비는 운용 중 전파혼신 가능성이 있음 This radio equipment has possibility of radio interference during operation.<br>송신기 주파수: 21.85 MHz<br>Frequency: 21.85 MHz |
| United Arab Emirates (UAE) | TRA<br>Registered No: ER67463/18<br>Dealer No: 0018994/09                                                                                                                                                                                                                                                                                                       |
| United Kingdom             |                                                                                                                                                                                                                                                                              |

## Technical data

### Radio regulations

#### Radio type approval for radar sensor system

#### Radio equipment approval numbers SRR2T L2C0055TR

Name and address of the manufacturer:  
Aptiv Services Deutschland GmbH, 42367  
Wuppertal, Germany

| Country or region                                                                                            | Radio approval number                                                                            |
|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Eurasian Economic Union<br> | —                                                                                                |
| Morocco                                                                                                      | AGREE PAR L'ANRT MAROC<br>Numéro d'agrément: MR 9929<br>ANTR 2014<br>Date d'agrément: 2014_12_26 |

| Country or region                                                                                      | Radio approval number |
|--------------------------------------------------------------------------------------------------------|-----------------------|
| Serbia<br><br>M011 15 | 34540-456/18-3        |

#### Radio equipment approval numbers ARS3 and ARS4

#### Radio equipment approval numbers ARS3-A

| Country or region                                                                              | Radio approval number |
|------------------------------------------------------------------------------------------------|-----------------------|
| Argentina<br> | C-22787               |

#### Radio equipment approval numbers ARS4-A

| Country or region                                                                               | Radio approval number |
|-------------------------------------------------------------------------------------------------|-----------------------|
| Argentina<br> | C-18005               |

| Country or region                                                                                | Radio approval number                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indonesia<br> | 94710/SDPPI/2023<br>9338<br>Dilarang melakukan perubahan spesifikasi yang dapat menimbulkan gangguan fisik dan/atau elektromagnetik terhadap lingkungan sekitarnya<br><a href="https://sertifikasi.postel.go.id/sertifikat/88460">https://sertifikasi.postel.go.id/sertifikat/88460</a> |

#### Radio approval numbers ARS4-B

| Country or region                                                                                 | Radio approval number |
|---------------------------------------------------------------------------------------------------|-----------------------|
| Argentina<br> | C-17908               |

### For Paraguay only:

Name and address of the manufacturer:

ADC Automotive Distance Control Systems GmbH Peter-Dornier-Strasse 10, 88131, Lindau, Germany

|                                                                                   |                |
|-----------------------------------------------------------------------------------|----------------|
|  |                |
| ARS4-A is registered with:                                                        | 2019-05-I-0271 |
| ARS4-B is registered with:                                                        | 2019-07-I-0353 |

### For Thailand only:

| Country or region | Radio approval number                                                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thailand          | <br>Missing values were not available at the time of going to print:<br>Radio equipment approval information: NBTC ID. A57007-14-XXXX |

### For Ukraine only:

The original (ADC Automotive Distance Control Systems GmbH) states that the two-way radio type fulfils the technical regulations for two-way radios; The full text of the declaration of conformity is available on the website at the following address:

<https://continental.automotive-approvals.com/>

Related EU declaration of conformity:  
 ADC Automotive Distance Control Systems GmbH hereby declares that two-way radio types ARS3-A/ARS4-A/ARS4-B conform to the requirements of Directive

2014/53/EU. The full text of the EU declarations of conformity are available at the following internet address:  
<https://continental.automotive-approvals.com/>

Name and address of the manufacturer:

ADC Automotive Distance Control Systems GmbH Peter-Dornier-Strasse 10, 88131, Lindau, Germany

|                                                                                     |             |
|-------------------------------------------------------------------------------------|-------------|
|  |             |
| Frequency band(s) in which the two-way radio is operated:                           | 76 - 77 GHz |

## Technical data

### Radio regulations

|                                                                                                                                          |                                                |
|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Maximum output of the radio frequency signal which is transmitted in the frequency range/ ranges in which the two-way radio is operated: | 3.16 W (35 dBm RMS effective radiation output) |
| ARS3-A is registered with:                                                                                                               | UA RF: 1CONT 0002                              |
| ARS4-A is registered with:                                                                                                               | UA RF: 1CONT 0004                              |
| ARS4-B is registered with:                                                                                                               | UA RF: 1CONT 0001                              |

#### Radio equipment approval numbers CR5TPCC

Name and address of the manufacturer:

Robert Bosch GmbH, Postfach 1661,  
71226 Leonberg Germany

| Country or region | Radio approval number                                                                                                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Algeria           | 029/H/ANF/2022<br>030/H/ANF/2022                                                                                                                                                                                                               |
| Argentina         | <br>H-27132                                                                                                                                                   |
| Armenia           | <br>AST-016/S.A-0029 -2023                                                                                                                                    |
| Botswana          | BOCRA/TA/2023/8050                                                                                                                                                                                                                             |
| Brazil            | 01303-22-03745<br>Este equipamento opera em caráter secundário, isto é, não tem direito a proteção contra interferência prejudicial, mesmo de estações do mesmo tipo, e não pode causar interferência a sistemas operando em caráter primário. |
| China             | CMIIT ID: 2022LJ7418                                                                                                                                                                                                                           |
| Israel            | 51-90540                                                                                                                                                                                                                                       |

| Country or region | Radio approval number                                                                                                                                                                                                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Japan             | 219-210020<br>Japanese Radio Law Compliance. This device is granted pursuant to the Japanese Radio Law (電波法) 日本 の電波法の遵守。この装置は、日本 の電波法 (電波法) に基づいて認可されています。 This device should not be modified (otherwise the granted designation number will become invalid) 本製品の改造は禁止されています。(適合証 明番号などが無効となります。) |

| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Canada            | 3887A-CR5TPCC<br>This device contains licence-exempt transmitter(s)/receiver(s) that comply with Innovation, Science and Economic Development Canada's licence-exempt RSS(s). Operation is subject to the following two conditions:(1) This device may not cause interference. (2) This device must accept any interference, including interference that may cause undesired operation of the device. |

| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Canada            | 3887A-CR5TPCC<br>L'émetteur/récepteur exempt de licence contenu dans le présent appareil est conforme aux CNR d'Innovation, Sciences et Développement économique Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes: (1) L'appareil ne doit pas produire de brouillage; (2) L'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement. |
| Malaysia          | RGLN/21A/1221/S(21-5761)<br>RGLN/42A/0323/S(23-1338)                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Morocco           | MR 00030786 ANRT 2021                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mexico            | RCPBOCR22-1798<br>La operación de este equipo está sujeta a las siguientes dos condiciones: (1) es posible que este equipo o dispositivo no cause interferencia perjudicial y (2) este equipo o dispositivo debe aceptar cualquier interferencia, incluyendo la que pueda causar su operación no deseada.                                                                                                                                                            |
| Moldova           | <br>MD OC TIP 024 CET7415-22<br>Prin prezenta, Robert Bosch GmbH declară că tipul de echipamente radio CR5TP-CC este în conformitate cu Reglementarea tehnică „Punerea la dispoziție pe piață a echipamentelor radio”. Ex-tul integral al declarației de conformitate este disponibil la următoarea adresă de Internet: <a href="https://ita.bosch.com">https://ita.bosch.com</a> |
| Nigeria           | CERT/00081/2023                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## Technical data

### Radio regulations

| Country or region | Radio approval number                                                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|
| Oman              | TRA/TA-R/12714/21                                                                                                     |
| Paraguay          | 2021-12-I-0738                                                                                                        |
| Zambia            | <br>ZMB / ZICTA / TA / 2023 / 1 / 39 |
| Serbia            | <br>I038 21<br>00814 01236           |
| Singapore         | N2149-23<br>Complies with IMDA Standards DA 108791                                                                    |
| South Africa      | <br>TA-2021/2953                     |
| South Korea       | R-C-BO2-CR5TPCC                                                                                                       |

| Country or region | Radio approval number                                                                                                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Taiwan            | CCAH22LP2620T1<br>警語 取得審驗證明之低功率射頻器材，非經核准，公司、商號或使用者均不得擅自變更頻率、加大功率或變更原設計之特性及功能。低功率射頻器材之使用不得影響飛航安全及干擾合法通信；經發現有干擾現象時，應立即停用，並改善至無干擾時方得繼續使用。前述合法通信，指依電信管理法規定作業之無線電通信。低功率射頻器材須忍受合法通信或工業、科學及醫療用電波輻射性電機設備之干擾。                                                                         |
| Thailand          | A57016-21<br>1) เครื่องโทรคมนาคมและอุปกรณ์นี้ มีความสอดคล้องตามมาตรฐานหรือข้อกำหนดของ กสทช. 2) เครื่องวิทยุคมนาคมนี้มีระดับการแผ่คลื่นแม่เหล็กไฟฟ้าสอดคล้องตามมาตรฐานความปลอดภัยต่อสุขภาพของมนุษย์จากการใช้เครื่องวิทยุคมนาคมที่คณะกรรมการกิจการโทรคมนาคมแห่งชาติประกาศกำหนด |

| Country or region | Radio approval number                                                                                                                                                                                                                                                                        |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ukraine           | 1BOSC0015<br>справжнім (Robert Bosch GmbH) заявляє, що тип радіобладнання (CR5TP-CC) відповідає Технічному регламенту радіобладнання; повний текст декларації про відповідність доступний на веб-сайті за такою адресою: <a href="http://ita.bosch.com/radar">http://ita.bosch.com/radar</a> |



| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| USA               | <p>NF3-CR5TPCC</p> <p>This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:</p> <ol style="list-style-type: none"> <li>1. this device may not cause harmful interference, and</li> <li>2. this device must accept any interference received, including interference that may cause undesired operation.</li> </ol> <p>Changes or modifications made to this equipment not expressly approved by Robert BOSCH GmbH may void the FCC authorization to operate this equipment.</p> |

| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| USA               | <p>NF3-CR5TPCC</p> <p>This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.</p> |

| Country or region | Radio approval number                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| USA               | <p>NF3-CR5TPCC</p> <p>Radiofrequency radiation exposure Information: This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with minimum distance of 20 cm between the radiator and your body. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.</p> |
| United Kingdom    | <p><b>UK</b><br/><b>CA</b><br/>U1177-210984<br/>1177</p>                                                                                                                                                                                                                                                                                                                                                     |

## Technical data

### Radio regulations

#### Radio approval for telematics platform

#### Radio equipment approval numbers CTP2019

| Country or region | Radio approval numbers (if available)                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------------|
| Argentina         | <br>C-25874                      |
| Australia         |                                  |
| Belarus           | <br>BY/112 11.01.TP024 030 01244 |

| Country or region | Radio approval numbers (if available)                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brazil            | <br><br>081817-20-12739<br>Este equipamento não tem direito á proteção contra interferencia prejudicial a não pode causar interferencia em sistemas devidamente autorizados. Para maiores informações, consulte o site da ANATEL: <a href="https://www.gov.br/anatel/pt-br">https://www.gov.br/anatel/pt-br</a> |
| Kazakhstan        |                                                                                                                                                                                                                                                                                                                 |
| Kyrgyzstan        |                                                                                                                                                                                                                                                                                                                 |
| Morocco           | AGREE PAR L'ANRT MAROC<br>Number of agreement: MR_21614 ANRT_2019<br>Date of agreement: 2019.12.03                                                                                                                                                                                                                                                                                               |

| Country or region    | Radio approval numbers (if available)                                                                         |
|----------------------|---------------------------------------------------------------------------------------------------------------|
| Mexico               | <br>IFETEL: NYC-CT176220C0 |
| Paraguay             | <br>RN:2020-01-I-0040      |
| Russia               |                            |
| Serbia               |                            |
| South Africa         | <br>TA-2020-7696           |
| Ukraine              |                            |
| United Arab Emirates |                            |
| United Kingdom       |                            |

### Radio equipment approval numbers CTP2019 from 10/2022

| Country or region | Radio approval numbers (if available)                                                                                                                                                                                                                                                                                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Argentina         | Approval ongoing                                                                                                                                                                                                                                                                                                                                                                                  |
| Australia         | Approval ongoing                                                                                                                                                                                                                                                                                                                                                                                  |
| Brazil            | <br>11523-23-12739<br>Este equipamento não tem direito á proteção contra interferencia prejudicial a não pode causar interferencia em sistemas devidamente autorizados.<br>Para maiores informações, consulte o site da ANATEL:<br><a href="https://www.gov.br/anatel/pt-br">https://www.gov.br/anatel/pt-br</a> |
| Morocco           | AGREE PAR L'ANRT MAROC<br>Number of agreement: MR00036129ANRT2023<br>Date of agreement 06/01/2023                                                                                                                                                                                                                                                                                                 |

| Country or region    | Radio approval numbers (if available)                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------|
| Serbia               | <br>RD11 23                   |
| South Africa         | <br>Approval activity ongoing |
| Ukraine              |                               |
| United Arab Emirates |                               |
| United Kingdom       |                               |

### Fluids and lubricants

#### Notes on operating fluids



**Danger**

#### Risk of injury from hazardous operating fluids

Operating fluids may be poisonous and harmful to your health.

- Observe the text on the original containers when using, storing or disposing of operating fluids.
- Always store operating fluids in their closed original containers.
- Always keep operating fluids away from children.



#### NOTE

#### Damage to major assemblies caused by additives

Using additives in approved operating fluids can damage major assemblies.

- Do not mix additives with the operating fluids.



#### ENVIRONMENTAL NOTE

#### Pollution of the environment due to irresponsible disposal of operating fluids

Incorrectly disposing of operating fluids can cause considerable damage to the environment.

- Dispose of operating fluids in an environmentally responsible manner.

## Technical data

### Fluids and lubricants

Operating fluids include:

- Lubricants
- Coolant
- Windscreen washer fluid
- Air conditioning system refrigerants

Only use products approved by Daimler Truck AG. Damage caused by the use of unsuitable operating fluids is not covered by the warranty.

Products approved by Daimler Truck AG are clearly marked on the containers.

You can obtain information about approved operating fluids and container designations from an authorised Mercedes-Benz Trucks Service Centre or on the Internet at:

<https://bevo.mercedes-benz-trucks.com>

#### Transmission oils



##### NOTE

##### **Damage to the drive axle caused by non-approved transmission oils**

- ▶ Only use approved transmission oils that comply with the Mercedes-Benz Trucks Specifications for Operating Fluids.

Observe the notes on operating fluids (→ Page 289).

At the factory, the transmission and the differential of the drive axle are filled with a high-quality synthetic oil.

Information on the usability of transmission oils can be found in the Mercedes-Benz Trucks Specifications for Operating Fluids.

#### Notes on coolant



##### **Danger**

##### **Risk of fire and injury from antifreeze agent**

If antifreeze agent comes into contact with hot component parts in the engine compartment, it may ignite.

- ▶ Allow the drive system to cool down before you top up the antifreeze agent.
- ▶ Make sure that no antifreeze agent spills out next to the filler opening.
- ▶ Thoroughly clean the antifreeze agent from component parts before starting the vehicle.

Observe the notes on operating fluids (→ Page 289).

A coolant that ensures anti-corrosion/antifreeze protection and other important protective effects is added at the factory.

Coolant is a mixture of water and anti-corrosion/antifreeze agent.

The anti-corrosion/antifreeze agent in the coolant has the following properties:

- Heat dissipation
- Anti-corrosion protection
- Cavity seal (protection against pitting)
- Antifreeze protection
- Increases the boiling point

Leave the coolant in the drive cooling system all year round - even in countries with high outside temperatures.

Check the anti-corrosion/antifreeze agent concentration in the coolant every six months.

Only use approved anti-corrosion/antifreeze agents.

If you replace the coolant, make sure that the coolant contains 50% anti-corrosion/antifreeze by volume. This provides anti-freeze protection down to temperatures of -37 °C.

Do not exceed a concentration of 55% by volume (antifreeze protection down to a temperature of around -45 °C). Heat

dissipation and antifreeze protection may otherwise deteriorate.

In the event of coolant loss, do not just refill with water, but also proportionally with an approved anti-corrosion/antifreeze agent.

The water used in coolant must meet specific requirements, which are often met by drinking water. Have the water treated if the water quality is not sufficient.

Mix the water and the anti-corrosion/antifreeze agent together before filling into the coolant circuit.

Avoid mixing different anti-corrosion/antifreeze agents.

### Notes on refrigerant

Your vehicle's climate control system is filled with refrigerant R-134a and contains fluorinated greenhouse gas.

The warning label for the refrigerant type used is located behind the maintenance flap.



### NOTE

#### Damage due to incorrect refrigerant or refrigerant compressor oil

If the incorrect refrigerant or refrigerant compressor oil (PAG oil) is used, this can damage the climate control system.

- ▶ Only use refrigerant R-134a and the PAG oil approved by Daimler Truck AG for your vehicle.
- ▶ Do not mix the approved PAG oil with another PAG oil.

Maintenance work, such as topping up the refrigerant or replacing components, must be carried out only by a qualified specialist workshop. All applicable regulations and SAE standard J639 must be complied with.

## Technical data

### Operating data

#### Refrigerant information label



- ① Symbols for hazard and service information
- ② Refrigerant filling capacity
- ③ CO<sub>2</sub> equivalent of the refrigerant used
- ④ Applicable standards
- ⑤ PAG oil part number
- ⑥ GWP (global warming potential) of the refrigerant used
- ⑦ Refrigerant type

The symbols ① indicate the following:

- Possible dangers
- The performance of maintenance work at a qualified specialist workshop

#### Operating data

##### Compressed-air system

| Minimum pressures                             | Pressure |
|-----------------------------------------------|----------|
| Brake circuit 1                               | 6,8 bar  |
| Brake circuit 2                               | 6,8 bar  |
| Brake circuit 3                               | 5,5 bar  |
| Transmission circuit                          | 5,5 bar  |
| Spring-loaded brake cylinder release pressure | 6,5 bar  |

| Supply pressures                                                       | Pressure      |
|------------------------------------------------------------------------|---------------|
| Service brake                                                          | 10.5–13,6 bar |
| Air suspension                                                         | 10.5–15,5 bar |
| External source of compressed air (charging the compressed-air system) | 11.0–12,5 bar |
| Remaining pressure circuits                                            | 7.0–8,7 bar   |

#### Drive system

##### Technical data of the drive axle

| Drive axle                 | Power output |
|----------------------------|--------------|
| Maximum output (temporary) | 400 kW       |
| Constant output            | 330 kW       |

##### Operating data (coolant temperature)

| Operating mode                                           | Coolant temperature, assemblies | Coolant temperature, high-voltage battery |
|----------------------------------------------------------|---------------------------------|-------------------------------------------|
| Normal operation                                         | approx. 60 °C                   | approx. 25 °C                             |
| Maximum permissible coolant temperature during operation | to 65 °C                        | to 50 °C                                  |

#### Notes on the energy supply for vehicles with a high-voltage battery

In accordance with European standard EN 17186:2019, vehicle identifiers are

located on the vehicle in the following places:

- On the inside of the socket flap
- On the charging cable connector

In addition, charging stations in European countries are equipped with energy supply identifiers. You can recharge your vehicle at charging stations when the charging station identifier corresponds to the following vehicle identifier.



- ① Supply type: DC  
Standard: EN 62196-3  
Design: FF  
Connector type: Charging cable socket and vehicle plug  
Voltage range: 50 V to 500 V
- ② Supply type: DC

Standard: EN 62196-3

Design: FF

Connector type: Charging cable socket and vehicle plug

Voltage range: 200 V to 920 V

For further information on charging the high-voltage battery, refer to the "Charging the high-voltage battery" section (→ Page 201).

### Technical data of the high-voltage battery

Your vehicle is equipped with three high-voltage batteries.

| High-voltage battery                    | 3 high-voltage batteries                   |
|-----------------------------------------|--------------------------------------------|
| Type                                    | Lithium-ion                                |
| Installed battery capacity <sup>1</sup> | 336 kWh (112 kWh per high-voltage battery) |
| Useable battery capacity <sup>2</sup>   | 291 kWh (97 kWh per high-voltage battery)  |

| High-voltage battery                                    | 3 high-voltage batteries |
|---------------------------------------------------------|--------------------------|
| Charging time – mode 4<br>With 80 kW charging capacity  | Approx. 2 h 30 min       |
| Charging time – mode 4<br>With 160 kW charging capacity | Approx. 1 h 15 min       |

<sup>1</sup> Installed battery capacity: Nominal capacity of a new battery, based on internally defined framework conditions. This can vary depending on the application and ambient conditions.

<sup>2</sup> Useable battery capacity: Useable energy content for regular truck operation with new batteries. This is based on internally defined conditions and may vary depending on the application and conditions.

Charging time – mode 4 applies to direct current charging from 20% to 80% of the usable energy content.

## Technical data

### Compressed-air reservoir

Low or high outside temperatures can increase charging times. In addition to the ambient temperature and the battery charge status, the charging capacity determines the charging time. Charging capacity depends on supply voltage, current intensity, and the type of power supply.

#### Tyre pressure

| Tyres                                                       | Pressure                           |
|-------------------------------------------------------------|------------------------------------|
| Tyre pressure table for single tyres                        | See the "Wheels and tyres" section |
| Tyre pressure table for twin tyres                          | See the "Wheels and tyres" section |
| Permissible difference in pressure between tyres on an axle | 20 kPa (0.2 bar/3 psi)             |
| Maximum permissible air pressure for inflating tyres        | 1000 kPa (10.0 bar/145 psi)        |

#### Wheel nut tightening torques

| Wheels                                                         | Torque |
|----------------------------------------------------------------|--------|
| Wheel nuts with pressure plate (hub centring) 10-hole mounting | 600 Nm |
| Wheel nuts with conical spring washers                         | 450 Nm |
| Connecting flange for twin tyres                               | 450 Nm |
| Wheel nut covers                                               | 60 Nm  |

#### Spring-loaded brake cylinder release torque/release pressure

| Spring-loaded brake cylinder                              | Torque/pressure |
|-----------------------------------------------------------|-----------------|
| Release torque of the release screw                       | Max. 70 Nm      |
| Tightening torque of the release screw                    | Max. 35 Nm      |
| Release pressure (with external source of compressed air) | Min. 6,5 bar    |

#### Level control system

|                                                                                 | Pressure      |
|---------------------------------------------------------------------------------|---------------|
| Fill the air suspension via connection 28 below the coolant expansion reservoir | Max. 12.5 bar |

### Compressed-air reservoir

#### Information on the compressed-air reservoir

#### Information for the initial purchaser and other users

Supporting documentation in accordance with Directive 2009/105/EC of the European Parliament and of the Council and in accordance with the technical standard EN 286-2.

|    |                                                                                                                                                                               |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a- | The reservoir is only intended for use in compressed-air systems and auxiliary equipment on motor vehicles and their trailers, and is only to be used to hold compressed air. |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



## Technical data

### Compressed-air reservoir

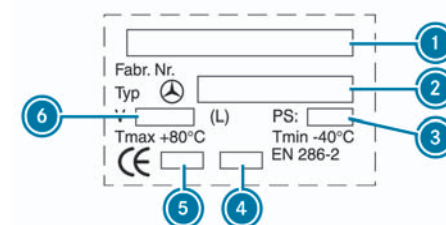
|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| b- | For the purposes of identification, the reservoir is marked with a factory no. and the manufacturer of the reservoir, as well as the key operating data and the CE mark, see the identification plate or inscriptions directly on the wall of the reservoir.                                                                                                                                                                                                                                                                                                                                                                                         |
| c- | The reservoir is manufactured with a "Declaration of conformity" in accordance with Article 12 of Directive 2009/105/EC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| d- | The reservoir is to be secured to the vehicle by tensioning straps (clamps). In the case of aluminium reservoirs, contact surfaces must be designed to inhibit corrosion or mechanical damage. Tensioning straps are to be positioned in such a way that they do not come in contact with the base connecting seams; the reservoir is not to be subjected to any stress that would jeopardise operating safety. Coatings applied to aluminium reservoirs must not contain lead, and the top coat of paint must only be applied over a suitable primer coat. Steel threaded connections for aluminium reservoirs must have a corrosion-proof coating. |

|    |                                                                                                                                                                    |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -  | Aluminium reservoirs are only to be cleaned using non-alkaline cleaning agents.                                                                                    |
| -  | The interior of the reservoir must be visible through the threaded connections.                                                                                    |
| -  | The reservoir must be emptied at regular intervals to prevent the accumulation of condensation (pull ring on drain valve at the lowest point of the reservoir).    |
| e- | The reservoir requires no maintenance if item d is complied with.                                                                                                  |
| f- | No welding, heat treatment or other operation relevant to safety is to be performed on the pressure-bearing walls of the reservoir (casing, base, threaded rings). |
| g- | The internal pressure of the reservoir may exceed the maximum operating pressure Ps by no more than 10% for a brief period.                                        |

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### Compressed-air reservoir identification plate

#### Aluminium reservoir

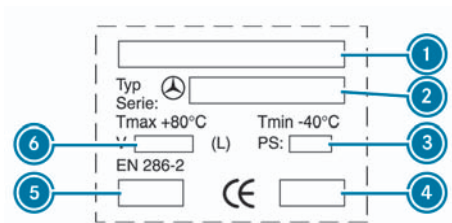


- ① Manufacturer
- ② -MB part number
- ③ Maximum operating pressure (bar)
- ④ Year of construction
- ⑤ Testing establishment code number
- ⑥ Volume (litres)

## Technical data

### Compressed-air reservoir

#### Steel reservoir



- ① Manufacturer
- ② -MB part number
- ③ Maximum operating pressure (bar)
- ④ Testing establishment code number
- ⑤ Year of construction
- ⑥ Volume (litres)

### Notes on display messages

#### Safety instructions

If you ignore warning and indicator lamps or display messages, you will not be able to identify failures and malfunctions in components or systems. Driving/braking characteristics may be affected and the reliability and road safety of your vehicle may be impaired. Have the affected system checked and repaired at a qualified workshop. Always observe warning lamps and display messages and take the corresponding action.

Operating information, fault messages and warnings will automatically be displayed in a display message by the instrument cluster. As well as the display message, an indicator lamp may light up in the status area (→ Page 83).

Depending on the priority of the event, the instrument cluster will display the display message in different colours:

- For a malfunction/message with low priority, the instrument cluster will display a **grey** display message. Ob-

serve the instructions in the display message.

Continuing the journey is possible.

- For a malfunction/message with medium priority, the instrument cluster will display a **yellow** display message.

The instrument cluster will also display a yellow display message for special operating states, e.g. if the tyre pressure is too low in one of the tyres.

Observe the instructions in the display message.

- For a high-priority malfunction, the instrument cluster will display a **red** display message.

The instrument cluster will show a red display message if the brake supply pressure is low, for example.

Observe the instructions in the display message.

Park the vehicle in a safe place as soon as possible and notify a qualified specialist workshop.

## Display messages

### Drive system

| Drive system                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Display messages                                                                                                                             | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                        |
|  <p>Charging cable inserted Start-off protection active</p> | <p>You have switched on the vehicle with the charging cable connector inserted. Start-off protection is active. The drive system cannot be started.</p> <p>► End the charging process ( → Page 208).</p>                                                                                                                                                            |
| Display messages                                                                                                                             | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                        |
|  <p>Charging cable inserted Start-off protection active</p> | <p>You have tried to release the parking brake with the charging cable connector inserted. Start-off protection is active. It is not possible to release the parking brake.</p> <p>► End the charging process ( → Page 208).</p>                                                                                                                                    |
| Display messages                                                                                                                             | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                        |
|  <p>Remaining range low (equipment-dependent)</p>           | <p>Additional text : Charge traction battery</p> <p>The yellow warning lamp  on the instrument cluster also lights up.</p> <p>► Charge the high-voltage battery ( → Page 201).</p>                |
| Display messages                                                                                                                             | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                        |
|  <p>Remaining range very low</p>                           | <p>Additional text : Charge traction battery</p> <p>The yellow warning lamp  on the instrument cluster also lights up.</p> <p>► Charge the high-voltage battery immediately ( → Page 201).</p> |

| Display messages                                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>High-voltage battery ageing advanced</p>              | <p>Additional text: <a href="#">Visit workshop.</a></p> <p>The high-voltage batteries are near the end of their life.</p> <ul style="list-style-type: none"> <li>► Have the high-voltage batteries checked at a qualified specialist workshop.</li> </ul>                                                                                                                                                                                                                                                                                                                                       |
| Display messages                                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  <p>Charge level of drive battery low</p>                 | <p>The yellow warning lamp  on the instrument cluster also lights up.</p> <p>The charge level of the high-voltage battery has dropped into the reserve range.</p> <ul style="list-style-type: none"> <li>► Charge the high-voltage battery ( → Page 201).</li> </ul>                                                                                                                                                                                                                                           |
| Display messages                                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  <p>Charge level of drive battery too low</p>             | <p>Additional text : <a href="#">Drive power greatly reduced.</a></p> <p>The yellow warning lamp  on the instrument cluster also lights up.</p> <p>The high-voltage battery is not charged sufficiently.</p> <p>Drive system power output is severely reduced.</p> <ul style="list-style-type: none"> <li>► Continue your journey carefully.</li> <li>► Charge the high-voltage battery immediately ( → Page 201).</li> </ul> |
| Display messages                                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  <p>High-voltage battery fault Limp Home mode active</p> | <p>Additional text : <a href="#">Visit workshop</a></p> <p>The yellow warning lamp  on the instrument cluster also lights up.</p> <p>High-voltage battery fault.</p> <p>The power output and range of the vehicle may be reduced.</p> <ul style="list-style-type: none"> <li>► Consult a qualified specialist workshop immediately.</li> </ul>                                                                                |

## Display messages

### Drive system

| Display messages                                                                                                                                       | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>High-voltage battery ageing critical Remaining time: XXX days</p> | <p>Additional text: Visit workshop immediately If ignored, veh. cannot be started in XXX days for safety reasons</p> <p>The yellow indicator lamp  in the instrument cluster also lights up.</p> <p>You have exceeded the maintenance due date of the high-voltage batteries.</p> <p>If you do not have the maintenance work carried out, it will no longer be possible to start the vehicle after the displayed calendar days have elapsed for safety reasons.</p> <ul style="list-style-type: none"> <li>► Have the high-voltage batteries checked at a qualified specialist workshop right away.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Display messages                                                                                                                                       | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  <p>High-voltage battery faulty Restart not possible</p>              | <p>Additional text : Consult a workshop without restarting Do not switch the gear to neutral.</p> <p>The red warning lamp  on the instrument cluster also lights up.</p> <p>A severe malfunction has occurred in the high-voltage battery.</p> <ul style="list-style-type: none"> <li>► Consult a qualified specialist workshop immediately; do not shift the transmission to position <b>N</b> in the process.</li> </ul> <p>or</p> <ul style="list-style-type: none"> <li>► Stop the vehicle immediately in accordance with the traffic conditions.</li> <li>► Contact a qualified specialist workshop.</li> </ul> <p>If the red warning lamp  on the instrument cluster also lights up:</p> <ul style="list-style-type: none"> <li>► Stop the vehicle immediately in accordance with the traffic conditions. Do not continue driving!</li> <li>► Contact a qualified specialist workshop.</li> </ul> |

| Display messages                                                                                                                                                | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>High-voltage battery faulty Driving mode not possible</p>                  | <p>Additional text : Stop the vehicle and switch off the engine Contact Service.</p> <p>The red warning lamp  on the instrument cluster also lights up.</p> <p>A severe malfunction has occurred in the high-voltage battery.</p> <p>Driving mode is not possible.</p> <ul style="list-style-type: none"> <li>► Contact a qualified specialist workshop.</li> </ul>                                                                                                                                                           |
|  <p>High-voltage battery fault Stop vehicle immediately</p>                    | <p>Additional text : Stop the vehicle and switch off the engine Contact Service.</p> <p>The red warning lamp  on the instrument cluster also lights up.</p> <p>A severe malfunction has occurred in the high-voltage battery.</p> <ul style="list-style-type: none"> <li>► Immediately stop safely and switch off the drive system. Do not continue driving!</li> <li>► Contact a qualified specialist workshop.</li> </ul>                                                                                                   |
|  <p>High-voltage system fault Immediately stop vehicle Get out of vehicle!</p> | <p>Additional text : Get out immediately! Remain outdoors.</p> <p>The high-voltage battery has overheated. There is a risk of fire!</p> <ul style="list-style-type: none"> <li>► Stop the vehicle immediately in accordance with the traffic conditions.</li> <li>► If possible, stop the vehicle in the open air and ensure that all vehicle occupants get out.</li> <li>► Do not continue driving.</li> <li>► If smoke is present, leave the danger zone and call the fire service immediately.</li> <li>► Consult a qualified specialist workshop even if there are no external signs of a fire.</li> </ul> |

## Display messages

### Drive system

| Display messages                                                                                                                         | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Power take-off fault</p>                            | <p>Additional text : <a href="#">Visit workshop</a></p> <p>The yellow warning lamp  on the instrument cluster also lights up.</p> <p>Electric power take-off system fault.</p> <ul style="list-style-type: none"> <li>► Consult a qualified specialist workshop immediately.</li> </ul>                                                                                                                                                                                                                                                                                                                     |
|  <p>High-voltage system faulty Restart not possible</p> | <p>Additional text : <a href="#">Consult a workshop without restarting Do not switch the gear to neutral.</a></p> <p>The yellow warning lamp  on the instrument cluster also lights up.</p> <p>The high-voltage system is faulty.</p> <ul style="list-style-type: none"> <li>► Consult a qualified specialist workshop immediately; do not shift the transmission to position <b>N</b> in the process.</li> </ul> <p>or</p> <ul style="list-style-type: none"> <li>► Stop the vehicle immediately in accordance with the traffic conditions.</li> <li>► Contact a qualified specialist workshop.</li> </ul> |
|  <p>High-voltage system malfunction</p>                 | <p>Additional text : <a href="#">Visit workshop</a></p> <p>The yellow warning lamp  on the instrument cluster also lights up.</p> <p>The high-voltage system is faulty.</p> <ul style="list-style-type: none"> <li>► Consult a qualified specialist workshop immediately.</li> </ul>                                                                                                                                                                                                                                                                                                                        |



| Display messages                                                                                                            | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>High-voltage system malfunction</p>    | <p>Additional text : <a href="#">Stop vehicle Contact Service Unable to continue driving.</a></p> <p>The red warning lamp  on the instrument cluster also lights up.</p> <p>The high-voltage system is faulty.</p> <p>Driving mode is not possible.</p> <p>► Contact a qualified specialist workshop.</p>                             |
| Display messages                                                                                                            | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|  <p>Electric drive fault</p>               | <p>Additional text : <a href="#">Visit workshop</a></p> <p>The yellow warning lamp  on the instrument cluster also lights up.</p> <p>The drive system is malfunctioning.</p> <p>The vehicle's power output may be reduced.</p> <p>► Consult a qualified specialist workshop immediately.</p>                                          |
| Display messages                                                                                                            | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|  <p>Electric drive fault</p>               | <p>Additional text : <a href="#">Stop vehicle Contact Service.</a></p> <p>The red warning lamp  on the instrument cluster also lights up.</p> <p>The drive system is malfunctioning.</p> <p>► Immediately stop safely and switch off the drive system. Do not continue driving!</p> <p>► Contact a qualified specialist workshop.</p> |
| Display messages                                                                                                            | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|  <p>Coolant levelDrive batteryToo low</p> | <p>The coolant level has dropped below the normal fill level.</p> <p>► Check the coolant level and top up with coolant if necessary ( → Page 225).</p> <p>► If the coolant level drops again, have the vehicle checked immediately at a qualified specialist workshop.</p>                                                                                                                                                                                                                              |

## Display messages

### Drive system

| Display messages                                                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Coolant levelElectric driveToo low                                    | <p>The coolant level has dropped below the normal fill level.</p> <ul style="list-style-type: none"><li>► Check the coolant level and top up with coolant if necessary ( → Page 225).</li><li>► If the coolant level drops again, have the vehicle checked immediately at a qualified specialist workshop.</li></ul>                                                                                                                                                                                                                                  |
| Display messages                                                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <br>High-voltage battery ageing exceeded Vehicle can no longer be started | <p>Additional text: <a href="#">Cannot start veh. for safety reasons</a> <a href="#">Contact customer service</a></p> <p>The red indicator lamp in the instrument cluster also lights up  .</p> <p>You have exceeded a due maintenance date of the high-voltage batteries by a significant amount of time.</p> <p>The vehicle cannot be started for safety reasons.</p> <ul style="list-style-type: none"><li>► Contact a qualified specialist workshop.</li></ul> |

### Operating fluids and maintenance

| Display messages                                                                  | Possible causes/consequences and ► solutions                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>The washer fluid level in the washer fluid reservoir for the windscreen washer/headlamp cleaning system has fallen to approximately one litre.</p> <ul style="list-style-type: none"> <li>► Top up the washer fluid reservoir ( → Page 226).</li> </ul> |

| Display messages                                                                                         | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Maintenance due</p> | <p>In addition to the display message, the indicator lamp  in the status area turns grey.</p> <p>A maintenance appointment is due.</p> <ul style="list-style-type: none"> <li>► Have the maintenance work carried out at a qualified specialist workshop.</li> </ul> |

| Display messages                                                                                                     | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Maintenance due immediately</p> | <p>In addition to the display message, the  indicator lamp in the status area also lights up yellow.</p> <p>A maintenance due date has been exceeded by a significant amount of time. This may result in damage to the vehicle and its major assemblies. It could also result in increased wear.</p> <ul style="list-style-type: none"> <li>► Have the maintenance work carried out immediately at a qualified specialist workshop.</li> </ul> |

## Display messages

### Brakes

#### Brakes

| Display messages                                                                                                                         | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Handling and braking characteristics affected</p>   | <p>Additional text : <a href="#">Go to workshop</a></p> <p>In addition, the  warning lamp on the instrument cluster lights up yellow.</p> <p>The vehicle's brake system is malfunctioning.</p> <div>  <b>Danger</b> <p><b>Risk of an accident due to a brake system malfunction</b></p> <p>If the brake system is malfunctioning, braking characteristics may be impaired.</p> <ul style="list-style-type: none"> <li>► Continue your journey carefully.</li> <li>► Have the brake system checked at a qualified specialist workshop immediately.</li> </ul> </div> |
|  <p>Braking force increased, pedal travel increased</p> | <p>Additional text : <a href="#">Stop vehicle Notify Service</a></p> <p>In addition, the warning lamp  lights up red on the instrument cluster.</p> <p>The vehicle's brake system is malfunctioning.</p> <p>Full braking power may not be available.</p> <p><b>WARNING!</b></p> <p>Driving and braking behaviour has changed. The operating and road safety of the vehicle is jeopardised.</p> <p>There is a risk of accident!</p> <ul style="list-style-type: none"> <li>► Carefully bring the vehicle to a standstill and park it safely.</li> <li>► Apply the parking brake.</li> <li>► Consult a qualified specialist workshop.</li> </ul>       |

| Display messages                                                                                                                              | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Continuous brake power reduced</p>                       | <p>The vehicle's recuperation power is reduced due to a high charge level in the high-voltage battery.</p> <ul style="list-style-type: none"> <li>► If necessary, use the service brake to brake the vehicle and adapt your driving style.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  <p>Continuous brake out of order</p>                        | <p>Continuous brake is out of order. Adequate recuperation no longer available.</p> <ul style="list-style-type: none"> <li>► Immediately have the continuous brakes checked at a qualified specialist workshop.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  <p>Brakes, axle 1°Maintenance due immediately°(example)</p> | <p>In addition to the display message, the  indicator lamp, shown with , lights up yellow in the status area.</p> <p>The service work due has not been performed.</p> <p>The wear limits of the brake linings and/or brake disks have been exceeded.</p> <div>  <b>Danger</b> </div> <p><b>Risk of accident due to impaired braking power</b></p> <p>When the brake pads have reached their wear limit, the braking power may be impaired.</p> <ul style="list-style-type: none"> <li>► Continue your journey carefully.</li> <li>► Have the brake system checked at a qualified specialist workshop immediately.</li> <li>► Visit a qualified workshop.</li> </ul> |

## Display messages

### Brakes

| Display messages                                                                                                                                                                                  | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   <p>Limited braking effect</p> | <p>Additional text : <a href="#">Adapt your driving style</a></p> <p>The temperature of one of the disk brakes on the towing vehicle is too high.</p> <p><b>WARNING!</b></p> <p>The disk brake may overheat. Driving/braking behaviour may change.</p> <p>There is a risk of accident!</p> <ul style="list-style-type: none"> <li>► Drive on carefully.</li> <li>► Shift to a lower gear.</li> <li>► Brake the vehicle with the continuous brake.</li> <li>► Depress the brake pedal in addition only if the braking power of the continuous brake is not sufficient.</li> </ul> |
| Display messages                                                                                                                                                                                  | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|  <p>Engage the parking brake</p>                                                                                 | <p>Vehicles with parametrisable special module (PSM): You have not applied the parking brake.</p> <p>You have not applied the parking brake before switching on the electric power take-off system.</p> <p><b>WARNING!</b></p> <p>The parked vehicle may roll away. You may endanger yourself and others.</p> <p>There is a risk of accident!</p> <ul style="list-style-type: none"> <li>► Apply the parking brake before switching on the electric power take-off system.</li> </ul>                                                                                                                                                                             |

### Compressed-air system and drive control

| Display messages                                                                                                               | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><a href="#">Drive control malfunction</a> | <p>Additional text : <a href="#">Go to workshop</a></p> <p>The electronic drive control is malfunctioning.</p> <ul style="list-style-type: none"> <li>► Observe the instructions in the display message.</li> </ul> |

| Display messages                                                                                                                                  | Possible causes/consequences and ► solutions                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><a href="#">Condensation in the compressed-air reservoir</a> | <p>The compressed-air dryer is malfunctioning.</p> <ul style="list-style-type: none"> <li>► Have the compressed-air dryer checked at a qualified specialist workshop.</li> </ul> |

## Display messages

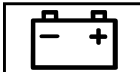
### Compressed-air system and drive control

| Display messages                                                                                                                                 | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Brake reservoir pressure in circuit 1 too low (example)</p> | <p>In addition, the warning lamp  lights up red on the instrument cluster.</p> <p>The supply pressure in brake circuit 1  or 2  is too low.</p> <p>If the supply pressure in the spring-loaded brake circuit is too low, the display message shows the  symbol.</p> <p>Possible causes:</p> <ul style="list-style-type: none"> <li>• Too much compressed air has been consumed.</li> <li>• There is a leak in the compressed-air system.</li> </ul> <p><b>WARNING!</b></p> <p>The operating and road safety of the vehicle is jeopardised.</p> <p>There is a risk of accident!</p> <ul style="list-style-type: none"> <li>► Park the vehicle safely as soon as possible.</li> <li>► Apply the parking brake.</li> <li>► Start the drive system. <ul style="list-style-type: none"> <li>◁ The compressed-air system is charged.</li> </ul> </li> </ul> <p>If the warning lamp  in the instrument cluster goes out:</p> <ul style="list-style-type: none"> <li>► Continue the journey.</li> </ul> <p>If the warning lamp  in the instrument cluster does not go out:</p> <ul style="list-style-type: none"> <li>► Check the compressed-air brake system for leaks ( → Page 129).</li> <li>► If the compressed-air brake system is not leaking, but the warning lamp  does not go out: Have the compressed-air brake system checked at a qualified specialist workshop.</li> </ul> |



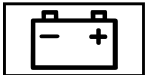
### Electrical system

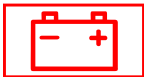
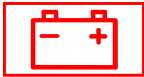
| Display messages                                                                                                                              | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Instrument cluster display and operation malfunction</p> | <p>The CAN connection to the instrument cluster has failed.</p> <p>The instrument cluster can no longer display information important for the operational and road safety of the vehicle.</p> <ul style="list-style-type: none"> <li>► Drive on carefully.</li> <li>► Have the instrument cluster checked at a qualified specialist workshop.</li> </ul> |

| Display messages                                                                                                   | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Energy-saving mode active</p> | <p>Supplementary text : <a href="#">Electrical consumers are switched off</a></p> <p>In addition to the display message, the indicator lamp  in the status area turns grey.</p> <p>The vehicle is switched off and the charge level of the 24 --V on-board electrical system battery is low</p> <p>Depending on the charge level of the 24 --V on-board electrical system battery, the following electrical consumers may be automatically switched off:</p> <ul style="list-style-type: none"> <li>• Audio device/subwoofer</li> <li>• 24 --V sockets/12 V sockets</li> <li>• Switch 1 for non-MB body (e.g. load compartment lamp)</li> <li>• Reading lamps/night lighting/rear stowage compartment lighting</li> <li>• Seat heating/USB charging function</li> <li>• Heating for outside mirrors/heating for compressed-air dryer</li> </ul> <p>For the following functions, the duration of actuation will be shortened:</p> <ul style="list-style-type: none"> <li>• Automatic activation of the interior lighting</li> <li>• Switch-off delay time of the exterior lighting</li> </ul> <p>► If necessary, start the vehicle.</p> <p>◁ When you start the vehicle, the electrical consumers will be available again.</p> |

## Display messages

### Electrical system

| Display messages                                                                                      | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Undervoltage</p> | <p>Supplementary text while driving : Stop vehicle, the driving characteristics have changed.</p> <p>Supplementary text if the vehicle is stationary : Charge the battery if possible, contact Service if the fault persists.</p> <p>The red indicator lamp  on the instrument cluster also lights up.</p> <p><b>WARNING!</b></p> <p>The vehicle's driving and braking characteristics may change.</p> <p>There is a risk of an accident if you do not adapt your driving style!</p> <ul style="list-style-type: none"> <li>► Stop vehicle.</li> <li>► Charge the battery ( → Page 232).</li> </ul> <p>If the malfunction persists:</p> <ul style="list-style-type: none"> <li>► Contact a qualified specialist workshop.</li> </ul> |
|  <p>Overvoltage</p>  | <p>Additional text : Stop the vehicle Switch off the engine Contact Service</p> <p>The voltage in the 24 V-on-board electrical system batteries is too high.</p> <p><b>WARNING!</b></p> <p>The vehicle's driving characteristics may change. The operating and road safety of the vehicle is jeopardised.</p> <p>There is a risk of accident!</p> <ul style="list-style-type: none"> <li>► Stop the vehicle safely as soon as possible.</li> <li>► Switch off the vehicle.</li> <li>► Contact a qualified specialist workshop.</li> </ul>                                                                                                                                                                                                                                                                                                                                                               |

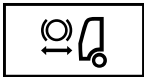
| Display messages                                                                                                                | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Unable to detect starter battery fault</p> | <p>Additional text : Visit workshop Starter battery fault Electric motor may potentially no longer start</p> <p>The red indicator lamp  on the instrument cluster also lights up.</p> <p>24 V on-board electrical system battery fault or state of charge of 24 V on-board electrical system battery very low. You may no longer be able to switch on the drive system if you switch it off.</p> <ul style="list-style-type: none"> <li>► Stop the vehicle immediately in accordance with the traffic conditions. Do not continue driving!</li> <li>► Contact a qualified specialist workshop.</li> </ul> |
|  <p>Starter battery failed</p>                 | <p>Additional text : Visit workshop Starter battery unavailable Unable to start electric motor</p> <p>The red indicator lamp  on the instrument cluster also lights up.</p> <p>24 V on-board electrical system battery fault or state of charge of 24 V on-board electrical system battery very low. You may no longer be able to switch on the drive system if you switch it off.</p> <ul style="list-style-type: none"> <li>► Stop the vehicle immediately in accordance with the traffic conditions. Do not continue driving!</li> <li>► Contact a qualified specialist workshop.</li> </ul>           |
|  <p>Starter battery not charging</p>           | <p>Additional text : Discharging starter battery Stop vehicle Contact Service</p> <p>The red indicator lamp  on the instrument cluster also lights up.</p> <p>Not charging 24 V on-board electrical system battery or high-voltage on-board electrical system fault.</p> <ul style="list-style-type: none"> <li>► Immediately stop safely and switch off the drive system. Do not continue driving!</li> <li>► Contact a qualified specialist workshop.</li> </ul>                                                                                                                                        |

## Display messages

### Driving systems and steering

#### Driving systems and steering

| Display messages                                                                                                         | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>ESP unavailable</p>                 | <p>Additional text : <a href="#">Go to workshop</a></p> <p>In addition, the warning lamp  lights up on the instrument cluster.</p> <p>Stability Control Assist is malfunctioning.</p> <div style="background-color: #f0f0f0; padding: 10px; margin-top: 10px;"> <p> <b>Danger</b></p> <p><b>Risk of skidding if ESP is malfunctioning</b></p> <p>If ESP® is malfunctioning, ESP® cannot perform vehicle stabilisation.</p> <ul style="list-style-type: none"> <li>► Continue your journey carefully.</li> <li>► Have ESP® checked at a qualified specialist workshop.</li> </ul> </div> |
| Display messages                                                                                                         | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  <p>ESP deactivatedSet normal level</p> | <p>In addition, the warning lamp  lights up on the instrument cluster. If the chassis is not at driving level while driving, Stability Control Assist is deactivated.</p> <p><b>WARNING!</b></p> <p>Driving/braking behaviour may change.</p> <p>There is a risk of an accident if you do not adapt your driving style!</p> <ul style="list-style-type: none"> <li>► Set driving level ( → Page 193).</li> </ul>                                                                                                                                                                                                                                                                                                                                          |

| Display messages                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Emergency braking completed</p>       | <p>Active Brake Assist has automatically triggered an emergency braking manoeuvre (maximum full-stop braking) and the emergency braking manoeuvre has been completed.</p> <ul style="list-style-type: none"> <li>► Remove the vehicle as quickly as possible from the danger zone, paying attention to road and traffic conditions as you do so.</li> <li>► Stop the vehicle, paying attention to road and traffic conditions.</li> <li>► Make sure that the vehicle is in proper working order and that the load is secured properly.</li> </ul>              |
|  <p>Active Brake Assist not available</p> | <p>Additional text: <a href="#">Visit workshop Frequent emergency braking</a></p> <p><b>WARNING!</b></p> <p>If Active Brake Assist is not available, you will not receive any collision warnings. In critical situations, the vehicle will not brake automatically.</p> <p>There is a risk of an accident if you do not adapt your driving style!</p> <ul style="list-style-type: none"> <li>► If necessary, use the service brake to brake the vehicle.</li> <li>► Have the Active Brake Assist system checked at a qualified specialist workshop.</li> </ul> |
|  <p>Assistance systems malfunction</p>    | <p>Additionally, the indicator lamp in the instrument cluster lights up yellow .</p> <p><b>WARNING!</b></p> <p>Driver assistance/driving safety systems may be malfunctioning or temporarily unavailable.</p> <p>There is a risk of accident!</p> <ul style="list-style-type: none"> <li>► Drive on carefully.</li> <li>► Have the vehicle checked by a qualified specialist workshop.</li> </ul>                                                                           |

## Display messages

### Driving systems and steering

| Display messages                                                                                                                                             | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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|  <p>Assistance systems malfunction Distance sensor viewing area blocked</p> | <p>Additional text: <a href="#">Clean sensors</a></p> <p>In addition, the corresponding indicator lamps of malfunctioning driver assistance/driving safety systems in the instrument cluster may light up.</p> <p>There is a malfunction in the distance sensors.</p> <p>Possible causes:</p> <ul style="list-style-type: none"> <li>• Contamination of the sensor system</li> <li>• Heavy rainfall</li> <li>• Longer journey next to uniform structures close to the carriageway, e.g. tunnel walls</li> </ul> <p><b>WARNING!</b></p> <p>Driver assistance/driving safety systems may be malfunctioning or temporarily unavailable.</p> <p>There is a risk of accident!</p> <p>► Drive on carefully.</p> <p>If the causes are no longer present, the driver assistance and driving safety systems are activated again. The corresponding indicator lamps and the display message in the instrument cluster go out.</p> <p>If the display message does not disappear:</p> <p>► Stop the vehicle in accordance with the traffic conditions.</p> <p>► Clean the covers of the distance sensors on the front bumper and the side mud guards with water ( → Page 219).</p> <p>◁ Do not use any dry, rough or hard cloths and do not scrub or scratch.</p> <p>► Restart the vehicle.</p> <p>If the display message does not disappear:</p> <p>► Have the vehicle checked at a qualified specialist workshop immediately.</p> |

| Display messages                                                                                                                                          | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Assistance systems malfunction Front camera viewing area blocked</p> | <p>Additional text: <a href="#">Clean windscreen</a></p> <p>In addition, the corresponding indicator lamps of malfunctioning driver assistance/driving safety systems in the instrument cluster may light up.</p> <p>The view of the multifunction camera is reduced.</p> <p>Possible causes:</p> <ul style="list-style-type: none"> <li>• Dirt on windscreen in field of view of multifunction camera</li> <li>• Heavy rain or fog</li> <li>• The windscreen is steamed up on the inside: Under certain weather conditions, condensation may form on the inside of the windscreen, especially during the winter months.</li> </ul> <p><b>WARNING!</b></p> <p>Driver assistance/driving safety systems may be malfunctioning or temporarily unavailable.</p> <p>There is a risk of accident!</p> <p>► Drive on carefully.</p> <p>If the causes are no longer present, the driver assistance and driving safety systems are activated again. The corresponding indicator lamps and the display message in the instrument cluster go out.</p> <p>If the display message does not disappear:</p> <p>► Stop the vehicle in accordance with the traffic conditions.</p> <p>► Clean the area of the windscreen where the camera is located ( → Page 219).</p> <p>► Restart the vehicle.</p> <p>If the display message does not disappear:</p> <p>► Have the vehicle checked at a qualified specialist workshop immediately.</p> |

## Display messages

### Driving systems and steering

| Display messages                                                                                                                                      | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Assistance systems malfunction Front camera temperature too high | <p>Additional text: Reduce interior temperature</p> <p>Multifunction camera temperature too high</p> <p><b>If the display message appears while the vehicle is in motion:</b></p> <p>The multifunction camera is malfunctioning.</p> <p>In addition, the corresponding indicator lamps of malfunctioning driver assistance/driving safety systems in the instrument cluster may light up.</p> <p><b>WARNING!</b></p> <p>Driver assistance/driving safety systems may be malfunctioning or temporarily unavailable.</p> <p>There is a risk of accident!</p> <ul style="list-style-type: none"><li>► Drive on carefully.</li><li>► Have the multifunction camera checked at a qualified specialist workshop immediately.</li></ul> <p><b>If the display message appears when the vehicle is parked:</b></p> <ul style="list-style-type: none"><li>► Reduce the interior temperature, e.g. by ventilating.</li><li>► Wait until the multifunction camera has cooled down and the display message disappears.</li></ul> |



| Display messages                                                                                                                     | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Lane Keeping Assist not available (example)</p> | <p>Additional text : <a href="#">Visit workshop Camera calibration faulty</a></p> <p><b>WARNING!</b></p> <p>The following driver assistance systems/driving safety systems could be impaired or no longer operational:</p> <ul style="list-style-type: none"> <li>• Lane Keeping Assist</li> <li>• Attention Assist</li> <li>• Traffic Sign Assist</li> <li>• Active Brake Assist</li> <li>• Proximity Control Assist</li> <li>• Sideguard Assist</li> <li>• Frontguard Assist</li> </ul> <p>There is a risk of accident!</p> <p>► Have the affected driver assistance system/driving safety system checked at a qualified specialist workshop.</p> |

| Display messages                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Lane Keeping Assist not available</p> | <p>Additional text : <a href="#">Visit workshop, Lane Keeping Assist and Attention Assist unavailable</a></p> <p><b>WARNING!</b></p> <p>If Lane Keeping Assist and Attention Assist are not available, you will not receive a warning from the systems.</p> <p>There is a risk of accident!</p> <p>► Have Lane Keeping Assist and Attention Assist checked at a qualified specialist workshop.</p> |

## Display messages

### Driving systems and steering

| Display messages                                                                                                         | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Lane Keeping Assist malfunction</p> | <p>Additional text : <a href="#">Visit workshop, Lane Keeping Assist and Attention Assist unavailable</a></p> <p><b>WARNING!</b></p> <p>If Lane Keeping Assist and Attention Assist are not available, you will not receive a warning from the systems. There is a risk of accident!</p> <p>► Have Lane Keeping Assist and Attention Assist checked at a qualified specialist workshop.</p>                                                                                                                                                                                                                                                                           |
|  <p>Lane Keeping Assist malfunction</p> | <p>Additional text : <a href="#">Consult a workshop Left Lane Keeping Assist speaker malfunction</a> or <a href="#">Consult a workshop Right Lane Keeping Assist speaker malfunction</a></p> <p><b>WARNING!</b></p> <p>The left or right speaker is malfunctioning.</p> <p>If a speaker is malfunctioning on one side, Lane Keeping Assist will issue a warning tone from the intact speaker, regardless of the side. It is possible that in this situation, a warning tone may sound on the side opposite to the affected side of the vehicle.</p> <p>There is a risk of accident!</p> <p>► Have Lane Keeping Assist checked at a qualified specialist workshop.</p> |
|  <p>Attention Assist: Take a break?</p> | <p>Attention Assist has detected tiredness or increasing lapses in concentration.</p> <p>A warning tone will also sound.</p> <p>► If necessary, take a break.</p> <p>During long journeys, take regular breaks in good time.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Display messages                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Setting the driving level</p>         | <p>The chassis frame is not at driving level. Active Brake Assist is inoperative.</p> <ul style="list-style-type: none"> <li>► Set the driving level ( → Page 193).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Display messages                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|  <p>Steering characteristics affected</p> | <p>Additional text : <a href="#">Visit workshop</a>.</p> <p>The power steering has detected a malfunction. The steering behaviour may change.</p> <ul style="list-style-type: none"> <li>► Continue your journey carefully.</li> <li>► Stop the vehicle in a secure manner at the next opportunity.</li> <li>► Switch off the vehicle.</li> <li>► Restart the vehicle after ten seconds. <ul style="list-style-type: none"> <li>◁ The display message will disappear.</li> </ul> </li> </ul> <p>When the display message is shown again:</p> <ul style="list-style-type: none"> <li>► Have the power steering checked at a qualified specialist workshop.</li> </ul> |

## Display messages

### Driving systems and steering

| Display messages                                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Additional axle steering characteristics affected</p> | <p>In addition, the  warning lamp on the instrument cluster lights up yellow.</p> <p>The steered additional axle is malfunctioning and will no longer steer actively.</p> <p><b>WARNING!</b></p> <p>Driving/steering characteristics may change.</p> <p>There is a risk of an accident if you do not adapt your driving style!</p> <ul style="list-style-type: none"> <li>► Drive on carefully and stop at the next available opportunity. Adjust your driving style to the changed handling and steering characteristics.</li> <li>► Stop the vehicle and switch it off.</li> <li>► Restart the vehicle after approximately ten seconds. <ul style="list-style-type: none"> <li>◁ The display message will disappear. The steered additional axle is reactivated.</li> </ul> </li> <li>► If the display message does not disappear: Have the steered additional axle checked at a qualified specialist workshop.</li> </ul> |

| Display messages                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Steering characteristics affected</p> | <p>Additional text : Do not steer when the vehicle is at standstill Steer carefully when driving at walking pace or Have fault repaired at next service</p> <p>The power steering has detected a malfunction. The feel of the steering may change.</p> <p>If you steer abruptly when manoeuvring, the steering may go into emergency operation mode.</p> <ul style="list-style-type: none"> <li>► Stop the vehicle, paying attention to road and traffic conditions.</li> <li>► Switch off the vehicle.</li> <li>► Restart the vehicle.</li> <li>► If the error is displayed again or still displayed: Have the vehicle checked by a qualified specialist workshop.</li> </ul> |

| Display messages                                                                                                                    | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Steering characteristics affected</p>          | <p>Additional text : <a href="#">Visit workshop</a>.</p> <p>The power steering has detected a malfunction. The steering behaviour may change.</p> <ul style="list-style-type: none"> <li>► Continue your journey carefully.</li> <li>► Stop the vehicle in a secure manner at the next opportunity.</li> <li>► Switch off the vehicle.</li> <li>► Restart the vehicle after ten seconds. <ul style="list-style-type: none"> <li>◁ The display message will disappear.</li> </ul> </li> </ul> <p>When the display message is shown again:</p> <ul style="list-style-type: none"> <li>► Have the power steering checked at a qualified specialist workshop.</li> </ul>                                                                             |
|  <p>Do not turn the steering wheel to the stop</p> | <p>Additional text : <a href="#">Active steering wheel return out of order</a> or <a href="#">Have fault repaired at next service</a></p> <p>A fault has occurred in the steering control unit. The vehicle's driving characteristics (feel) may change. The steering wheel will not return to the straight-ahead position.</p> <p>Do not turn the steering wheel to the stop. Otherwise, the steering could overheat.</p> <ul style="list-style-type: none"> <li>► Stop the vehicle, paying attention to road and traffic conditions.</li> <li>► Switch off the vehicle.</li> <li>► Restart the vehicle.</li> <li>► If the error is displayed again or still displayed: Have the vehicle checked by a qualified specialist workshop.</li> </ul> |

## Display messages

### Driving systems and steering

| Display messages                                                                                                       | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Leading/trailing axle centred</p> | <p>The steered additional axle is malfunctioning and has been centred automatically. It is deactivated and no longer steers actively.</p> <ul style="list-style-type: none"> <li>► Drive on carefully and stop at the next available opportunity. Adjust your driving style to the changed handling and steering characteristics.</li> <li>► Stop the vehicle and switch it off.</li> <li>► Restart the vehicle after approximately ten seconds. <ul style="list-style-type: none"> <li>◄ The display message will disappear. The steered additional axle is reactivated.</li> </ul> </li> <li>► If the display message does not disappear: Have the steered additional axle checked at a qualified specialist workshop.</li> </ul> |

| Display messages                                                                                                              | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Power steering: Function not ensured</p> | <p>Additional text : <a href="#">Visit workshop.</a></p> <p>Power steering has failed. Only reduced power steering is available.</p> <ul style="list-style-type: none"> <li>► Continue your journey carefully.</li> <li>► Stop the vehicle in a secure manner at the next opportunity.</li> <li>► Switch off the vehicle.</li> <li>► Restart the vehicle after ten seconds. <ul style="list-style-type: none"> <li>◄ The display message will disappear.</li> </ul> </li> </ul> <p>When the display message is shown again:</p> <ul style="list-style-type: none"> <li>► Have the power steering checked at a qualified specialist workshop.</li> </ul> |

| Display messages                                                                                                                     | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Additional axle steering oil level too low</p> | <p>Additional text : <a href="#">Visit workshop.</a></p> <p>The steering oil level is too low for the steered additional axle.</p> <ul style="list-style-type: none"> <li>► Have the steering oil level for the steered additional axle checked in a qualified specialist workshop and, if necessary, corrected.</li> </ul> |

| Display messages                                                                                                              | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Power steering: Function not ensured</p> | <p>Additional text : Stop vehicle or Start engine.</p> <p>In addition, the warning lamp  lights up red on the instrument cluster.</p> <ul style="list-style-type: none"> <li>► Stop vehicle.</li> </ul> <p>or</p> <ul style="list-style-type: none"> <li>► Start the vehicle.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                     |
|  <p>Power steering: Function not ensured</p> | <p>Additional text : Stop vehicle Check hydraulic steering when manoeuvring Contact Service.</p> <p>In addition, the warning lamp  lights up red on the instrument cluster.</p> <p>Steering operates in emergency operation mode.</p> <p><b>WARNING!</b></p> <p>The hydraulic power steering has failed.</p> <p>There is a risk of accident!</p> <ul style="list-style-type: none"> <li>► Stop the vehicle safely as soon as possible.</li> <li>► Switch off the vehicle.</li> <li>► Restart the vehicle.</li> </ul> <p>If the error is displayed again or still displayed:</p> <ul style="list-style-type: none"> <li>► Contact a qualified specialist workshop.</li> </ul> |

## Display messages

### Driving systems and steering

| Display messages                                                                                                              | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Power steering: Function not ensured</p> | <p>Additional text : <a href="#">Stop vehicle Contact Service</a>.</p> <p>In addition, the warning lamp  lights up red on the instrument cluster.</p> <p>A fault has occurred in the electric power steering.</p> <p><b>WARNING!</b></p> <p>Power steering is not ensured continuously.</p> <p>There is a risk of accident!</p> <ul style="list-style-type: none"> <li>► Stop the vehicle safely as soon as possible.</li> <li>► Switch off the vehicle.</li> <li>► Restart the vehicle.</li> </ul> <p>If the error is displayed again or still displayed:</p> <ul style="list-style-type: none"> <li>► Contact a qualified specialist workshop.</li> </ul> |
|  <p>Steering characteristics affected</p>    | <p>Additional text : <a href="#">Visit workshop</a>.</p> <p>The power steering has detected a malfunction. The steering behaviour may change.</p> <ul style="list-style-type: none"> <li>► Continue your journey carefully.</li> <li>► Stop the vehicle in a secure manner at the next opportunity.</li> <li>► Switch off the vehicle.</li> <li>► Restart the vehicle after ten seconds. <ul style="list-style-type: none"> <li>◁ The display message will disappear.</li> </ul> </li> </ul> <p>When the display message is shown again:</p> <ul style="list-style-type: none"> <li>► Have the power steering checked at a qualified specialist workshop.</li> </ul>                                                                         |



| Display messages                                                                                                   | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Power steering overheated</p> | <p>Additional text : Stop vehicle Switch off engine.</p> <p>In addition, the warning lamp  lights up red on the instrument cluster.</p> <p>The steering gear has overheated.</p> <ul style="list-style-type: none"> <li>► Stop the vehicle safely as soon as possible.</li> <li>► Switch off the vehicle.</li> <li>► Let the steering gear cool down.</li> <li>► Restart the vehicle.</li> </ul> <p>If the fault is displayed again:</p> <ul style="list-style-type: none"> <li>► Contact a qualified specialist workshop.</li> </ul> |

## Display messages

### Driving systems and steering

| Display messages                                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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|  <p>Additional axle steering characteristics affected</p> | <p>Additional text : <a href="#">Visit workshop</a>.</p> <p>The steered additional axle is malfunctioning and is only steering passively. Stability Control Assist may also have been deactivated as a result. A yellow display message indicating  <a href="#">ESP unavailable</a> shows that Stability Control Assist has been switched off.</p> <p>In extreme driving conditions, e.g. when braking suddenly on a slippery or uneven carriageway, the steered additional axle may deactivate itself. It will then only steer passively. On vehicles with a height-adjustable trailing axle, it is no longer possible to lower the additional axle while the vehicle is moving.</p> <p><b>WARNING!</b></p> <p>If Stability Control Assist is deactivated due to a faulty steered additional axle, the vehicle's stability is decreased. There is a risk of accident!</p> <ul style="list-style-type: none"> <li>► Drive on carefully and stop at the next opportunity.             <ul style="list-style-type: none"> <li>◁ Adjust your driving style to the changed handling and steering characteristics.</li> </ul> </li> <li>► Stop the vehicle and switch it off.</li> <li>► Restart the vehicle after approximately ten seconds.             <ul style="list-style-type: none"> <li>◁ The display message will disappear. The steered additional axle is reactivated.</li> </ul> </li> <li>► If the display message does not disappear: Have the steered additional axle checked at a qualified specialist workshop.</li> </ul> |

### Tyres

| Display messages                                                                                                                  | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Sp. wheel tyre press. sens.: battery low</p> | <p>Additional text: <a href="#">Replace tyre pressure sensor.</a></p> <p>In addition to the display message, the grey indicator lamp  in the instrument cluster lights up.</p> <p>The battery of the tyre pressure sensor in the spare wheel is flat.</p> <ul style="list-style-type: none"> <li>► Have the tyre pressure sensor replaced at a qualified specialist workshop.</li> </ul>                                                                                              |
|  <p>Tyre pressure in spare wheel too low</p>     | <p>Additional text: <a href="#">Check and correct tyre pressure..</a></p> <p>In addition to the display message, the grey indicator lamp  in the instrument cluster lights up.</p> <p>The tyre pressure in the spare wheel is too low.</p> <ul style="list-style-type: none"> <li>► Check the tyre pressure and correct it if required.</li> </ul>                                                                                                                                    |
|  <p>Spare wheel tyre pressure monitor faulty</p> | <p>Additional text: <a href="#">Spare wheel tyre pressure monitor failure.</a></p> <p>In addition to the display message, the grey indicator lamp  in the instrument cluster lights up.</p> <p>The tyre pressure monitor of the spare wheel is malfunctioning.</p> <ul style="list-style-type: none"> <li>► Have the tyre pressure sensor replaced at a qualified specialist workshop.</li> </ul>                                                                                     |
|  <p>Tyre pressure too low</p>                   | <p>Additional text: <a href="#">Check and correct tyre pressure..</a></p> <p>The yellow  indicator lamp in the instrument cluster also lights up.</p> <p>The tyre pressure in one or more of the tyres is too low.</p> <p>The corresponding tyres are displayed in yellow or red in the <a href="#">Status</a> menu window of <a href="#">Tyres</a> the multimedia system.</p> <ul style="list-style-type: none"> <li>► Check the tyre pressure and correct it if required.</li> </ul> |

## Display messages

### Tyres

| Display messages                                                                                                   | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Tyre pressure too high</p>    | <p>Additional text: <a href="#">Check and correct tyre pressure..</a></p> <p>The yellow  indicator lamp in the instrument cluster also lights up.</p> <p>The pressure in one or more tyres has risen significantly while driving.</p> <p>The corresponding tyres are displayed in colour in the <a href="#">Status</a> menu window of <a href="#">Tyres</a> the multimedia system.</p> <ul style="list-style-type: none"> <li>► Stop the vehicle without steering or braking suddenly. Pay attention to the traffic conditions.</li> <li>► Check the tyre pressure and correct it if required.</li> </ul>                                                                                                                                                                                                        |
|  <p>Tyre temperature too high</p> | <p>Additional text: <a href="#">Reduce speed drastically..</a></p> <p>The yellow  indicator lamp in the instrument cluster also lights up.</p> <p><b>WARNING!</b></p> <p>The temperature in one or more tyres has risen significantly while driving.</p> <ul style="list-style-type: none"> <li>• The brakes may have overheated.</li> <li>• The tyre pressure may be too low.</li> </ul> <p>Driving/braking behaviour may change.</p> <p>There is a risk of accident!</p> <ul style="list-style-type: none"> <li>► Drive on slowly. <ul style="list-style-type: none"> <li>◁ The tyre temperature is lowered by the airflow.</li> </ul> </li> <li>► If the tyre temperature has fallen, have the tyres and brakes checked and the cause of the problem rectified at a qualified specialist workshop.</li> </ul> |

| Display messages                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Tyre pressure sensor: battery low</p> | <p>Additional text: <b>Replace tyre pressure sensor.</b></p> <p>The yellow  indicator lamp in the instrument cluster also lights up.</p> <p>The battery of a tyre pressure sensor is discharged.</p> <ul style="list-style-type: none"> <li>► Have the tyre pressure sensor replaced at a qualified specialist workshop.</li> </ul>                                                                                                                                                                                                                                                                                         |
|  <p>Tyre pressure sensor faulty</p>       | <p>Additional text: <b>Replace tyre pressure sensor</b> or <b>Failure, tyre pressure monitor for one/multiple tyre(s).</b></p> <p>The tyre pressure sensor is malfunctioning on one of the tyres.</p> <ul style="list-style-type: none"> <li>► Have the tyre pressure sensor replaced at a qualified specialist workshop.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                         |
|  <p>Tyre pressure monitor faulty</p>      | <p>Additional text: <b>Failure, tyre pressure monitor for one/multiple tyre(s).</b></p> <p>The tyre pressure monitoring system on one or more tyres has failed. No signal is received by one or more tyre pressure sensors, e.g. due to a source of radio interference.</p> <ul style="list-style-type: none"> <li>► Drive on. <ul style="list-style-type: none"> <li>◁ As soon as the cause is rectified, the tyre pressure monitor is reactivated and the tyre pressure level is displayed again.</li> </ul> </li> <li>► If the tyre pressure monitoring system is not automatically activated after a long journey, have the tyre pressure monitor checked at a qualified specialist workshop.</li> </ul> |

## Display messages

### Tyres

| Display messages                                                                                                      | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Tyre pressure monitor faulty</p> | <p>The tyre pressure monitoring system is faulty.</p> <ul style="list-style-type: none"> <li>► Drive on. <ul style="list-style-type: none"> <li>◄ As soon as the cause is rectified, the tyre pressure monitor is reactivated and the tyre pressure level is displayed again.</li> </ul> </li> <li>► If the tyre pressure monitoring system is not automatically activated after a long journey, have the tyre pressure monitor checked at a qualified specialist workshop.</li> </ul> |
| Display messages                                                                                                      | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|  <p>Tyre detection not possible</p>  | <p>The wheel detection on the tyre pressure monitor is malfunctioning.</p> <ul style="list-style-type: none"> <li>► Drive the vehicle at over 30 km/h for at least ten minutes.</li> </ul> <p>If the display message continues to be displayed:</p> <ul style="list-style-type: none"> <li>► Have the tyre pressure control checked by a qualified specialist workshop.</li> </ul>                                                                                                     |
| Display messages                                                                                                      | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|  <p>Tyre detection not possible</p>  | <p>Additional text: <a href="#">Lower additional axle.</a></p> <p>The trailing axle is raised while the tyre pressure monitor performs automatic wheel detection.</p> <ul style="list-style-type: none"> <li>► Lower the trailing axle ( → Page 197).</li> </ul>                                                                                                                                                                                                                       |
| Display messages                                                                                                      | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|  <p>Tyre detection not possible</p> | <p>Additional text: <a href="#">Unlock differential.</a></p> <ul style="list-style-type: none"> <li>► The differential locks are activated. As a result, wheel detection is not possible on the locked axle.</li> <li>► Disengage the differential locks ( → Page 148).</li> </ul>                                                                                                                                                                                                     |

| Display messages                                                                                                           | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Tyre pressure monitor inoperative</p> | <p>In addition to the display message, the yellow  indicator lamp in the instrument cluster flashes for 75 seconds and then remains lit.</p> <p>The tyre pressure monitoring system is faulty.</p> <ul style="list-style-type: none"> <li>► Visit a qualified workshop.</li> </ul> |

| Display messages                                                                                   | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Flat tyre</p> | <p>Additional text: <a href="#">Change tyre..</a></p> <p>The yellow  indicator lamp in the instrument cluster also lights up.</p> <p>The tyre pressure has suddenly dropped in one or more tyres. A warning tone will also sound.</p> <p>The corresponding tyre is displayed in red in the <a href="#">Status</a> menu window of <a href="#">Tyres</a> the multimedia system.</p> <p><b>WARNING!</b></p> <p>The driving and braking characteristics are affected.</p> <p>There is a risk of accident!</p> <ul style="list-style-type: none"> <li>► Stop the vehicle without steering or braking suddenly. Pay attention to the traffic conditions.</li> <li>► If necessary, change the wheel ( → Page 240).</li> </ul> <p><b>Concrete mixer vehicle with single tyres:</b> The vehicle is designed in such a way that, in the case of a flat tyre on a rear axle, the vehicle can continue to travel a short distance at reduced speed. This allows you to find a safe place to stop the vehicle for the purposes of changing a wheel.</p> <ul style="list-style-type: none"> <li>► Check the scale of the damage to the wheel (tyre and disk wheel).</li> <li>► <b>If the defective tyre constitutes a hazard to road users:</b> Do not continue driving.</li> <li>► Replace the wheel immediately ( → Page 240).</li> </ul> <p>or</p> <ul style="list-style-type: none"> <li>► <b>If the defective tyre does not constitute a hazard to road users:</b> Continue the journey to the next opportunity to stop.</li> <li>► Drive at a maximum speed of 40 km/h when driving straight.</li> <li>► Drive at a maximum speed of 15 km/h when cornering.</li> </ul> |

## Display messages

### Tyres

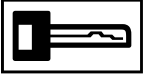
| Display messages                                                                                 | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Fixed wheel | <p>Additional text: <a href="#">Stop vehicle and release locked wheel</a>.</p> <p>The yellow  indicator lamp in the instrument cluster also lights up.</p> <p>At least one wheel is blocked. A warning tone will also sound.</p> <p>The corresponding tyre is displayed in red in the <a href="#">Status</a> menu window of <a href="#">Tyres</a> the multimedia system.</p> <p><b>WARNING!</b></p> <p>The driving characteristics are affected.</p> <p>There is a risk of accident!</p> <ul style="list-style-type: none"><li>► Stop the vehicle. Pay attention to the traffic conditions.</li><li>► Check the affected wheel position and release the blocking if necessary.</li></ul> |



| Key                                                                                                                |                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Display messages                                                                                                   | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                |
|  <p>Using the replacement key</p> | <ul style="list-style-type: none"> <li>► Use the replacement key.</li> </ul>                                                                                                                                                                                                                                                                |
| Display messages                                                                                                   | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                |
|  <p>Key not detected</p>          | <p>The key is currently not detected.<br/>The key must be in the cab.</p> <ul style="list-style-type: none"> <li>► Change the location of the key in the vehicle.</li> </ul> <p>If the key is still not detected:</p> <ul style="list-style-type: none"> <li>► Hold the key next to the start/stop button and start the vehicle.</li> </ul> |

## Display messages

### Key

| Display messages                                                                                                                                                        | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Press and hold the emergency stop button for 3 s</p>                               | <p>If you have briefly pressed the start/stop button while the vehicle is in motion, the display message will appear. In an emergency situation, you can switch off the vehicle while the vehicle is in motion by pressing and holding the start/stop button.</p> <p> <b>Danger</b></p> <p><b>Risk of accident when the vehicle is switched off while driving</b></p> <p>If you switch off the vehicle while driving in an emergency, safety-relevant functions are restricted or no longer available.</p> <p>This applies, for example, to the compressed-air system, which is still supplied from the compressed-air reservoirs but is no longer filled.</p> <p>You must then stop immediately in accordance with traffic regulations. If the reservoir pressure is too low, it is no longer possible to brake the vehicle.</p> <ul style="list-style-type: none"> <li>► Do not switch off the vehicle while driving.</li> <li>► If you want to switch the vehicle off, press and hold the start/stop button for roughly three seconds.</li> </ul> |
| Display messages                                                                                                                                                        | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|  <p>Hold the key against the contact point according to the operating instructions</p> | <p>Key detection is malfunctioning.</p> <ul style="list-style-type: none"> <li>► Change the location of the key in the vehicle.</li> <li>► Hold the key to the lining on the right-hand side next to the multifunction lever and start the vehicle ( → Page 125).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Display messages                                                                                                                 | Possible causes/consequences and ► solutions                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Incorrect key</p>                           | <p>You are using an incorrect key.</p> <ul style="list-style-type: none"> <li>► Use the correct key.</li> </ul>                                                                                                                                                                                                                                   |
|  <p>Replace key</p>                             | <p>Additional text : Please consult a workshop and have the key replaced</p> <p>The key must be replaced.</p> <ul style="list-style-type: none"> <li>► Visit a qualified workshop.</li> </ul>                                                                    |
|  <p>Replace the battery in the ignition key</p> | <p>The key's battery is flat.</p> <ul style="list-style-type: none"> <li>► Replace the battery ( → Page 42).</li> </ul>                                                                                                                                                                                                                           |
|  <p>Key not detected</p>                        | <p>The key cannot be detected and may no longer be in the vehicle.</p> <p>If the key is no longer in the vehicle and you switch the vehicle off:</p> <ul style="list-style-type: none"> <li>• The vehicle can no longer be started</li> <li>• The vehicle cannot be centrally locked</li> <li>► Ensure that the key is in the vehicle.</li> </ul> |

Confirmations

Vehicle data

Vehicle data

|                                |               |               |
|--------------------------------|---------------|---------------|
| Type:                          |               |               |
| Vehicle identification number: |               |               |
| Initial registration:          |               |               |
| Paintwork:                     |               |               |
| Licence plate                  | Licence plate | Licence plate |
| Licence plate                  | Licence plate | Licence plate |

## Confirmations

### Body manufacturer

#### Body manufacturer

#### Body manufacturer

The following work or modifications have been carried out, e.g. to the chassis, cab, body (box), or additional assemblies/add-on equipment has been fitted:

- ▶
- ▶
- ▶
- ▶
- ▶
- ▶
- ▶

**The vehicle has been subjected to a quality control check and/or final inspection with regard to the work or modifications carried out on the vehicle. No faults or malfunctions were found.**

Please tick:

☐ Work and execution were carried out in accordance with the Mercedes-Benz Trucks body/mounting directives.

☐ The certificate of non-objection was obtained from Daimler Truck AG, at the department responsible for body/equipment approval (see Mercedes-Benz Trucks body/equipment mounting directives).

☐ Operating instructions and maintenance instructions for the work, modifications or the assemblies fitted have been placed in the vehicle.

City/town

▶

Date

▶

Mileage (km)

▶

Operating hours (OH)

▶

Signature

▶

Signature/Stamp of the body manufacturer

## Confirmations

### Body manufacturer

#### Body manufacturer

The following work or modifications have been carried out, e.g. to the chassis, cab, body (box), or additional assemblies/add-on equipment has been fitted:

- ▶
- ▶
- ▶
- ▶
- ▶
- ▶
- ▶

**The vehicle has been subjected to a quality control check and/or final inspection with regard to the work or modifications carried out on the vehicle. No faults or malfunctions were found.**

Please tick:

☐ Work and execution were carried out in accordance with the Mercedes-Benz Trucks body/mounting directives.

☐ The certificate of non-objection was obtained from Daimler Truck AG, at the department responsible for body/equipment approval (see Mercedes-Benz Trucks body/equipment mounting directives).

☐ Operating instructions and maintenance instructions for the work, modifications or the assemblies fitted have been placed in the vehicle.

City/town

▶

Date

▶

Mileage (km)

▶

Operating hours (OH)

▶

Signature

▶

Signature/Stamp of the body manufacturer

## Confirmations

### Body manufacturer

#### Body manufacturer

The following work or modifications have been carried out, e.g. to the chassis, cab, body (box), or additional assemblies/add-on equipment has been fitted:

- ▶
- ▶
- ▶
- ▶
- ▶
- ▶
- ▶

**The vehicle has been subjected to a quality control check and/or final inspection with regard to the work or modifications carried out on the vehicle. No faults or malfunctions were found.**

Please tick:

☐ Work and execution were carried out in accordance with the Mercedes-Benz Trucks body/mounting directives.

☐ The certificate of non-objection was obtained from Daimler Truck AG, at the department responsible for body/equipment approval (see Mercedes-Benz Trucks body/equipment mounting directives).

☐ Operating instructions and maintenance instructions for the work, modifications or the assemblies fitted have been placed in the vehicle.

City/town

- ▶

Date

- ▶

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▶

Date

▶

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▶

Operating hours (OH)

▶

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▶

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